ST7796

320RGB x 480 dot 262K Color with Frame Memory Single-Chip TFT Controller/Driver

Data sheet

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Sitronix Technology Corporation

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1 GENERAL DESCRIPTION

The ST7796 is a single-chip controller/driver for 262K-color, graphic type TFT-LCD. It consists of 960 source lines and 480 gate lines driving circuits. The ST7796 is capable of connecting directly to an external microprocessor, and accepts 8-bit/9-bit/16-bit/18-bit parallel interface, SPI, and the ST7796 also provides MDDI and MIPI interface. Display data can be stored in the on-chip display data RAM of 320x480x18 bits. It can perform display data RAM read-/write-operation with no external clock to minimize power consumption. In addition, because of the integrated power supply circuit necessary to drive liquid crystal; it is possible to make a display system with fewest components.

2 FEATURES

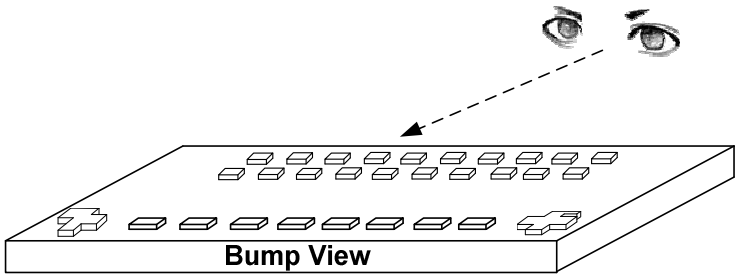
- Single chip TFT-LCD Controller/Driver with On-chip Frame Memory
- Display Resolution: 320*RGB (H) *480(V)
- Frame Memory Size: 320 x 480 x 18-bit = 2,764,800 bits
- LCD Driver Output Circuits
  - Source Outputs: 320 RGB Channels
  - Gate Outputs: 480 Channels
  - Common Electrode Output
- Display Colors (Color Mode)
  - Full Color: 16M, RGB=(888) max., Idle Mode Off
  - Color Reduce: 8-color, RGB=(111), Idle Mode On
- Programmable Pixel Color Format (Color Depth) for Various Display Data Input Format
  - 16-bit/pixel: RGB=(565) 65K color
  - 18-bit/pixel: RGB=(666) 262K color
  - 24-bit/pixel: RGB=(888) 16M color
- Interface
  - Parallel 8080-series MCU Interface (8-bit, 9-bit, 16-bit, and 18-bit)
  - 16/18 RGB Interface (VSYNCX, HSYNCX, DOTCLK, ENABLE, DB[17:0])
  - Serial Peripheral Interface (SPI Interface)
  - MDDI
  - MIPI
- Display Features
  - Partial Display Function
  - 8-color Display Function
  - Vertical Scroll Function
- Support LC Type Option
  - MVA LC Type
  - Transflective LC Type
  - Transmissive LC Type
- On Chip Build-In Circuits
  - DC/DC Converter
  - Non-Volatile (NV) Memory to Store Initial Register Setting and Factory Default Value (Module ID, Module Version, etc)
  - Internal Oscillator for Display Clock Generation
  - Timing Controller
- Build-In NV Memory for LCD Initial Register Setting
  - 8-bit for ID1

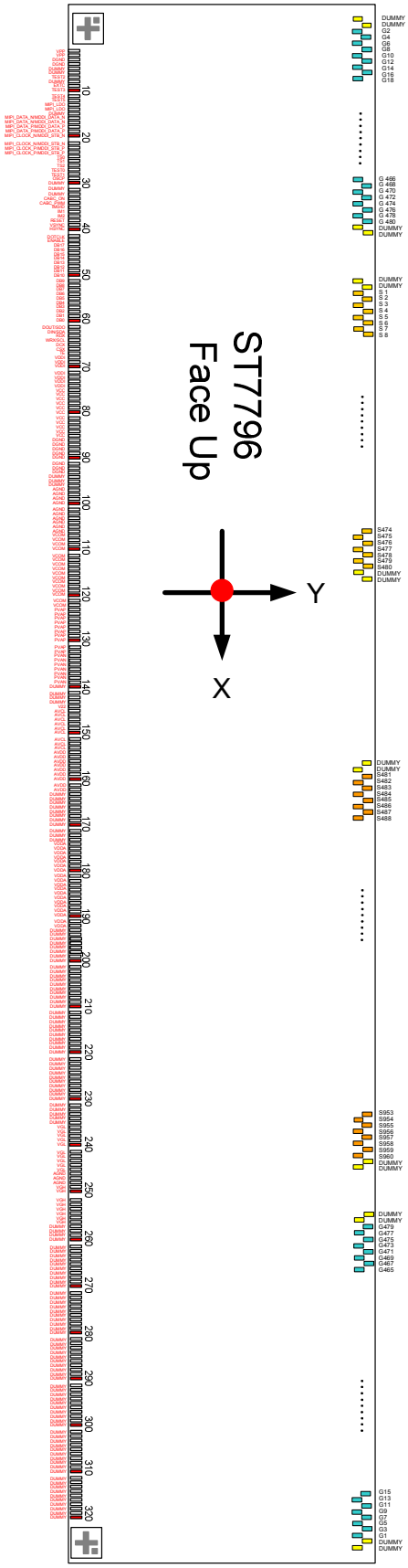
  - 8-bit for ID2
  - 8-bit for ID3
  - 6-bit for flicker adjustment
- Driving Algorithm
  - Dot Inversion
  - Column Inversion
- Wide Supply Voltage Range
  - I/O Voltage (VDDI to DGND): 1.65V ~ 3.3V (VDDI≦VDD)
  - Voltage for Digital Circuit (VDD to DGND): 2.5V ~ 3.3V
  - Voltage for Analog Circuit (VDDA to AGND): 2.5V ~ 3.3V
- On-Chip Power System
  - VCOM Level: AGND
- Optimized layout for COG Assembly
- Operate temperature range: −30ºC to +85 ºC
- Lower Power Consumption

3 PAD ARRANGEMENT

3.1.. Output Bump Dimension

Au bump height	12μm
Au bump size	15μmx100μm Gate : G1~G480 Source : S1~S960
	50μmx80μm (Pad1 to Pad320)

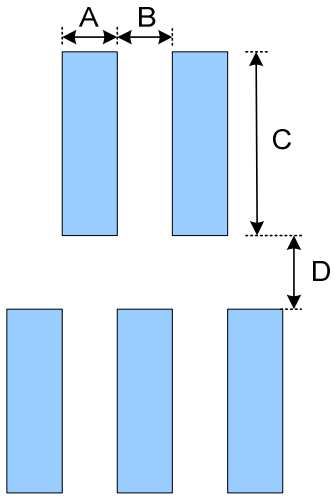




3.2.. Bump Dimension

- **Output Pads**

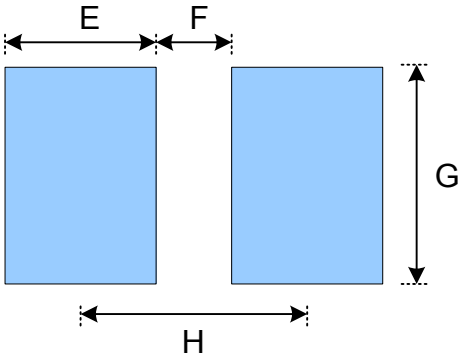
Pad No.



Symbol	Item	Size
A	Bump Width	15 um
B	Bump Gap 1 (Horizontal)	15 um
C	Bump Height	100um
D	Bump Gap 2 (Vertical)	25um

- **Input Pads**

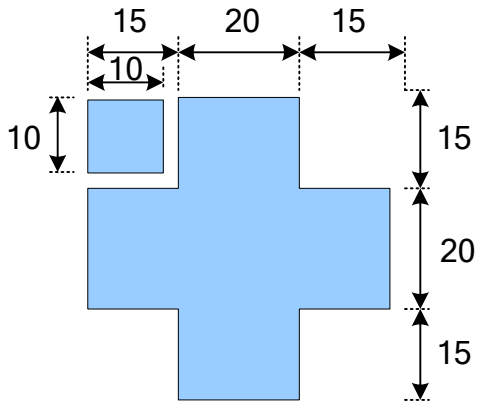
Pad No.1~320



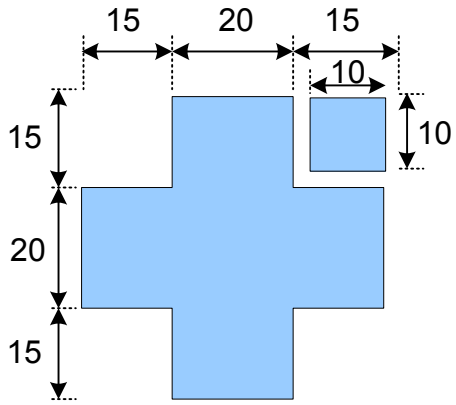
Symbol	Item	Size
E	Bump Width	50 um
F	Bump Gap	20 um
G	Bump Height	80um
H	Bump Pitch	70 um

3.3.. Alignment Mark Dimension

- **Alignment Mark Left: L(X,Y)=(-11300, -270)**



- **Alignment Mark Right: R(X,Y)= (11300, -270)**



3.4.. Chip Information

Chip size	22850um x 820um
Chip thickness	280um
Pad Location	Pad center
Coordinate Origin	Chip center

4 PAD CENTER COORDINATES

PAD No.	PIN Name	X	Y	PAD No.	PIN Name	X	Y
1	VPP	-11165	-279	34	CABC_PWN	-8855	-279
2	VPP	-11095	-279	35	IM0/ID	-8785	-279
3	DGND	-11025	-279	36	IM1	-8715	-279
4	DGND	-10955	-279	37	IM2	-8645	-279
5	DUMMY	-10885	-279	38	RESET	-8575	-279
6	DUMMY	-10815	-279	39	VSYNC	-8505	-279
7	TEST2	-10745	-279	40	HSYNC	-8435	-279
8	DUMMY	-10675	-279	41	DOTCLK	-8365	-279
9	EXTC	-10605	-279	42	ENABLE	-8295	-279
10	TEST3	-10535	-279	43	DB17	-8225	-279
11	TEST4	-10465	-279	44	DB16	-8155	-279
12	TEST5	-10395	-279	45	DB15	-8085	-279
13	MIPI_LDO	-10325	-279	46	DB14	-8015	-279
14	MIPI_LDO	-10255	-279	47	DB13	-7945	-279
15	DUMMY	-10185	-279	48	DB12	-7875	-279
16	MIPI_DATA_N	-10115	-279	49	DB11	-7805	-279
17	MIPI_DATA_N	-10045	-279	50	DB10	-7735	-279
18	MIPI_DATA_P	-9975	-279	51	DB9	-7665	-279
19	MIPI_DATA_P	-9905	-279	52	DB8	-7595	-279
20	MIPI_CLOCK_N	-9835	-279	53	DB7	-7525	-279
21	MIPI_CLOCK_N	-9765	-279	54	DB6	-7455	-279
22	MIPI_CLOCK_P	-9695	-279	55	DB5	-7385	-279
23	MIPI_CLOCK_P	-9625	-279	56	DB4	-7315	-279
24	TS0	-9555	-279	57	DB3	-7245	-279
25	TS1	-9485	-279	58	DB2	-7175	-279
26	TS2	-9415	-279	59	DB1	-7105	-279
27	TEST0	-9345	-279	60	DB0	-7035	-279
28	TEST1	-9275	-279	61	DOUT	-6965	-279
29	OSCP	-9205	-279	62	DIN/SDA	-6895	-279
30	DUMMY	-9135	-279	63	RDX	-6825	-279
31	DUMMY	-9065	-279	64	WRX/SCL	-6755	-279
32	DUMMY	-8995	-279	65	DCX	-6685	-279
33	CABC_ON	-8925	-279	66	CSX	-6615	-279

PAD No.	PIN Name	X	Y	PAD No.	PIN Name	X	Y
67	TE	-6545	-279	101	AGND	-4165	-279
68	VDDI	-6475	-279	102	AGND	-4095	-279
69	VDDI	-6405	-279	103	AGND	-4025	-279
70	VDDI	-6335	-279	104	AGND	-3955	-279
71	VDDI	-6265	-279	105	AGND	-3885	-279
72	VDDI	-6195	-279	106	AGND	-3815	-279
73	VDDI	-6125	-279	107	VCOM	-3745	-279
74	VDDI	-6055	-279	108	VCOM	-3675	-279
75	VCC	-5985	-279	109	VCOM	-3605	-279
76	VCC	-5915	-279	110	VCOM	-3535	-279
77	VCC	-5845	-279	111	VCOM	-3465	-279
78	VCC	-5775	-279	112	VCOM	-3395	-279
79	VCC	-5705	-279	113	VCOM	-3325	-279
80	VCC	-5635	-279	114	VCOM	-3255	-279
81	VCC	-5565	-279	115	VCOM	-3185	-279
82	VCC	-5495	-279	116	VCOM	-3115	-279
83	VCC	-5425	-279	117	VCOM	-3045	-279
84	VCC	-5355	-279	118	VCOM	-2975	-279
85	VCC	-5285	-279	119	VCOM	-2905	-279
86	DGND	-5215	-279	120	VCOM	-2835	-279
87	DGND	-5145	-279	121	VCOM	-2765	-279
88	DGND	-5075	-279	122	VCOM	-2695	-279
89	DGND	-5005	-279	123	PVAP	-2625	-279
90	DGND	-4935	-279	124	PVAP	-2555	-279
91	DGND	-4865	-279	125	PVAP	-2485	-279
92	DGND	-4795	-279	126	PVAP	-2415	-279
93	DGND	-4725	-279	127	PVAP	-2345	-279
94	DUMMY	-4655	-279	128	PVAP	-2275	-279
95	DUMMY	-4585	-279	129	PVAP	-2205	-279
96	DUMMY	-4515	-279	130	PVAP	-2135	-279
97	AGND	-4445	-279	131	PVAP	-2065	-279
98	AGND	-4375	-279	132	PVAP	-1995	-279
99	AGND	-4305	-279	133	PVAN	-1925	-279
100	AGND	-4235	-279	134	PVAN	-1855	-279

PAD No.	PIN Name	X	Y	PAD No.	PIN Name	X	Y
135	PVAN	-1785	-279	169	DUMMY	595	-279
136	PVAN	-1715	-279	170	DUMMY	665	-279
137	PVAN	-1645	-279	171	DUMMY	735	-279
138	PVAN	-1575	-279	172	DUMMY	805	-279
139	PVAN	-1505	-279	173	DUMMY	875	-279
140	DUMMY	-1435	-279	174	VDDA	945	-279
141	DUMMY	-1365	-279	175	VDDA	1015	-279
142	DUMMY	-1295	-279	176	VDDA	1085	-279
143	DUMMY	-1225	-279	177	VDDA	1155	-279
144	V22	-1155	-279	178	VDDA	1225	-279
145	AVCL	-1085	-279	179	VDDA	1295	-279
146	AVCL	-1015	-279	180	VDDA	1365	-279
147	AVCL	-945	-279	181	VDDA	1435	-279
148	AVCL	-875	-279	182	VDDA	1505	-279
149	AVCL	-805	-279	183	VDDA	1575	-279
150	AVCL	-735	-279	184	VDDA	1645	-279
151	AVCL	-665	-279	185	VDDA	1715	-279
152	AVCL	-595	-279	186	VDDA	1785	-279
153	AVCL	-525	-279	187	VDDA	1855	-279
154	AVDD	-455	-279	188	VDDA	1925	-279
155	AVDD	-385	-279	189	VDDA	1995	-279
156	AVDD	-315	-279	190	VDDA	2065	-279
157	AVDD	-245	-279	191	VDDA	2135	-279
158	AVDD	-175	-279	192	VDDA	2205	-279
159	AVDD	-105	-279	193	DUMMY	2275	-279
160	AVDD	-35	-279	194	DUMMY	2345	-279
161	AVDD	35	-279	195	DUMMY	2415	-279
162	AVDD	105	-279	196	DUMMY	2485	-279
163	DUMMY	175	-279	197	DUMMY	2555	-279
164	DUMMY	245	-279	198	DUMMY	2625	-279
165	DUMMY	315	-279	199	DUMMY	2695	-279
166	DUMMY	385	-279	200	DUMMY	2765	-279
167	DUMMY	455	-279	201	DUMMY	2835	-279
168	DUMMY	525	-279	202	DUMMY	2905	-279

PAD No.	PIN Name	X	Y	PAD No.	PIN Name	X	Y
203	DUMMY	2975	-279	237	VGL	5355	-279
204	DUMMY	3045	-279	238	VGL	5425	-279
205	DUMMY	3115	-279	239	VGL	5495	-279
206	DUMMY	3185	-279	240	VGL	5565	-279
207	DUMMY	3255	-279	241	VGL	5635	-279
208	DUMMY	3325	-279	242	VGL	5705	-279
209	DUMMY	3395	-279	243	VGL	5775	-279
210	DUMMY	3465	-279	244	VGL	5845	-279
211	DUMMY	3535	-279	245	VGL	5915	-279
212	DUMMY	3605	-279	246	AGND	5985	-279
213	DUMMY	3675	-279	247	AGND	6055	-279
214	DUMMY	3745	-279	248	AGND	6125	-279
215	DUMMY	3815	-279	249	VGH	6195	-279
216	DUMMY	3885	-279	250	VGH	6265	-279
217	DUMMY	3955	-279	251	VGH	6335	-279
218	DUMMY	4025	-279	252	VGH	6405	-279
219	DUMMY	4095	-279	253	VGH	6475	-279
220	DUMMY	4165	-279	254	VGH	6545	-279
221	DUMMY	4235	-279	255	VGH	6615	-279
222	DUMMY	4305	-279	256	VGH	6685	-279
223	DUMMY	4375	-279	257	DUMMY	6755	-279
224	DUMMY	4445	-279	258	DUMMY	6825	-279
225	DUMMY	4515	-279	259	DUMMY	6895	-279
226	DUMMY	4585	-279	260	DUMMY	6965	-279
227	DUMMY	4655	-279	261	DUMMY	7035	-279
228	DUMMY	4725	-279	262	DUMMY	7105	-279
229	DUMMY	4795	-279	263	DUMMY	7175	-279
230	DUMMY	4865	-279	264	DUMMY	7245	-279
231	DUMMY	4935	-279	265	DUMMY	7315	-279
232	DUMMY	5005	-279	266	DUMMY	7385	-279
233	DUMMY	5075	-279	267	DUMMY	7455	-279
234	DUMMY	5145	-279	268	DUMMY	7525	-279
235	DUMMY	5215	-279	269	DUMMY	7595	-279
236	VGL	5285	-279	270	DUMMY	7665	-279

PAD No.	PIN Name	X	Y	PAD No.	PIN Name	X	Y
271	DUMMY	7735	-279	305	DUMMY	10115	-279
272	DUMMY	7805	-279	306	DUMMY	10185	-279
273	DUMMY	7875	-279	307	DUMMY	10255	-279
274	DUMMY	7945	-279	308	DUMMY	10325	-279
275	DUMMY	8015	-279	309	DUMMY	10395	-279
276	DUMMY	8085	-279	310	DUMMY	10465	-279
277	DUMMY	8155	-279	311	DUMMY	10535	-279
278	DUMMY	8225	-279	312	DUMMY	10605	-279
279	DUMMY	8295	-279	313	DUMMY	10675	-279
280	DUMMY	8365	-279	314	DUMMY	10745	-279
281	DUMMY	8435	-279	315	DUMMY	10815	-279
282	DUMMY	8505	-279	316	DUMMY	10885	-279
283	DUMMY	8575	-279	317	DUMMY	10955	-279
284	DUMMY	8645	-279	318	DUMMY	11025	-279
285	DUMMY	8715	-279	319	DUMMY	11095	-279
286	DUMMY	8785	-279	320	DUMMY	11165	-279
287	DUMMY	8855	-279	321	ALIGN_R	11300	-270
288	DUMMY	8925	-279	322	DUMMY	11205	164
289	DUMMY	8995	-279	323	DUMMY	11190	289
290	DUMMY	9065	-279	324	G1	11175	164
291	DUMMY	9135	-279	325	G3	11160	289
292	DUMMY	9205	-279	326	G5	11145	164
293	DUMMY	9275	-279	327	G7	11130	289
294	DUMMY	9345	-279	328	G9	11115	164
295	DUMMY	9415	-279	329	G11	11100	289
296	DUMMY	9485	-279	330	G13	11085	164
297	DUMMY	9555	-279	331	G15	11070	289
298	DUMMY	9625	-279	332	G17	11055	164
299	DUMMY	9695	-279	333	G19	11040	289
300	DUMMY	9765	-279	334	G21	11025	164
301	DUMMY	9835	-279	335	G23	11010	289
302	DUMMY	9905	-279	336	G25	10995	164
303	DUMMY	9975	-279	337	G27	10980	289
304	DUMMY	10045	-279	338	G29	10965	164

PAD No.	PIN Name	X	Y	PAD No.	PIN Name	X	Y
339	G31	10950	289	373	G99	10440	289
340	G33	10935	164	374	G101	10425	164
341	G35	10920	289	375	G103	10410	289
342	G37	10905	164	376	G105	10395	164
343	G39	10890	289	377	G107	10380	289
344	G41	10875	164	378	G109	10365	164
345	G43	10860	289	379	G111	10350	289
346	G45	10845	164	380	G113	10335	164
347	G47	10830	289	381	G115	10320	289
348	G49	10815	164	382	G117	10305	164
349	G51	10800	289	383	G119	10290	289
350	G53	10785	164	384	G121	10275	164
351	G55	10770	289	385	G123	10260	289
352	G57	10755	164	386	G125	10245	164
353	G59	10740	289	387	G127	10230	289
354	G61	10725	164	388	G129	10215	164
355	G63	10710	289	389	G131	10200	289
356	G65	10695	164	390	G133	10185	164
357	G67	10680	289	391	G135	10170	289
358	G69	10665	164	392	G137	10155	164
359	G71	10650	289	393	G139	10140	289
360	G73	10635	164	394	G141	10125	164
361	G75	10620	289	395	G143	10110	289
362	G77	10605	164	396	G145	10095	164
363	G79	10590	289	397	G147	10080	289
364	G81	10575	164	398	G149	10065	164
365	G83	10560	289	399	G151	10050	289
366	G85	10545	164	400	G153	10035	164
367	G87	10530	289	401	G155	10020	289
368	G89	10515	164	402	G157	10005	164
369	G91	10500	289	403	G159	9990	289
370	G93	10485	164	404	G161	9975	164
371	G95	10470	289	405	G163	9960	289
372	G97	10455	164	406	G165	9945	164

PAD No.	PIN Name	X	Y	PAD No.	PIN Name	X	Y
407	G167	9930	289	441	G235	9420	289
408	G169	9915	164	442	G237	9405	164
409	G171	9900	289	443	G239	9390	289
410	G173	9885	164	444	G241	9375	164
411	G175	9870	289	445	G243	9360	289
412	G177	9855	164	446	G245	9345	164
413	G179	9840	289	447	G247	9330	289
414	G181	9825	164	448	G249	9315	164
415	G183	9810	289	449	G251	9300	289
416	G185	9795	164	450	G253	9285	164
417	G187	9780	289	451	G255	9270	289
418	G189	9765	164	452	G257	9255	164
419	G191	9750	289	453	G259	9240	289
420	G193	9735	164	454	G261	9225	164
421	G195	9720	289	455	G263	9210	289
422	G197	9705	164	456	G265	9195	164
423	G199	9690	289	457	G267	9180	289
424	G201	9675	164	458	G269	9165	164
425	G203	9660	289	459	G271	9150	289
426	G205	9645	164	460	G273	9135	164
427	G207	9630	289	461	G275	9120	289
428	G209	9615	164	462	G277	9105	164
429	G211	9600	289	463	G279	9090	289
430	G213	9585	164	464	G281	9075	164
431	G215	9570	289	465	G283	9060	289
432	G217	9555	164	466	G285	9045	164
433	G219	9540	289	467	G287	9030	289
434	G221	9525	164	468	G289	9015	164
435	G223	9510	289	469	G291	9000	289
436	G225	9495	164	470	G293	8985	164
437	G227	9480	289	471	G295	8970	289
438	G229	9465	164	472	G297	8955	164
439	G231	9450	289	473	G299	8940	289
440	G233	9435	164	474	G301	8925	164

PAD No.	PIN Name	X	Y	PAD No.	PIN Name	X	Y
475	G303	8910	289	509	G371	8400	289
476	G305	8895	164	510	G373	8385	164
477	G307	8880	289	511	G375	8370	289
478	G309	8865	164	512	G377	8355	164
479	G311	8850	289	513	G379	8340	289
480	G313	8835	164	514	G381	8325	164
481	G315	8820	289	515	G383	8310	289
482	G317	8805	164	516	G385	8295	164
483	G319	8790	289	517	G387	8280	289
484	G321	8775	164	518	G389	8265	164
485	G323	8760	289	519	G391	8250	289
486	G325	8745	164	520	G393	8235	164
487	G327	8730	289	521	G395	8220	289
488	G329	8715	164	522	G397	8205	164
489	G331	8700	289	523	G399	8190	289
490	G333	8685	164	524	G401	8175	164
491	G335	8670	289	525	G403	8160	289
492	G337	8655	164	526	G405	8145	164
493	G339	8640	289	527	G407	8130	289
494	G341	8625	164	528	G409	8115	164
495	G343	8610	289	529	G411	8100	289
496	G345	8595	164	530	G413	8085	164
497	G347	8580	289	531	G415	8070	289
498	G349	8565	164	532	G417	8055	164
499	G351	8550	289	533	G419	8040	289
500	G353	8535	164	534	G421	8025	164
501	G355	8520	289	535	G423	8010	289
502	G357	8505	164	536	G425	7995	164
503	G359	8490	289	537	G427	7980	289
504	G361	8475	164	538	G429	7965	164
505	G363	8460	289	539	G431	7950	289
506	G365	8445	164	540	G433	7935	164
507	G367	8430	289	541	G435	7920	289
508	G369	8415	164	542	G437	7905	164

PAD No.	PIN Name	X	Y	PAD No.	PIN Name	X	Y
543	G439	7890	289	577	S951	7230	289
544	G441	7875	164	578	S950	7215	164
545	G443	7860	289	579	S949	7200	289
546	G445	7845	164	580	S948	7185	164
547	G447	7830	289	581	S947	7170	289
548	G449	7815	164	582	S946	7155	164
549	G451	7800	289	583	S945	7140	289
550	G453	7785	164	584	S944	7125	164
551	G455	7770	289	585	S943	7110	289
552	G457	7755	164	586	S942	7095	164
553	G459	7740	289	587	S941	7080	289
554	G461	7725	164	588	S940	7065	164
555	G463	7710	289	589	S939	7050	289
556	G465	7695	164	590	S938	7035	164
557	G467	7680	289	591	S937	7020	289
558	G469	7665	164	592	S936	7005	164
559	G471	7650	289	593	S935	6990	289
560	G473	7635	164	594	S934	6975	164
561	G475	7620	289	595	S933	6960	289
562	G477	7605	164	596	S932	6945	164
563	G479	7590	289	597	S931	6930	289
564	DUMMY	7575	164	598	S930	6915	164
565	DUMMY	7560	289	599	S929	6900	289
566	DUMMY	7395	164	600	S928	6885	164
567	DUMMY	7380	289	601	S927	6870	289
568	S960	7365	164	602	S926	6855	164
569	S959	7350	289	603	S925	6840	289
570	S958	7335	164	604	S924	6825	164
571	S957	7320	289	605	S923	6810	289
572	S956	7305	164	606	S922	6795	164
573	S955	7290	289	607	S921	6780	289
574	S954	7275	164	608	S920	6765	164
575	S953	7260	289	609	S919	6750	289
576	S952	7245	164	610	S918	6735	164

PAD No.	PIN Name	X	Y	PAD No.	PIN Name	X	Y
611	S917	6720	289	645	S883	6210	289
612	S916	6705	164	646	S882	6195	164
613	S915	6690	289	647	S881	6180	289
614	S914	6675	164	648	S880	6165	164
615	S913	6660	289	649	S879	6150	289
616	S912	6645	164	650	S878	6135	164
617	S911	6630	289	651	S877	6120	289
618	S910	6615	164	652	S876	6105	164
619	S909	6600	289	653	S875	6090	289
620	S908	6585	164	654	S874	6075	164
621	S907	6570	289	655	S873	6060	289
622	S906	6555	164	656	S872	6045	164
623	S905	6540	289	657	S871	6030	289
624	S904	6525	164	658	S870	6015	164
625	S903	6510	289	659	S869	6000	289
626	S902	6495	164	660	S868	5985	164
627	S901	6480	289	661	S867	5970	289
628	S900	6465	164	662	S866	5955	164
629	S899	6450	289	663	S865	5940	289
630	S898	6435	164	664	S864	5925	164
631	S897	6420	289	665	S863	5910	289
632	S896	6405	164	666	S862	5895	164
633	S895	6390	289	667	S861	5880	289
634	S894	6375	164	668	S860	5865	164
635	S893	6360	289	669	S859	5850	289
636	S892	6345	164	670	S858	5835	164
637	S891	6330	289	671	S857	5820	289
638	S890	6315	164	672	S856	5805	164
639	S889	6300	289	673	S855	5790	289
640	S888	6285	164	674	S854	5775	164
641	S887	6270	289	675	S853	5760	289
642	S886	6255	164	676	S852	5745	164
643	S885	6240	289	677	S851	5730	289
644	S884	6225	164	678	S850	5715	164

PAD No.	PIN Name	X	Y	PAD No.	PIN Name	X	Y
679	S849	5700	289	713	S815	5190	289
680	S848	5685	164	714	S814	5175	164
681	S847	5670	289	715	S813	5160	289
682	S846	5655	164	716	S812	5145	164
683	S845	5640	289	717	S811	5130	289
684	S844	5625	164	718	S810	5115	164
685	S843	5610	289	719	S809	5100	289
686	S842	5595	164	720	S808	5085	164
687	S841	5580	289	721	S807	5070	289
688	S840	5565	164	722	S806	5055	164
689	S839	5550	289	723	S805	5040	289
690	S838	5535	164	724	S804	5025	164
691	S837	5520	289	725	S803	5010	289
692	S836	5505	164	726	S802	4995	164
693	S835	5490	289	727	S801	4980	289
694	S834	5475	164	728	S800	4965	164
695	S833	5460	289	729	S799	4950	289
696	S832	5445	164	730	S798	4935	164
697	S831	5430	289	731	S797	4920	289
698	S830	5415	164	732	S796	4905	164
699	S829	5400	289	733	S795	4890	289
700	S828	5385	164	734	S794	4875	164
701	S827	5370	289	735	S793	4860	289
702	S826	5355	164	736	S792	4845	164
703	S825	5340	289	737	S791	4830	289
704	S824	5325	164	738	S790	4815	164
705	S823	5310	289	739	S789	4800	289
706	S822	5295	164	740	S788	4785	164
707	S821	5280	289	741	S787	4770	289
708	S820	5265	164	742	S786	4755	164
709	S819	5250	289	743	S785	4740	289
710	S818	5235	164	744	S784	4725	164
711	S817	5220	289	745	S783	4710	289
712	S816	5205	164	746	S782	4695	164

PAD No.	PIN Name	X	Y	PAD No.	PIN Name	X	Y
747	S781	4680	289	781	S747	4170	289
748	S780	4665	164	782	S746	4155	164
749	S779	4650	289	783	S745	4140	289
750	S778	4635	164	784	S744	4125	164
751	S777	4620	289	785	S743	4110	289
752	S776	4605	164	786	S742	4095	164
753	S775	4590	289	787	S741	4080	289
754	S774	4575	164	788	S740	4065	164
755	S773	4560	289	789	S739	4050	289
756	S772	4545	164	790	S738	4035	164
757	S771	4530	289	791	S737	4020	289
758	S770	4515	164	792	S736	4005	164
759	S769	4500	289	793	S735	3990	289
760	S768	4485	164	794	S734	3975	164
761	S767	4470	289	795	S733	3960	289
762	S766	4455	164	796	S732	3945	164
763	S765	4440	289	797	S731	3930	289
764	S764	4425	164	798	S730	3915	164
765	S763	4410	289	799	S729	3900	289
766	S762	4395	164	800	S728	3885	164
767	S761	4380	289	801	S727	3870	289
768	S760	4365	164	802	S726	3855	164
769	S759	4350	289	803	S725	3840	289
770	S758	4335	164	804	S724	3825	164
771	S757	4320	289	805	S723	3810	289
772	S756	4305	164	806	S722	3795	164
773	S755	4290	289	807	S721	3780	289
774	S754	4275	164	808	S720	3765	164
775	S753	4260	289	809	S719	3750	289
776	S752	4245	164	810	S718	3735	164
777	S751	4230	289	811	S717	3720	289
778	S750	4215	164	812	S716	3705	164
779	S749	4200	289	813	S715	3690	289
780	S748	4185	164	814	S714	3675	164

PAD No.	PIN Name	X	Y	PAD No.	PIN Name	X	Y
815	S713	3660	289	849	S679	3150	289
816	S712	3645	164	850	S678	3135	164
817	S711	3630	289	851	S677	3120	289
818	S710	3615	164	852	S676	3105	164
819	S709	3600	289	853	S675	3090	289
820	S708	3585	164	854	S674	3075	164
821	S707	3570	289	855	S673	3060	289
822	S706	3555	164	856	S672	3045	164
823	S705	3540	289	857	S671	3030	289
824	S704	3525	164	858	S670	3015	164
825	S703	3510	289	859	S669	3000	289
826	S702	3495	164	860	S668	2985	164
827	S701	3480	289	861	S667	2970	289
828	S700	3465	164	862	S666	2955	164
829	S699	3450	289	863	S665	2940	289
830	S698	3435	164	864	S664	2925	164
831	S697	3420	289	865	S663	2910	289
832	S696	3405	164	866	S662	2895	164
833	S695	3390	289	867	S661	2880	289
834	S694	3375	164	868	S660	2865	164
835	S693	3360	289	869	S659	2850	289
836	S692	3345	164	870	S658	2835	164
837	S691	3330	289	871	S657	2820	289
838	S690	3315	164	872	S656	2805	164
839	S689	3300	289	873	S655	2790	289
840	S688	3285	164	874	S654	2775	164
841	S687	3270	289	875	S653	2760	289
842	S686	3255	164	876	S652	2745	164
843	S685	3240	289	877	S651	2730	289
844	S684	3225	164	878	S650	2715	164
845	S683	3210	289	879	S649	2700	289
846	S682	3195	164	880	S648	2685	164
847	S681	3180	289	881	S647	2670	289
848	S680	3165	164	882	S646	2655	164

PAD No.	PIN Name	X	Y	PAD No.	PIN Name	X	Y
883	S645	2640	289	917	S611	2130	289
884	S644	2625	164	918	S610	2115	164
885	S643	2610	289	919	S609	2100	289
886	S642	2595	164	920	S608	2085	164
887	S641	2580	289	921	S607	2070	289
888	S640	2565	164	922	S606	2055	164
889	S639	2550	289	923	S605	2040	289
890	S638	2535	164	924	S604	2025	164
891	S637	2520	289	925	S603	2010	289
892	S636	2505	164	926	S602	1995	164
893	S635	2490	289	927	S601	1980	289
894	S634	2475	164	928	S600	1965	164
895	S633	2460	289	929	S599	1950	289
896	S632	2445	164	930	S598	1935	164
897	S631	2430	289	931	S597	1920	289
898	S630	2415	164	932	S596	1905	164
899	S629	2400	289	933	S595	1890	289
900	S628	2385	164	934	S594	1875	164
901	S627	2370	289	935	S593	1860	289
902	S626	2355	164	936	S592	1845	164
903	S625	2340	289	937	S591	1830	289
904	S624	2325	164	938	S590	1815	164
905	S623	2310	289	939	S589	1800	289
906	S622	2295	164	940	S588	1785	164
907	S621	2280	289	941	S587	1770	289
908	S620	2265	164	942	S586	1755	164
909	S619	2250	289	943	S585	1740	289
910	S618	2235	164	944	S584	1725	164
911	S617	2220	289	945	S583	1710	289
912	S616	2205	164	946	S582	1695	164
913	S615	2190	289	947	S581	1680	289
914	S614	2175	164	948	S580	1665	164
915	S613	2160	289	949	S579	1650	289
916	S612	2145	164	950	S578	1635	164

PAD No.	PIN Name	X	Y	PAD No.	PIN Name	X	Y
951	S577	1620	289	985	S543	1110	289
952	S576	1605	164	986	S542	1095	164
953	S575	1590	289	987	S541	1080	289
954	S574	1575	164	988	S540	1065	164
955	S573	1560	289	989	S539	1050	289
956	S572	1545	164	990	S538	1035	164
957	S571	1530	289	991	S537	1020	289
958	S570	1515	164	992	S536	1005	164
959	S569	1500	289	993	S535	990	289
960	S568	1485	164	994	S534	975	164
961	S567	1470	289	995	S533	960	289
962	S566	1455	164	996	S532	945	164
963	S565	1440	289	997	S531	930	289
964	S564	1425	164	998	S530	915	164
965	S563	1410	289	999	S529	900	289
966	S562	1395	164	1000	S528	885	164
967	S561	1380	289	1001	S527	870	289
968	S560	1365	164	1002	S526	855	164
969	S559	1350	289	1003	S525	840	289
970	S558	1335	164	1004	S524	825	164
971	S557	1320	289	1005	S523	810	289
972	S556	1305	164	1006	S522	795	164
973	S555	1290	289	1007	S521	780	289
974	S554	1275	164	1008	S520	765	164
975	S553	1260	289	1009	S519	750	289
976	S552	1245	164	1010	S518	735	164
977	S551	1230	289	1011	S517	720	289
978	S550	1215	164	1012	S516	705	164
979	S549	1200	289	1013	S515	690	289
980	S548	1185	164	1014	S514	675	164
981	S547	1170	289	1015	S513	660	289
982	S546	1155	164	1016	S512	645	164
983	S545	1140	289	1017	S511	630	289
984	S544	1125	164	1018	S510	615	164

PAD No.	PIN Name	X	Y	PAD No.	PIN Name	X	Y
1019	S509	600	289	1053	S479	-195	164
1020	S508	585	164	1054	S478	-210	289
1021	S507	570	289	1055	S477	-225	164
1022	S506	555	164	1056	S476	-240	289
1023	S505	540	289	1057	S475	-255	164
1024	S504	525	164	1058	S474	-270	289
1025	S503	510	289	1059	S473	-285	164
1026	S502	495	164	1060	S472	-300	289
1027	S501	480	289	1061	S471	-315	164
1028	S500	465	164	1062	S470	-330	289
1029	S499	450	289	1063	S469	-345	164
1030	S498	435	164	1064	S468	-360	289
1031	S497	420	289	1065	S467	-375	164
1032	S496	405	164	1066	S466	-390	289
1033	S495	390	289	1067	S465	-405	164
1034	S494	375	164	1068	S464	-420	289
1035	S493	360	289	1069	S463	-435	164
1036	S492	345	164	1070	S462	-450	289
1037	S491	330	289	1071	S461	-465	164
1038	S490	315	164	1072	S460	-480	289
1039	S489	300	289	1073	S459	-495	164
1040	S488	285	164	1074	S458	-510	289
1041	S487	270	289	1075	S457	-525	164
1042	S486	255	164	1076	S456	-540	289
1043	S485	240	289	1077	S455	-555	164
1044	S484	225	164	1078	S454	-570	289
1045	S483	210	289	1079	S453	-585	164
1046	S482	195	164	1080	S452	-600	289
1047	S481	180	289	1081	S451	-615	164
1048	DUMMY	165	164	1082	S450	-630	289
1049	DUMMY	150	289	1083	S449	-645	164
1050	DUMMY	-150	289	1084	S448	-660	289
1051	DUMMY	-165	164	1085	S447	-675	164
1052	S480	-180	289	1086	S446	-690	289

PAD No.	PIN Name	X	Y	PAD No.	PIN Name	X	Y
1087	S445	-705	164	1121	S411	-1215	164
1088	S444	-720	289	1122	S410	-1230	289
1089	S443	-735	164	1123	S409	-1245	164
1090	S442	-750	289	1124	S408	-1260	289
1091	S441	-765	164	1125	S407	-1275	164
1092	S440	-780	289	1126	S406	-1290	289
1093	S439	-795	164	1127	S405	-1305	164
1094	S438	-810	289	1128	S404	-1320	289
1095	S437	-825	164	1129	S403	-1335	164
1096	S436	-840	289	1130	S402	-1350	289
1097	S435	-855	164	1131	S401	-1365	164
1098	S434	-870	289	1132	S400	-1380	289
1099	S433	-885	164	1133	S399	-1395	164
1100	S432	-900	289	1134	S398	-1410	289
1101	S431	-915	164	1135	S397	-1425	164
1102	S430	-930	289	1136	S396	-1440	289
1103	S429	-945	164	1137	S395	-1455	164
1104	S428	-960	289	1138	S394	-1470	289
1105	S427	-975	164	1139	S393	-1485	164
1106	S426	-990	289	1140	S392	-1500	289
1107	S425	-1005	164	1141	S391	-1515	164
1108	S424	-1020	289	1142	S390	-1530	289
1109	S423	-1035	164	1143	S389	-1545	164
1110	S422	-1050	289	1144	S388	-1560	289
1111	S421	-1065	164	1145	S387	-1575	164
1112	S420	-1080	289	1146	S386	-1590	289
1113	S419	-1095	164	1147	S385	-1605	164
1114	S418	-1110	289	1148	S384	-1620	289
1115	S417	-1125	164	1149	S383	-1635	164
1116	S416	-1140	289	1150	S382	-1650	289
1117	S415	-1155	164	1151	S381	-1665	164
1118	S414	-1170	289	1152	S380	-1680	289
1119	S413	-1185	164	1153	S379	-1695	164
1120	S412	-1200	289	1154	S378	-1710	289

PAD No.	PIN Name	X	Y	PAD No.	PIN Name	X	Y
1155	S377	-1725	164	1189	S343	-2235	164
1156	S376	-1740	289	1190	S342	-2250	289
1157	S375	-1755	164	1191	S341	-2265	164
1158	S374	-1770	289	1192	S340	-2280	289
1159	S373	-1785	164	1193	S339	-2295	164
1160	S372	-1800	289	1194	S338	-2310	289
1161	S371	-1815	164	1195	S337	-2325	164
1162	S370	-1830	289	1196	S336	-2340	289
1163	S369	-1845	164	1197	S335	-2355	164
1164	S368	-1860	289	1198	S334	-2370	289
1165	S367	-1875	164	1199	S333	-2385	164
1166	S366	-1890	289	1200	S332	-2400	289
1167	S365	-1905	164	1201	S331	-2415	164
1168	S364	-1920	289	1202	S330	-2430	289
1169	S363	-1935	164	1203	S329	-2445	164
1170	S362	-1950	289	1204	S328	-2460	289
1171	S361	-1965	164	1205	S327	-2475	164
1172	S360	-1980	289	1206	S326	-2490	289
1173	S359	-1995	164	1207	S325	-2505	164
1174	S358	-2010	289	1208	S324	-2520	289
1175	S357	-2025	164	1209	S323	-2535	164
1176	S356	-2040	289	1210	S322	-2550	289
1177	S355	-2055	164	1211	S321	-2565	164
1178	S354	-2070	289	1212	S320	-2580	289
1179	S353	-2085	164	1213	S319	-2595	164
1180	S352	-2100	289	1214	S318	-2610	289
1181	S351	-2115	164	1215	S317	-2625	164
1182	S350	-2130	289	1216	S316	-2640	289
1183	S349	-2145	164	1217	S315	-2655	164
1184	S348	-2160	289	1218	S314	-2670	289
1185	S347	-2175	164	1219	S313	-2685	164
1186	S346	-2190	289	1220	S312	-2700	289
1187	S345	-2205	164	1221	S311	-2715	164
1188	S344	-2220	289	1222	S310	-2730	289

PAD No.	PIN Name	X	Y	PAD No.	PIN Name	X	Y
1223	S309	-2745	164	1257	S275	-3255	164
1224	S308	-2760	289	1258	S274	-3270	289
1225	S307	-2775	164	1259	S273	-3285	164
1226	S306	-2790	289	1260	S272	-3300	289
1227	S305	-2805	164	1261	S271	-3315	164
1228	S304	-2820	289	1262	S270	-3330	289
1229	S303	-2835	164	1263	S269	-3345	164
1230	S302	-2850	289	1264	S268	-3360	289
1231	S301	-2865	164	1265	S267	-3375	164
1232	S300	-2880	289	1266	S266	-3390	289
1233	S299	-2895	164	1267	S265	-3405	164
1234	S298	-2910	289	1268	S264	-3420	289
1235	S297	-2925	164	1269	S263	-3435	164
1236	S296	-2940	289	1270	S262	-3450	289
1237	S295	-2955	164	1271	S261	-3465	164
1238	S294	-2970	289	1272	S260	-3480	289
1239	S293	-2985	164	1273	S259	-3495	164
1240	S292	-3000	289	1274	S258	-3510	289
1241	S291	-3015	164	1275	S257	-3525	164
1242	S290	-3030	289	1276	S256	-3540	289
1243	S289	-3045	164	1277	S255	-3555	164
1244	S288	-3060	289	1278	S254	-3570	289
1245	S287	-3075	164	1279	S253	-3585	164
1246	S286	-3090	289	1280	S252	-3600	289
1247	S285	-3105	164	1281	S251	-3615	164
1248	S284	-3120	289	1282	S250	-3630	289
1249	S283	-3135	164	1283	S249	-3645	164
1250	S282	-3150	289	1284	S248	-3660	289
1251	S281	-3165	164	1285	S247	-3675	164
1252	S280	-3180	289	1286	S246	-3690	289
1253	S279	-3195	164	1287	S245	-3705	164
1254	S278	-3210	289	1288	S244	-3720	289
1255	S277	-3225	164	1289	S243	-3735	164
1256	S276	-3240	289	1290	S242	-3750	289

PAD No.	PIN Name	X	Y	PAD No.	PIN Name	X	Y
1291	S241	-3765	164	1325	S207	-4275	164
1292	S240	-3780	289	1326	S206	-4290	289
1293	S239	-3795	164	1327	S205	-4305	164
1294	S238	-3810	289	1328	S204	-4320	289
1295	S237	-3825	164	1329	S203	-4335	164
1296	S236	-3840	289	1330	S202	-4350	289
1297	S235	-3855	164	1331	S201	-4365	164
1298	S234	-3870	289	1332	S200	-4380	289
1299	S233	-3885	164	1333	S199	-4395	164
1300	S232	-3900	289	1334	S198	-4410	289
1301	S231	-3915	164	1335	S197	-4425	164
1302	S230	-3930	289	1336	S196	-4440	289
1303	S229	-3945	164	1337	S195	-4455	164
1304	S228	-3960	289	1338	S194	-4470	289
1305	S227	-3975	164	1339	S193	-4485	164
1306	S226	-3990	289	1340	S192	-4500	289
1307	S225	-4005	164	1341	S191	-4515	164
1308	S224	-4020	289	1342	S190	-4530	289
1309	S223	-4035	164	1343	S189	-4545	164
1310	S222	-4050	289	1344	S188	-4560	289
1311	S221	-4065	164	1345	S187	-4575	164
1312	S220	-4080	289	1346	S186	-4590	289
1313	S219	-4095	164	1347	S185	-4605	164
1314	S218	-4110	289	1348	S184	-4620	289
1315	S217	-4125	164	1349	S183	-4635	164
1316	S216	-4140	289	1350	S182	-4650	289
1317	S215	-4155	164	1351	S181	-4665	164
1318	S214	-4170	289	1352	S180	-4680	289
1319	S213	-4185	164	1353	S179	-4695	164
1320	S212	-4200	289	1354	S178	-4710	289
1321	S211	-4215	164	1355	S177	-4725	164
1322	S210	-4230	289	1356	S176	-4740	289
1323	S209	-4245	164	1357	S175	-4755	164
1324	S208	-4260	289	1358	S174	-4770	289

PAD No.	PIN Name	X	Y	PAD No.	PIN Name	X	Y
1359	S173	-4785	164	1393	S139	-5295	164
1360	S172	-4800	289	1394	S138	-5310	289
1361	S171	-4815	164	1395	S137	-5325	164
1362	S170	-4830	289	1396	S136	-5340	289
1363	S169	-4845	164	1397	S135	-5355	164
1364	S168	-4860	289	1398	S134	-5370	289
1365	S167	-4875	164	1399	S133	-5385	164
1366	S166	-4890	289	1400	S132	-5400	289
1367	S165	-4905	164	1401	S131	-5415	164
1368	S164	-4920	289	1402	S130	-5430	289
1369	S163	-4935	164	1403	S129	-5445	164
1370	S162	-4950	289	1404	S128	-5460	289
1371	S161	-4965	164	1405	S127	-5475	164
1372	S160	-4980	289	1406	S126	-5490	289
1373	S159	-4995	164	1407	S125	-5505	164
1374	S158	-5010	289	1408	S124	-5520	289
1375	S157	-5025	164	1409	S123	-5535	164
1376	S156	-5040	289	1410	S122	-5550	289
1377	S155	-5055	164	1411	S121	-5565	164
1378	S154	-5070	289	1412	S120	-5580	289
1379	S153	-5085	164	1413	S119	-5595	164
1380	S152	-5100	289	1414	S118	-5610	289
1381	S151	-5115	164	1415	S117	-5625	164
1382	S150	-5130	289	1416	S116	-5640	289
1383	S149	-5145	164	1417	S115	-5655	164
1384	S148	-5160	289	1418	S114	-5670	289
1385	S147	-5175	164	1419	S113	-5685	164
1386	S146	-5190	289	1420	S112	-5700	289
1387	S145	-5205	164	1421	S111	-5715	164
1388	S144	-5220	289	1422	S110	-5730	289
1389	S143	-5235	164	1423	S109	-5745	164
1390	S142	-5250	289	1424	S108	-5760	289
1391	S141	-5265	164	1393	S139	-5295	164
1392	S140	-5280	289	1394	S138	-5310	289

PAD No.	PIN Name	X	Y	PAD No.	PIN Name	X	Y
1425	S107	-5775	164	1459	S73	-6285	164
1426	S106	-5790	289	1460	S72	-6300	289
1427	S105	-5805	164	1461	S71	-6315	164
1428	S104	-5820	289	1462	S70	-6330	289
1429	S103	-5835	164	1463	S69	-6345	164
1430	S102	-5850	289	1464	S68	-6360	289
1431	S101	-5865	164	1465	S67	-6375	164
1432	S100	-5880	289	1466	S66	-6390	289
1433	S99	-5895	164	1467	S65	-6405	164
1434	S98	-5910	289	1468	S64	-6420	289
1435	S97	-5925	164	1469	S63	-6435	164
1436	S96	-5940	289	1470	S62	-6450	289
1437	S95	-5955	164	1471	S61	-6465	164
1438	S94	-5970	289	1472	S60	-6480	289
1439	S93	-5985	164	1473	S59	-6495	164
1440	S92	-6000	289	1474	S58	-6510	289
1441	S91	-6015	164	1475	S57	-6525	164
1442	S90	-6030	289	1476	S56	-6540	289
1443	S89	-6045	164	1477	S55	-6555	164
1444	S88	-6060	289	1478	S54	-6570	289
1445	S87	-6075	164	1479	S53	-6585	164
1446	S86	-6090	289	1480	S52	-6600	289
1447	S85	-6105	164	1481	S51	-6615	164
1448	S84	-6120	289	1482	S50	-6630	289
1449	S83	-6135	164	1483	S49	-6645	164
1450	S82	-6150	289	1484	S48	-6660	289
1451	S81	-6165	164	1485	S47	-6675	164
1452	S80	-6180	289	1486	S46	-6690	289
1453	S79	-6195	164	1487	S45	-6705	164
1454	S78	-6210	289	1488	S44	-6720	289
1455	S77	-6225	164	1489	S43	-6735	164
1456	S76	-6240	289	1490	S42	-6750	289
1457	S75	-6255	164	1491	S41	-6765	164
1458	S74	-6270	289	1492	S40	-6780	289

PAD No.	PIN Name	X	Y	PAD No.	PIN Name	X	Y
1493	S39	-6795	164	1527	S5	-7305	164
1494	S38	-6810	289	1528	S4	-7320	289
1495	S37	-6825	164	1529	S3	-7335	164
1496	S36	-6840	289	1530	S2	-7350	289
1497	S35	-6855	164	1531	S1	-7365	164
1498	S34	-6870	289	1532	DUMMY	-7380	289
1499	S33	-6885	164	1533	DUMMY	-7395	164
1500	S32	-6900	289	1534	DUMMY	-7560	289
1501	S31	-6915	164	1535	DUMMY	-7575	164
1502	S30	-6930	289	1536	G480	-7590	289
1503	S29	-6945	164	1537	G478	-7605	164
1504	S28	-6960	289	1538	G476	-7620	289
1505	S27	-6975	164	1539	G474	-7635	164
1506	S26	-6990	289	1540	G472	-7650	289
1507	S25	-7005	164	1541	G470	-7665	164
1508	S24	-7020	289	1542	G468	-7680	289
1509	S23	-7035	164	1543	G466	-7695	164
1510	S22	-7050	289	1544	G464	-7710	289
1511	S21	-7065	164	1545	G462	-7725	164
1512	S20	-7080	289	1546	G460	-7740	289
1513	S19	-7095	164	1547	G458	-7755	164
1514	S18	-7110	289	1548	G456	-7770	289
1515	S17	-7125	164	1549	G454	-7785	164
1516	S16	-7140	289	1550	G452	-7800	289
1517	S15	-7155	164	1551	G450	-7815	164
1518	S14	-7170	289	1552	G448	-7830	289
1519	S13	-7185	164	1553	G446	-7845	164
1520	S12	-7200	289	1554	G444	-7860	289
1521	S11	-7215	164	1555	G442	-7875	164
1522	S10	-7230	289	1556	G440	-7890	289
1523	S9	-7245	164	1557	G438	-7905	164
1524	S8	-7260	289	1558	G436	-7920	289
1525	S7	-7275	164	1559	G434	-7935	164
1526	S6	-7290	289	1560	G432	-7950	289

PAD No.	PIN Name	X	Y	PAD No.	PIN Name	X	Y
1561	G430	-7965	164	1595	G362	-8475	164
1562	G428	-7980	289	1596	G360	-8490	289
1563	G426	-7995	164	1597	G358	-8505	164
1564	G424	-8010	289	1598	G356	-8520	289
1565	G422	-8025	164	1599	G354	-8535	164
1566	G420	-8040	289	1600	G352	-8550	289
1567	G418	-8055	164	1601	G350	-8565	164
1568	G416	-8070	289	1602	G348	-8580	289
1569	G414	-8085	164	1603	G346	-8595	164
1570	G412	-8100	289	1604	G344	-8610	289
1571	G410	-8115	164	1605	G342	-8625	164
1572	G408	-8130	289	1606	G340	-8640	289
1573	G406	-8145	164	1607	G338	-8655	164
1574	G404	-8160	289	1608	G336	-8670	289
1575	G402	-8175	164	1609	G334	-8685	164
1576	G400	-8190	289	1610	G332	-8700	289
1577	G398	-8205	164	1611	G330	-8715	164
1578	G396	-8220	289	1612	G328	-8730	289
1579	G394	-8235	164	1613	G326	-8745	164
1580	G392	-8250	289	1614	G324	-8760	289
1581	G390	-8265	164	1615	G322	-8775	164
1582	G388	-8280	289	1616	G320	-8790	289
1583	G386	-8295	164	1617	G318	-8805	164
1584	G384	-8310	289	1618	G316	-8820	289
1585	G382	-8325	164	1619	G314	-8835	164
1586	G380	-8340	289	1620	G312	-8850	289
1587	G378	-8355	164	1621	G310	-8865	164
1588	G376	-8370	289	1622	G308	-8880	289
1589	G374	-8385	164	1623	G306	-8895	164
1590	G372	-8400	289	1624	G304	-8910	289
1591	G370	-8415	164	1625	G302	-8925	164
1592	G368	-8430	289	1626	G300	-8940	289
1593	G366	-8445	164	1627	G298	-8955	164
1594	G364	-8460	289	1628	G296	-8970	289

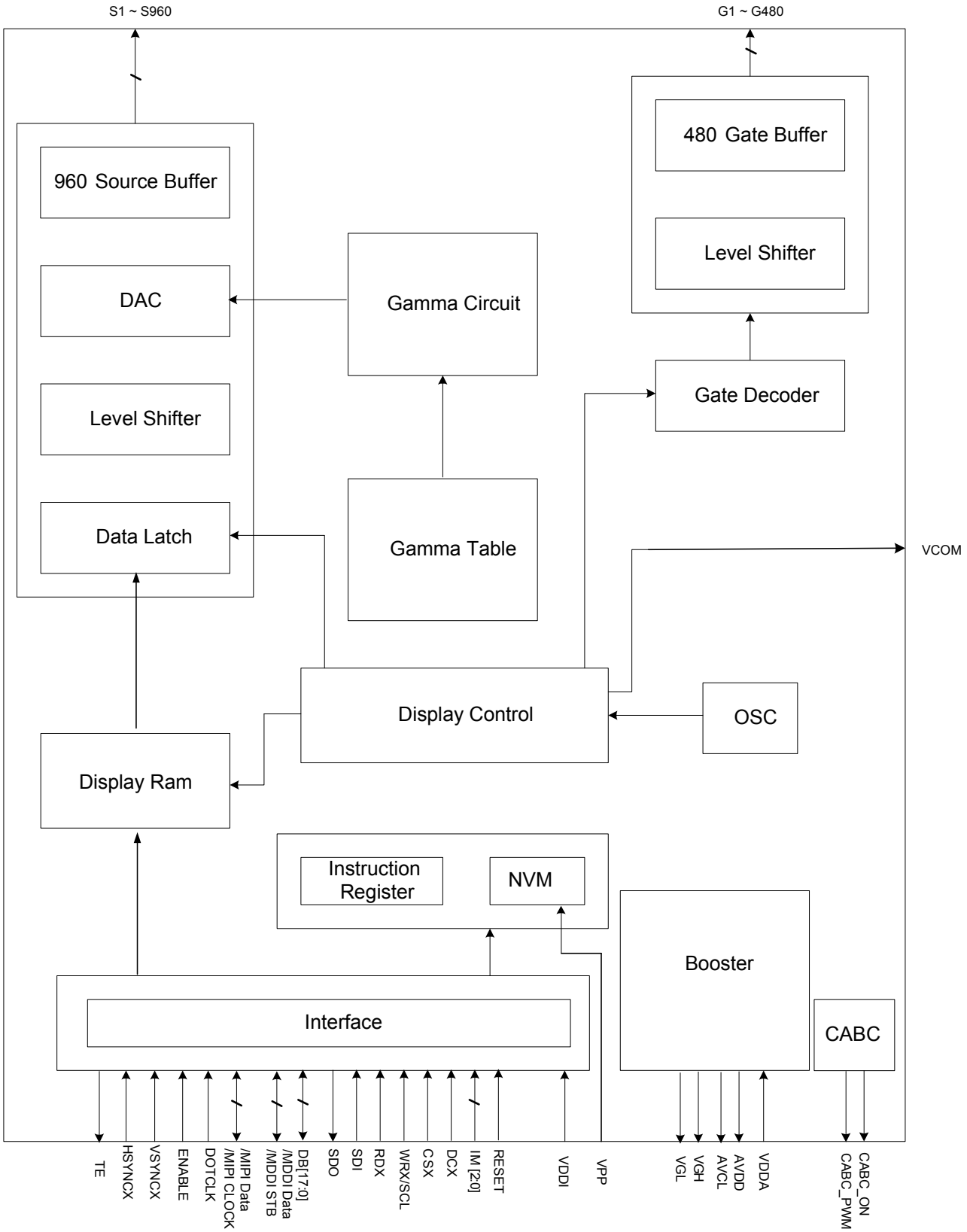
PAD No.	PIN Name	X	Y	PAD No.	PIN Name	X	Y
1629	G294	-8985	164	1663	G226	-9495	164
1630	G292	-9000	289	1664	G224	-9510	289
1631	G290	-9015	164	1665	G222	-9525	164
1632	G288	-9030	289	1666	G220	-9540	289
1633	G286	-9045	164	1667	G218	-9555	164
1634	G284	-9060	289	1668	G216	-9570	289
1635	G282	-9075	164	1669	G214	-9585	164
1636	G280	-9090	289	1670	G212	-9600	289
1637	G278	-9105	164	1671	G210	-9615	164
1638	G276	-9120	289	1672	G208	-9630	289
1639	G274	-9135	164	1673	G206	-9645	164
1640	G272	-9150	289	1674	G204	-9660	289
1641	G270	-9165	164	1675	G202	-9675	164
1642	G268	-9180	289	1676	G200	-9690	289
1643	G266	-9195	164	1677	G198	-9705	164
1644	G264	-9210	289	1678	G196	-9720	289
1645	G262	-9225	164	1679	G194	-9735	164
1646	G260	-9240	289	1680	G192	-9750	289
1647	G258	-9255	164	1681	G190	-9765	164
1648	G256	-9270	289	1682	G188	-9780	289
1649	G254	-9285	164	1683	G186	-9795	164
1650	G252	-9300	289	1684	G184	-9810	289
1651	G250	-9315	164	1685	G182	-9825	164
1652	G248	-9330	289	1686	G180	-9840	289
1653	G246	-9345	164	1687	G178	-9855	164
1654	G244	-9360	289	1688	G176	-9870	289
1655	G242	-9375	164	1689	G174	-9885	164
1656	G240	-9390	289	1690	G172	-9900	289
1657	G238	-9405	164	1691	G170	-9915	164
1658	G236	-9420	289	1692	G168	-9930	289
1659	G234	-9435	164	1693	G166	-9945	164
1660	G232	-9450	289	1694	G164	-9960	289
1661	G230	-9465	164	1695	G162	-9975	164
1662	G228	-9480	289	1696	G160	-9990	289

PAD No.	PIN Name	X	Y	PAD No.	PIN Name	X	Y
1697	G158	-10005	164	1731	G90	-10515	164
1698	G156	-10020	289	1732	G88	-10530	289
1699	G154	-10035	164	1733	G86	-10545	164
1700	G152	-10050	289	1734	G84	-10560	289
1701	G150	-10065	164	1735	G82	-10575	164
1702	G148	-10080	289	1736	G80	-10590	289
1703	G146	-10095	164	1737	G78	-10605	164
1704	G144	-10110	289	1738	G76	-10620	289
1705	G142	-10125	164	1739	G74	-10635	164
1706	G140	-10140	289	1740	G72	-10650	289
1707	G138	-10155	164	1741	G70	-10665	164
1708	G136	-10170	289	1742	G68	-10680	289
1709	G134	-10185	164	1743	G66	-10695	164
1710	G132	-10200	289	1744	G64	-10710	289
1711	G130	-10215	164	1745	G62	-10725	164
1712	G128	-10230	289	1746	G60	-10740	289
1713	G126	-10245	164	1747	G58	-10755	164
1714	G124	-10260	289	1748	G56	-10770	289
1715	G122	-10275	164	1749	G54	-10785	164
1716	G120	-10290	289	1750	G52	-10800	289
1717	G118	-10305	164	1751	G50	-10815	164
1718	G116	-10320	289	1752	G48	-10830	289
1719	G114	-10335	164	1753	G46	-10845	164
1720	G112	-10350	289	1754	G44	-10860	289
1721	G110	-10365	164	1755	G42	-10875	164
1722	G108	-10380	289	1756	G40	-10890	289
1723	G106	-10395	164	1757	G38	-10905	164
1724	G104	-10410	289	1758	G36	-10920	289
1725	G102	-10425	164	1759	G34	-10935	164
1726	G100	-10440	289	1760	G32	-10950	289
1727	G98	-10455	164	1761	G30	-10965	164
1728	G96	-10470	289	1762	G28	-10980	289
1729	G94	-10485	164	1763	G26	-10995	164
1730	G92	-10500	289	1764	G24	-11010	289

PAD No.	PIN Name	X	Y
1765	G22	-11025	164
1766	G20	-11040	289
1767	G18	-11055	164
1768	G16	-11070	289
1769	G14	-11085	164
1770	G12	-11100	289
1771	G10	-11115	164
1772	G8	-11130	289
1773	G6	-11145	164
1774	G4	-11160	289
1776	G2	-11175	164
1775	DUMMY	-11190	289
1777	DUMMY	-11205	164
1778	ALIGN_L	-11300	-270

Unit: μm

5 BLOCK DIAGRAM



6 PIN DESCRIPTION

6.1.. Power Supply Pins

Name	I/O	Description	Connect Pin
VDDI	I	- Power supply for I/O system. - VDDI must be lower than or equal to VDD.	VDDI
VDDA	I	- Power supply for analog and booster circuits. Input voltage level should be the same as VDD.	VDDA
AGND	I	- System ground for analog system and booster circuit.	GND
DGND	I	- System ground for I/O system and digital system.	GND
VPP	I	- Power supply for internal NVM. - Writing NVM needs external power supply voltage with 8.25V. - The current of Ivpp must be more than 10mA. - Leaves these pins open if not used.	-

6.2.. Interface Logic Pins

<table>
<tr><th>Name</th><th>I/O</th><th>Description</th><th>Connect Pin</th></tr>
<tr><td>IM2, IM1,IM0</td><td>I</td><td>- The MCU interface mode select.
<table>
<tr><th>IM2</th><th>IM1</th><th>IM0</th><th>MPU Interface Mode</th><th>Data pin</th></tr>
<tr><td>0</td><td>0</td><td>0</td><td>8080 18-bit Interface</td><td>DB[17:0]</td></tr>
<tr><td>0</td><td>0</td><td>1</td><td>8080 9-bit Interface</td><td>DB[8:0]</td></tr>
<tr><td>0</td><td>1</td><td>0</td><td>8080 16-bit Interface</td><td>DB[15:0]</td></tr>
<tr><td>0</td><td>1</td><td>1</td><td>8080 8-bit Interface</td><td>DB[7:0],</td></tr>
<tr><td>1</td><td>0</td><td>0</td><td>MDDI</td><td>MDDI_DATA
MDDI_STB</td></tr>
<tr><td>1</td><td>0</td><td>1</td><td>Reserve</td><td></td></tr>
<tr><td>1</td><td>1</td><td>0</td><td>MIPI</td><td>MIPI_DATA
MIPI_CLOCK</td></tr>
<tr><td>1</td><td>1</td><td>1</td><td>4Line SPI</td><td>SDA, SDO</td></tr>
</table></td><td>GND / VDDI</td></tr>
<tr><td>RESET</td><td>I</td><td>- This signal will reset the device and it must be applied to properly initialize the chip.
- Signal is active low.</td><td>MCU</td></tr>
<tr><td>CSX</td><td>I</td><td>- Chip selection pin. Low-active.
- If not used, please fix this pin at VDDI level.</td><td>MCU</td></tr>
<tr><td>DCX</td><td>I</td><td>- Display data/command selection (RS) pin in MCU interface.
DCX=’1’: display data or parameter.
DCX=’0’: register index / command.
- In MDDI mode, connect this pin to DGND.
- If not used, please fix this pin at VDDI or DGND level.</td><td>MCU / GND</td></tr>
<tr><td>RDX</td><td>I</td><td>- Read enable in 8080 MCU parallel interface. Low-active.
- In MDDI mode, connect this pin to DGND.
- If not used, please fix this pin at VDDI or DGND level.</td><td>MCU / GND</td></tr>
<tr><td>WRX/SCL</td><td>I</td><td>- Write enable in MCU parallel interface.
- In SPI mode, this pin is used as SCL.
- If not used, please fix this pin at VDDI or DGND level.</td><td>MCU</td></tr>
<tr><td>VSYNC</td><td>I</td><td>-Vertical synchronizing input signal for RGB interface operation.
-If not used, please fix to the VDDI or DGND.</td><td>MCU</td></tr>
<tr><td>HSYNC</td><td>I</td><td>-Horizontal synchronizing input signal for RGB interface operation.
- If not used, please fix to VDDI or DGND.</td><td>MCU</td></tr>
<tr><td>ENABLE</td><td>I</td><td>-Data enable signal for RGB interface operation.</td><td>MCU</td></tr>
</table>

Name	I/O	Description	Connect Pin
		-If not used, please fix this pin at VDDI or DGND.	
DOTCLK	I	-Dot clock signal for RGB interface operation. -If not used, please fix this pin at VDDI or DGND.	MCU
SDA	I/O	- SPI interface input/output pin. - The data is latched on the rising edge of the SCL signal. - If not used, please fix this pin at VDDI or DGND level.	MCU
SDO	O	- SPI interface output pin. - The data is outputted on the falling edge of the SCL signal. - If not used, please fix this pin at floating.	MCU
MIPI_CLK_P /MDDI_STB_P	I	-Positive polarity of low voltage differential clock signal -Leave the pin to open when not in use.	MDDI/MIPI
MIPI_CLK_N /MDDI_STB_N	I	-Negative polarity of low voltage differential clock signal -Leave the pin to open when not in use.	MDDI/MIPI
MIPI_DATA_P /MDDI_DATA_P	I/O	-Positive polarity of low voltage differential data signal -Leave the pin to open when not in use.	MDDI/MIPI
MIPI_DATA_N MDDI_DATA_N	I/O	-Negative polarity of low voltage differential data signal -Leave the pin to open when not in use.	MDDI/MIPI
DB[17:0]	I/O	- In MCU 8080 parallel interface, DB[17:0] are used as data bus. 8-bit I/F: DB[7:0] is used. 9-bit I/F: DB[8:0] is used. 16-bit I/F: DB[15:0] is used. 18-bit I/F: DB[17:0] is used. - In RGB interface, DB[17:0] are used as data bus. 16-bit RGB I/F: DB[15:0] are used. 18-bit RGB I/F: DB[17:0] are used. - If not used, please fix this pin at VDDI or DGND level.	MCU / DGND /VDDI
TE	O	- Tearing effect output. - If not used, leave this pin open	MCU

Note1. "1" = VDDI level, "0" = DGND level.

Note2. When in parallel mode, unused data pins must be connected to "1" or "0".

Note3. When CSX="1", there is no influence to the parallel and serial interface.

6.3.. Driver Output Pins

Name	I/O	Description	Connect pin
S1 to S960	O	Source output voltage signals applied to liquid crystal.	LCD
G1 to G480	O	- Gate driver output pins. VGH: Selecting Gate Lines Level. VGL: Non-selecting Gate Lines Level.	LCD
AVDD	O	- Power output pin for monitoring analogy circuit. - Leave open when not in use.	-
AVCL	O	- Power output pin for monitoring analogy circuit. - Leave open when not in use.	-
PVAP	O	- Used for monitoring - Leave open.	
PVAN	O	- Used for monitoring - Leave open.	
V22	O	- Used for monitoring - Leave open.	
VGH	O	- Power output pin for gate driver - Leave open when not in use.	-
VGL	O	- Power output (Negative) pin for gate driver - Leave open when not in use.	-
VCC	O	- Monitoring pin of internal digital reference voltage. - Leave open when not in use.	-
VCOM	O	- A power supply for the TFT-LCD common electrode.	Common Electrode
CABC_PWM	O	-Output pad for PWM output signal to driving LED. -If not used, keep it open.	-
CABC_ON	O	-Output pad for enabling LED. -If not used, keep it open.	-

6.4.. Test and Other Pins

Name	I/O	Description	Connect pin
Dummy	-	- These pins are dummy - Leave open	-

7 DRIVER ELECTRICAL CHARACTERISTICS

7.1.. Absolute Operation Range

Item	Symbol	Range	Unit
Supply Voltage (Analog)	VDDA	- 0.3 ~ +4.6	V
Supply Voltage (I/O)	VDDI	- 0.3 ~ +4.6	V
Supply Voltage (Logic)	VCC	-0.3 ~ +2	V
Driver Supply Voltage	VGH-VGL	-0.3 ~ +30.0	V
Logic Input Voltage Range	VIN	0.5 ~ VDDI + 0.5	V
Logic Output Voltage Range	VO	0.5 ~ VDDI + 0.5	V
Operating Temperature Range	TOPR	-30 ~ +85	℃
Storage Temperature Range	TSTG	-40 ~ +125	℃

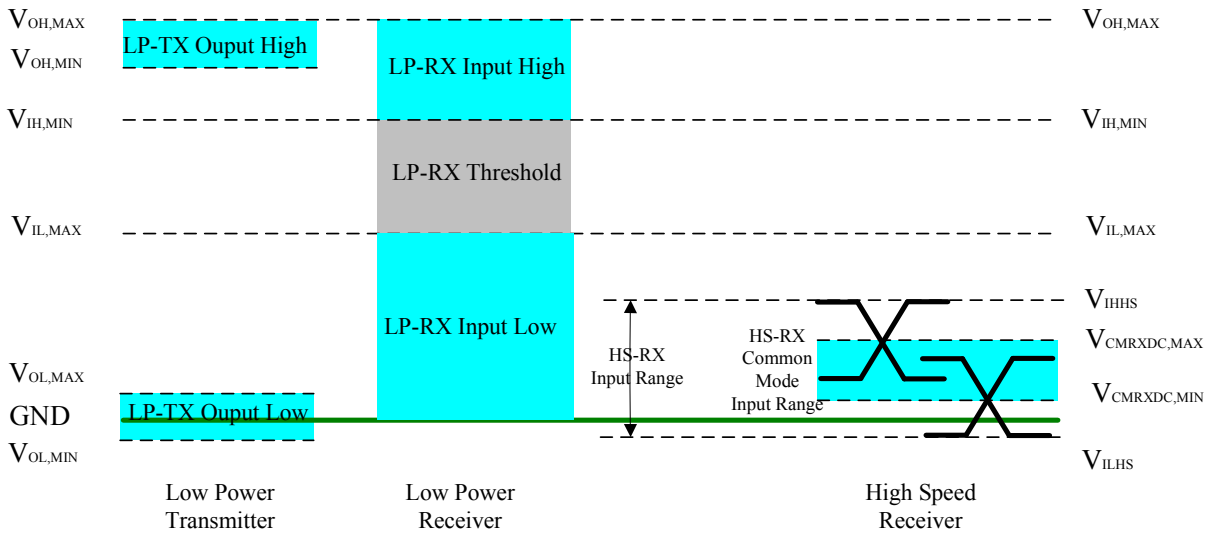
Absolute Operation Range

Note: If one of the above items is exceeded its maximum limitation momentarily, the quality of the product may be degraded. Absolute maximum limitation, therefore, specify the values exceeding which the product may be physically damaged. Be sure to use the product within the recommend range.

7.2.. DC Characteristics

7.2.1 DC characteristics for MIPI DSI

- MIPI Signaling Voltage Levels



- MIPI DC characteristics

Parameter	Symbol	Specification			Unit
		MIN	TYP	MAX	
Operation Voltage for MIPI Receiver					
Low power mode operating voltage	V_{LPH}	1.1	1.2	1.3	V
MIPI Characteristics for High Speed Receiver					
Single-endedl input low voltage	V_{ILHS}	-40	-	-	mV
Single-endedl input high voltage	V_{IHHS}	-	-	460	mV
Common-mode voltage	V_{CMRXDC}	70	-	330	mV
Differential input impedance	Z_{ID}	80	100	125	ohm
MIPI Characteristics for Low Power Mode					
Pad signal voltage range	V_I	-50	-	1350	mV
Differential Input Low Level Threshold Voltage for Clock	$V_{THLCLK-}$	-70	-	-	mV
Logic 0 input threshold	V_{IL}	0	-	550	mV
Logic 1 input threshold	V_{IH}	880	-	1350	mV
Output low level	V_{OL}	-50	-	50	mV
Output high level	V_{OH}	1.1	1.2	1.3	V

7.2.2 DC Characteristics for Panel Driving

Parameter	Symbol	Condition	Specification			Unit	Related Pins
			MIN.	TYP.	MAX.		
Power & Operation Voltage							
System Voltage	VDD / VDDA	Operating voltage	2.5	2.75	3.3	V	
Interface Operation Voltage	VDDI	I/O Supply Voltage	1.65	1.8	3.3	V	
Gate Driver High Voltage	VGH		12.7		15.7	V	
Gate Driver Low Voltage	VGL		-13.47		-8.41	V	
Gate Driver Supply Voltage		| VGH-VGL |			29.17	V	
Input / Output							
Logic-High Input Voltage	VIH		0.7VDDI		VDDI	V	Note 1
Logic-Low Input Voltage	VIL		VSS		0.3VDDI	V	Note 1
Differential Input High Threshold Voltage	VIT+			0	50	mV	MDDI_STB, MDDI_DATA
Differential Input Low Threshold Voltage	VIT-		-50	0		mV	
Single-ended Receiver Input Operation Voltage Range	VIR		0.5		1.2	V	
Logic-High Output Voltage	VOH	IOH = -1.0mA	0.8VDDI		VDDI	V	Note 1
Logic-Low Output Voltage	VOL	IOL = +1.0mA	VSS		0.2VDDI	V	Note 1
Logic-High Input Current	IIH	VIN = VDDI			1	uA	Note 1
Logic-Low Input Current	IIL	VIN = VSS	-1			uA	Note 1
Input Leakage Current	ILI	IOH = -1.0mA	-0.1		+0.1	uA	Note 1
VCOM Voltage							
VCOM Voltage	VCOM			VSS		V	
Source Driver							
Gamma Reference Voltage(Positive)	PVAP		4.45		6.4	V	
Gamma Reference Voltage(Negative)	PVAN		-4.6		-2.65		
Source Output Settling Time	Tr	Below with 99% precision			20	us	Note 2

Output Offset Voltage	VOFFSET				35	mV	Note 3

Basic DC Characteristics

Notes:

1. TA= -30 to 85℃.

2. Source channel loading= 2KΩ+12pF/channel, Gate channel loading=5KΩ+40pF/channel.

3. The max. value is between measured point of source output and gamma setting value.

7.3.. Power Consumption

Ta=25℃, Frame rate = 60Hz, Registers setting are IC default setting.

Operation Mode	Image	Current Consumption			
		Typical		Maximum	
		IDDI (mA)	IDD (mA)	IDDI (mA)	IDD (mA)
Normal Mode	Note 1	0.005	16	0.01	18
Stand by Mode	Note 1	0.005	0.015	0.01	0.03

Power Consumption

Notes:

1. All pixels black.

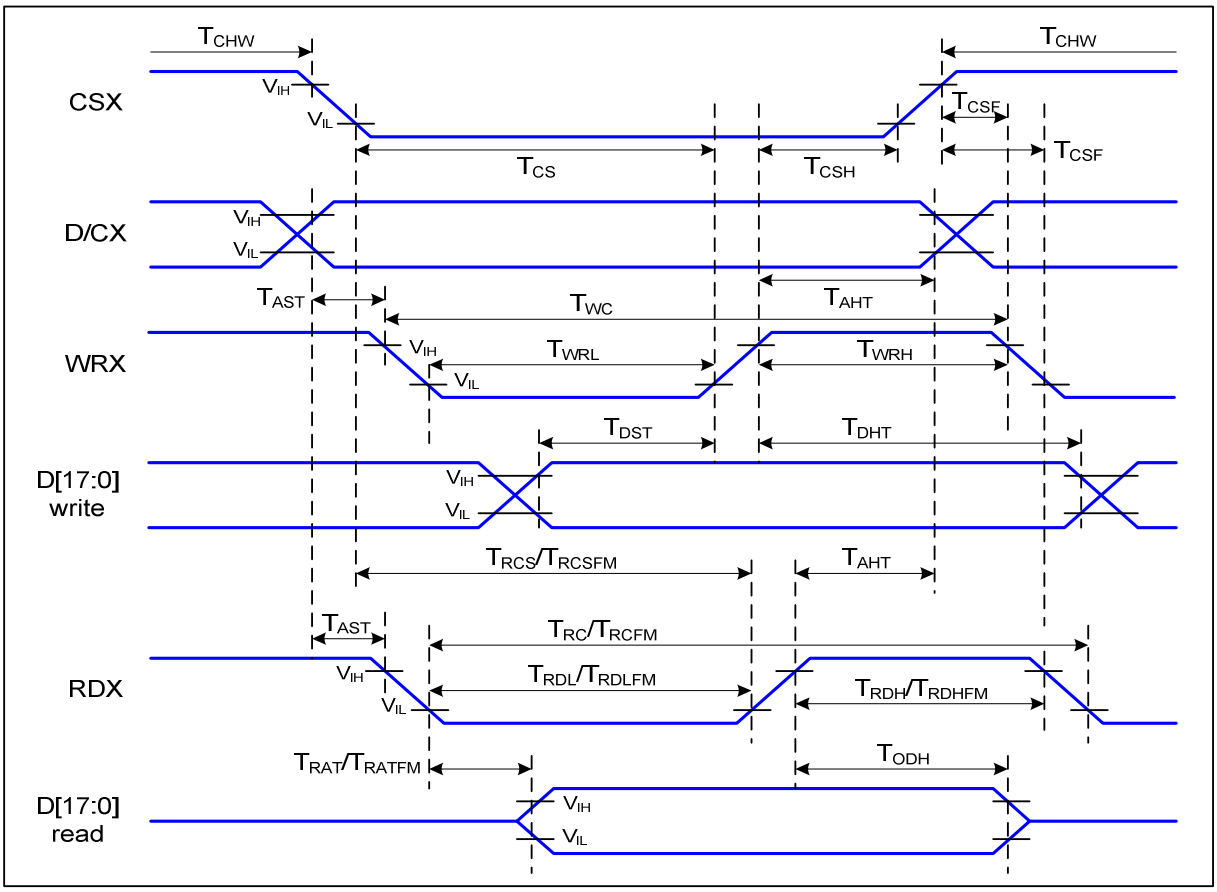
2. All pixels white.

3. The Current Consumption is DC characteristics of ST7796.

4. Typical: VDDI=1.8V, VDD=VDDA=2.75V; Maximum: VDDI=1.65 to VDD, VDD=2.5 to 3.3V, VDDA=2.5 to 3.3V

7.4.. AC Characteristics

7.4.1 8080 Series MCU Parallel Interface Characteristics: 18/16/9/8-bit Bus



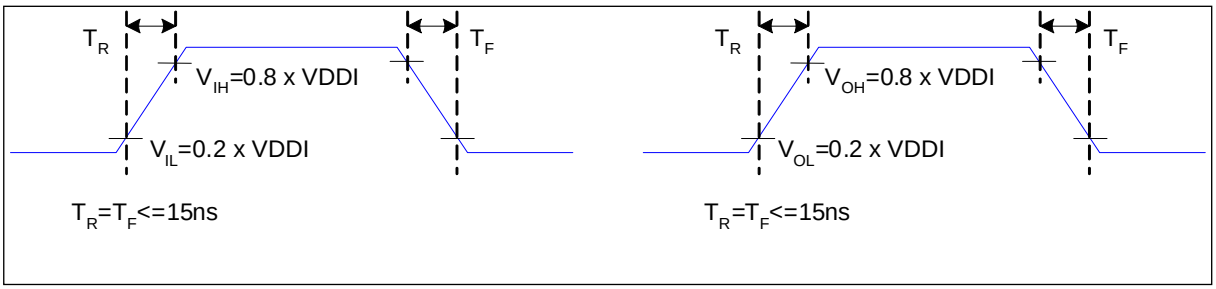
Parallel Interface Timing Characteristics (8080-Series MCU Interface)

VDDI=1.65 to VDD, VDD=2.5 to 3.3V, AGND=DGND=0V, Ta=25 ℃

Signal	Symbol	Parameter	Min	Max	Unit	Description
D/CX	T_{AST}	Address setup time	0		ns	-
	T_{AHT}	Address hold time (Write/Read)	10		ns	
CSX	T_{CHW}	Chip select “H” pulse width	0		ns	-
	T_{CS}	Chip select setup time (Write)	15		ns	
	T_{RCS}	Chip select setup time (Read ID)	45		ns	
	T_{RCSFM}	Chip select setup time (Read FM)	355		ns	
	T_{CSF}	Chip select wait time (Write/Read)	10		ns	
	T_{CSH}	Chip select hold time	10		ns	
WRX	T_{WC}	Write cycle	66		ns	
	T_{WRH}	Control pulse “H” duration	15		ns	

	T_{WRL}	Control pulse “L” duration	15		ns	
RDX (ID)	T_{RC}	Read cycle (ID)	160		ns	When read ID data
	T_{RDH}	Control pulse “H” duration (ID)	90		ns	
	T_{RDL}	Control pulse “L” duration (ID)	45		ns	
RDX (FM)	T_{RCFM}	Read cycle (FM)	450		ns	When read from frame memory
	T_{RDHFM}	Control pulse “H” duration (FM)	90		ns	
	T_{RDLFM}	Control pulse “L” duration (FM)	355		ns	
D[17:0]	T_{DST}	Data setup time	10		ns	For CL=30pF
	T_{DHT}	Data hold time	10		ns	
	T_{RAT}	Read access time (ID)		40	ns	
	T_{RATFM}	Read access time (FM)		340	ns	
	T_{ODH}	Output disable time	20	80	ns	

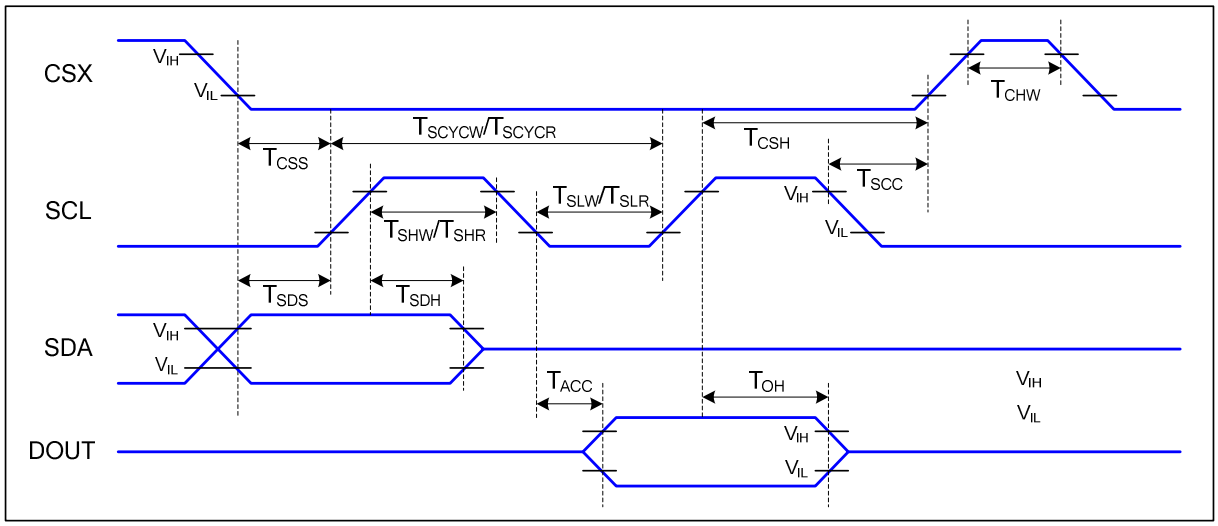
8080 Parallel Interface Characteristics



Rising and Falling Timing for I/O Signal

Note: The rising time and falling time (Tr, Tf) of input signal and fall time are specified at 15 ns or less. Logic high and low levels are specified as 20% and 80% of VDDI for Input signals.

7.4.2 Serial Data Transfer Interface Characteristics:



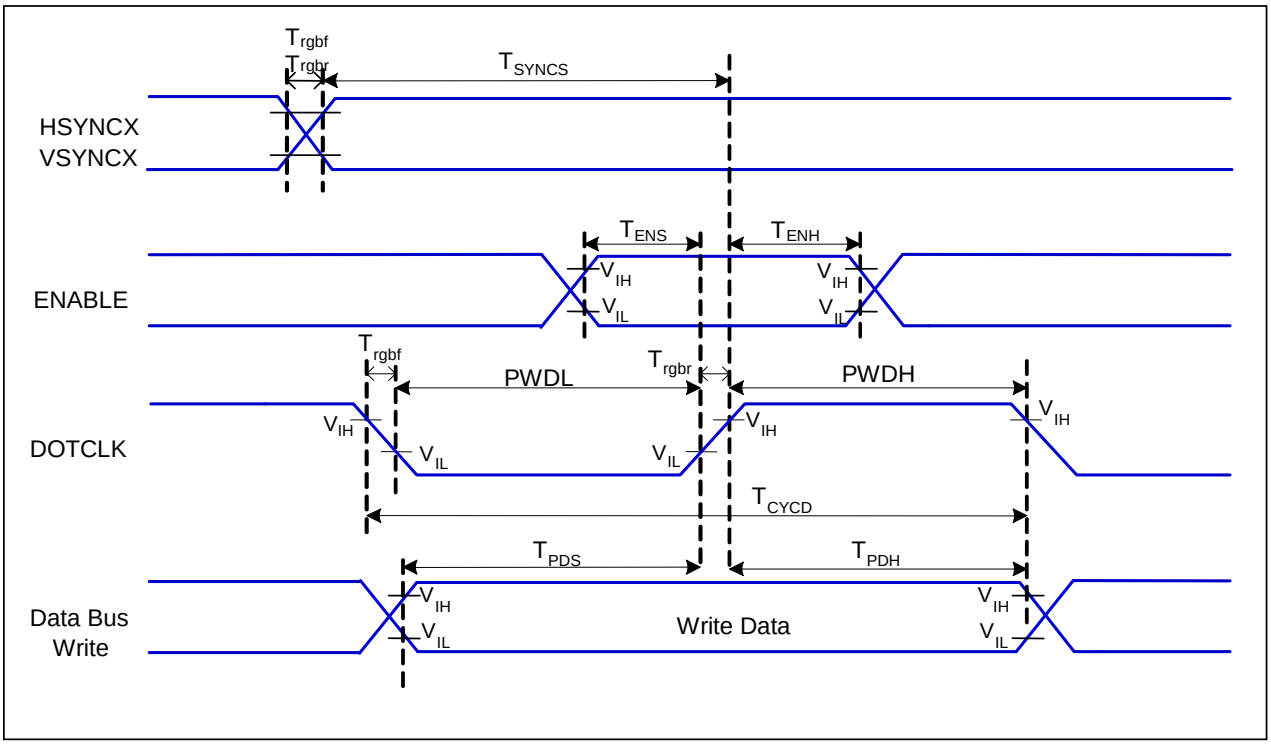
SPI Interface Timing Characteristics

VDDI=1.65 to VDD, VDD=2.5 to 3.3V, AGND=DGND=0V, Ta=25 ℃

Signal	Symbol	Parameter	Min	Max	Unit	Description
CSX	T_{CSS}	Chip select setup time (write)	15		ns	
	T_{CSH}	Chip select hold time (write)	15		ns	
	T_{CSS}	Chip select setup time (read)	60		ns	
	T_{SCC}	Chip select hold time (read)	65		ns	
	T_{CHW}	Chip select “H” pulse width	40		ns	
SCL	T_{SCYCW}	Serial clock cycle (Write)	66		ns	
	T_{SHW}	SCL “H” pulse width (Write)	15		ns	
	T_{SLW}	SCL “L” pulse width (Write)	15		ns	
	T_{SCYCR}	Serial clock cycle (Read)	150		ns	
	T_{SHR}	SCL “H” pulse width (Read)	60		ns	
	T_{SLR}	SCL “L” pulse width (Read)	60		ns	
SDA (DIN)	T_{SDS}	Data setup time	10		ns	
	T_{SDH}	Data hold time	10		ns	
DOUT	T_{ACC}	Access time	10	50	ns	For maximum CL=30pF
	T_{OH}	Output disable time	15	50	ns	For minimum CL=8pF

SPI Interface Characteristics

7.4.3 RGB Interface Characteristics:



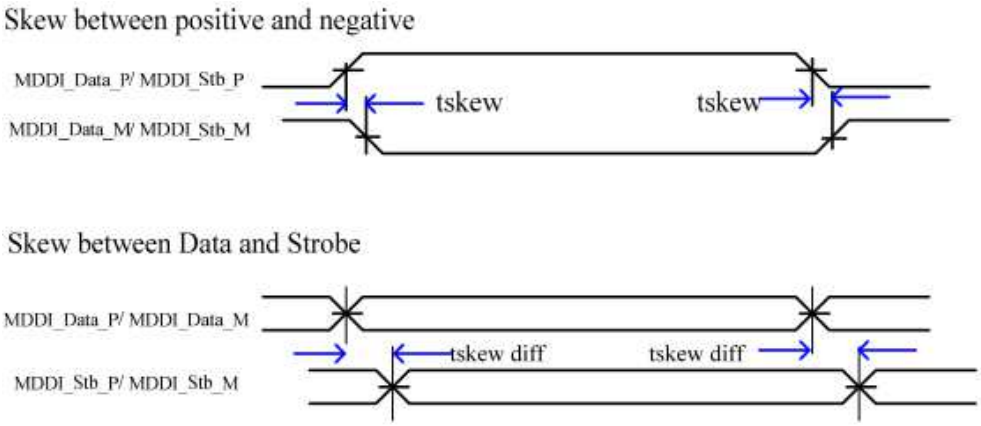
VDDI=1.65 to VDD, VDD=2.5 to 3.3V, AGND=DGND=0V, Ta=25 ℃

Signal	Symbol	Parameter	MIN	MAX	Unit	Description
HSYNCX VSYNCX	TSYNCS	VSYNC, HSYNC Setup Time	30	-	ns	
ENABLE	TENS	Enable Setup Time	25	-	ns	
	TENH	Enable Hold Time	25	-	ns	
DOTCLK	PWDH	DOTCLK High-level Pulse Width	60	-	ns	
	PWDL	DOTCLK Low-level Pulse Width	60	-	ns	
	TCYCD	DOTCLK Cycle Time	120	-	ns	
	Trghr, Trghf	DOTCLK Rise/Fall time		20	ns	
DB	TPDS	PD Data Setup Time	50	-	ns	
	TPDH	PD Data Hold Time	50	-	ns	

RGB Interface Timing Characteristics

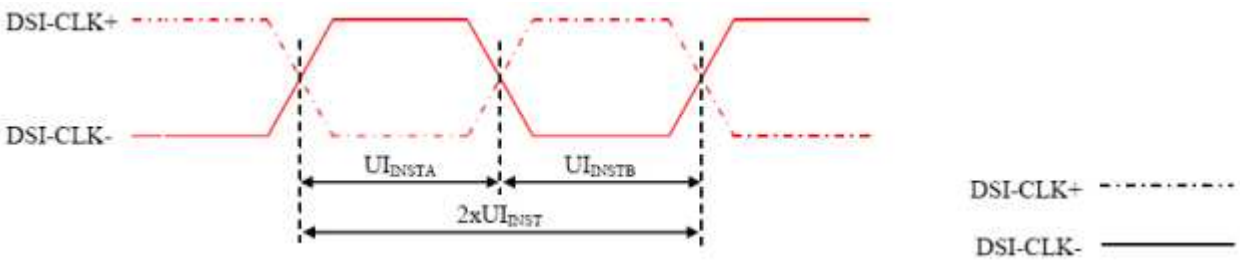
7.4.4 MDDI Receiver Timing Characteristics

Signal	Symbol	MIN	TYP	MAX	Unit	Description
Data transfer rate	1/t Bit	-	-	400	Mbps	
Differential Transfer Input Skew	| ± tskew|	-	-	0.25	ns	
Data_ Strobe Input Skew	| ± tskew diff|	-	-	2	ns	



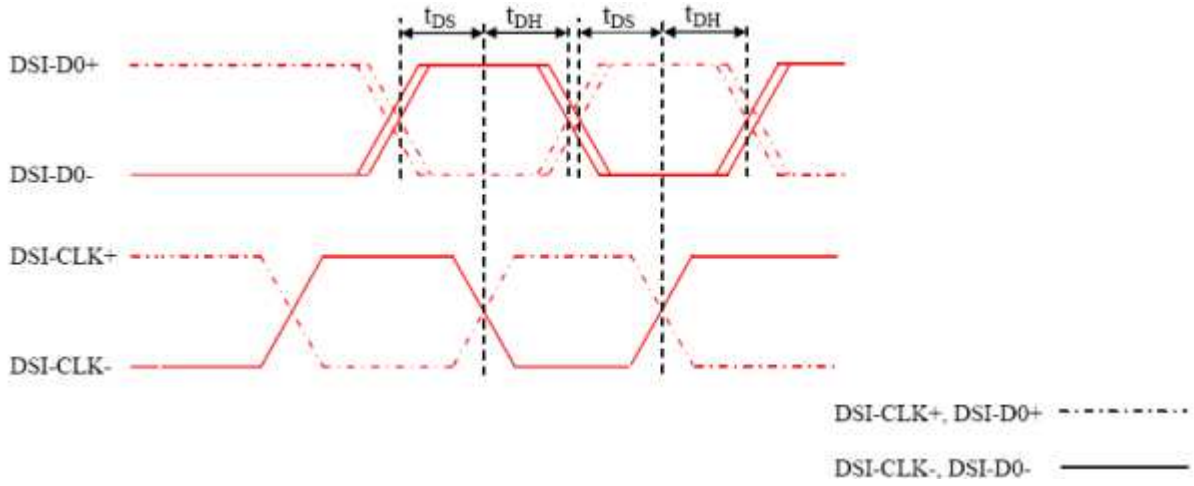
7.4.5 MIPI DSI Timing

High Speed Mode – Clock Channel Timing



Signal	Symbol	Parameter	MIN	MAX	Unit	Description
DSI-DATA_P/N	2xUI INST	Double UI instantaneous	4	25	ns	
DSI-DATA_P/N	UI INSTA ,UI INSTB	UI instantaneous Half	2	12.5	ns	

High Speed Mode – Data Clock Channel Timing



Signal	Symbol	Parameter	MIN	MAX	Unit	Description
DSI-DATA_P/N	tDS	Data to Clock Setup time	300	-	ps	
DSI-DATA_P/N	tDH	Clock to Data Hold Time	300	-	ps	

8 INTERFACE

8.1.. MPU Interface Type Selection

For communicating with MCU, ST7796 supports 8-/9-/16-/18-bit 8080-series interface, SPI, MIPI and MDDI. Selection of these interfaces are set by IM[2:0] pins as shown below.

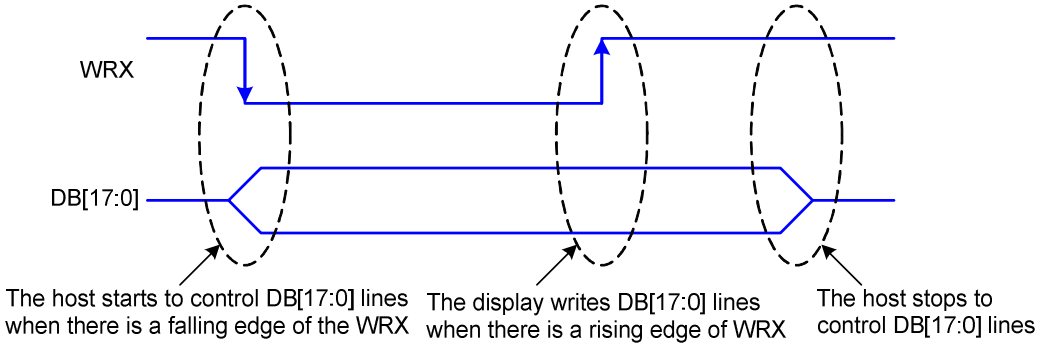
IM2	IM1	IM0	Interface	Read Back Data Bus Selection
0	0	0	8080-series MCU 18-bit	DB[17:0]
0	0	1	8080-series MCU 9-bit	DB[8:0]
0	1	0	8080-series MCU 16-bit	DB[15:0]
0	1	1	8080-series MCU 8-bit	DB[7:0]
1	0	0	MDDI	MDDI_DATA_P/N, MDDI_STB_P/N
1	0	1	Reserve	--
1	1	0	MIPI	MIPI_DATA_P/N, MIPI_CLOCK_P/N
1	1	1	Serial Peripheral Interface(4-Line SPI)	SDA, SDO

Interface Type Selection

8.2.. 8080-Series MCU Interface

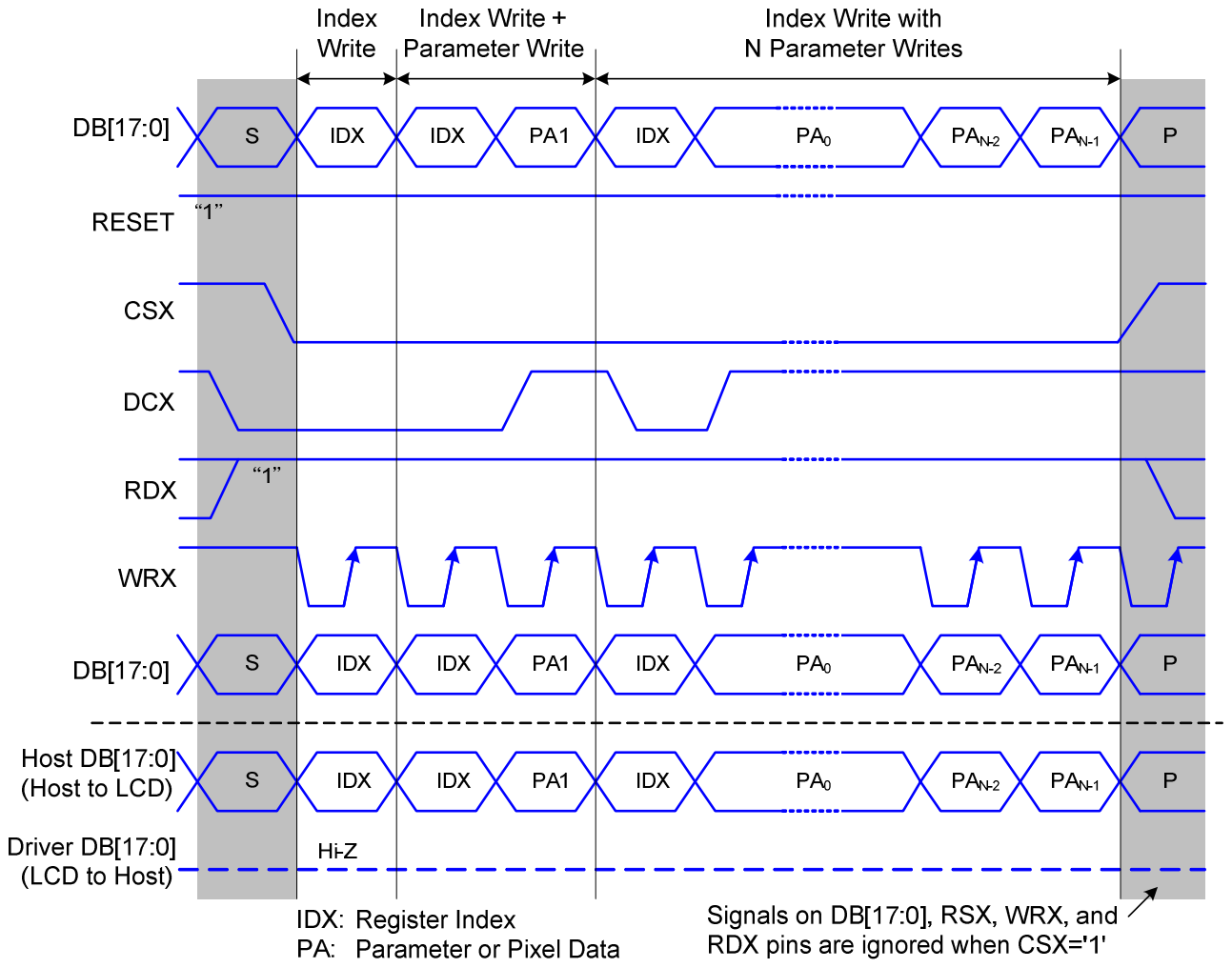
8.2.1 8080-Series MCU Write Cycle Sequence

The write cycle means that the host writes information (register index / parameter) to the display via the interface. Each write cycle (WRX high-low-high sequence) consists of 3 control signals (DCX, RDX, WRX) and data signals (DB[17:0]). DCX is a control signal, which tells if the data is an index or a parameter. The data signals represent index number if the signal is low (DCX='0') and vice versa the data signals represent parameter (DCX='1').



Note: WRX is an synchronized signal (It can be stopped).

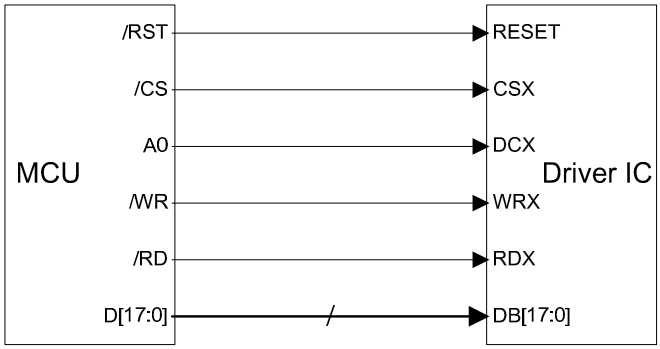
8080-Series WRX Protocol



8080-Series Parallel Bus Protocol, Write to Register or Display RAM

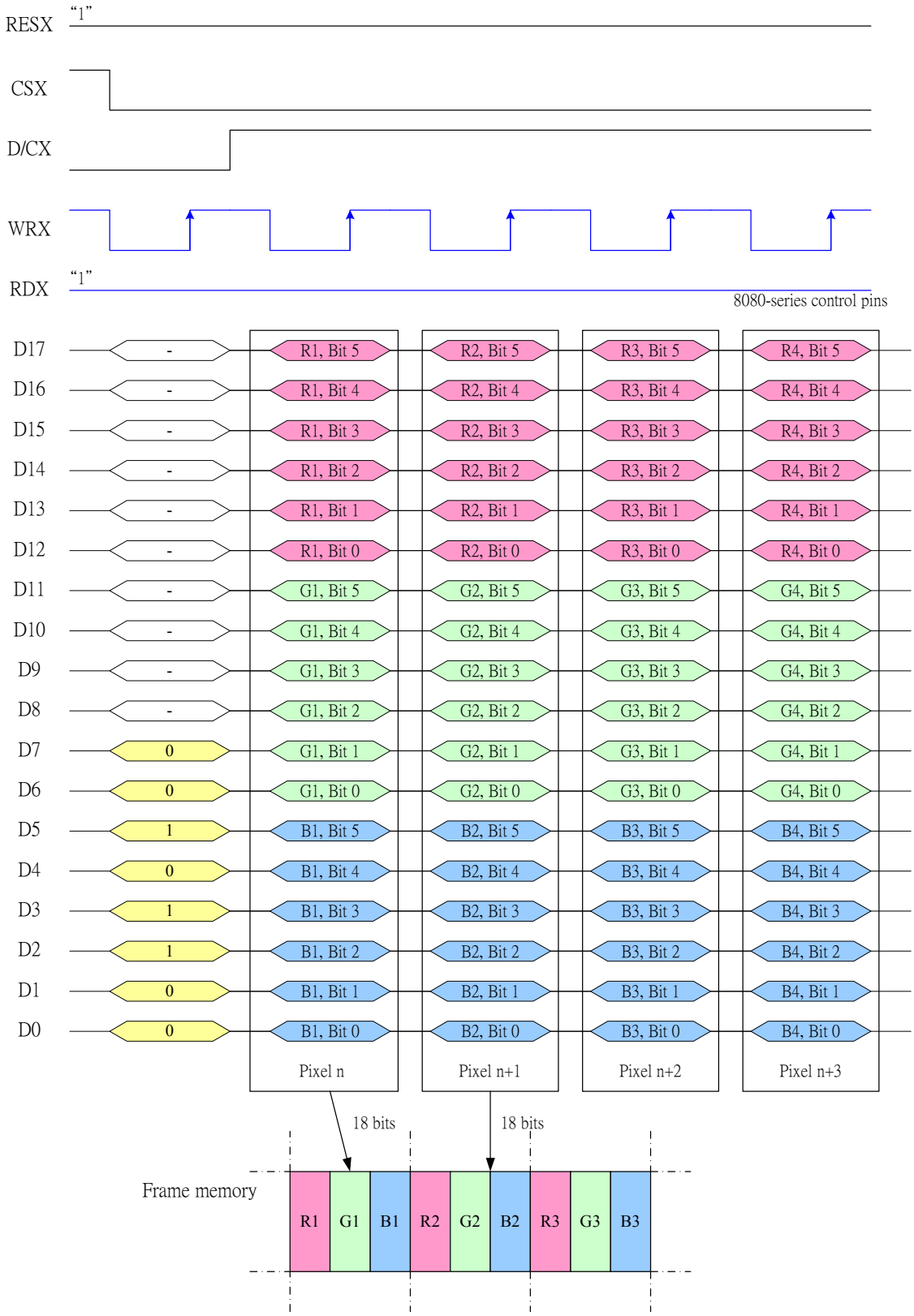
8.2.2 18-bit 8080-Series Interface Write Format

The 18-bit 8080-series interface is selected by setting the IM [2:0] ="000".



18-bit 8080-Series Interface Connection

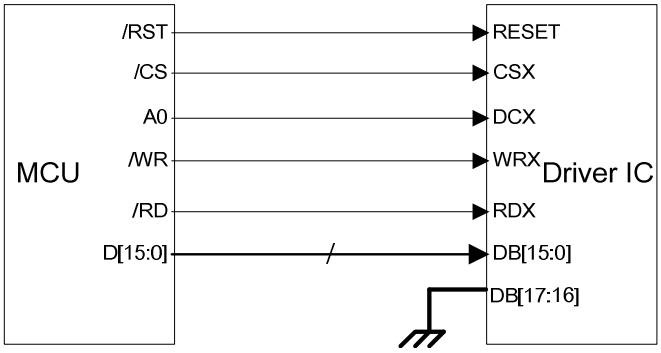
This mode accepts only 262k colors format in display. In this interface, index, parameter, and pixel-data should be written according to the following figures.



18-bit 8080-Series Interface Data Format (Index/Parameter/Pixel Data Write)

8.2.3 16-bit 8080-Series Interface Write Format

The 16-bit 8080-series interface is selected by setting IM[2:0]=”010”.

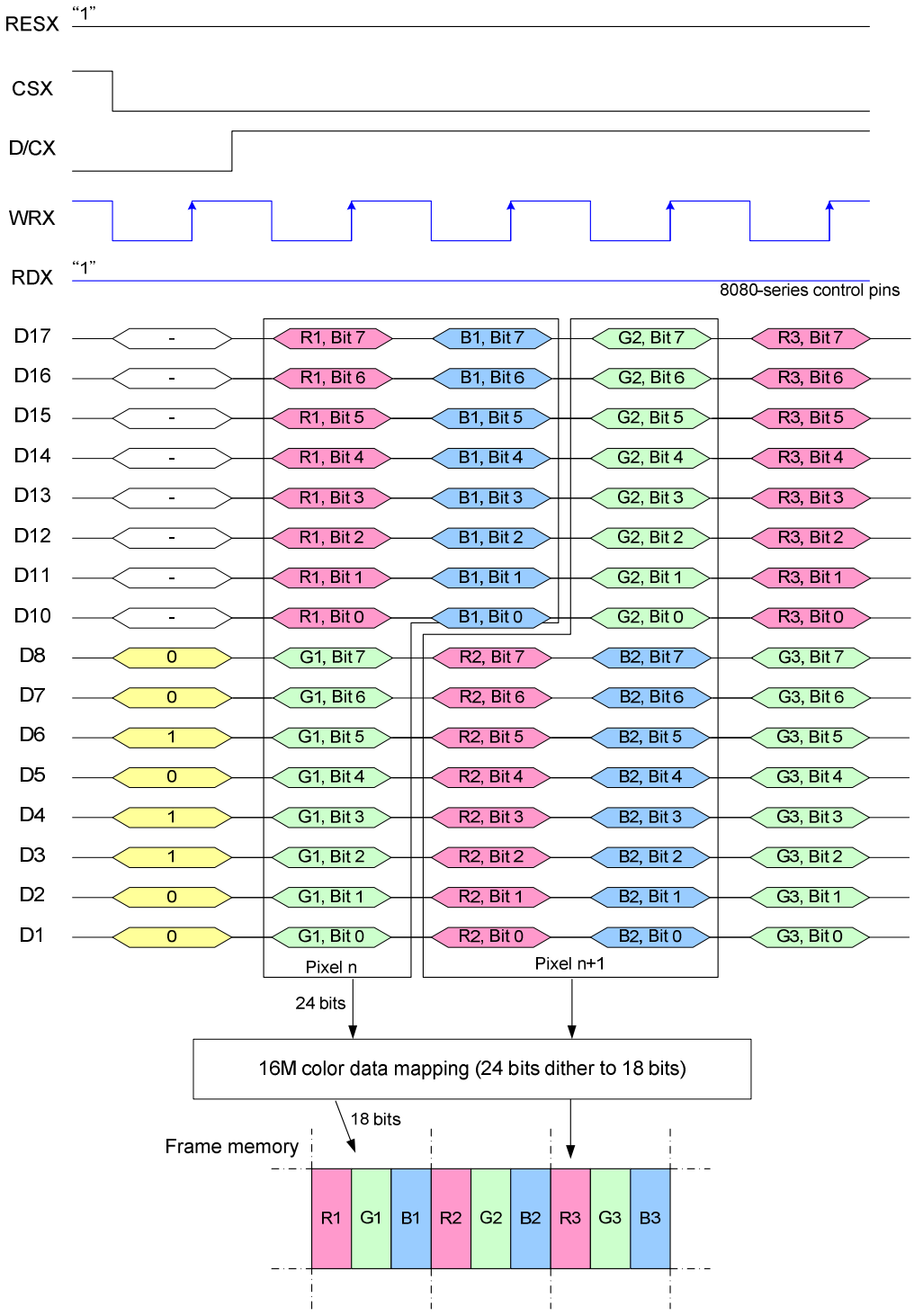


16-bit 8080-Series Interface Connection

ST7796 accepts 262k-color or 65k-color format in this mode. When the 262k-color format is used, two transfers for each pixel are required.

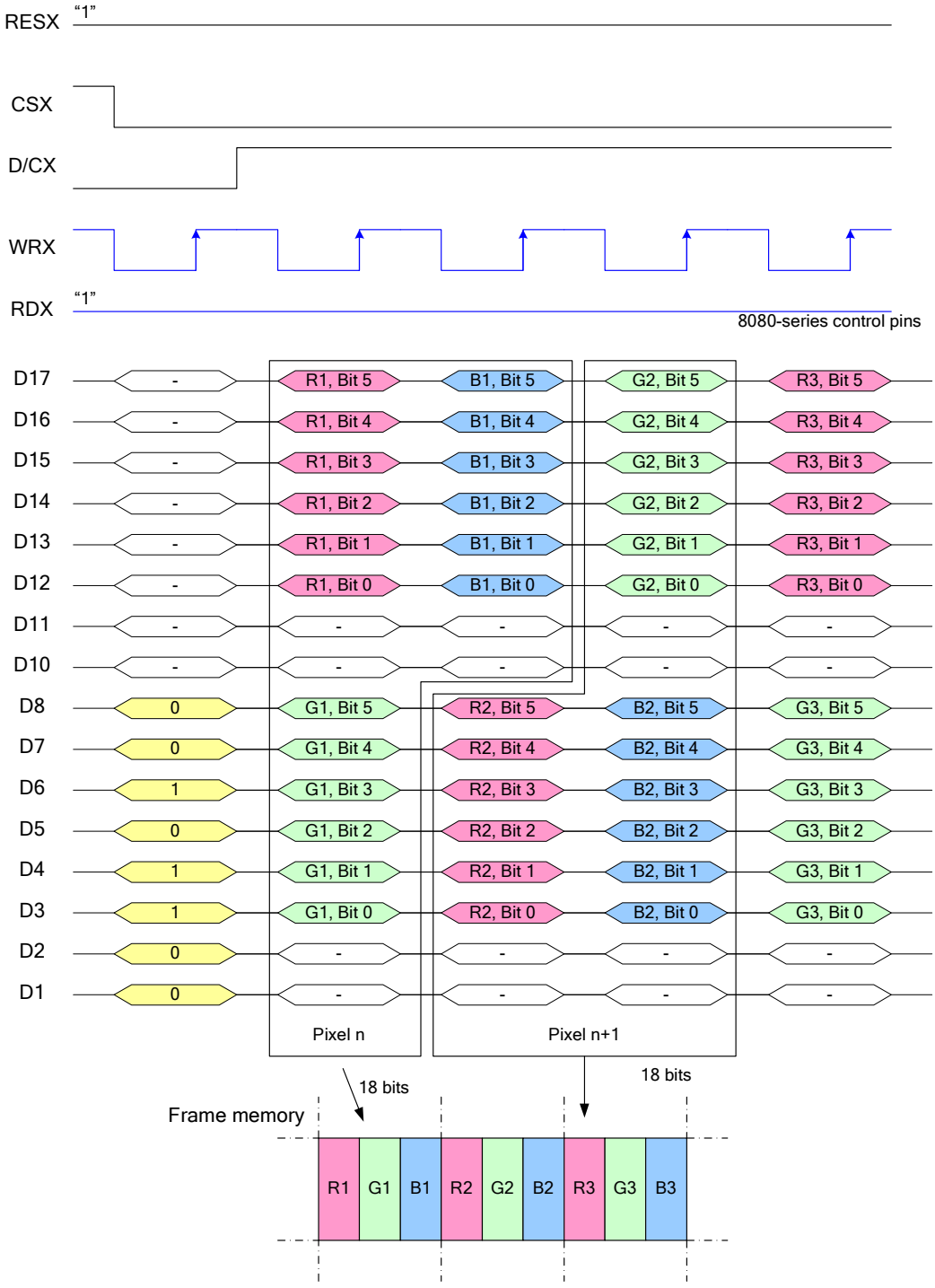
16-bit data bus for 24-bit/pixel (RGB-8-8-8-bit input), 16M-Colors, 3Ah="07h")

There are 2 pixels (6 sub-pixels) per 3 bytes



16-bit data bus for 18-bit/pixel (RGB-6-6-6-bit input), 262K-Colors, 3Ah="06h")

There are 2 pixels (6 sub-pixels) per 3 bytes

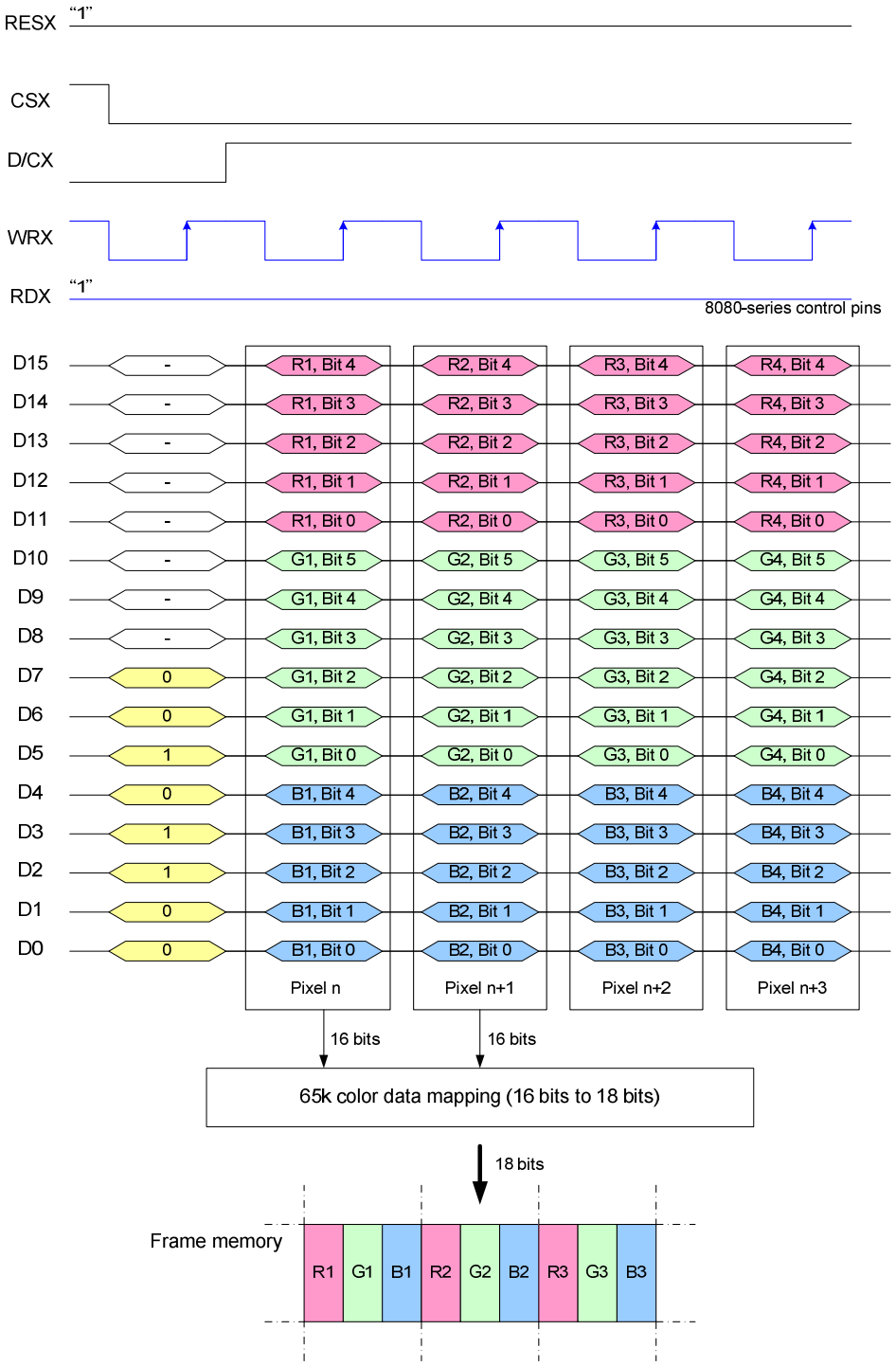


Note 1: The data order is as follows, MSB=D17, LSB=D1 and picture data is MSB=Bits 5, LSB=Bit 0 for Red, Green and Blue data.

Note 2: 3-times transfer is used to transmit 1 pixel data with the 18-bit color depth information.

Note 3: '-' = Don't care - Can be set to '0' or '1'

16-bit data bus for 16-bit/pixel (RGB 5-6-5-bit input) 65K-Color, 3Ah="05h)



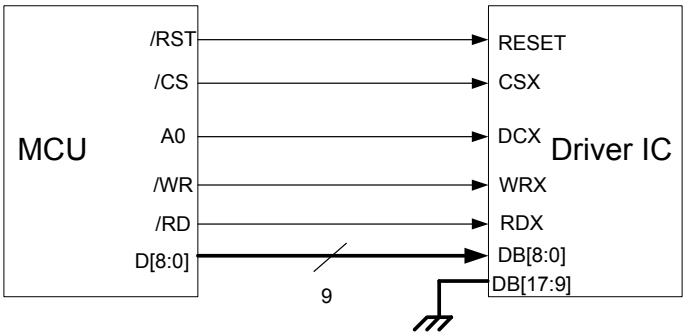
Note 1: The data order is as follows, MSB=D15, LSB=D0 and picture data is MSB=Bit 5, LSB=Bit 0 for Green, and MSB=Bit 4, LSB=Bit 0 for Red and Blue data.

Note 2: 1-times transfer (D15 to D0) is used to transmit 1 pixel data with the 16-bit color depth information.

Note 3: '-' = Don't care - Can be set to '0' or '1'

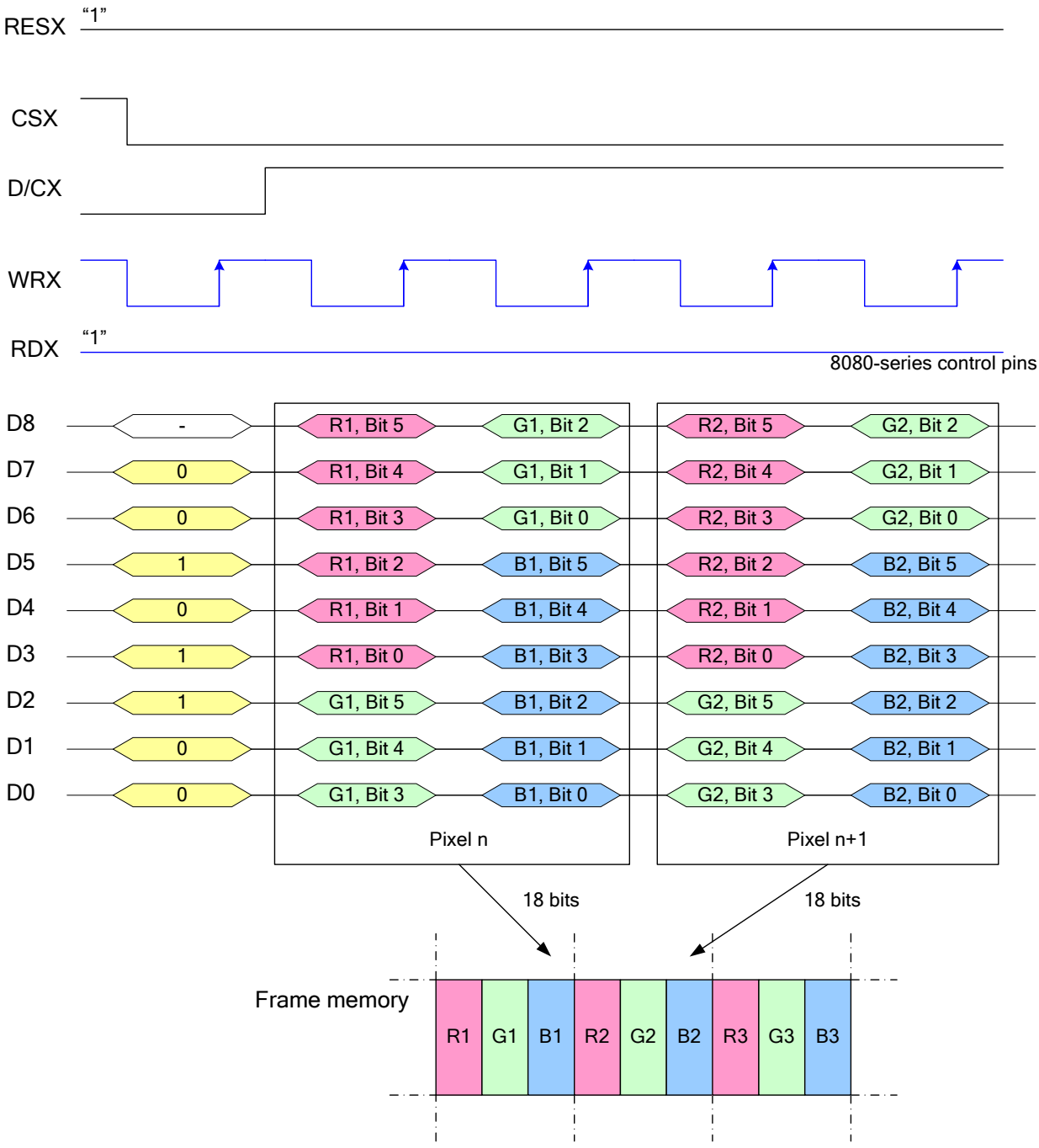
8.2.4 9-bit 8080-Series Interface Write Format

The 9-bit 8080-series interface is selected by setting the IM [2:0] = “011” and the DB [17:9] pins are used to transfer data. The display data is divided into upper part and lower part (9-bit for each part), and the upper part is transferred first. The unused DB [8:0] pins must be tied to either VDDI or DGND.



9-bit 8080-Series Interface Connection

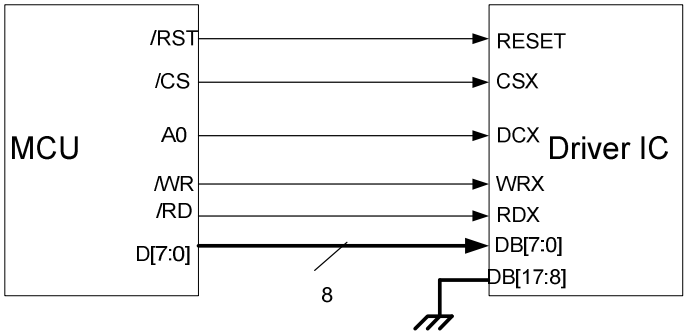
9-bit data bus for 18-bit/pixel (RGB 6-6-6-bit input), 262K-Colors, 3Ah="06h"



9-bit 8080-Series Interface Data Format (Index/Parameter/Pixel Data Write)

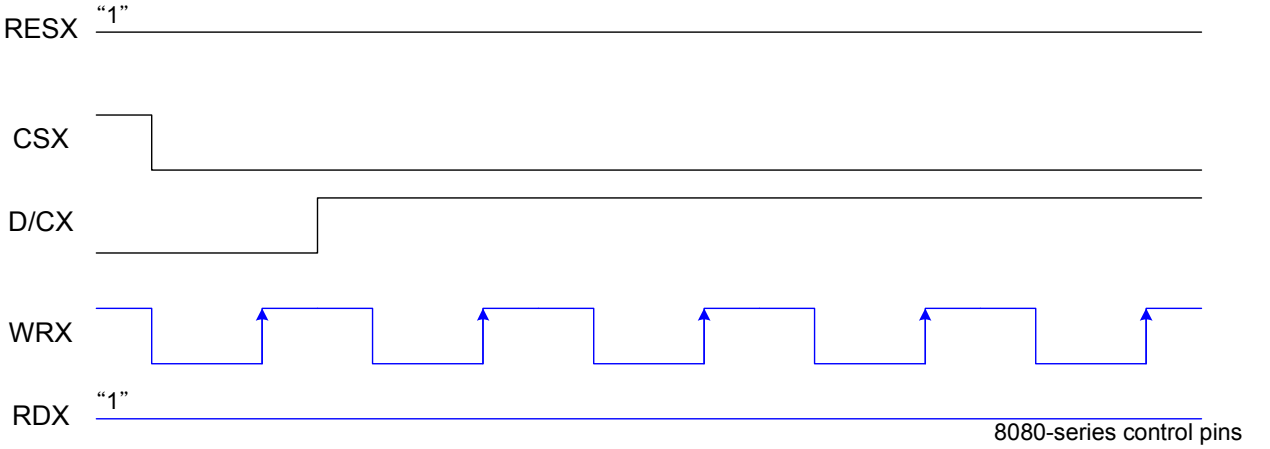
8.2.5 8-bit 8080-Series Interface Write Format

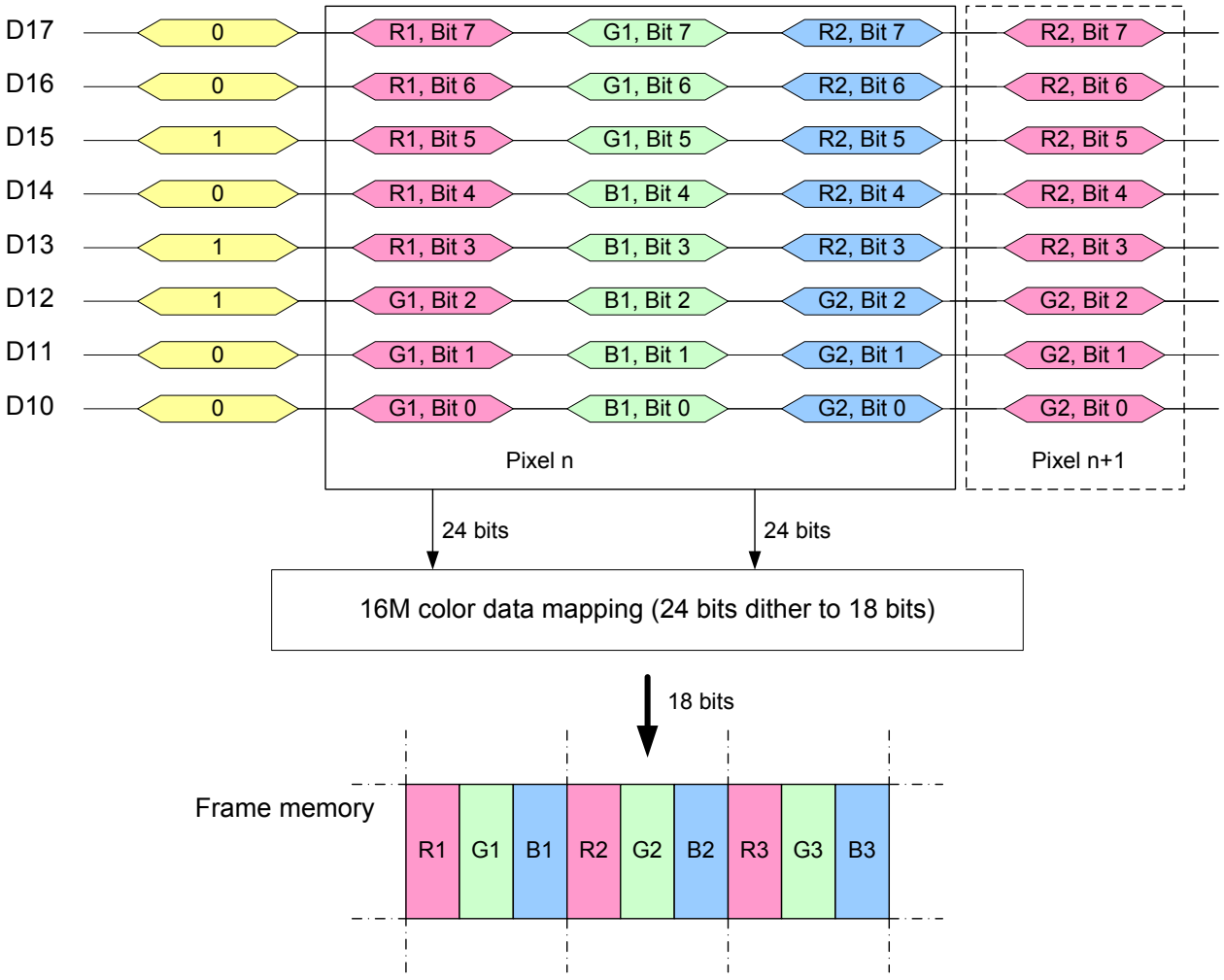
The 8080 8-bit interface is selected by setting the IM [2:0] as “011” and the DB [7:0] pins are used to transfer data. The mode accepts 262k-color or 65k-color format. The display data is divided into upper byte and lower byte, and the upper byte is transferred first. The written data is expanded into 18-bit internally (see the figure below) and then written into DRAM. The unused DB [17:8] pins must be tied to either VDDI or DGND.



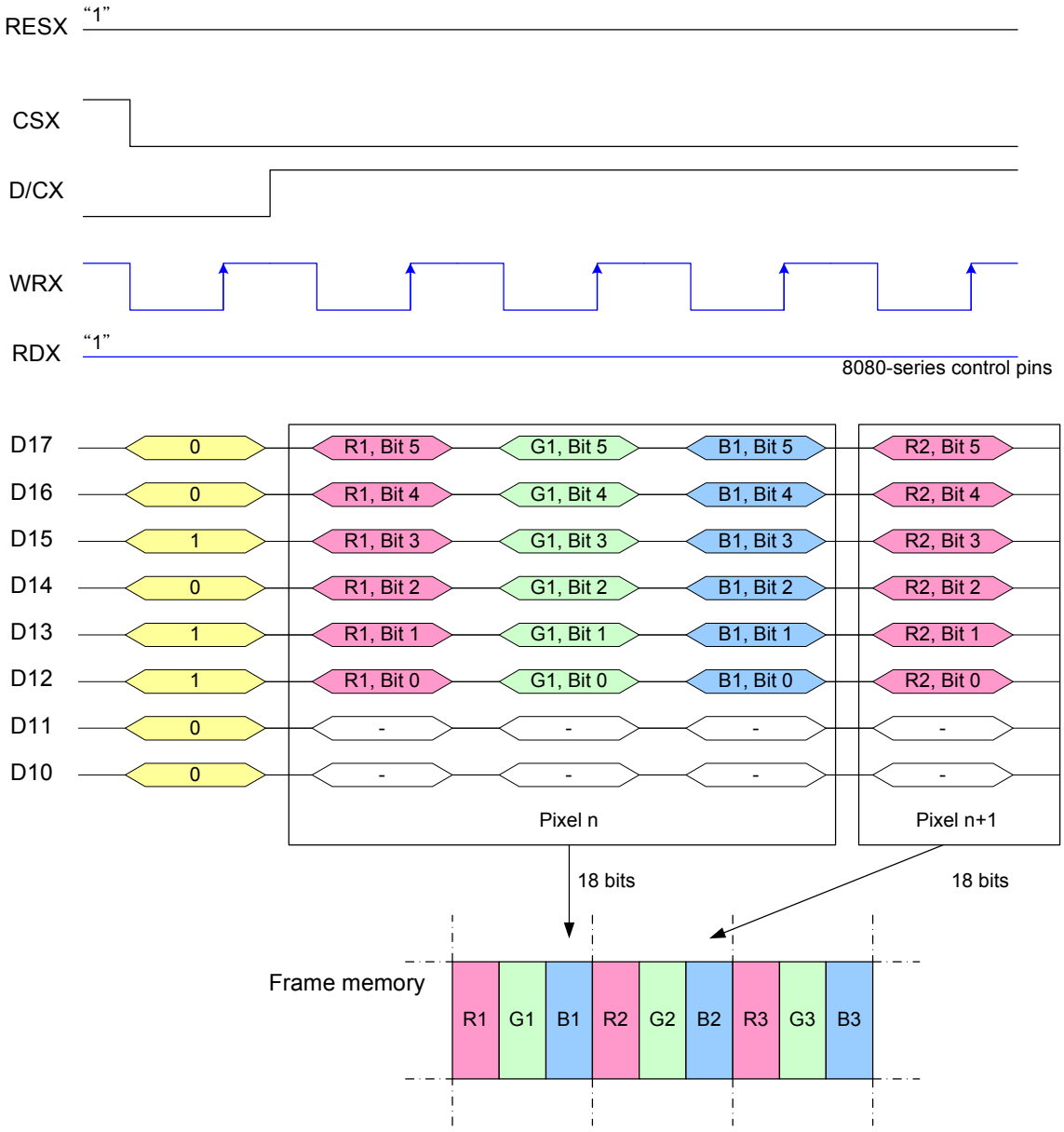
8-bit 8080-Series Interface Connection

8-bit data bus for 24-bit/pixel (RGB 8-8-8-bit input), 16M-Colors, 3Ah="07h"



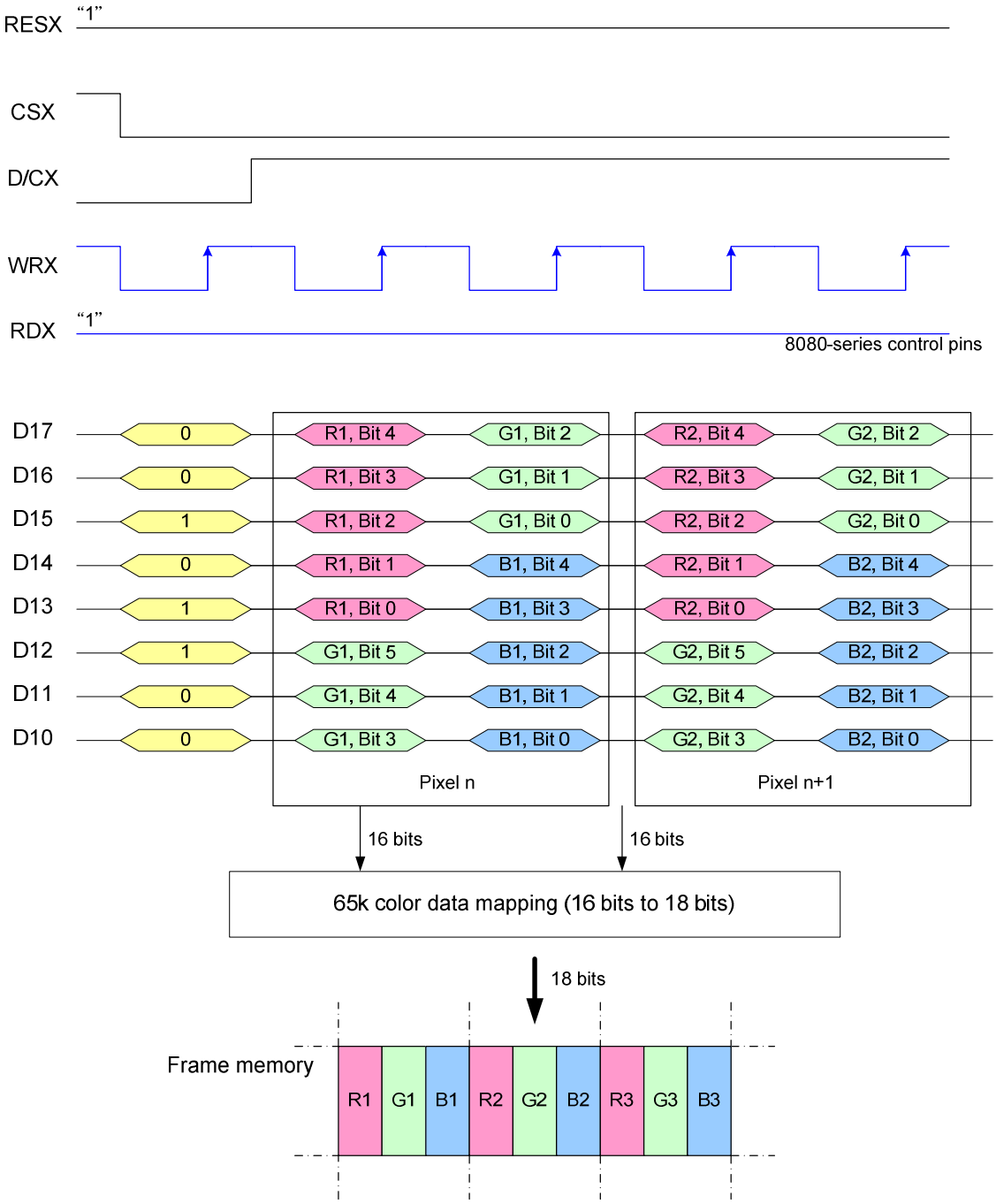


8-bit data bus for 18-bit/pixel (RGB 6-6-6-bit input), 262K-Colors, 3Ah="06h"



8-bit 8080-Series Interface Data Format (Index/Parameter/Pixel Data Write)

8-bit data bus for 16-bit/pixel (RGB 5-6-5-bit input), 65K-Colors, 3Ah="05h"



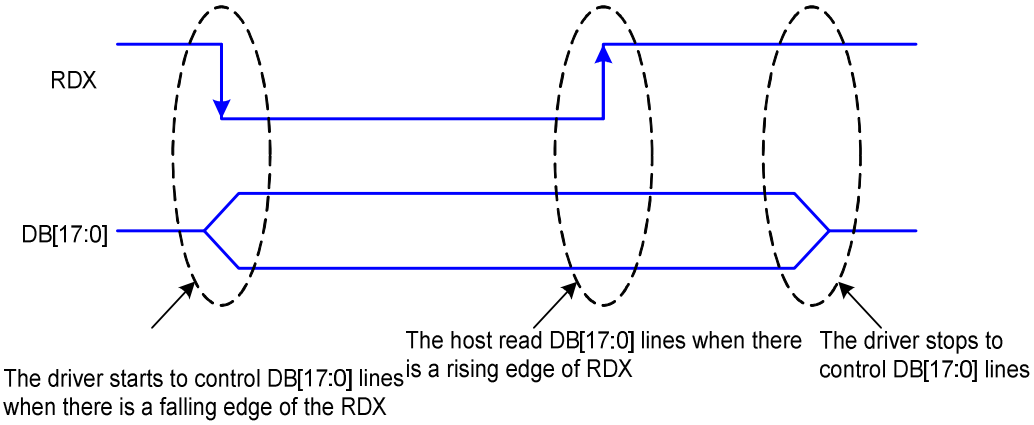
Note 1: The data order is as follows, MSB=D17, LSB=D10 and picture data is MSB=Bit 5, LSB=Bit 0 for Green, and MSB=Bit 4, LSB=Bit 0 for Red and Blue data.

Note 2: 2-times transfer transmit 1 pixel data with the 16-bit color depth information.

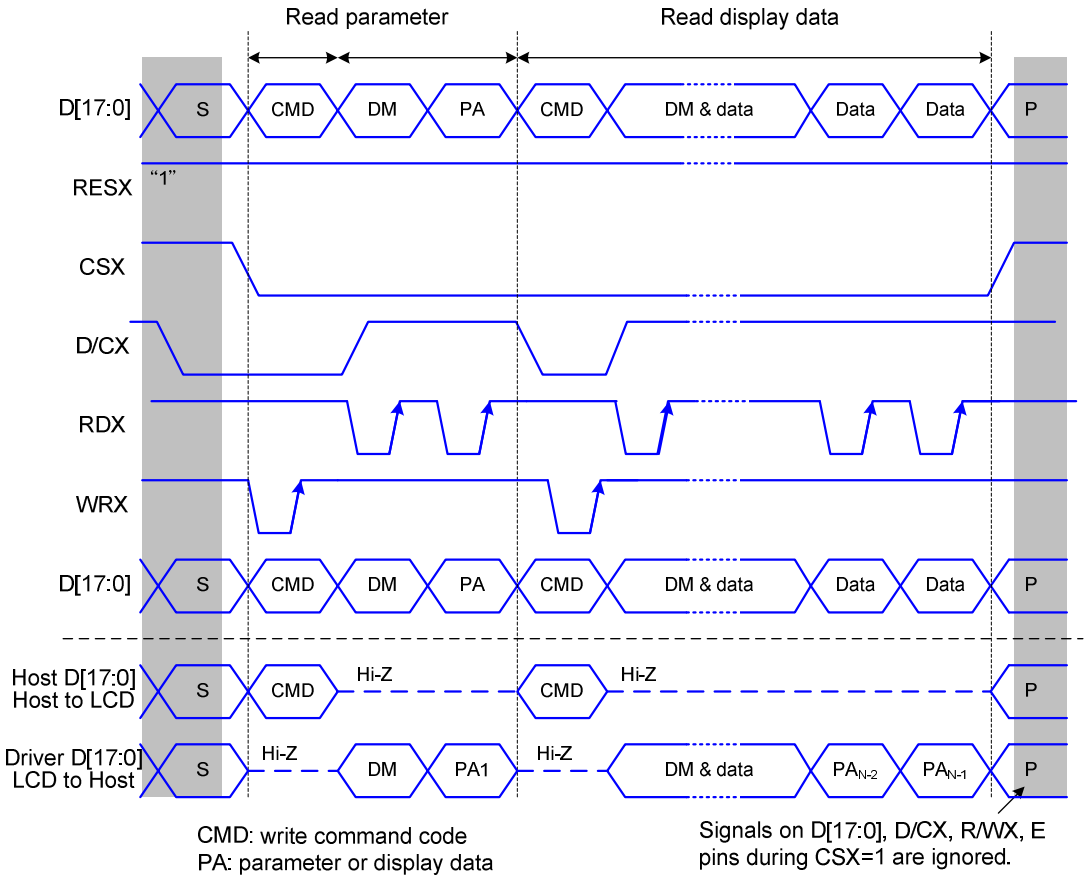
Note 3: '-' = Don't care - Can be set to '0' or '1

8.2.6 8080-series MCU Read Cycle Sequence

The read cycle (RDX high-low-high sequence) means that the host reads information from display via interface. The driver sends data (DB [17:0]) to the host when there is a falling edge of RDX and the host reads data when there is a rising edge of RDX.

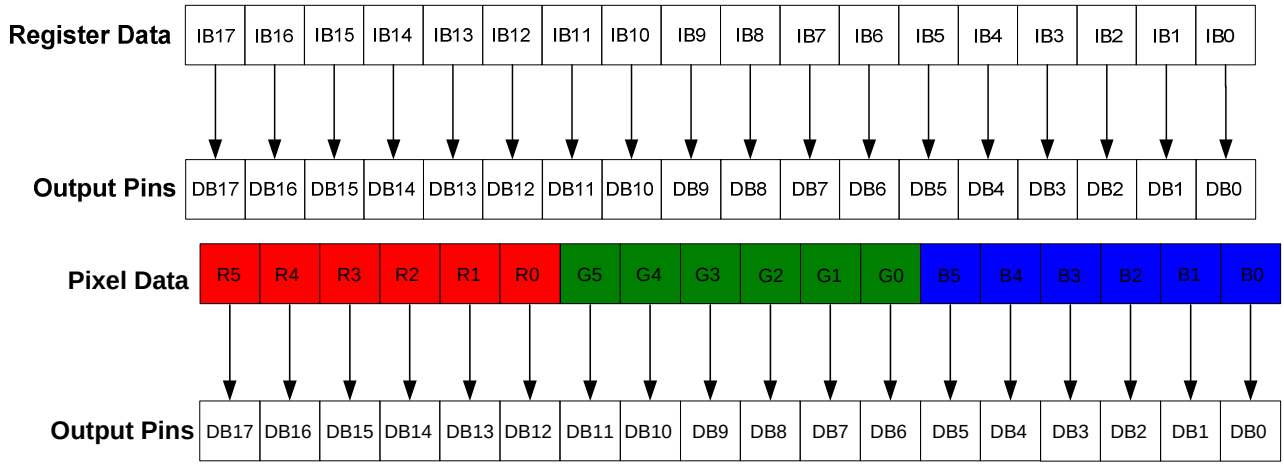


8080-Series RDX Protocol



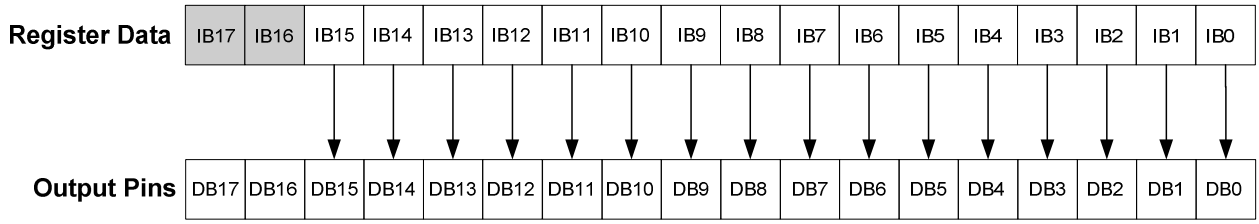
8080-Series Parallel Bus Protocol, Read from Register or Display RAM

8.2.7 18-bit 8080-Series Interface Read Format



18-bit 8080-Series Interface Data Format (Register/Pixel Data Read)

8.2.8 16-bit 8080-Series Interface Read Format



16-bit data bus for 18-bit/pixel, 262K-Colors, 3Ah="06h"

D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0	BUS width
X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	Dummy Read
R5	R4	R3	R2	R1	R0	X	X	G5	G4	G3	G2	G1	G0	X	X	3 Transfer 2 Pixel
B5	B4	B3	B2	B1	B0	X	X	R5	R4	R3	R2	R1	R0	X	X	
G5	G4	G3	G2	G1	G0	X	X	B5	B4	B3	B2	B1	B0	X	X	

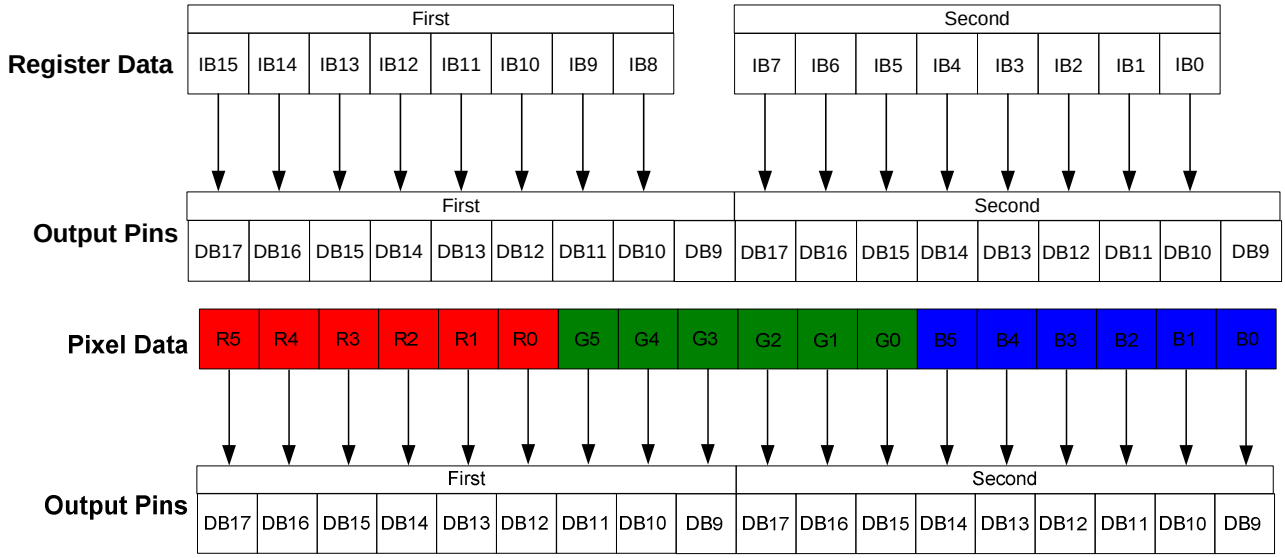
16-bit 8080-Series Interface Data Format 262K (Register/Pixel Data Read)

16-bit data bus for 16-bit/pixel, 65K-Colors, 3Ah="05h"

D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0	BUS width
X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	Dummy Read
R4	R3	R2	R1	R0	G5	G4	G3	G2	G1	G0	B4	B3	B2	B1	B0	1 Transfer 1 Pixel

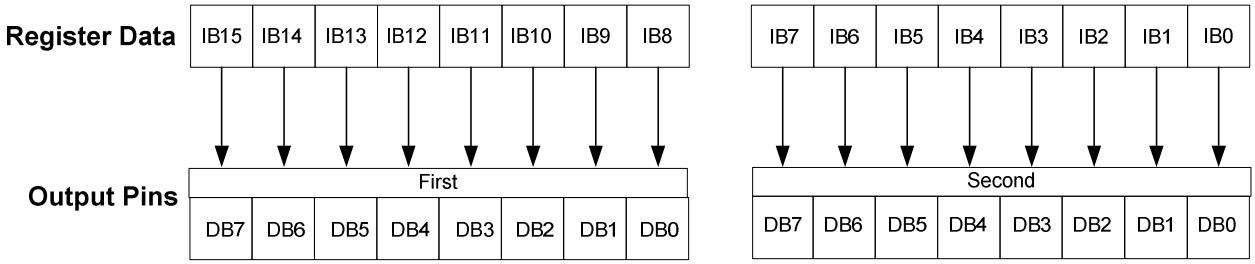
16-bit 8080-Series Interface Data Format 65K (Register/Pixel Data Read)

8.2.9 9-bit 8080-Series Interface Read Format



9-bit 8080-Series Interface Data Format (Register/Pixel Data Read)

8.2.10 8-bit 8080-Series Interface Read Format



8-bit data bus for 18-bit/pixel, 262K-Colors, 3Ah="06h"

D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0	BUS width
X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	Dummy Read
X	X	X	X	X	X	X	X	R5	R4	R3	R2	R1	R0	X	X	3 Transfer 1 Pixel
X	X	X	X	X	X	X	X	G5	G4	G3	G2	G1	G0	X	X	
X	X	X	X	X	X	X	X	B5	B4	B3	B2	B1	B0	X	X	

8-bit 8080-Series Interface Data Format (Register/Pixel Data Read)

8-bit data bus for 16-bit/pixel, 65K-Colors, 3Ah="05h"

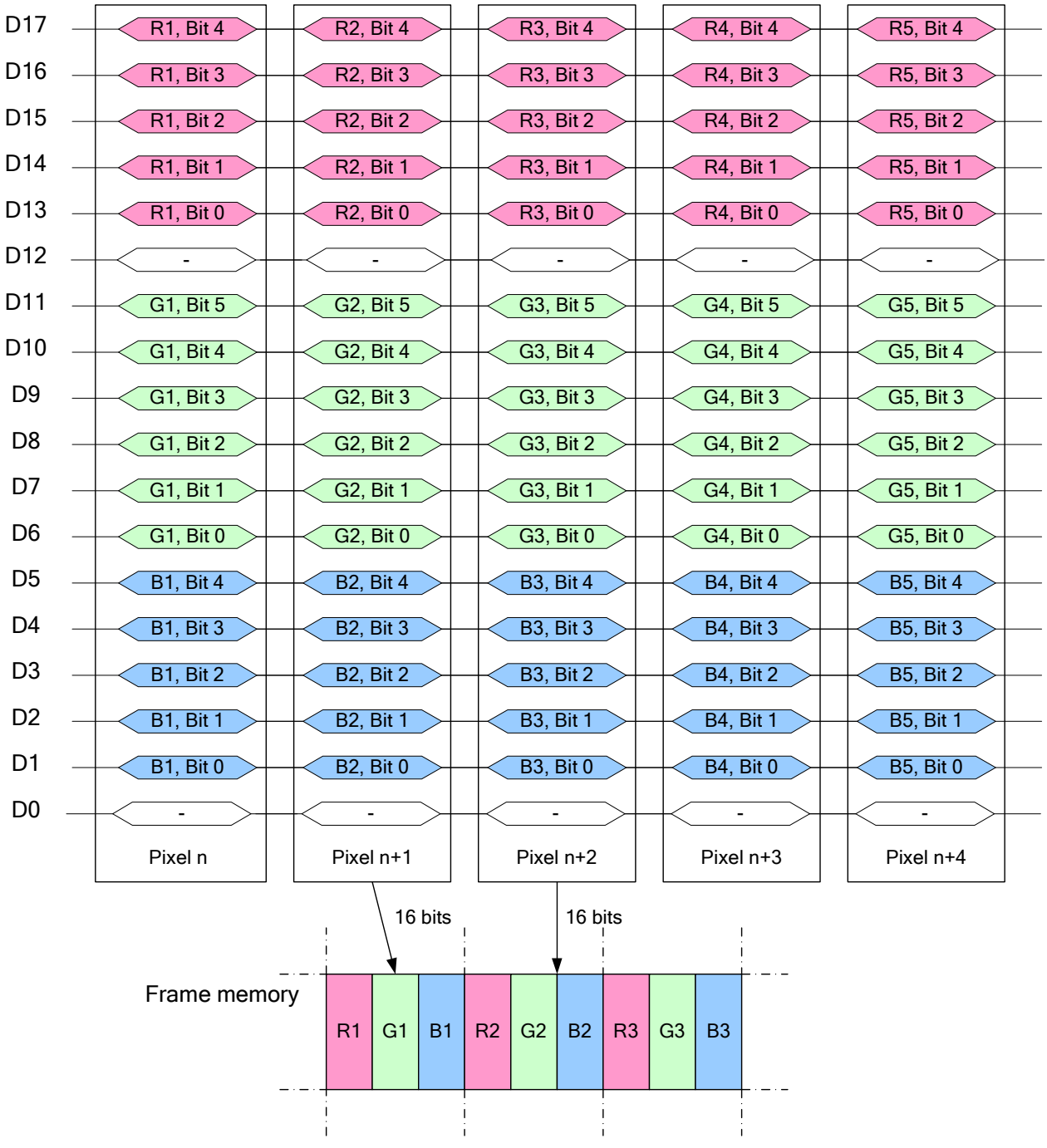
D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0	BUS width
X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	Dummy Read
X	X	X	X	X	X	X	X	R4	R3	R2	R1	R0	G5	G4	G3	2 Transfer 1 Pixel
X	X	X	X	X	X	X	X	G2	G1	G0	B4	B3	B2	B1	B0	

8.3.. RGB Interface

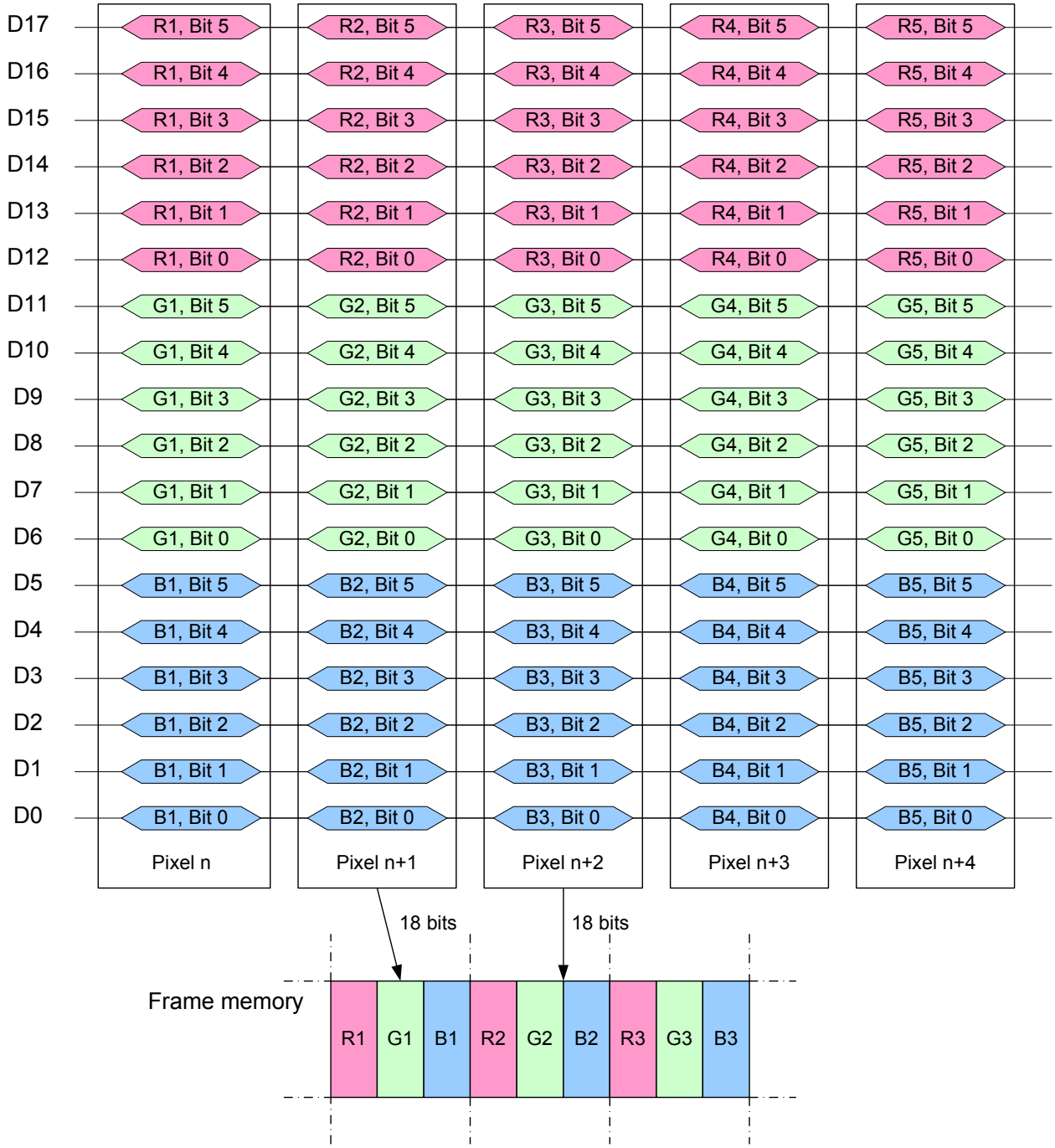
8.3.1 RGB Color Format

ST7796 supports two kinds of RGB interface, DE mode and HV mode, and 16bit/18bit data format. When DE mode is selected and the VSYNC, HSYNC, DOTCLK, DE, D[17:0] pins can be used; when HV mode is selected and the VSYNC, HSYNC, DOTCLK, D[17:0] pins can be used.When using RGB interface, only serial interface can be selected.

Write data for 16-bit/pixel (RGB 5-6-5-bit input), 65K-Colors

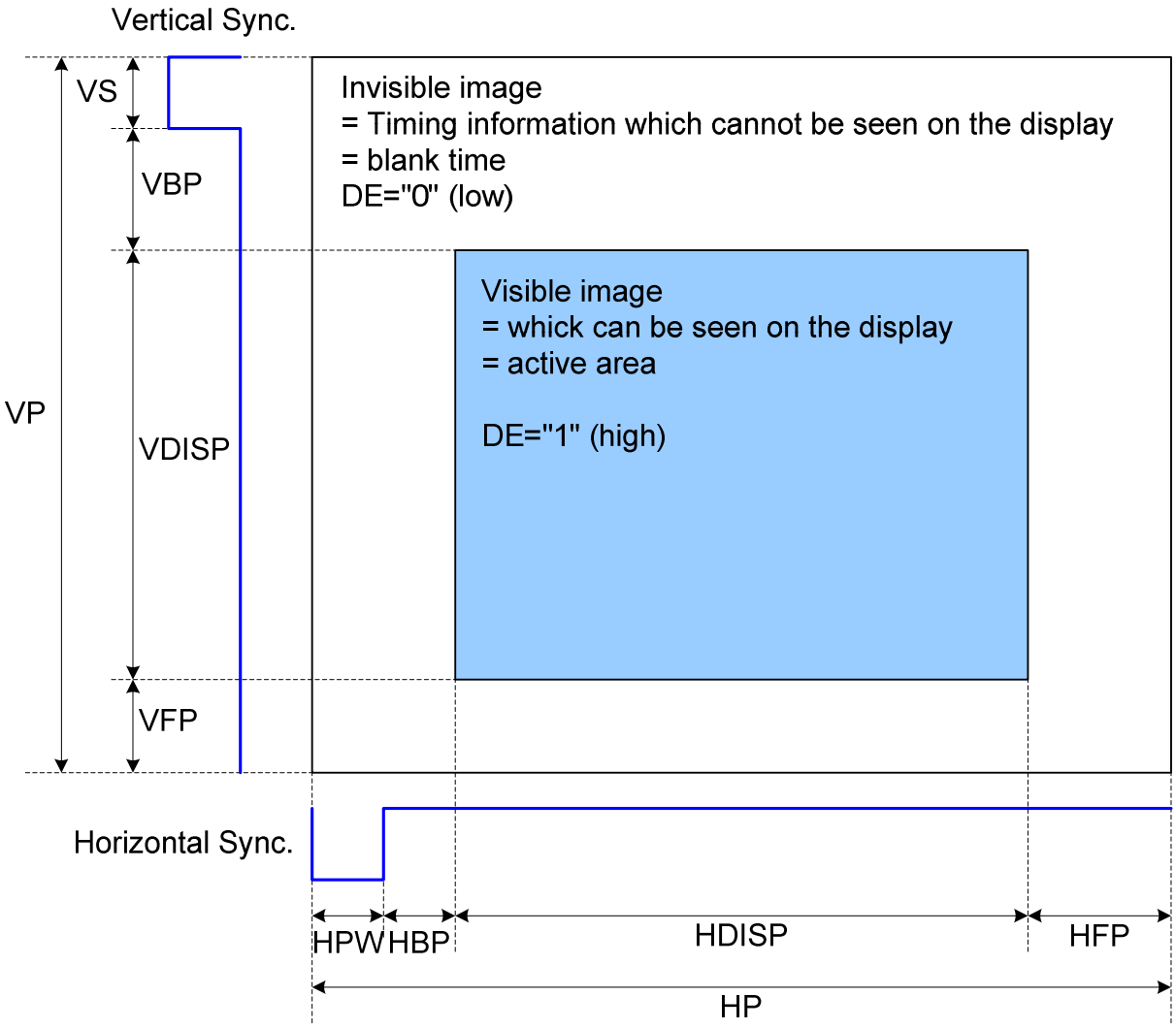


Write data for 18-bit/pixel (RGB 6-6-6-bit input), 262K-Colors



8.3.2 RGB Interface Definition

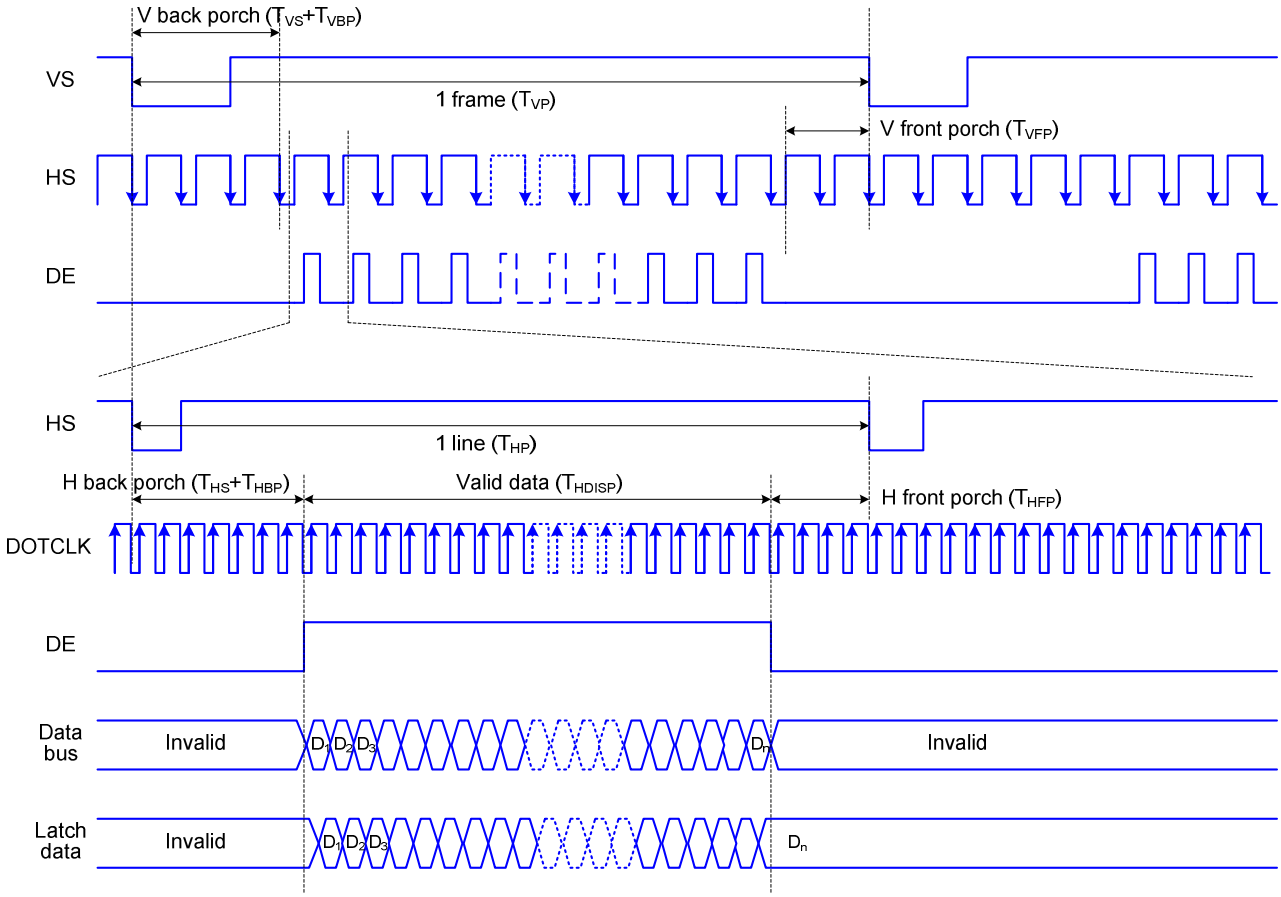
The display operation via the RGB interface is synchronized with the VSYNC, HSYNC, and DOTCLK signals. The data can be written only within the specified area with low power consumption by using window address function. The back porch and front porch are used to set the RGB interface timing.



DRAM Access Area by RGB Interface

8.3.3 RGB Interface Timing

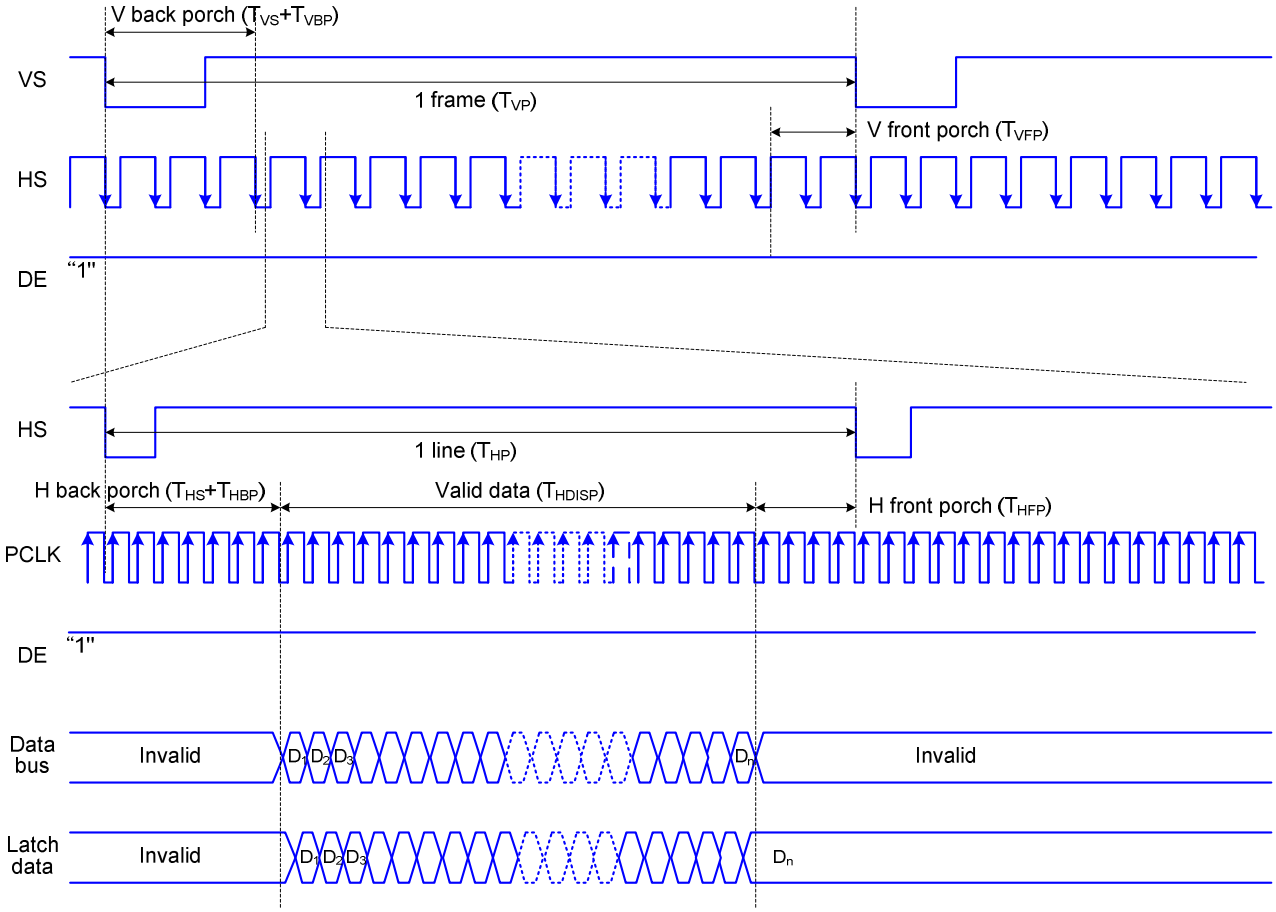
The timing chart of RGB interface DE mode is shown as follows.



Note: The setting of front porch and back porch in host must match that in IC as this mode.

Timing Chart of Signals in RGB Interface DE Mode

The timing chart of RGB interface HV mode is shown as follows.



Timing chart of RGB interface HV mod

The following are the functions not available in RGB Input Interface mode.

Function	RGB Interface	I80 System Interface
Partial display	Not available	Available
Scroll function	Not available	Available
Interlaced scan	Not available	Available
Graphics operation function	Not available	Available

VSYNC, HSYNC, and DOTCLK signals must be supplied during a display operation period. In RGB interface mode, the panel controlling signals are generated from DOTCLK, not the internal clock generated from the internal oscillator.

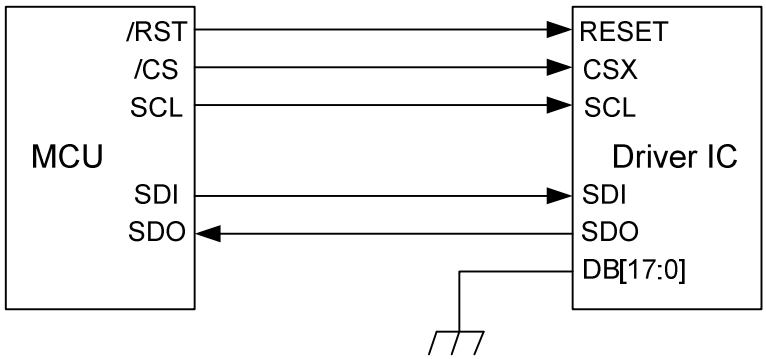
When switching between the internal operation mode and the external display interface operation mode, follow the sequences below in setting instruction.

In RGB interface mode, the front porch period continues until the next VSYNC input is detected after drawing one frame.

In RGB interface mode, a RAM address is set in the address counter every frame on the falling edge of VSYNC.

8.4.. Serial Peripheral Interface (SPI)

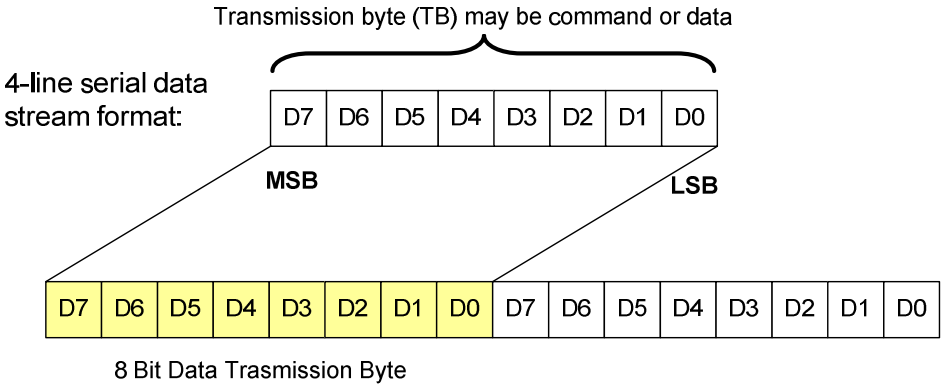
The Serial Peripheral Interface (SPI) is selected by setting the IM[2:0] pins as “111” level. The chip select pin (CSX), the serial transfer clock pin (SCL), the serial data input pin (SDI) and the serial data output pin (SDO) are used in SPI mode. The ID pin sets the least significant bit of the identification code. The DB[17:0] pins, which are not used, must be tied to GND.



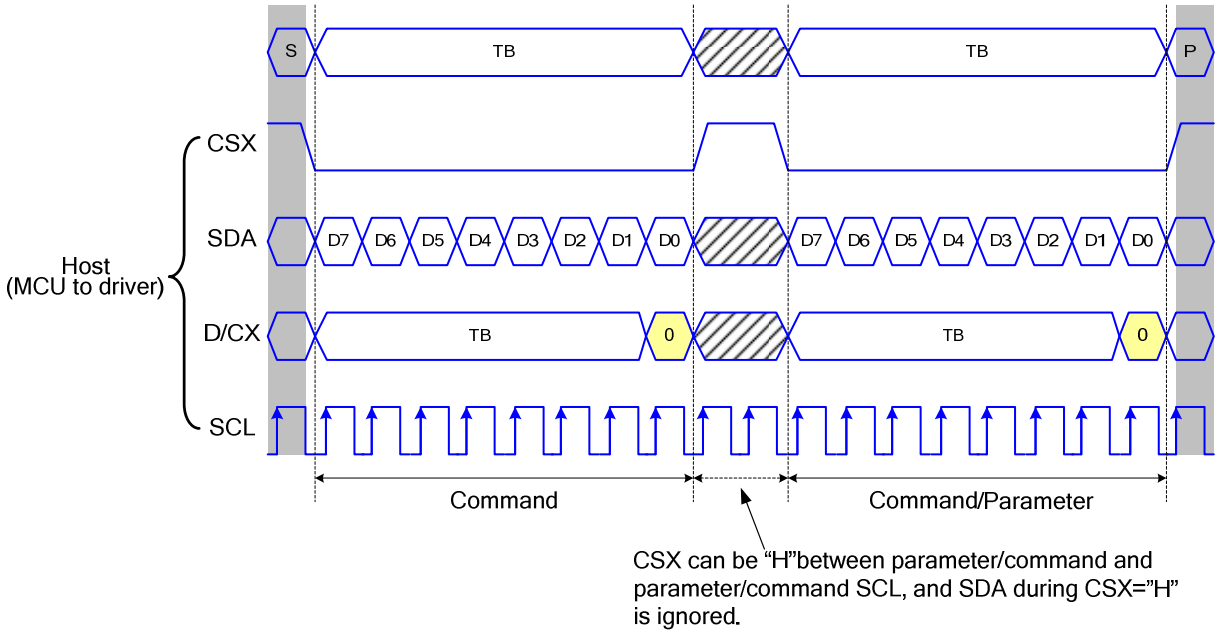
The SPI interface operation enables from the falling edge of CSX and ends of data transfer on the rising edge of CSX. The start byte is transferred to start the SPI interface and the read/write operation and RS information are also included in the start byte. When the start byte is matched, the subsequent data is received by ST7796.

8.4.1 Write Sequence

The write mode of the interface means the host writes commands and data to ST7796. The 4-lines serial data packet contains a data/command and a transmission byte. If D/CX is “low”, the transmission byte is interpreted as a command byte. If D/CX is “high”, the transmission byte is stored in the display data RAM (Memory write command), or command register as parameter.



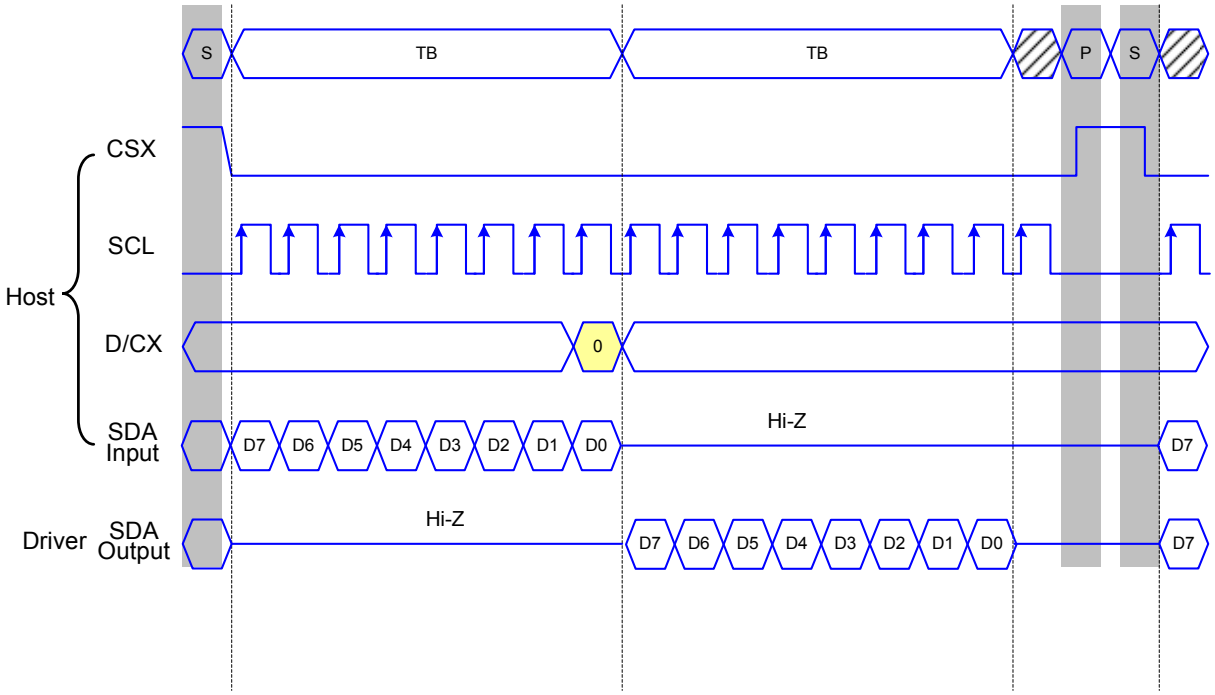
The host drives the CSX pin to low and the MSB data bit (D7) is set on SDA by the host. On the next falling edge of SCL the next bit (D6) is set on SDA. If the optional D/CX signal is used, a byte is eight read cycle long. The 4-line serial interface writes sequence described in the Figure as below.



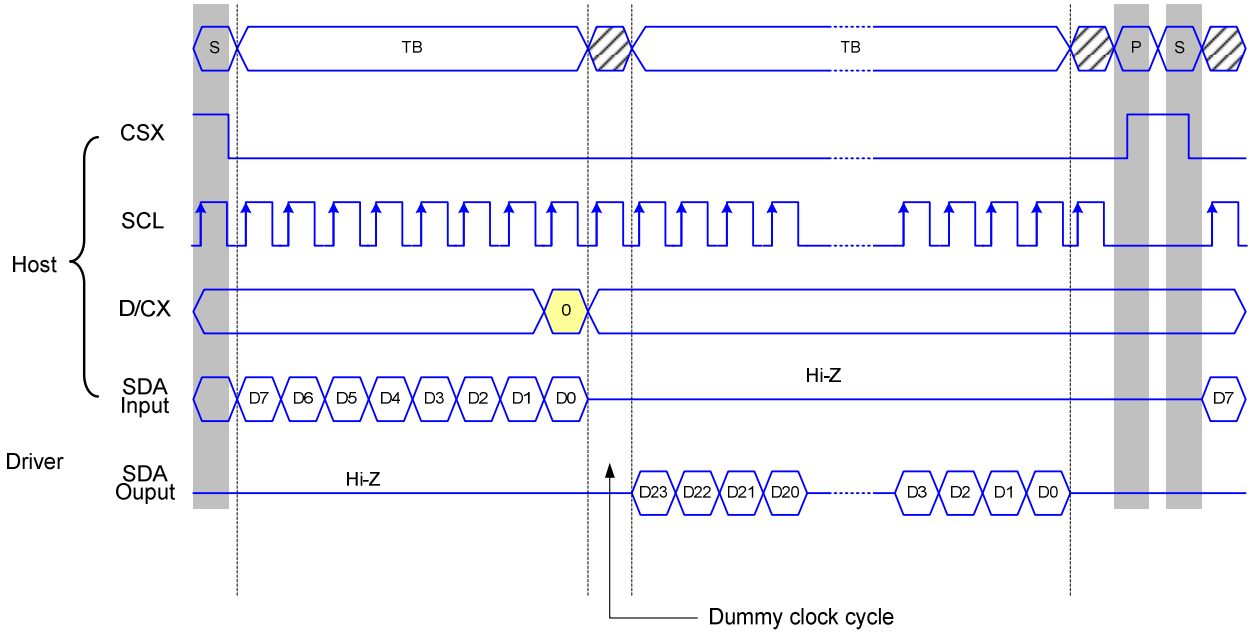
8.4.2 Read Sequence

The read mode of the interface means that the micro controller reads register value from the driver. To achieve read function, the micro controller first has to send a command (read ID or register command) and then the following byte is transmitted in the opposite direction. After that CSX is required to go to high before a new command is send (see the below figure). The driver samples the SDA (input data) at rising edge of SCL, but shifts SDA (output data) at the falling edge of SCL. Thus the micro controller is supported to read at the rising edge of SCL.

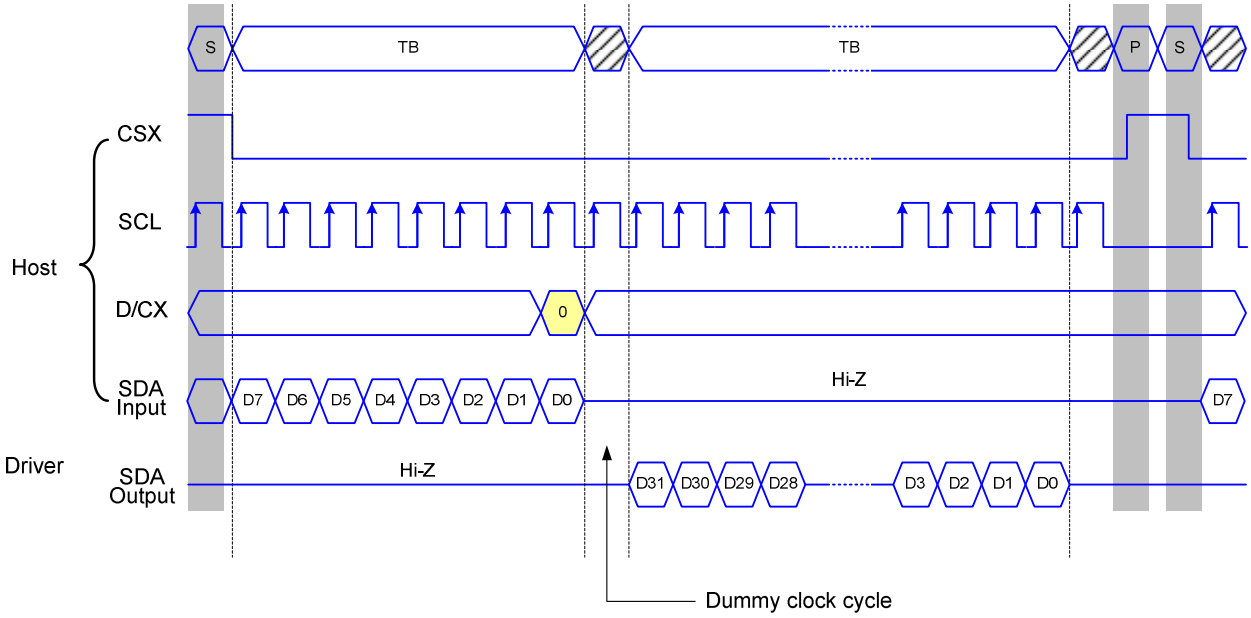
4-line serial protocol (for RDID1/RDID2/RDID3/0Ah/0Bh/0Ch/0Dh/0Eh/0Fh command: 8-bit read):



4-line serial protocol (for RDDID command: 24-bit read)



4-line serial protocol (for RDDST command: 32-bit read)



4-line serial interface read protocol

8.5.. Mobile Industry Processor Interface (MIPI)

8.5.1 Display Serial Interface (DSI)

8.5.1.1 GENERAL DESCRIPTION

The communication can be separated 2 different levels between the MCU and the display module:

1. Low level communication what is done on the interface level
2. High level communication what is done on the packet level

8.5.1.2 Interface Level Communication

The display module uses data and clock lane differential pairs for DSI (DSI-1M). Both differential lane pairs can be driven Low Power (LP) or High Speed (HS) mode.

Low Power mode means that each line of the differential pair is used in single end mode and a differential receiver is disable (A termination resistor of the receiver is disable) and it can be driven into a low power mode. High Speed mode means that differential pairs (The termination resistor of the receiver is enable) are not used in the single end mode. There are used different modes and protocols in each mode when there wanted to transfer information from the MCU to the display module and vice versa.

The State Codes of the High Speed (HS) and Low Power (LP) lane pair are defined below.

Lane Pair State	Line DC Voltage Levels		High Speed (HS)	Low Power	
	DATA_P	DATA_N	Burst Mode	CLOCK_P	CLOCK_N
HS-0	Low (HS)	High (HS)	Differential – 0	Note 1	Note1
HS-1	High (HS)	Low (HS)	Differential – 1	Note 1	Note 1
LP-00	Low (LP)	Low (LP)	Not Defined	Bridge	Space
LP-01	Low (LP)	High (LP)	Not Defined	HS – Request	Mark - 0
LP-10	High (LP)	Low (LP)	Not Defined	LP - Request	Mark - 1
LP-11	High (LP)	High (LP)	Not Defined	Stop	Note 2

Notes:

(1) Low-Power Receivers (LP-Rx) of the lane pair are checking the LP-00 state code, when the Lane Pair is in the High Speed (HS) mode.

(2 If Low-Power Receivers (LP-Rx) of the lane pair recognizes LP-11 state code, the lane pair returns to LP-11 of the Control Mode.

DSI-CLOCK Lanes

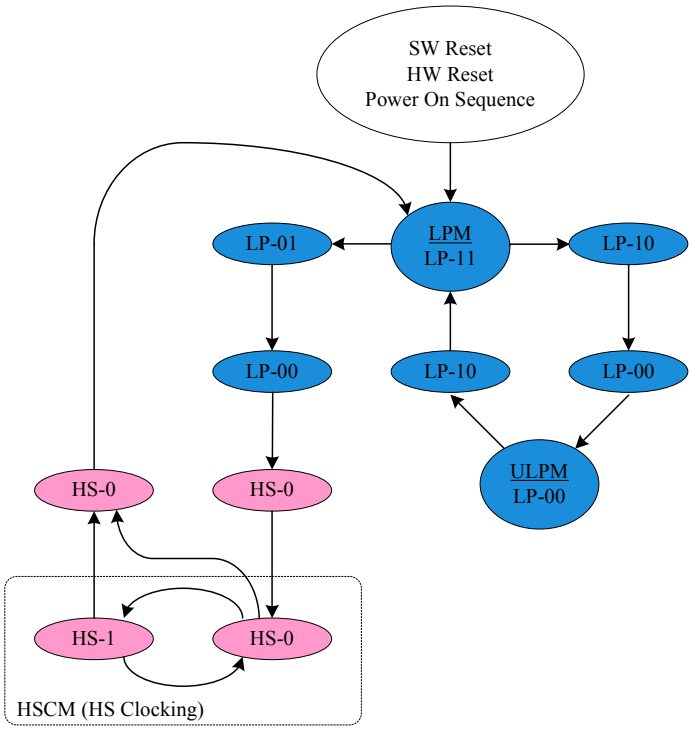
DSI-CLOCK_P/N lanes can be driven into three different power modes:

- Low Power Mode (LPM)
- Ultra Low Power Mode (ULPM)
- High Speed Clock Mode (HSCM)

Clock lanes are in a single end mode (LP = Low Power) when there is entering or leaving Low Power Mode (LPM) or Ultra Low Power Mode (ULPM).

Clock lanes are in the single end mode (LP = Low Power) when there is entering in or leaving out High Speed Clock Mode (HSCM). These entering and leaving protocols are using clock lanes in the single end mode to generate an entering or leaving sequences.

The principal flow chart of the different clock lanes power modes is illustrated below.

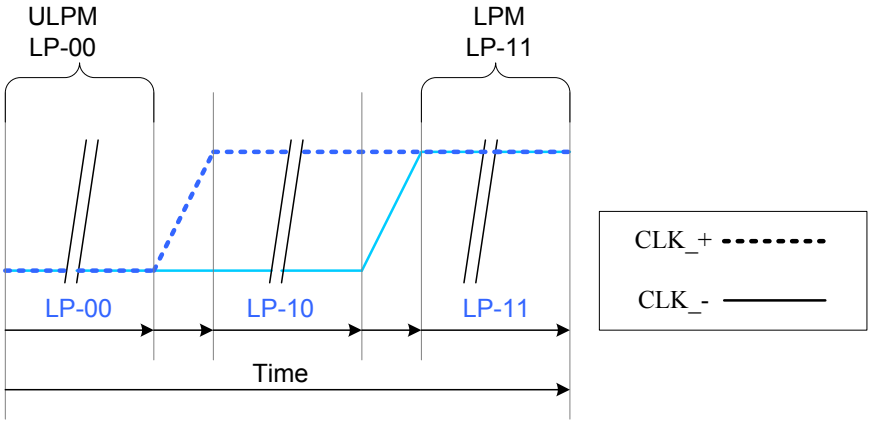


Flow chart of the different clock lanes

1. Low Power Mode (LPM)

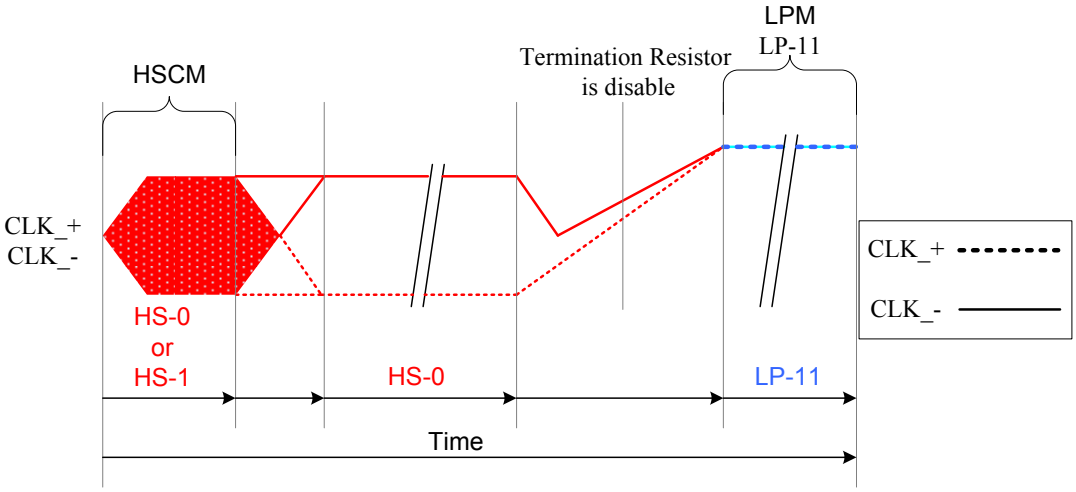
DSI-CLOCK_P/N lanes can be driven to the Low Power Mode (LPM), when DSI-CLOCK lanes are entering LP-11 State Code, in three different ways:

- After SW Reset, HW Reset or Power On Sequence =>LP-11After DSI-CLOCK_P/N lanes are leaving Ultra Low Power Mode (ULPM, LP-00 State Code) =>LP-10
- =>LP-11 (LPM). This sequence is illustrated below.



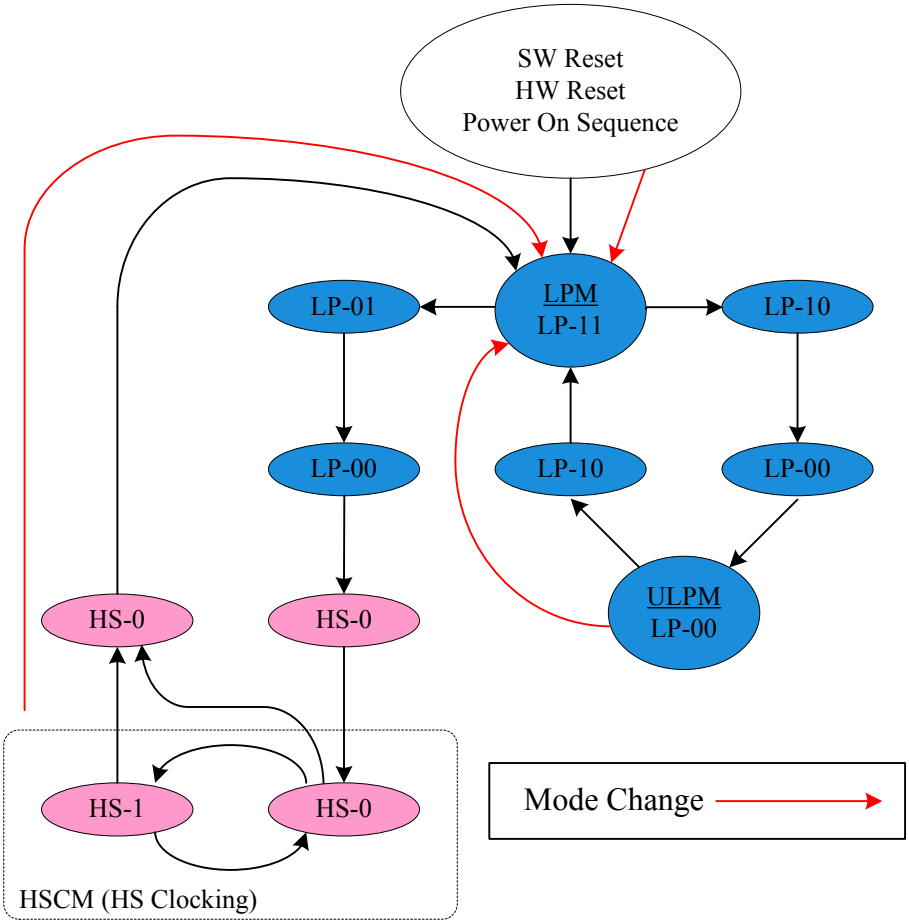
From ULPM to LPM

◆ After DSI-CLK+/- lanes are leaving High Speed Clock Mode (HSCM, HS-0 or HS-1 State Code) =>HS-0=>LP-11 (LPM). This sequence is illustrated below.



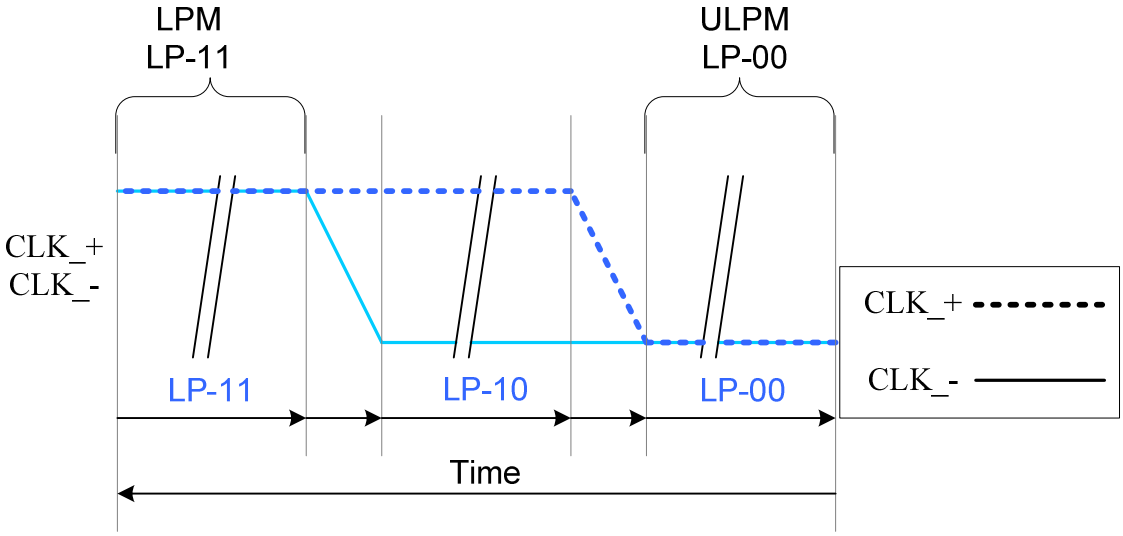
From High Speed Clock Mode (HSCM) to LPM

All three mode changes are illustrated a flow chart below.



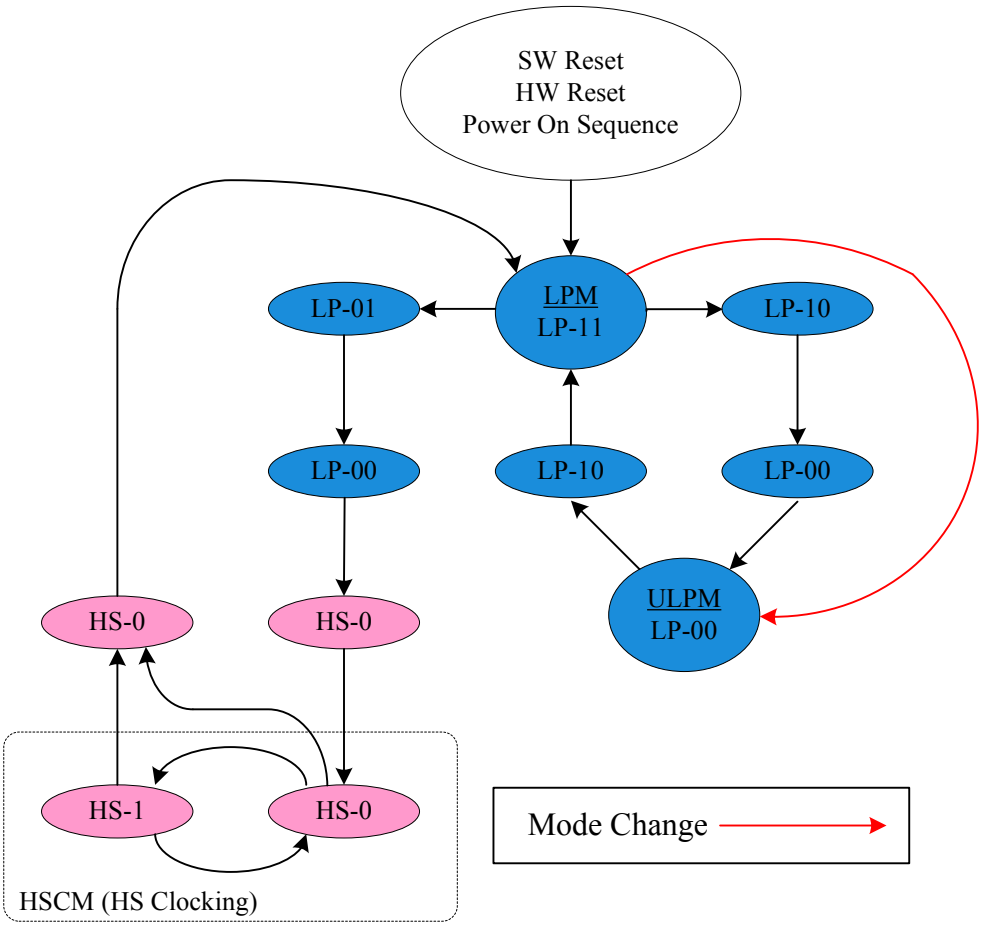
All Three Mode Changes to LPM on the Flow Chart

2. Ultra Low Power Mode (ULPM)



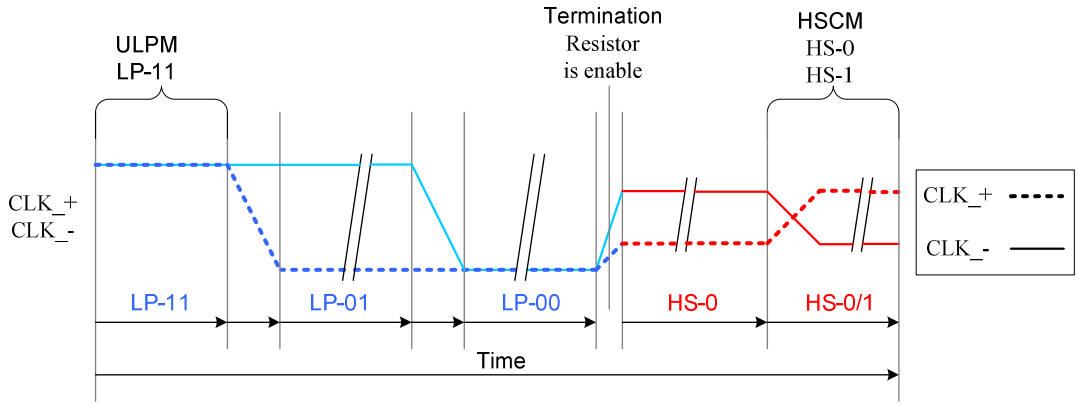
From LPM to ULPM

The mode change is also illustrated below.



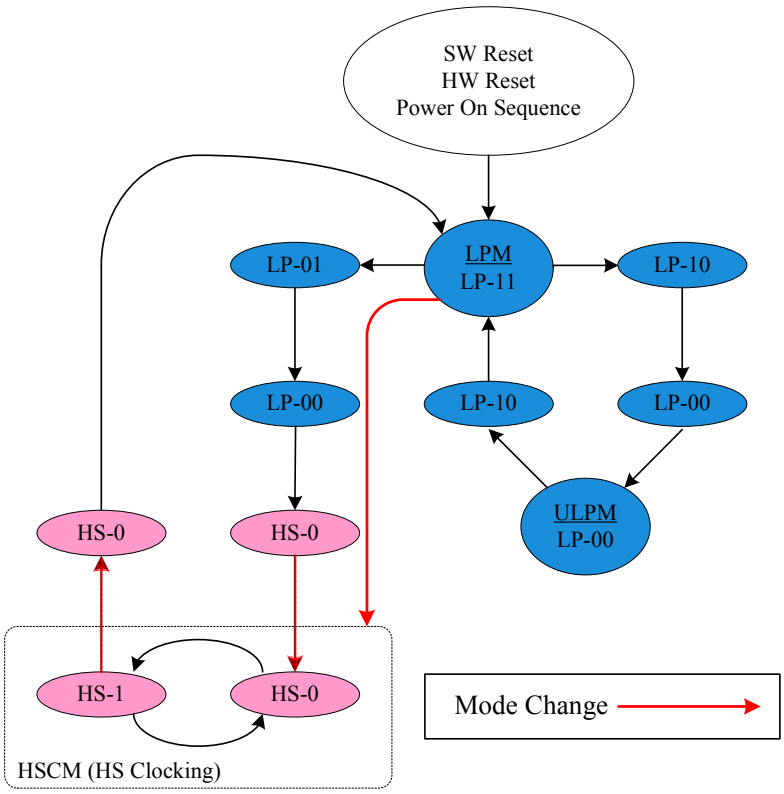
Mode Change from LPM to ULPM on the Flow Chart

3. High Speed Clock Mode (HSCM)



From LPM to HSCM

The mode change is also illustrated below.

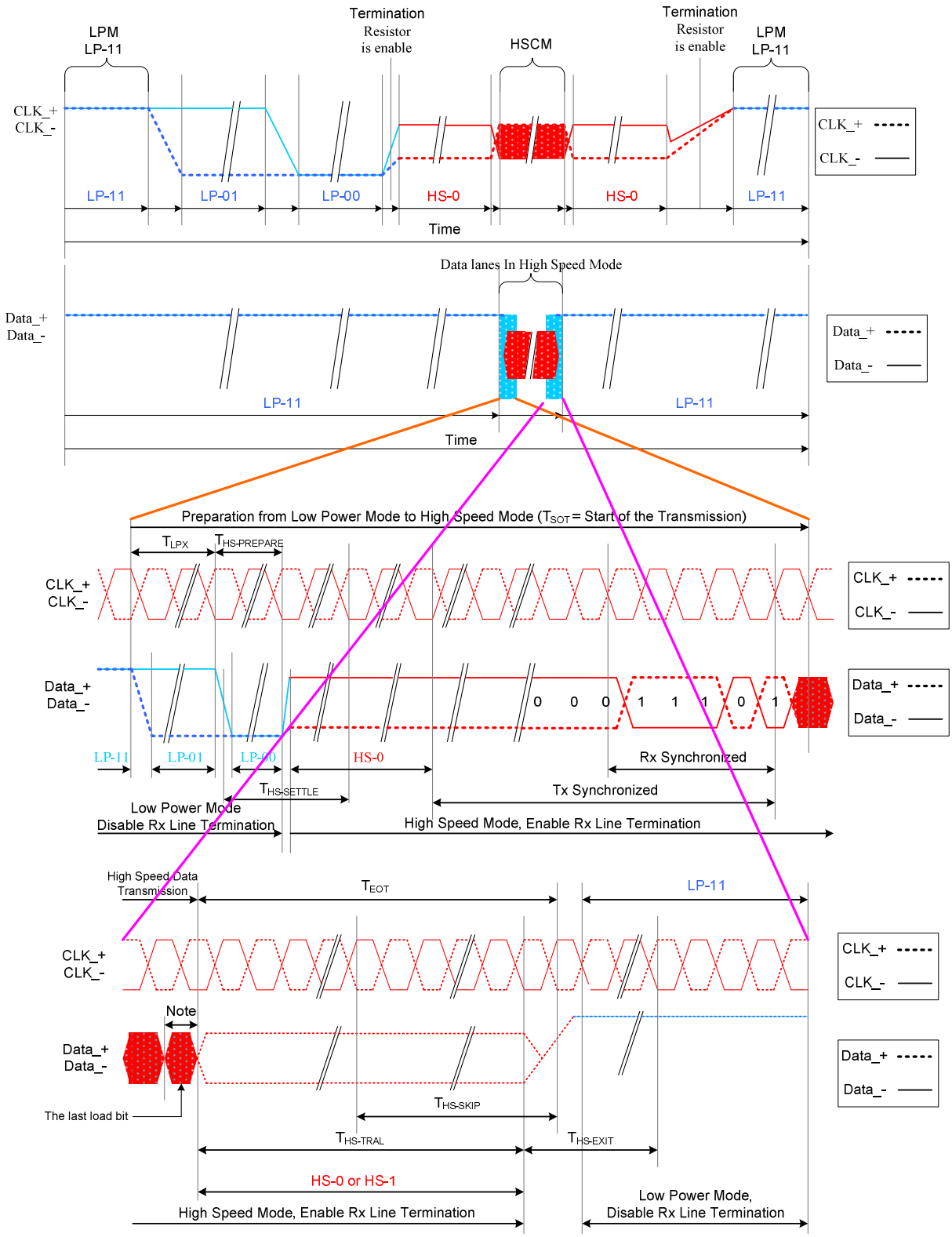


Mode Change from LPM to HSCM on the Flow Chart

The high speed clock (DSI-CLOCK_P/N) is started before high speed data is sent via DSI-DATA_P/N lanes. The high speed clock continues clocking after the high speed data sending has been stopped.

The burst of the high speed clock consists of:

- Even number of transitions
- Start state is HS-0
- End state is HS-0



High Speed Clock Burst

Note:

If the last load bit is HS-0, the transmitter changes from HS-0 to HS-1

If the last load bit is HS-1, the transmitter changes from HS-1 to HS-0

DSI-DATA Lanes

DSI-DATA_P/N Data Lanes can be driven in different modes which are:

- Escape Mode
- High-Speed Data Transmission
- Bus Turnaround Request

These modes and their entering codes are defined on the following table.

Mode	Entering Mode Sequence	Leaving Mode Sequence
Escape Mode	LP-11=>LP-10=>LP-00=>LP-01=>LP-00	LP-00=>LP-10=>LP-11(Mark-1)
High-Speed Data Transmission	LP-11=>LP-01=>LP-00=>HS-0	(HS-0 or HS-1) =>LP-11
Bus Turnaround Request	LP-11=>LP-10=>LP-00=>LP-10=>LP-00	High-Z, Note

1. Escape Mode

Data lanes (DSI-DATA_P/N) can be used in different Escape Modes when data lanes are in Low Power (LP) mode.

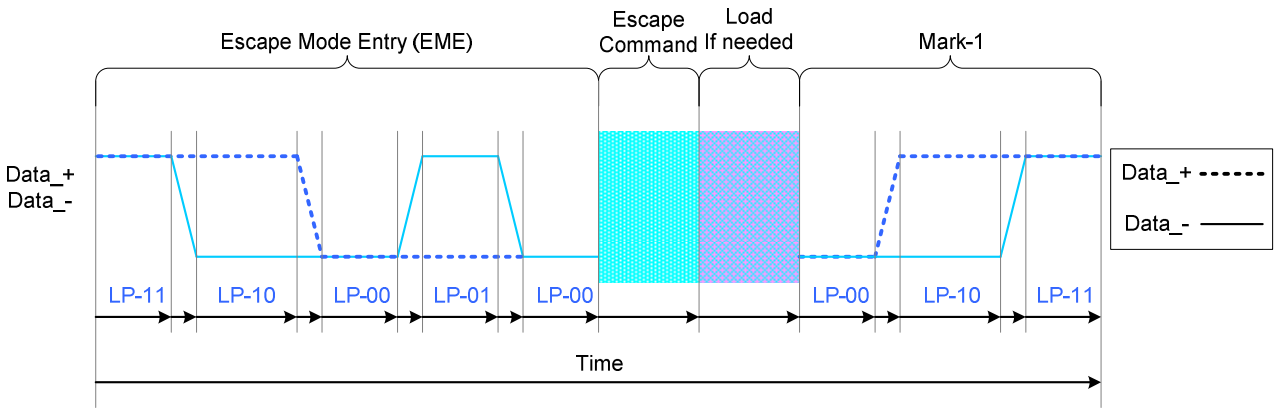
These Escape Modes are used to:

- Send “Low-Power Data Transmission” (LPDT) e.g. from the MCU to the display module
- Drive data lanes to “Ultra-Low Power State” (ULPS)
- Indicate “Remote Application Reset” (RAR), which is reset the display module
- Indicate “Tearing Effect”, which is used for a TE line event from the display module to the MCU
- Indicate (ACK), which is used for a non-error event from the display module to the MCU

The basic sequence of the Escape Mode is as follow

- Start: LP-11
- Escape Mode Entry (EME): LP-11 =>LP-10 =>LP-00 =>LP-01 =>LP-00
- Escape Command (EC), which is coded, when one of the data lanes is changing from low-to-high-to-low then this changed data lane is presenting a value of the current data bit (DSI-D0+ = 1, DSI-D0- = 0) e.g. when DSI-D0- is changing from low-to-high-to-low, the receiver is latching a data bit, which value is logical 0. The receiver is using this low-to-high-to-low transition for its internal clock.
- A load if it is needed
- Exit Escape (Mark-1) LP-00 =>LP-10 =>LP-11
- End: LP-11

This basic construction is illustrated below:



General Escape Mode Sequence

The number of the different Escape Commands (EC) is eight. These eight different Escape Commands (EC) can be divided 2 different groups: Mode or Trigger. The MCU is informing to the display module that it is controlling data lanes (RX_D0P/N) with the mode e.g. The MCU can inform to the display module that it can put data lanes in the low power mode. The MCU is waiting from the display module an event information, which has been set by the MCU, with the trigger e.g. when the display module reaches a new V-synch, the display module sent to the MCU a TE trigger (TEE), if the MCU has been requested it.

Escape commands are defined on the next table.

Escape command	Command Type Mode / Trigger	Entry command Pattern (First Last Bit Transmitted)
Low-Power Data	Mode	1110 0001 b
Ultra-Low Power Mode	Mode	0001 1110 b
Undefined-1, Note	Mode	1001 1111 b
Undefined-2, Note	Mode	1101 1110 b
Remote Application Reset	Trigger	0110 0010 b
Tearing Effect	Trigger	0101 1101 b
Acknowledge	Trigger	0010 0001 b
Uknown-5, Note	Trigger	1010 0000 b

Note: This Escape command support has not been implemented on the display module.

Low-Power Data Transmission (LPDT)

The MCU can send data to the display module in Low-Power Data Transmission (LPDT) mode when data lanes are entering in Escape Mode and Low-Power Data Transmission (LPDT) command has been sent to the display module. The display module is also using the same sequence when it is sending data to the MCU.

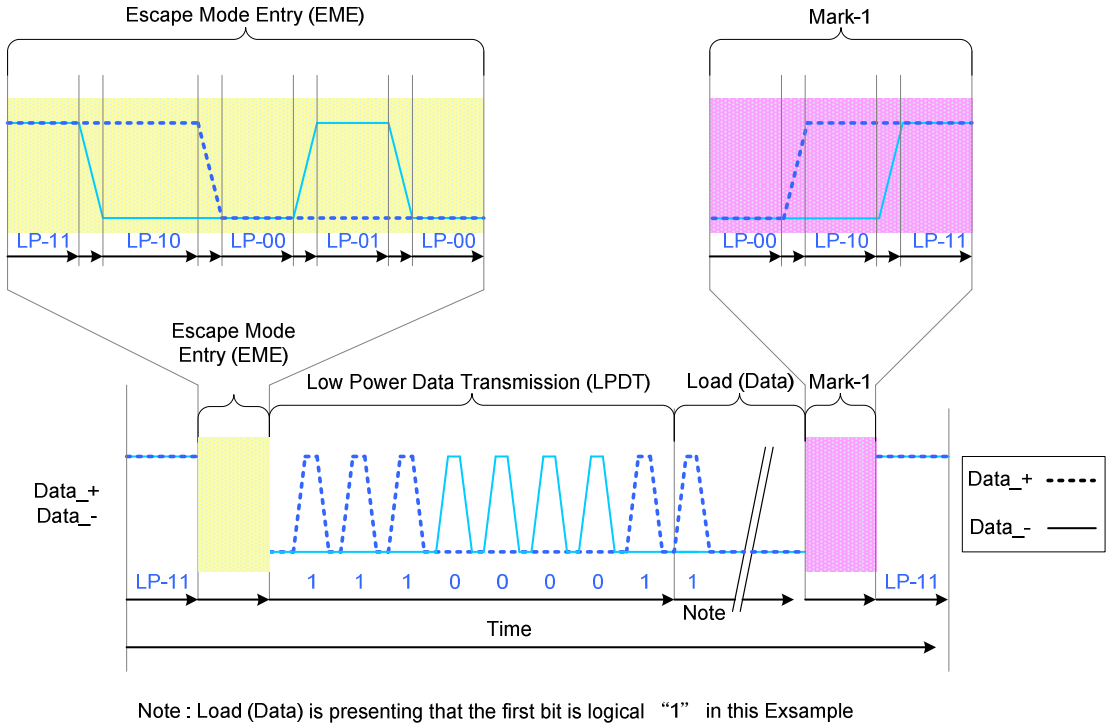
The Low Power Data Transmission (LPDT) is using a following sequence:

- Start: LP-11
- Escape Mode Entry (EME): LP-11 =>LP-10 =>LP-00 =>LP-01 =>LP-00
- Low-Power Data Transmission (LPDT) command in Escape Mode: 1110 0001 (First to Last bit)
- Load (Data):

  One or more bytes (8 bit)

  Data lanes are in pause mode when data lanes are stopped (Both lanes are low) between bytes
- Mark-1: LP-00 =>LP-10 =>LP-11
- End: LP-11

The Low-Power Data Transmission (LPDT) is as below,



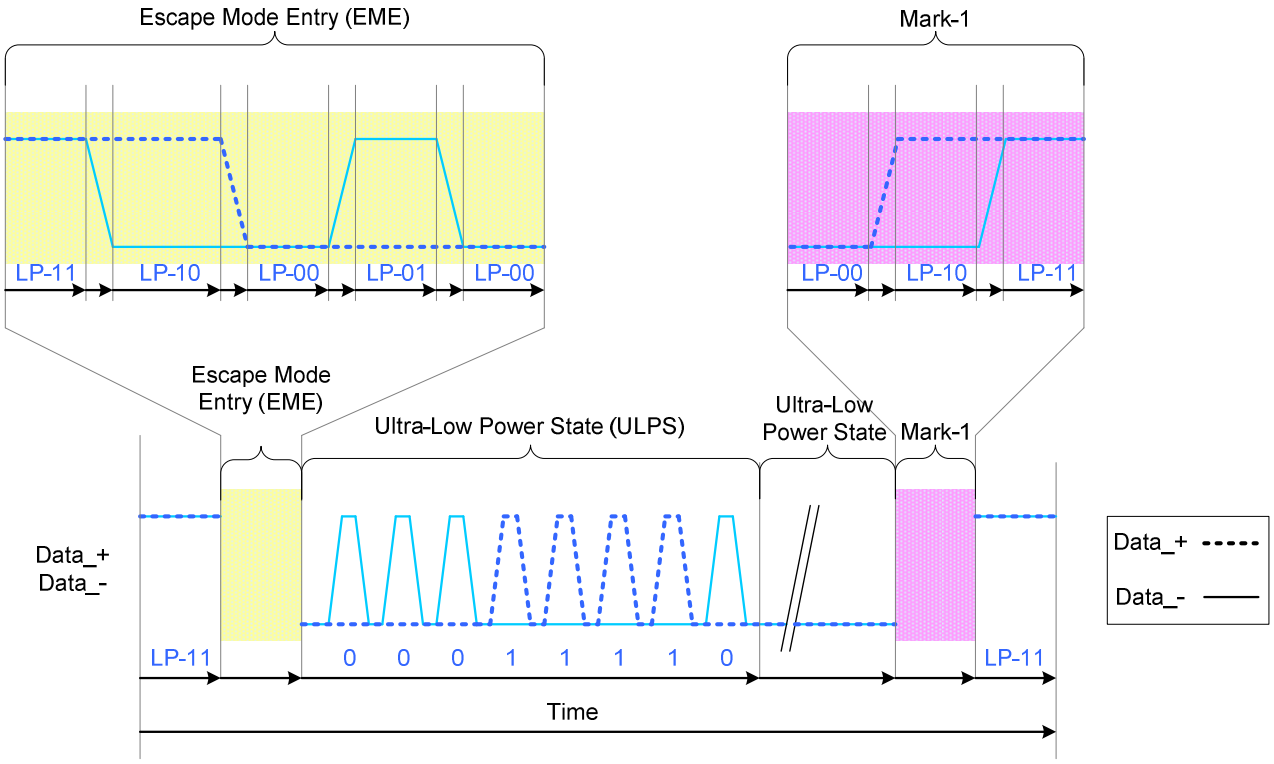
Low-Power Data Transmission (LPDT)

Ultra-Low Power State (ULPS)

The MCU can force data lanes in Ultra-Low Power State (ULPS) mode when data lanes are entering in Escape Mode. The Ultra-Low Power State (ULPS) is using a following sequence:

- Start: LP-11
- Escape Mode Entry (EME): LP-11 =>LP-10 =>LP-00 =>LP-01 =>LP-00
- Ultra-Low Power State (ULPS) command in Escape Mode: 0001 1110 (First to Last bit)
- Ultra-Low Power State (ULPS) when the MCU is keeping data lanes low
- Mark-1: LP-00 =>LP-10 =>LP-11
- End: LP-11

This sequence is illustrated for reference purposes below:



Ultra-Low Power State (ULPS)

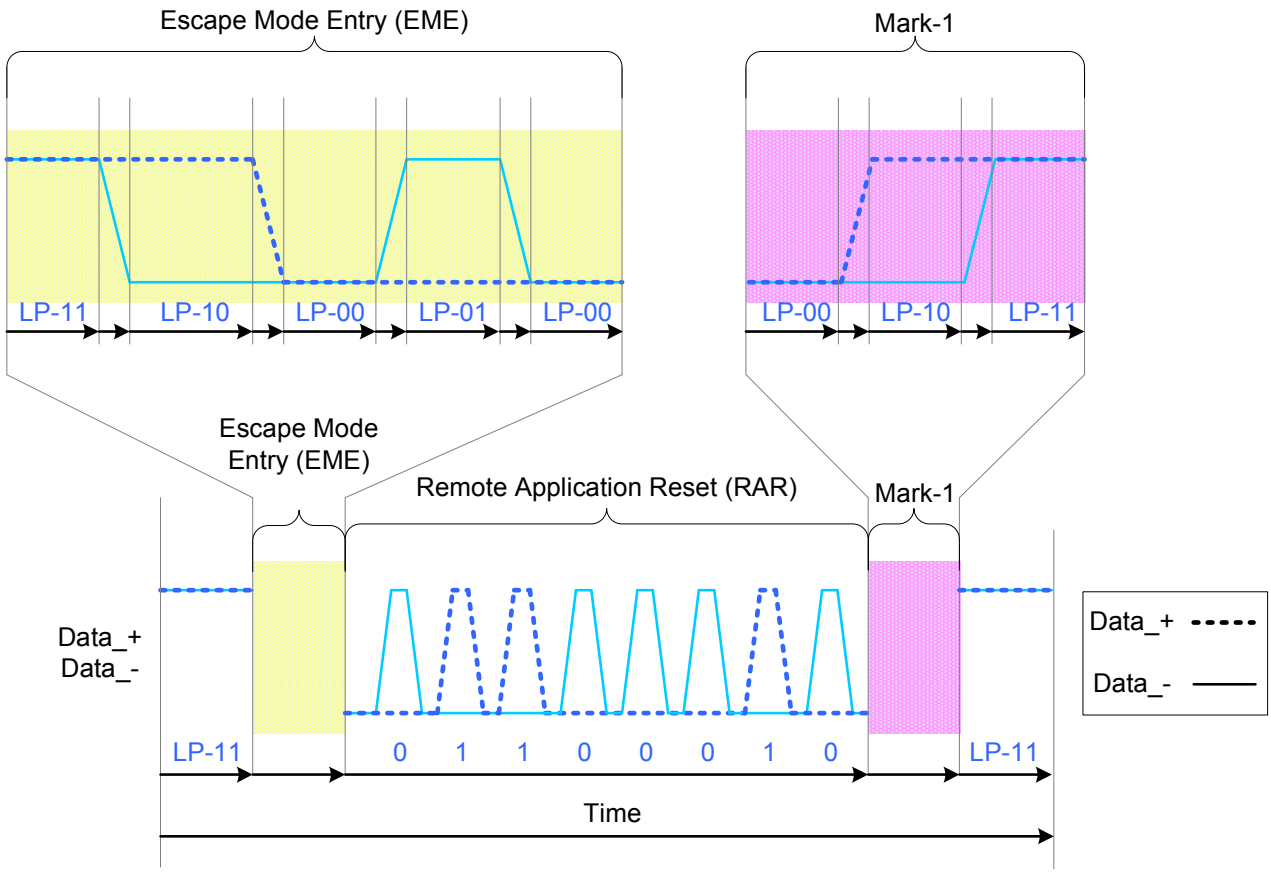
Remote Application Reset (RAR)

The MCU can inform to the display module that it should be reseted in Remote Application Reset (RAR) trigger when data lanes are entering in Escape Mode.

The Remote Application Reset (RAR) is using a following sequence:

- Start: LP-11
- Escape Mode Entry (EME): LP-11 =>LP-10 =>LP-00 =>LP-01 =>LP-00
- Remote Application Reset (RAR) command in Escape Mode: 0110 0010 (First to Last bit)
- Mark-1: LP-00 =>LP-10 =>LP-11
- End: LP-11

This sequence is illustrated for reference purposes below:



Remote Application Reset (RAR)

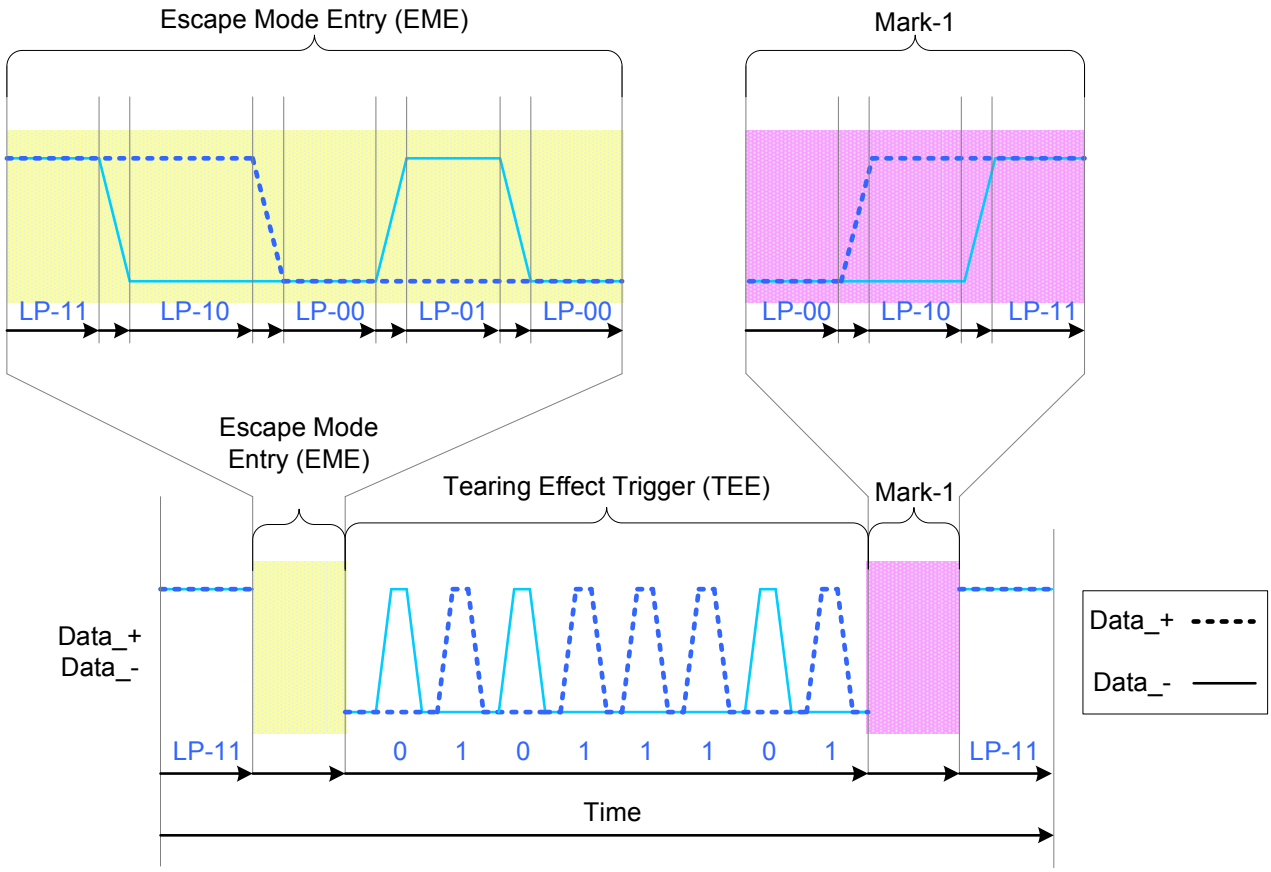
Tearing Effect (TEE)

The display module can inform to the MCU when a tearing effect event (New V-synch) has been happenned on the display module by Tearing Effect (TEE).

The display module is sending the Tearing Effect (TEE) what is a following sequence:

- Start: LP-11
- Escape Mode Entry (EME): LP-11 =>LP-10 =>LP-00 =>LP-01 =>LP-00
- Tearing Effect (TEE) trigger in Escape Mode: 0101 1101 (First to Last bit)
- Mark-1: LP-00 =>LP-10 =>LP-11
- End: LP-11

This sequence is illustrated for reference purposes below:



Tearing Effect (TEE)

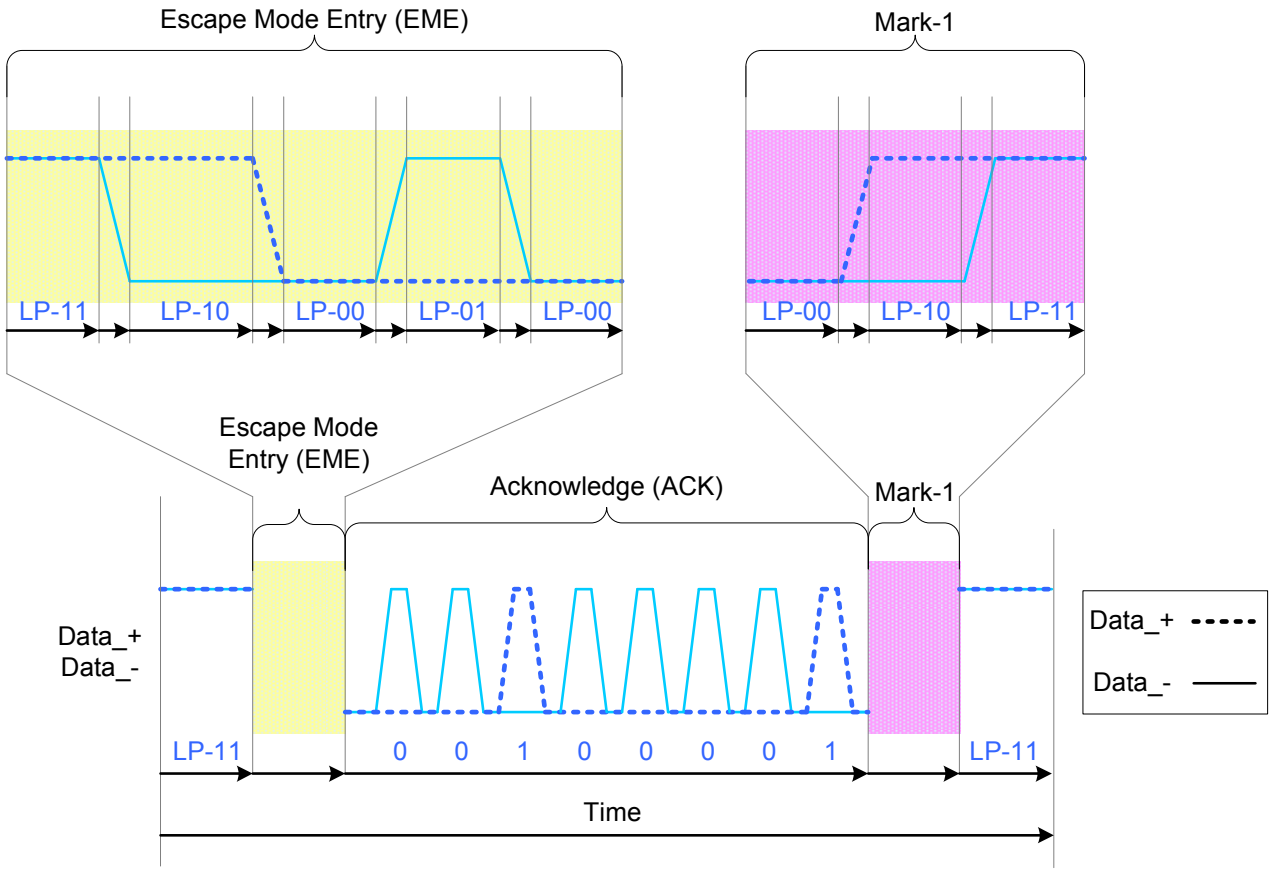
Acknowledge (ACK)

The display module can inform to the MCU when an error has not recognized on it by Acknowledge (ACK).

The display module is sending the Acknowledge (ACK) what is using a following sequence:

- Start: LP-11
- Escape Mode Entry (EME): LP-11 =>LP-10 =>LP-00 =>LP-01 =>LP-00
- Acknowledge (ACK) command in Escape Mode: 0010 0001 (First to Last bit)
- Mark-1: LP-00 =>LP-10 =>LP-11
- End: LP-11

This sequence is illustrated for reference purposes below:



Acknowledge (ACK)

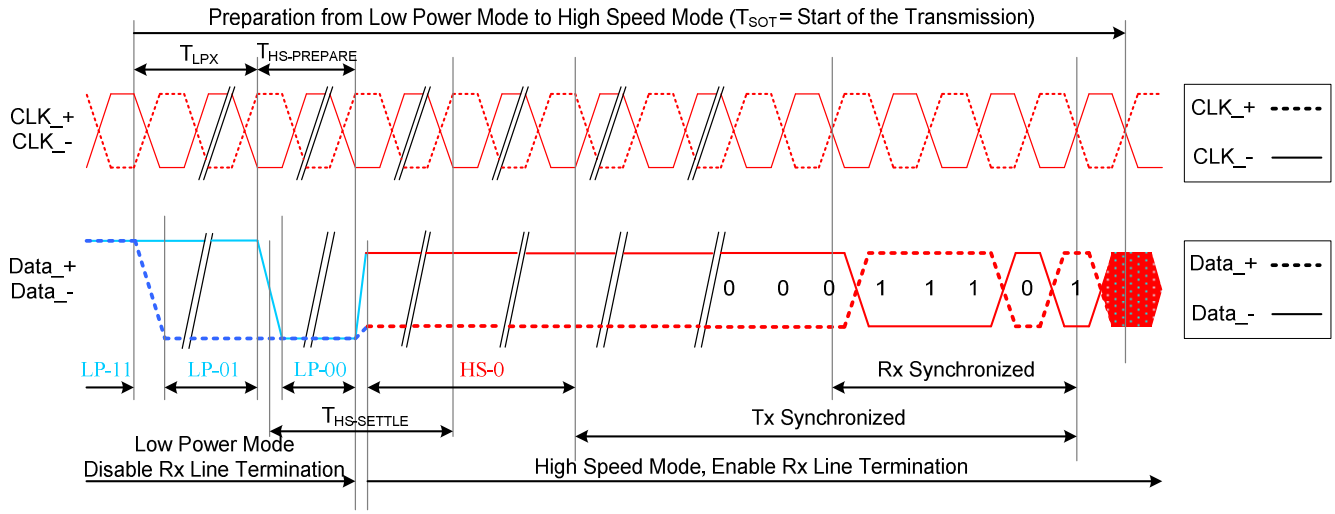
2. High-Speed Data Transmission

The display module is entering High-Speed Data Transmission (HSDT) when Clock lanes RX_CP/N have already been entered in the High-Speed Clock Mode (HSCM) by the MCU.

Data lanes of the display module are entering (T_{SOT}) in the High-Speed Data Transmission (HSDT) as follows:

- Start: LP-11
- HS-Request: LP-01
- HS-Settle: LP-00 => HS-0 (Rx: Lane Termination Enable)
- Rx Synchronization: 011101 (Tx (= MCU) Synchronization: 0001 1101)
- End: High-Speed Data Transmission (HSDT) – Ready to receive High-Speed Data Load

This same entering High-Speed Data Transmission (T_{SOT} of HSDT) sequence is illustrated below



Entering High-Speed Data Transmission (T_{SOT} of HSDT)

Leaving High-Speed Data Transmission

The display module is leaving the High-Speed Data Transmission (T_{EOT} of HSDT) when Clock lanes RX_CP/N are in the High-Speed Clock Mode (HSCM) by the MCU and this HSCM is kept until data lanes are in LP-11 mode.

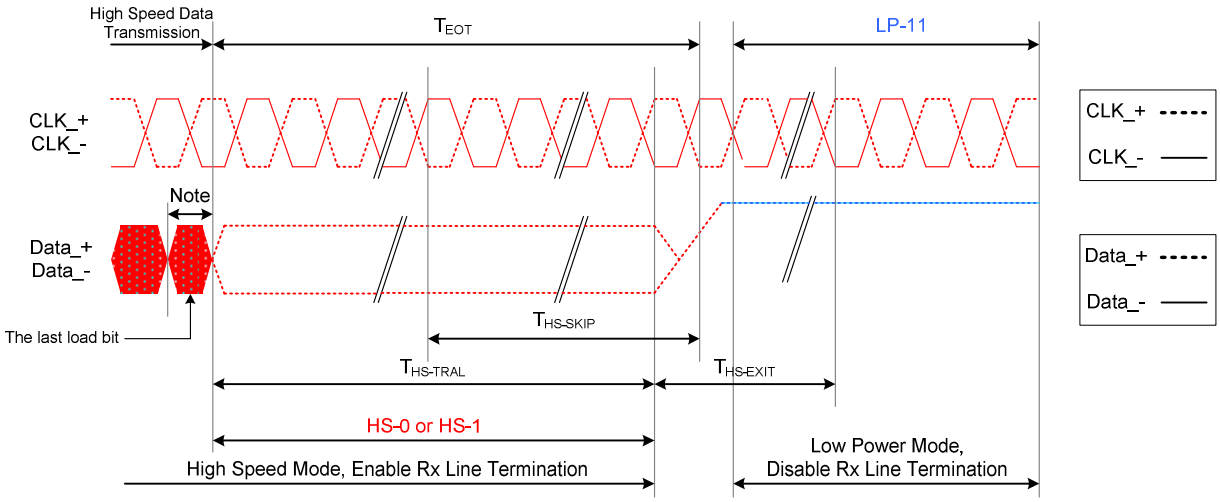
Data lanes of the display module are leaving from the High-Speed Data Transmission (T_{EOT} of HSDT) as follows:

- Start: High-Speed Data Transmission (HSDT)
- Stops High-Speed Data Transmission

  MCU changes to HS-1, if the last load bit is HS-0

  MCU changes to HS-0, if the last load bit is HS-1
- End: LP-11 (Rx: Lane Termination Disable)

This same leaving High-Speed Data Transmission (T_{EOT} of HSDT) sequence is illustrated below

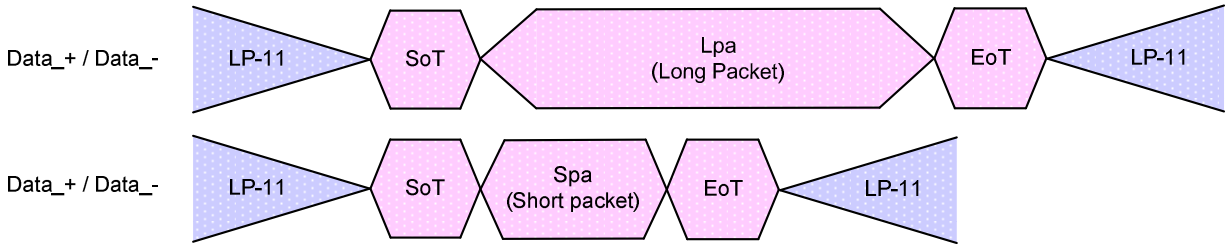


Leaving High-Speed Data Transmission (T_{EOT} of HSDT)

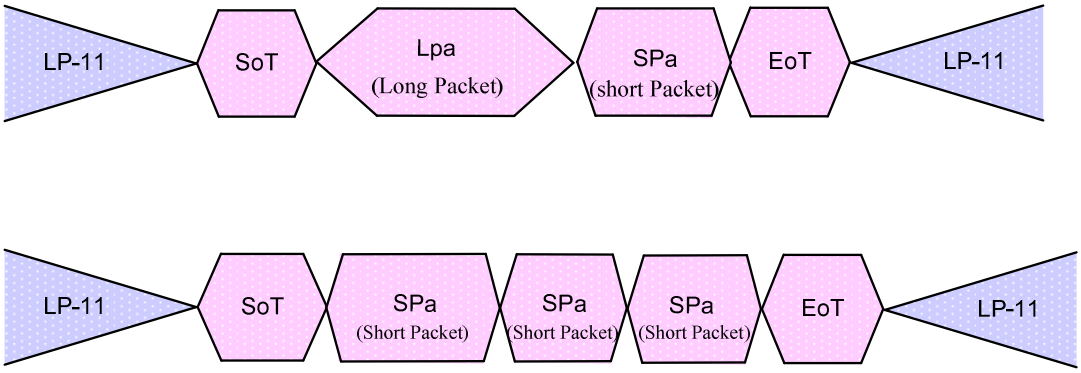
Burst of the High-Speed Data Transmission

The burst of the high-speed data transmission (HSDT) can consist of one data packet or several data packets. These data packets can be Long (LPa) or Short (SPa) packets.

The single packet in High-Speed Data Transmission is illustrated for reference purposes below:

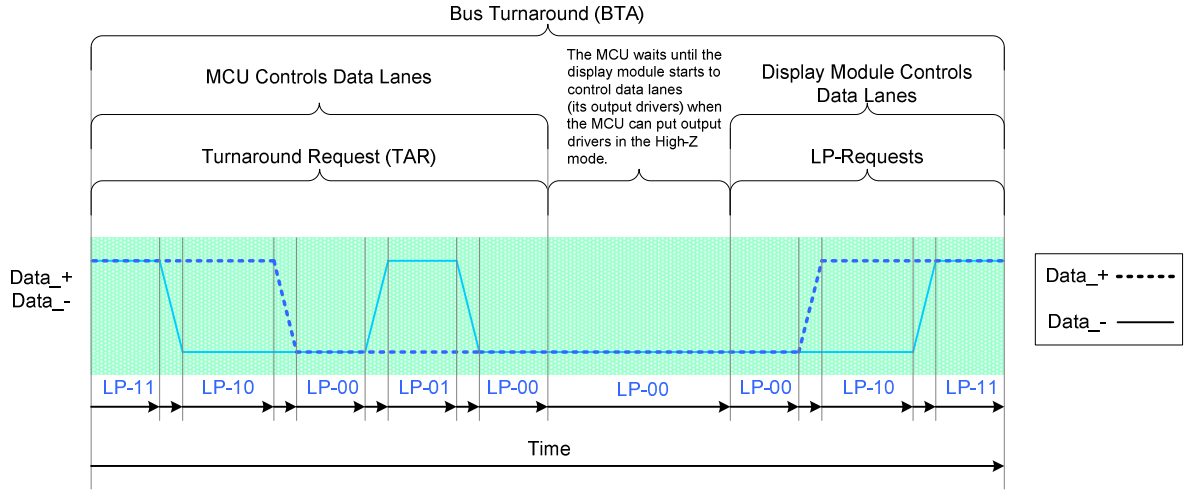


The multiple packets in High-Speed Data Transmission is illustrated for reference purposes below:



3. Bus Turnaround Request

The MCU which is controlling DSI-DATA_P/N Data Lanes, can start a bus turnaround procedure when it wants information from a receiver, which can be the MCU or Display Module. The MCU and Display Module are using the same sequence when this bus turnaround procedure is used. This sequence is described for reference purposes, when the MCU wants to do the bus turnaround procedure to Display Module, as follows.

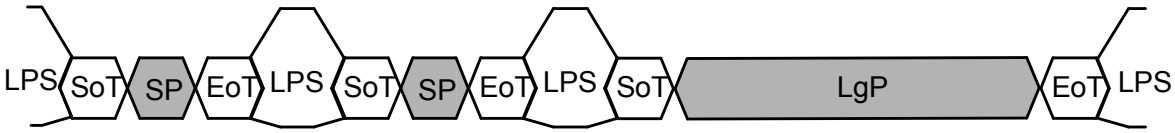


8.5.2 DSI protocol

The Protocol layer appends packet-protocol information and headers, and then sends complete bytes through the Lane Management layer to the PHY. Packets are serialized by the PHY and sent across the serial Link. The receiver side of a DSI Link performs the coverse of the transmitter side, decomposing the packet into parallel data, signal events and commands.

8.5.2.1 Multiple Packets per Transmission

There are two modes of data transmission, HS and LP transmission modes, at the PHY layer. Before a HS transmission can be started, the transmitter PHY issues a SoT sequence to the receiver. After that, data or command packets can be transmitted in HS mode. Multiple packets may exist within a single HS transmission and the end of transmission is always signaled at the PHY layer using a dedicated EoT sequence. In order to enhance the overall robustness of the system, DSI defines a dedicated EoT packet (EoTp) at the protocol layer for signaling the end of HS transmission. For backwards compatibility with earlier DSI systems, the capability of generating and interpreting this EoTp can be enabled or disabled.



Separate Transimssions

Key:

LPS -- Low power state
SoT -- Start of Transmission
EoT -- End of Transmission
SP -- Short Packet
LgP -- Long Packet



Single Transimssions

8.5.2.2 Packet Composition

The first byte of the packet, the Data Identifier (DI), includes information specifying the type of the packet. For example, in Video Mode systems in a display application the logical unit for a packet may be one horizontal display line. Command Mode systems send commands and an associated set of parameters, with the number of parameters depending on the command type.

Packet sizes fall into two categories:

- **Short packets** are four bytes in length including the ECC. Short packets are used for most Command Mode commands and associated parameters. Other Short packets convey events like H Sync and V Sync edges. Because they are Short packets they can convey accurate timing information to logic at the peripheral.

- **Long packets** specify the payload length using a two-byte Word Count field. Payloads may be from 0 to 2^{16}-1 bytes long. Therefore, a Long packet may be up to 65,541 bytes in length. Long packets permit transmission of large blocks of pixel or other data.

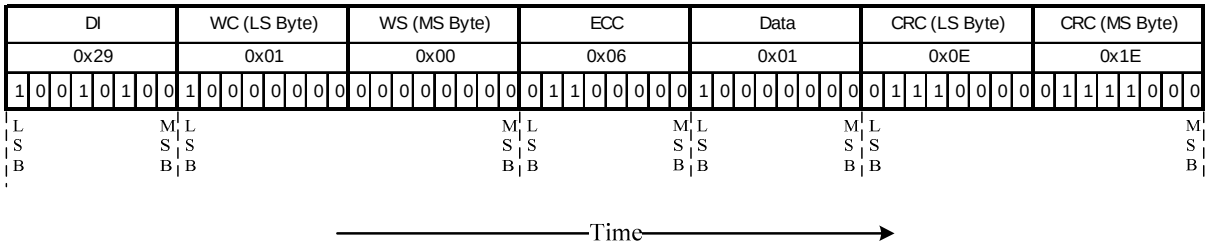
A special case of Command Mode operation is video-rate (update) streaming, which takes the form of an arbitrarily long stream of pixel or other data transmitted to the peripheral. As all DSI transactions use packets, the video stream shall be broken into separate packets. This “packetization” may be done by hardware or software. The peripheral may then reassemble the packets into a continuous video stream for display.

The Set Maximum Return Packet Size command allows the host processor to limit the size of response packets coming from a peripheral.

8.5.2.3 Endian Policy

All packet data traverses the interface as bytes. Sequentially, a transmitter shall send data LSB first, MSB last. For packets with multi-byte fields, the least significant byte shall be transmitted first unless otherwise specified.

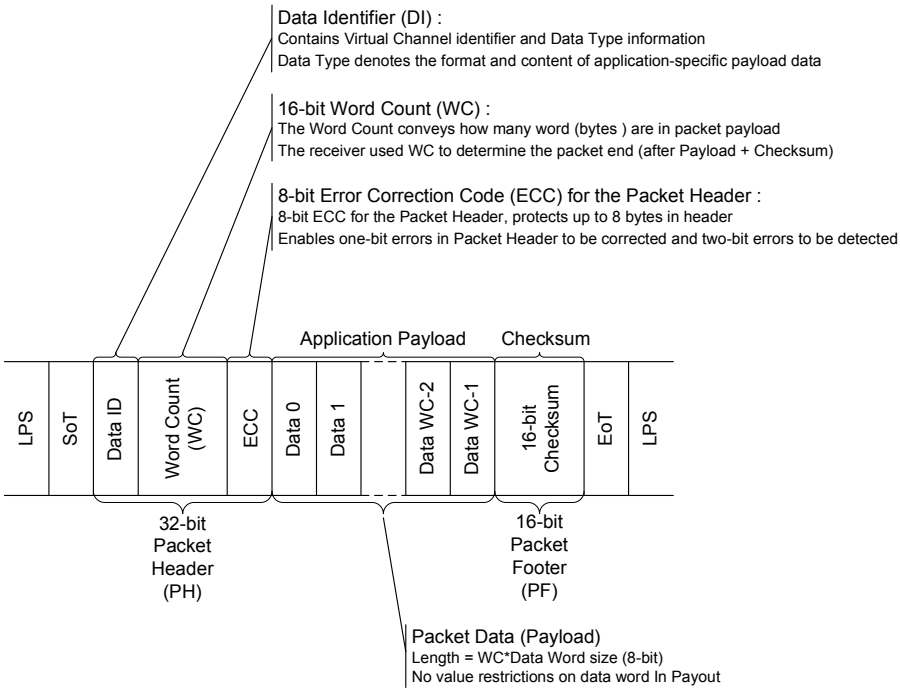
Figure 12 shows a complete Long packet data transmission. Note, the figure shows the byte values in standard positional notation, i.e. MSB on the left and LSB on the right, while the bits are shown in chronological order with the LSB on the left, the MSB on the right and time increasing left to right.



Endian Example (Long Packet)

8.5.2.4 General Packet Structure(Long Packet Format)

A Long packet shall consist of three elements: a 32-bit Packet Header (PH), an application-specific Data Payload with a variable number of bytes, and a 16-bit Packet Footer (PF). The Packet Header is further composed of three elements: an 8-bit Data Identifier, a 16-bit Word Count, and 8-bit ECC. The Packet Footer has one element, a 16-bit checksum. Long packets can be from 6 to 65,541 bytes in length.



Long Packet Structure

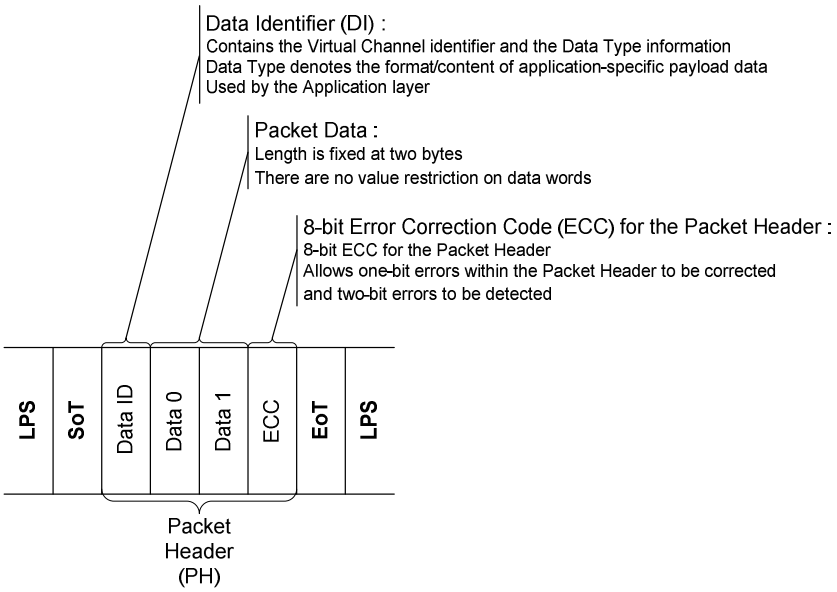
The Data Identifier defines the Virtual Channel for the data and the Data Type for the application specific payload data. See sections 8.8 through 8.10 for descriptions of Data Types. The Word Count defines the number of bytes in the Data Payload between the end of the Packet Header and the start of the Packet Footer. Neither the Packet Header nor the Packet Footer shall be included in the Word Count. The Error Correction Code (ECC) byte allows single-bit errors to be corrected and 2-bit errors to be detected in the Packet Header. This includes both the Data Identifier and Word Count fields. After the end of the Packet Header, the receiver reads the next Word Count * bytes of the Data Payload. Within the Data Payload block, there are no limitations on the value of a data word, i.e. no embedded codes are used. Once the receiver has read the Data Payload it reads the Checksum in the Packet Footer. The host processor shall always calculate and transmit a Checksum in the Packet Footer. Peripherals are not required to calculate a Checksum. Also note the special case of zero-byte Data Payload: if the payload has length 0, then the Checksum calculation results in (FFFFh). If the Checksum is not calculated, the Packet Footer shall consist of two bytes of all zeros (0000h). See section 9 for more information on calculating the Checksum. In the generic case, the length of the Data Payload shall be a multiple of bytes. In addition, each data format may impose additional restrictions on the length of the payload data, e.g. multiple of four bytes. Each byte shall be transmitted least significant bit first. Payload data may be transmitted in any byte order restricted only by data format requirements. Multi-byte elements such as Word Count and Checksum shall be transmitted least significant byte first

.

8.5.2.5 General Packet Structure(Short Packet Format)

A Short packet shall contain an 8-bit Data ID followed by two command or data bytes and an 8-bit ECC; a

Packet Footer shall not be present. Short packets shall be four bytes in length. The Error Correction Code (ECC) byte allows single-bit errors to be corrected and 2-bit errors to be detected in the Short packet.



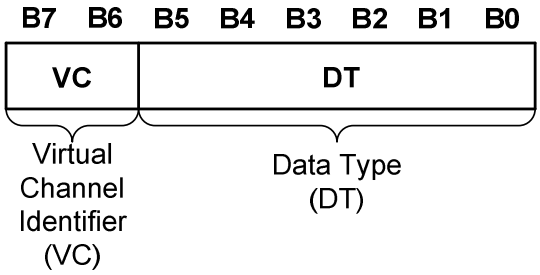
Short Packet Structure

8.5.2.6 Common Packet Elements

Long and Short packets have several common elements that are described in this section.

- Data Identifier Byte

The first byte of any packet is the DI (Data Identifier) byte. Figure 15 shows the composition of the Data Identifier (DI) byte. DI[7:6]: These two bits identify the data as directed to one of four virtual channels.DI[5:0]: These six bits specify the Data Type.

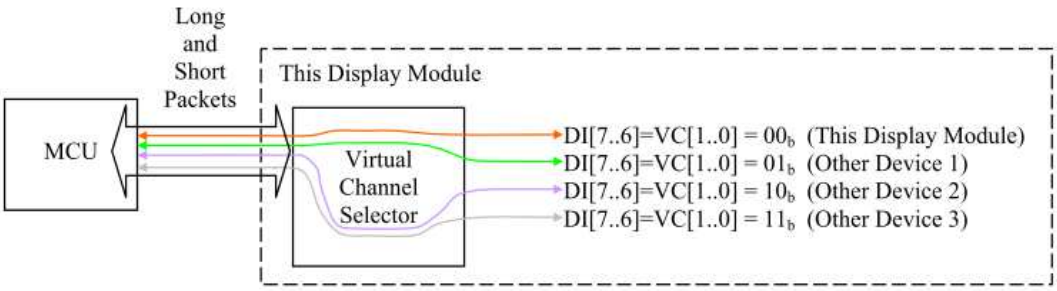


Data Identifier Byte

Virtual Channel Identifier – VC field, DI[7:6]

A processor may service up to four peripherals with tagged commands or blocks of data, using the Virtual Channel ID field of the header for packets targeted at different peripherals. The Virtual Channel ID enables one serial stream to service two or more virtual peripherals by multiplexing packets onto a common transmission channel. Note that packets sent in a single transmission each have their own Virtual Channel assignment and can be directed to different peripherals. Although the DSI protocol permits communication

with multiple peripherals, this specification only addresses the connection of a host processor to a single peripheral. Implementation details for connection to more than one physical peripheral are beyond the scope of this document.



Virtual Channel (VC) Configuration

Data Type Field DT[5:0]

The Data Type field specifies if the packet is a Long or Short packet type and the packet format. The Data Type field, along with the Word Count field for Long packets, informs the receiver of how many bytes to expect in the remainder of the packet. This is necessary because there are no special packet start / end sync codes to indicate the beginning and end of a packet. This permits packets to convey arbitrary data, but it also requires the packet header to explicitly specify the size of the packet. When the receiving logic has counted down to the end of a packet, it shall assume the next data is either the header of a new packet or the EoT (End of Transmission) sequence.

8.5.2.7 Error Correction Code

The Error Correction Code allows single-bit errors to be corrected and 2-bit errors to be detected in the Packet Header. The host processor shall always calculate and transmit an ECC byte. Peripherals shall support ECC in both forward- and reverse-direction communications.

Bits (P[7…0]) of the Error Correction Code (ECC) are defined, where the symbol '^' is presenting XOR function (Pn is '1' if there is odd number of '1's and Pn is '0' if there is even number of '1's), as follows.

P7 = 0

P6 = 0

P5 = D10^D11^D12^D13^D14^D15^D16^D17^D18^D19^D21^D22^D23

P4 = D4^D5^D6^D7^D8^D9^D16^D17^D18^D19^D20^D22^D23

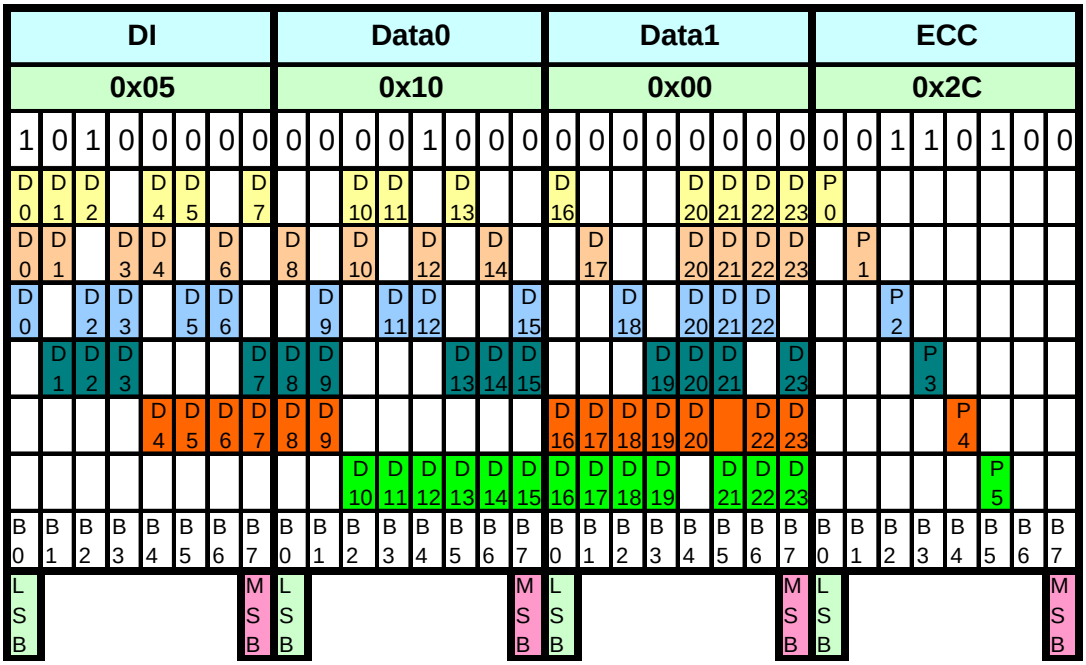
P3 = D1^D2^D3^D7^D8^D9^D13^D14^D15^D19^D20^D21^D23

P2 = D0^D2^D3^D5^D6^D9^D11^D12^D15^D18^D20^D21^D22

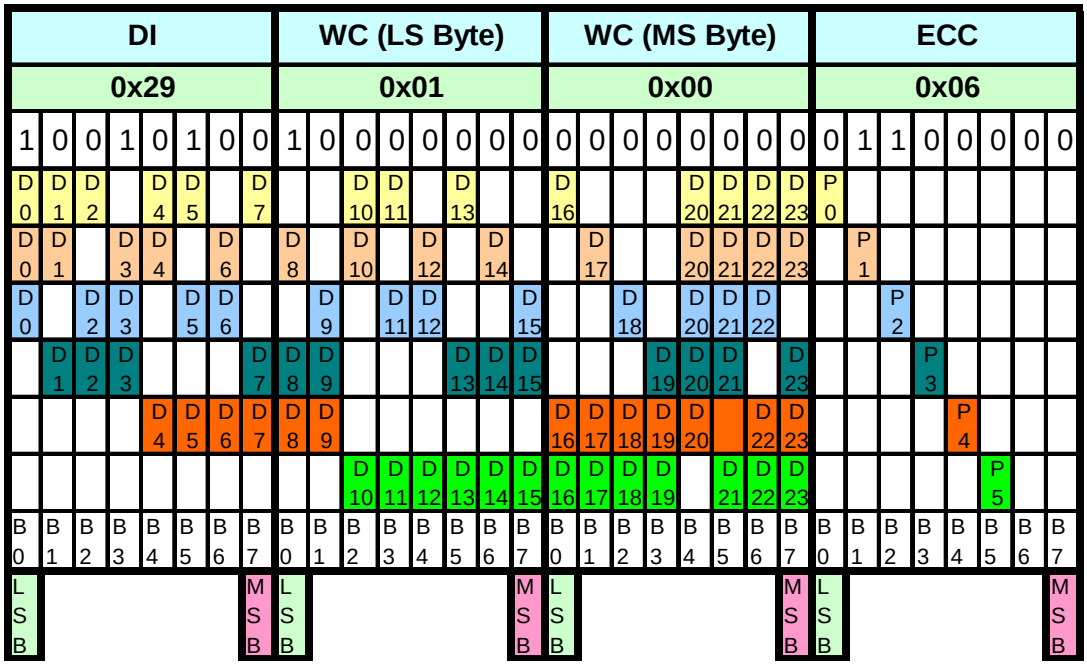
P1 = D0^D1^D3^D4^D6^D8^D10^D12^D14^D17^D20^D21^D22^D23

P0 = D0^D1^D2^D4^D5^D7^D10^D11^D13^D16^D20^D21^D22^D23

P7 and P6 are set to '0' because Error Correction Code (ECC) is based on 64 bit value ([D63…0]), but this implementation is based on 24 bit value (D [23…0]). Therefore, there is only needed 6 bits (P [5…0]) for Error Correction Code (ECC).



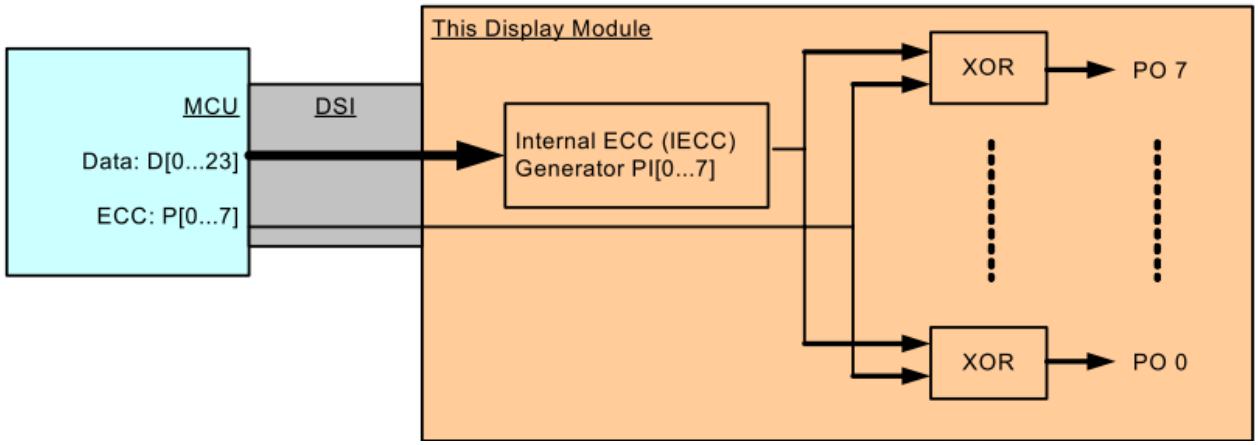
XOR Functionality on the Short Packet (SPa)



XOR Functionality on the Long Packet (LPa)

The transmitter (The MCU or the Display Module) is sending data bits D[23:0] and Error Correction Code (ECC) P[7:0]. The receiver (The Display module or the MCU) is calculate an Internal Error Correction

Code (IECC) and compares the received Error Correction Code (ECC) and the Internal Error Correction Code (IECC). This comparison is done when each power bit of ECC and IECC have been done XOR function. The result of this function is PO[7:0].



Internal Error Correction Code (IECC) on the Display Module (The Receiver)

The sent data bits (D[23:0]) and ECC (P[7:0]) are received correctly, if a value of the PO[7:0]) is 00h. The sent data bits (D[23:0]) and ECC (P[7:0]) are not received correctly, if a value of the PO[7:0]) is not 00h.

ECC P[7:0]	1	1	0	0	0	0	0	0	03h
IECC PI[7:0]	1	1	0	0	0	0	0	0	03h
XOR(ECC,IECC	0	0	0	0	0	0	0	0	=00h => No Error
	L							M	
	S							S	
	B							B	

Internal XOR Calculation between ECC and IECC Values – No Error

ECC P[7:0]	1	1	0	0	0	0	0	0	03h
IECC PI[7:0]	1	1	1	1	0	0	0	0	0Fh
XOR(ECC,IECC	0	0	1	1	0	0	0	0	=0Ch => Error
	L							M	
	S							S	
	B							B	

Internal XOR Calculation between ECC and IECC Values – Error

Data Bit	PO7	PO6	PO5	PO4	PO3	PO2	PO1	PO0	Hex
D[0]	0	0	0	0	0	1	1	1	07h
D[1]	0	0	0	0	1	0	1	1	0Bh
D[2]	0	0	0	0	1	1	0	1	0Dh
D[3]	0	0	0	0	1	1	1	0	0Eh
D[4]	0	0	0	1	0	0	1	1	13h
D[5]	0	0	0	1	0	1	0	1	15h
D[6]	0	0	0	1	0	1	1	0	16h
D[7]	0	0	0	1	1	0	0	1	19h
D[8]	0	0	0	1	1	0	1	0	1Ah
D[9]	0	0	0	1	1	1	0	0	1Ch
D[10]	0	0	1	0	0	0	1	1	23h
D[11]	0	0	1	0	0	1	0	1	25h
D[12]	0	0	1	0	0	1	1	0	26h
D[13]	0	0	1	0	1	0	0	1	29h
D[14]	0	0	1	0	1	0	1	0	2Ah
D[15]	0	0	1	0	1	1	0	0	2Ch
D[16]	0	0	1	1	0	0	0	1	31h
D[17]	0	0	1	1	0	0	1	0	32h
D[18]	0	0	1	1	0	1	0	0	34h
D[19]	0	0	1	1	1	0	0	0	38h
D[20]	0	0	0	1	1	1	1	1	1Fh
D[21]	0	0	1	0	1	1	1	1	2Fh
D[22]	0	0	1	1	0	1	1	1	37h
D[23]	0	0	1	1	1	0	1	1	3Bh

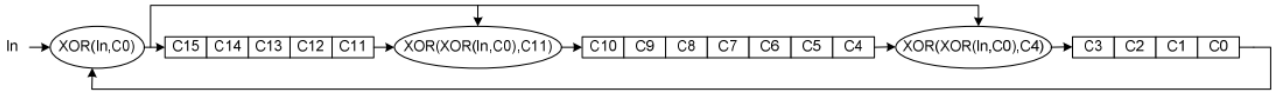
One error is detected if the value of the PO[7:0] is on the above table : One it Error Value of the Error Correction Code (ECC) and the receiver can correct this one bit error because this found value also defines what is a location of the corrupt bit e.g.

- PO [7…0] = 0Eh
- The bit of the data (D [23:0]), what is not correct, is D[3]

More than one error is detected if the value of the PO [7…0] is not on the above table: One Bit Error Value of the Error Correction Code (ECC) e.g. PO [7…0] = 0Ch.

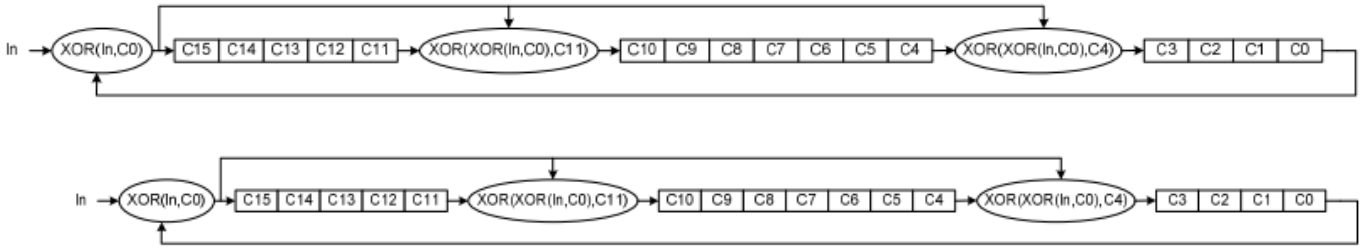
8.5.2.8 Packet Footer on the Long Packet

Packet Footer (PF) of the Long Packet (LPa) is defined after the Packet Data (PD) of the Long Packet (LPa). The Packet Footer (PF) is a checksum value what is calculated from the Packet Data of the Long Packet (LPa). The checksum is using a 16-bit Cyclic Redundancy Check (CRC) value which is generated with a polynomial $X_{16} + X_{12} + X_5 + X_0$ as it is illustrated below.



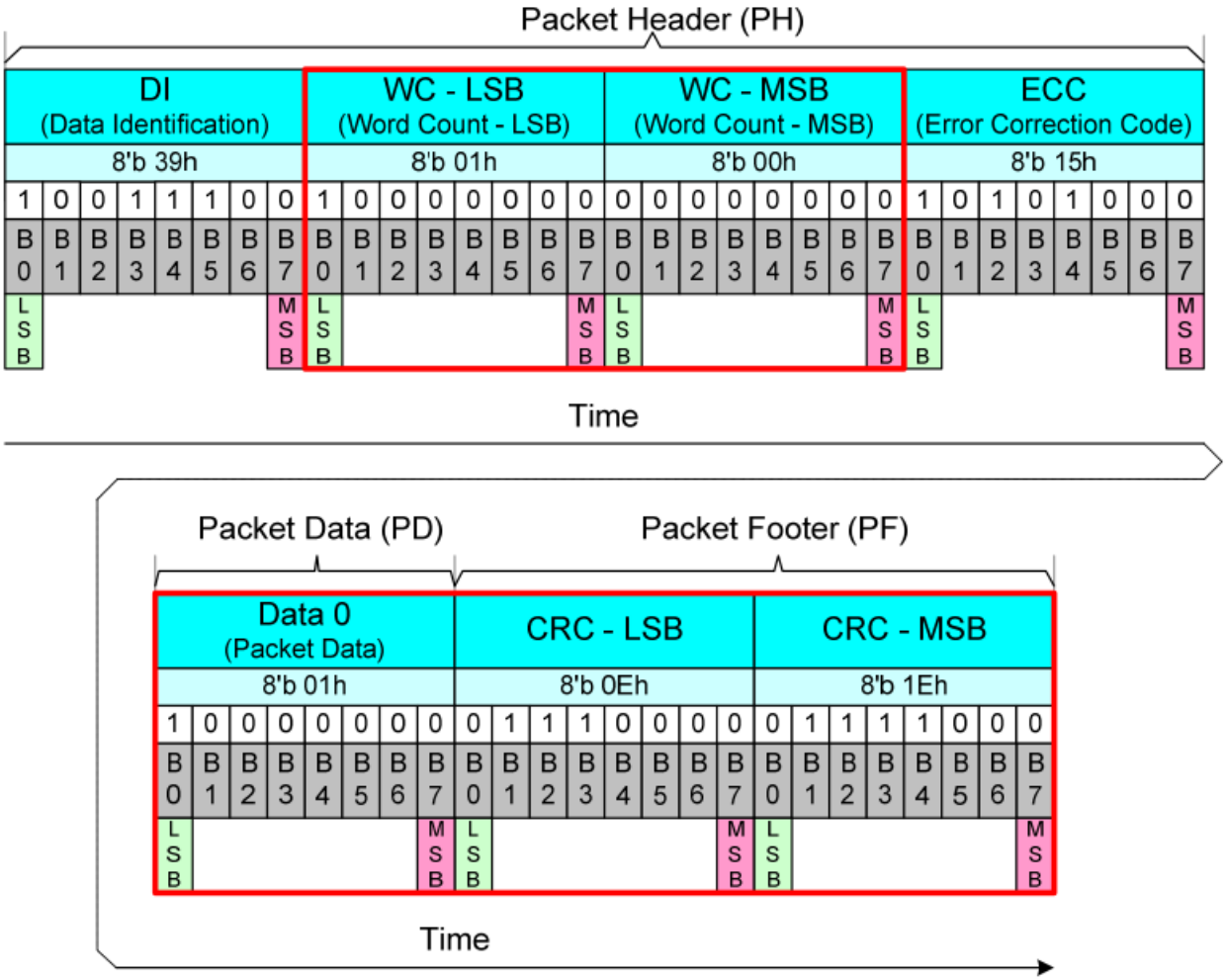
The 16-bit Cyclic Redundancy Check (CRC) generator is initialized to FFFFh before calculations. The Most Significant Bit (MSB) of the data byte of the Packet Data (PD) is the first bit what is inputted into the 16-bit Cyclic Redundancy Check (CRC).

An example of the 16-bit Cyclic Redundancy Check (CRC), where the Packet Data (PD) of the Long Packet (LPa) is 01h, is illustrated (step-by-step) below.



Step	In	XOR(In,C0)	C15	C14	C13	C12	C11	XOR(XOR(In,C0),C11(Step-1))	C10	C9	C8	C7	C6	C5	C4	XOR(XOR(In,C0),C4(Step-1))	C3	C2	C1	C0	C0
0	X	X	1	1	1	1	1	X	1	1	1	1	1	1	1	X	1	1	1	1	X
1	1(LSB)	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
2	0	1	1	0	1	1	1	0	0	1	1	1	1	1	1	0	0	1	1	1	1
3	0	1	1	1	0	1	1	0	0	0	1	1	1	1	1	0	0	0	1	1	1
4	0	1	1	1	1	0	1	0	0	0	0	1	1	1	1	0	0	0	0	1	1
5	0	1	1	1	1	1	0	0	0	0	0	0	1	1	1	0	0	0	0	0	0
6	0	0	0	1	1	1	1	0	0	0	0	0	0	1	1	1	1	0	0	0	0
7	0	0	0	0	1	1	1	1	1	0	0	0	0	0	1	1	1	1	0	0	0
8	0(MSB)	0	0	0	0	1	1	1	1	1	0	0	0	0	0	1	1	1	1	0	0
	1 byte	CRC result	0	0	0	1	1		1	1	0	0	0	0	0		1	1	1	0	
			MSB																	LSB	

A value of the Packet Footer (PF) is 1E0Eh in this example. This example (Command 01h has been sent) is illustrated below.



The receiver is calculated own checksum value from received Packet Data (PD). The receiver compares own checksum and the Packet Footer (PF) what the transmitter has sent. The received Packet Data (PD) and Packet Footer (PF) are correct if the own checksum of the receiver and Packet Footer (PF) is equal and vice versa the received Packet Data (PD) and Packet Footer(PF) are not correct if the own checksum of the receiver and Packet Footer (PF) are not equal.

8.5.2.9 Processor to Peripheral Direction Packet Data Types

The set of transaction types sent from the host processor to a peripheral, such as a display module, are show as below table.

Data type	Data type, binary	Description packet	Size
01h	00 0001	Sync Event, V Sync Start	Short
11h	01 0001	Sync Event, V Sync End	Short
21h	10 0001	Sync Event, H Sync Start	Short
31h	11 0001	Sync Event, H Sync End	Short
08h	00 1000	End of Transmission packet (EoTp)	Short
02h	00 0010	Color Mode (CM) Off Command	Short
12h	01 0010	Color Mode (CM) On Command	Short
22h	10 0010	Shut Down Peripheral Command	Short
32h	11 0010	Turn On Peripheral Command	Short
03h	00 0011	Generic Short WRITE, no parameters	Short
13h	01 0011	Generic Short WRITE, 1 parameter	Short
23h	10 0011	Generic Short WRITE, 2 parameters	Short
04h	00 0100	Generic READ, no parameters	Short
14h	01 0100	Generic READ, 1 parameter	Short
24h	10 0100	Generic READ, 2 parameters	Short
05h	00 0101	DCS Short WRITE, no parameters	Short
15h	01 0101	DCS Short WRITE, 1 parameter	Short
06h	00 0110	DCS READ, no parameters	Short
37h	11 0111	Set Maximum Return Packet Size	Short
09h	00 1001	Null Packet, no data	Long
19h	01 1001	Blanking Packet, no data	Long
39h	11 1001	DCS Long Write/write_LUT Command Packet	Long
0Eh	00 1110	Packed Pixel Stream, 16-bit RGB, 5-6-5 Format	Long
1Eh	01 1110	Packed Pixel Stream, 18-bit RGB, 6-6-6 Format	Long
2Eh	10 1110	Loosely Packed Pixel Stream, 18-bit RGB, 6-6-6 Format	Long
3Eh	11 1110	Packed Pixel Stream, 24-bit RGB, 8-8-8 Format	Long
X0h / XFh, unspecified	xx 0000 xx 1111	DO NOT USE All unspecified codes are reserved	

Data Types for Processor-sourced Packets

All detail function of data types is as below：

Sync event (H Start, H End, V Start, V End), Data Type = xx 0001 (x1h)

Sync event (H start, H end, V start, V end), data type=xx 0001 (x1h)		
Data type, hex	**Function description**	**Number of bytes**
01h	Sync Event, V Sync Start	Short
11h	Sync Event, V Sync End	Short
21h	Sync Event, H Sync Start	Short
31h	Sync Event, H Sync End	Short
Note: In order to represent timing information as accurately as possible a V Sync Start event represents the start of the VSA and also implies an H Sync Start event for the first line of the VSA. Similarly, a V Sync End event implies an H Sync Start event for the last line of the VSA.		

End of Transmission packet (EoTp)

Color mode status (Color Mode On, Color Mode Off)		
Data type, hex	**Function description**	**Number of bytes**
08h	Color Mode On that switches a Video Mode display module to a low-color mode for power saving.	Short
12h	Color Mode Off that switches a Video Mode display module from low-color display to normal display.	Short

Color mode status

Color mode status (Color Mode On, Color Mode Off)		
Data type, hex	**Function description**	**Number of bytes**
02h	Color Mode On that switches a Video Mode display module to a low-color mode for power saving.	Short
12h	Color Mode Off that switches a Video Mode display module from low-color display to normal display.	Short

Display status

Display status (shutdown command, turn-on command)		
Data type, hex	**Function description**	**Number of bytes**
22h	Shutdown Peripheral command that turns off the display in a Video Mode display for power saving.	Short
32h	Turn On Peripheral command that turns on the display in Video Mode display for normal display.	Short
Note: When use shutdown command, interface shall remain powered in order to receive the turn-on, or wake-up, command.		

DCS command setting

DCS command setting		
Data type, hex	**Function description**	**Number of bytes**
05/15h	DCS Short Write command is used to write a single data byte to a peripheral such as a display module. If a parameter is not required, the parameter byte shall be 00h.	Short
06h	DCS Read command, the returned data may be of Short or Long packet format.	Short
39h	DCS Long Write/ Write _ LUT Command is used to send larger blocks of data to a display module that implements the Display Command Set.	long

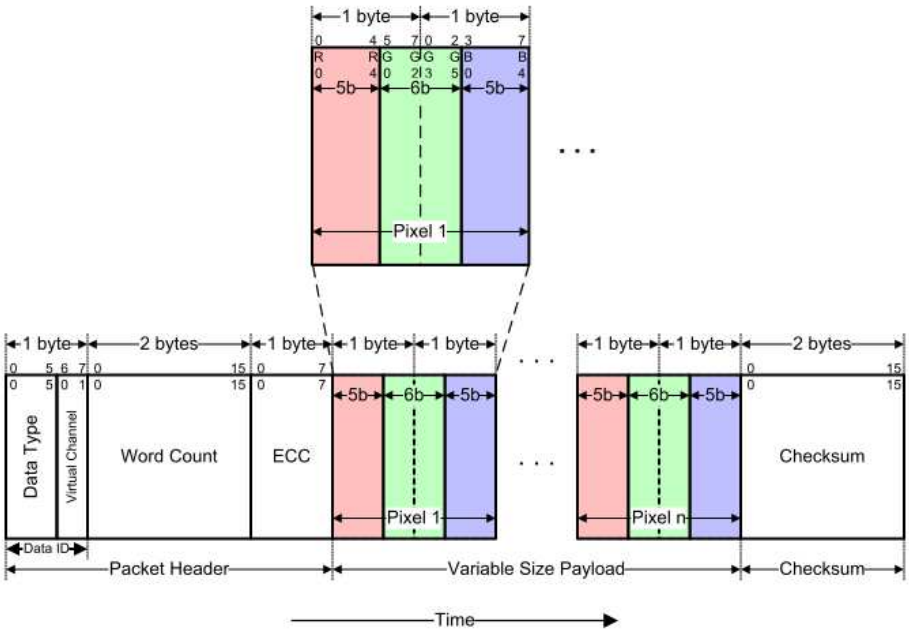
Return packet size setting

Color mode status (Color Mode On, Color Mode Off)		
Data type, hex	**Function description**	**Number of bytes**
37h	Set Maximum Return Packet Size that specifies the maximum size of the payload in a Long packet transmitted from peripheral back to the host processor.	Short
Note: The two-byte value is transmitted with LS byte first. And during a power-on or Reset sequence, the Maximum Return Packet Size shall be set by the peripheral to a default value of one.		

Variable data packet

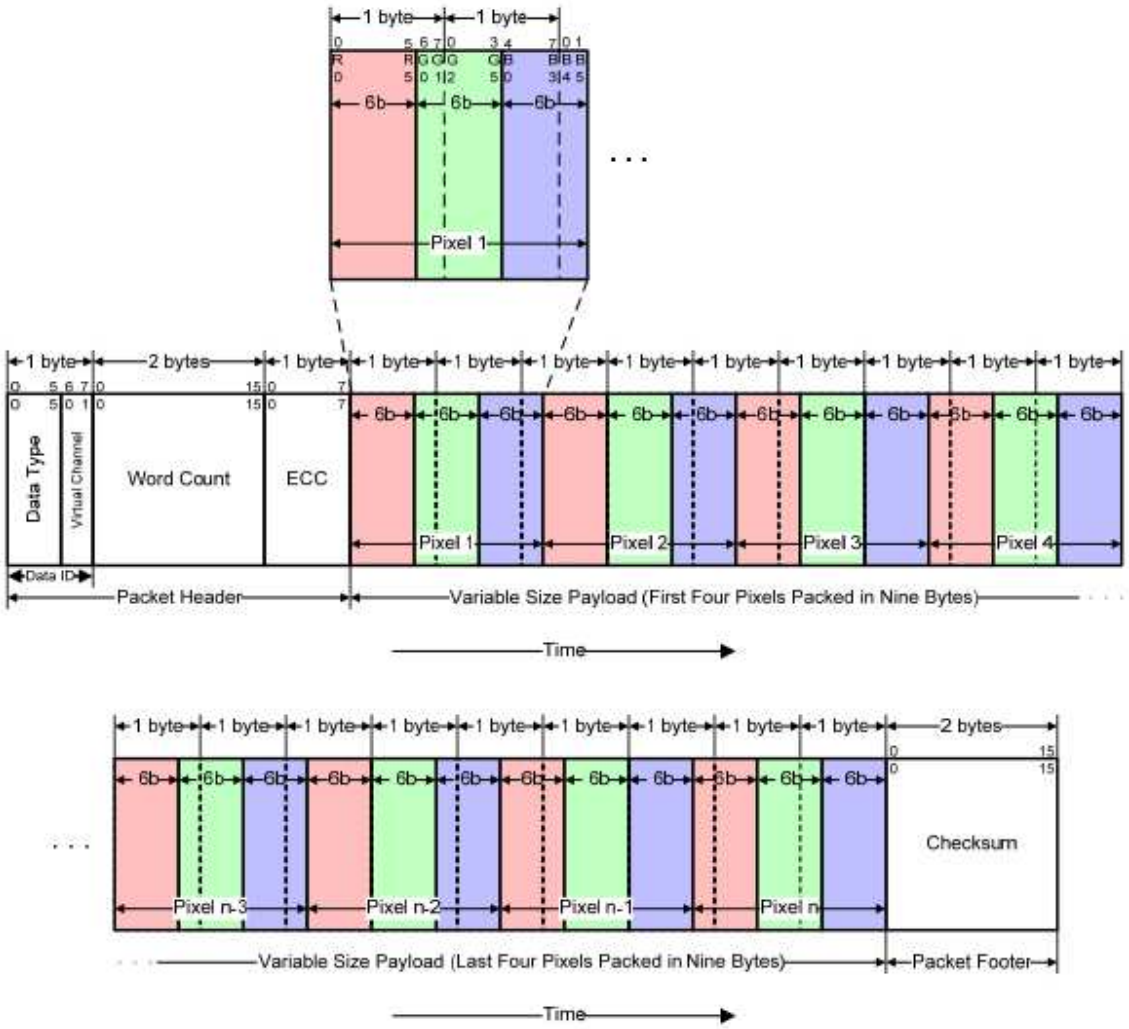
Display status (shutdown command, turn-on command)		
Data type, hex	**Function description**	**Number of bytes**
09h	Null Packet is a mechanism for keeping the serial Data Lane(s) in High-Speed mode while sending dummy data.	Short
19h	Blanking packet is used to convey blanking timing information in a Long packet.	Short
Note: (1) When Null Packet, the Payload Data belong “null” Data, actual data values sent are irrelevant because the peripheral does not capture or store the data. (2) When Blanking packet, the packet represents a period between active scan lines of a Video Mode display,		

Data stream format – 16bit Format

Data stream format – 16bit Format		
Data type, hex	**Function description**	**Number of bytes**
0Eh	Packed Pixel Stream 16-Bit Format is a Long packet used to transmit image data formatted as 16-bit pixels to a Video Mode display module. Pixel format is five bits red, six bits green, five bits blue, in that order.	long
		
Note: That the “Green” component is split across two bytes. Within a color component, the LSB is sent first, the MSB last.		

Data stream format – 18bit Format (mode1)

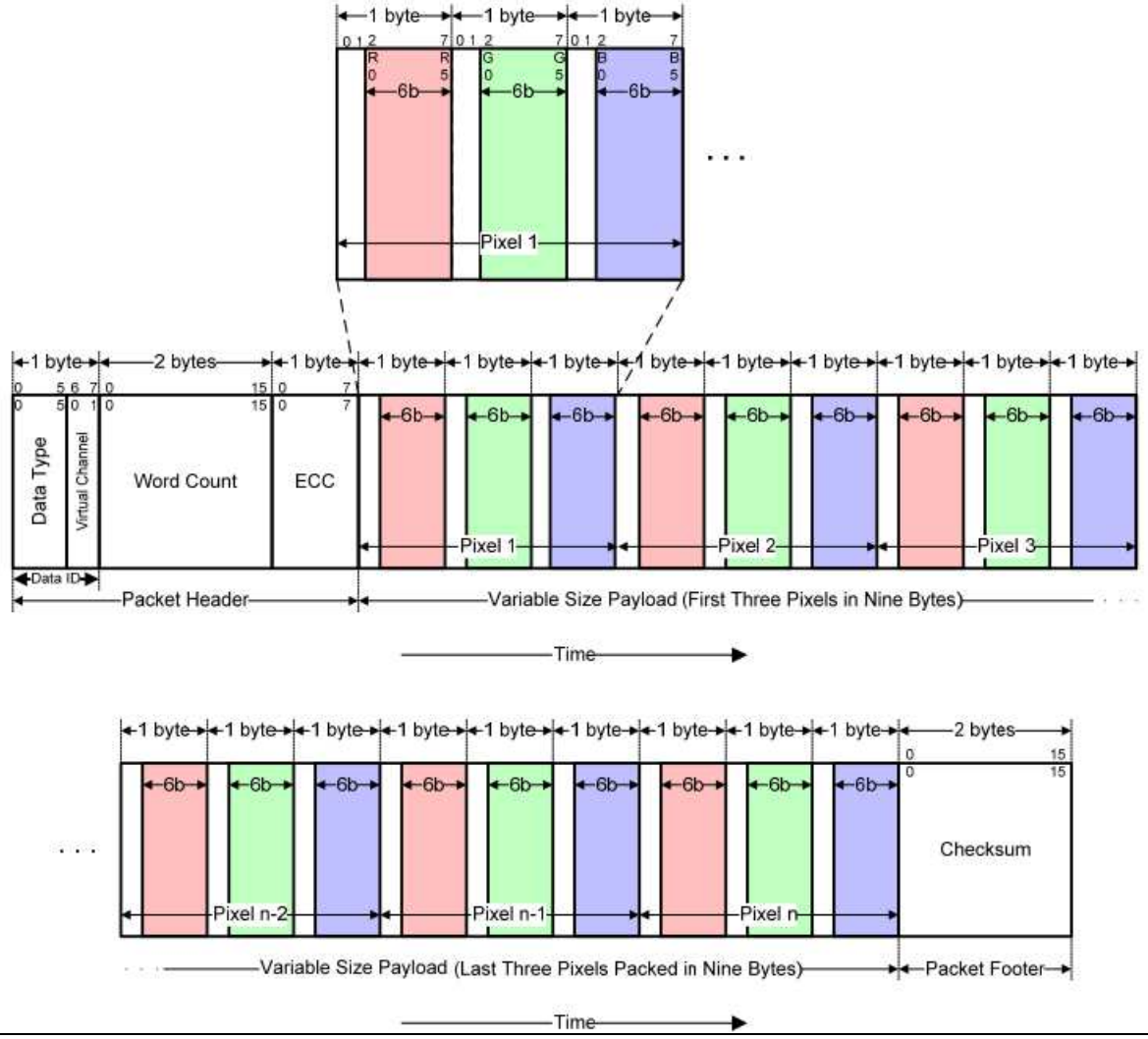
Data stream format – 18bit Format(Mode1)		
Data type, hex	**Function description**	**Number of bytes**
1Eh	Packed Pixel Stream 18-Bit Format is a Long packet used to transmit image data formatted as 18-bit pixels to a Video Mode display module. Pixel format is six bits red, six bits green, six bits blue, in that order.	long



Note: Within a color component, the LSB is sent first and the MSB last and pixel boundaries only line up with byte boundaries every four pixels (nine bytes). Preferably, display modules employing this format have a horizontal extent (width in pixels) evenly divisible by four, so no partial bytes remain at the end of the display line data. It is possible to send pixel data that represent a line width that is not a multiple of four pixels, but display logic on the receiver end shall dispose of the extra bits of the partial byte at the end of active display and ensure a “clean start” for the next line.

Data stream format – 18bit Format(mode2)

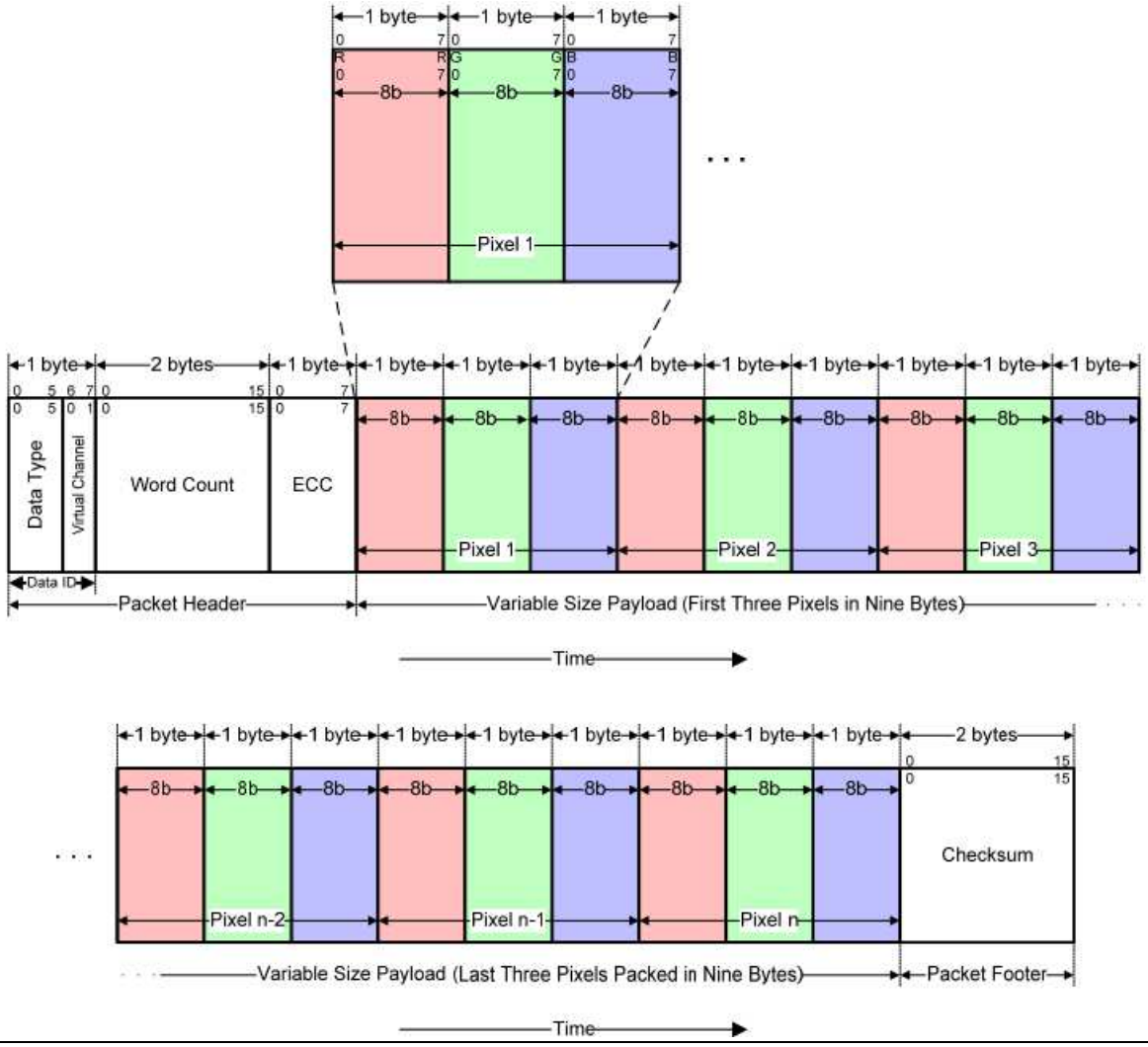
Data stream format – 18bit Format(Mode2)		
Data type, hex	**Function description**	**Number of bytes**
2Eh	In the 18-bit Pixel Loosely Packed format, each R, G, or B color component is six bits but is shifted to the upper bits of the byte, such that the valid pixel bits occupy bits [7:2] of each byte. Bits[1:0] of each payload byte representing active pixels are ignored.	long



Note: Within a color component, the LSB is sent first, the MSB last and With this format, pixel boundaries line up with byte boundaries every three bytes.

Data stream forma – 24bit Format

Data stream format – 18bit Format(Mode2)		
Data type, hex	**Function description**	**Number of bytes**
3Eh	Packed Pixel Stream 24-Bit Format is used to transmit image data formatted as 24-bit pixels to a Video Mode display module. Pixel format is (8 bits) red, (8 bits) green and (8 bits) blue.	long



Note: Within a color component, the LSB is sent first, the MSB last and With this format, pixel boundaries line up with byte boundaries every three bytes.

8.5.2.10 Peripheral-to-Processor (Reverse Direction) LP Transmissions

All Command Mode systems require bidirectional capability for returning READ data, acknowledge, or error information to the host processor. Multi-Lane systems shall use Lane 0 for all peripheral-to-processor transmissions; other Lanes shall be unidirectional. Reverse-direction signaling shall only use LP (Low Power) mode of transmission.

Peripheral-to-processor transactions are of four basic types:

- Tearing Effect is a Trigger message sent to convey display timing information to the host processor. Trigger messages ate signal byte packets sent by a peripheral's PHY layer in response to a signal form the DSI protocol layer.
- Acknowledge is a Trigger Message sent when the current transmission, as well as all preceding transmissions since the last peripheral to host communication.
- Acknowledge and Error Report is a Short packet sent if any errors were detected in preceding transmission from the host processor. Once reported, accumulated errors in the error register are cleared.
- Response to Read Request may be Short or Long packet that returns data requested by the preceding READ command from the processor.

In general, if the host processor completes a transmission to the peripheral with BTA asserted, the peripheral shall respond with one or more appropriate packet(s), and then return bus ownership to the host processor. If BTA is not asserted following a transmission from the host processor, the peripheral shall not communicate an Acknowledge or error information back to the host processor.

Interpretation of processor-to-peripheral transactions with BTA asserted, and the expected responses, are as follows:

- Following a non-Read command in which no error was detected, the peripheral shall respond with Acknowledge.
- Following a Read request in which no error was detected, the peripheral shall send the requested READ data.
- Following a Read request in which the ECC error was detected and corrected, the Peripheral shall send the requested READ data in a Long or Short packet, followed by a 4-byte (Acknowledge with Error Report) packet in the same LP transmission. The Error Report shall have the ECC Error flag set.
- Following a non-Read command in which the ECC error was detected and corrected, the peripheral shall proceed to execute the command, and shall respond to BTA by sending a 4-byte (Acknowledge with Error Report) packet, the Error Report shall have the ECC Error flag set.
- Following any command in which SoT Error, SoT Sync Error, EoT Sync Error, LP Transmit Sync Error, checksum error or DSI VC ID Invalid was detected, or the DSI command was not recognized, the peripheral shall send a 4-byte Acknowledge with Error Report response, with the appropriate error flags set in the two-byte error field. Only the ACK/Error Report packet shall be transmitted; no read or write

accesses shall take place on the peripheral in response.

An error report is a Short packet comprised of two bytes following the DI byte, with an ECC byte following the Error Report bytes. By convention, detection and reporting of each error type is signified by setting the corresponding bit to “1”. Table 18 shows the bit assignment for all error reporting.

Bit	Error Report Bit Description
0	SoT Error
1	SoT Sync Error
2	EoT Sync Error
3	Escape Mode Entry Command Error
4	Low-Power Transmit Sync Error
5	HS Receive Timeout Error
6	False Control Error
7	Reserved
8	ECC Error, single-bit (detected and corrected)
9	ECC Error, multi-bit (detected, not corrected)
10	Checksum Error (Long packet only)
11	DSI Data Type Not Recognized
12	DSI VC ID Invalid
13	Invalid Transmission Length
14	Reserved
15	DSI Protocol Violation

The table as below presents the complete set of peripheral-to-processor Data Types

Data type, hex	Data type, binary	Description packet	Size
02h	00 0010	Acknowledge and Error Report	Short
08h	00 1000	End of Transmission packet (EoTp)	Short
11h	01 0001	Generic Short READ Response, 1 byte returned	Short
12h	01 0010	Generic Short READ Response, 2 bytes returned	Short
1Ah	01 1010	Generic Long READ Response	Short
1Ch	01 1100	DCS Long READ Response	Short
21h	10 0001	DCS Short READ Response, 1 byte returned	Short
22h	10 0010	DCS Short READ Response, 2 bytes returned	Short

Data Types for Peripheral-sourced Packets

8.6.. Mobile Display Digital Interface (MDDI)

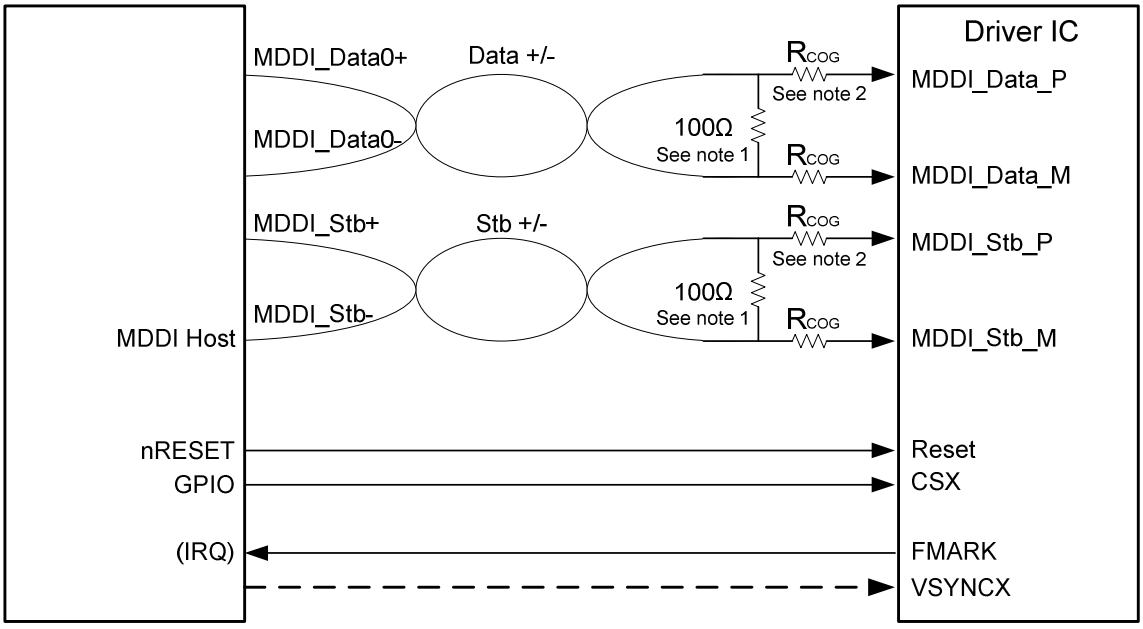
MDDI (Mobile display digital interface) is a differential small amplitude serial interface for high-speed data transfer via following 4 lines: Stb+/- (MDDI_STBP_B, MDDI_STB_M_B), Data+/- (MDDI_DATA_P_B, MDDI_DATA_M_B).

The specifications of MDDI supported by the ST7796 are compatible to the MDDI specifications disclosed by VESA, Video Electronics Standards Association. The following are the specifications particular to the ST7796 MDDI.

8.6.1 ST7796 MDDI Specification

- MDDI Type-I
- High-speed, differential, small-amplitude data transfer via Stb+/-, Data+/- lines
- MDDI client: the ST7796 enables direct connection to the base band (BB) chip without bridge chip
- Cost-performance optimized interface for mobile display systems

1. Only internal mode (one client) and Forward Link are supported
2. Hibernation mode to save power consumption
3. Tearing-free moving picture display via TE/VSYNC interface
4. Moving picture display with low power consumption, realized by the features 2 ~ 3
5. Shutdown mode for saving power consumption in the standby state



MDDI Interface Connection

Notes :

1. An external termination resistor of 100 ohm between Data+ and Data- lines is necessary.

2. Make the COG wiring resistances of Data+/-, Stb+/- lines as small as possible (RCOG < 10 ohm).

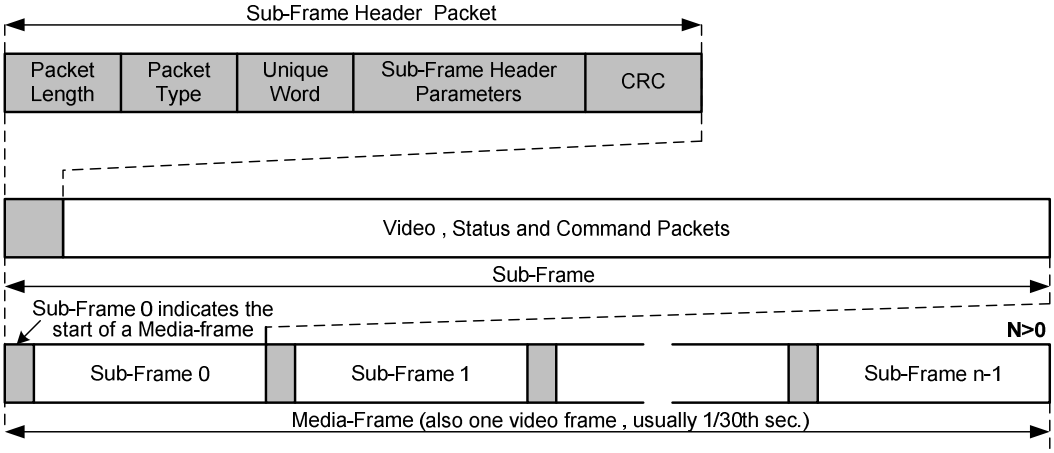
3. The max transmission rate is 130 Mbps!

8.6.2 Supported MDDI Packets

The MDDI Link Protocol of the ST7796 is in line with the MDDI specifications disclosed by VESA. See the MDDI specifications by VESA for details on the MDDI Link Protocol.

The following table describes 9 types of packet which is supported in ST7796. Do not send packets which are not supported by ST7796.

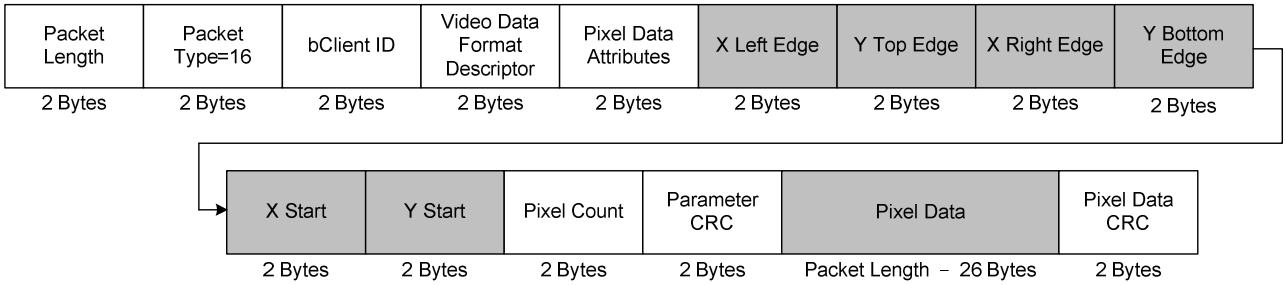
Packet	Function	Direction
Sub-frame header packet	Header of each sub frame	Forward
Register access packet	Register setting	Forward
Video stream packet	Video data transfer	Forward
Filler packet	Fill empty packet space	Forward
Link shut down packet	End of frame	Forward
Reverse link encapsulation packet	Reverse data packet	Reverse
Round-trip delay measurement packet	Host->client->host delay check	Forward/Reverse
Client capability packet	Capability of client check	Reverse
Client request and status packet	Information about client status	Reverse



MDDI Frame and Packet Structure

8.6.2.1 Video Stream Packet

The ST7796 writes pixel data to Display RAM via Video Stream Packet. The window and RAM addresses are set via Register Access Packet.



	0	1	2	3	4	5	6	7
1	Packet Length							
2								
3	Packet Type							
4	(0010h)							
5	bClient ID							
6	(0000)h							
7	Video Data Format Descriptor							
8								
9	Bit 0	Bit 1	Pixel Data Attributes					
10								
11	X Left Edge							
12								
13	Y Top Edge							
14								
15	X Right Edge							
16								
17	Y Bottom Edge							
18								
19	X Start							
20								
21	Y Start							
22								
23	Pixel Count							
24								
25	Parameter CRC							
26								
	Pixel Data (Packet Length – 26Bytes)							
	CRC							

Note:The parameters colored in gray are not supported by the ST7796.

Video Data Format Descriptor: Sets the pixel data format. ST7796 supports only the following format. Set the same pixel format (bpp) as selected by DSS[1:0] in Video Data Format Descriptor.

[15:13]	[12]	[11:8]	[7:4]	[3:0]	
“010”	1	5h	6h	5h	Packed 16bpp RGB Format (R:G:B = 5:6:5)
“010”	1	6h	6h	6h	Packed 18bpp RGB Format (R:G:B = 6:6:6)
Others					Setting Prohibited

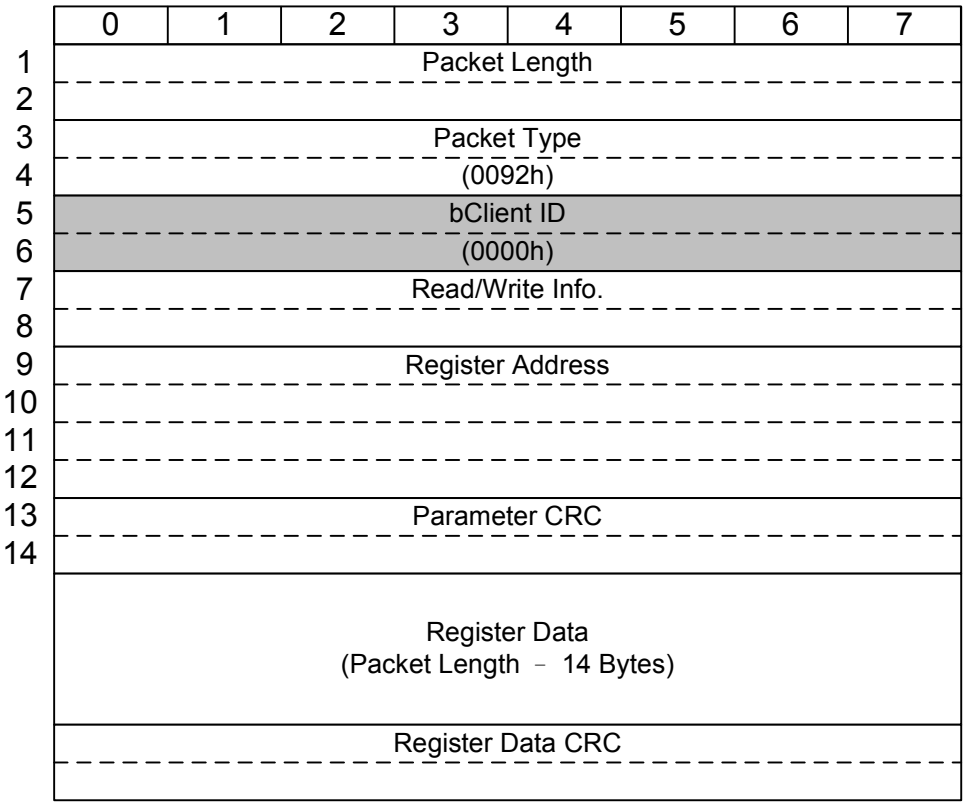
	MDDI Bytes n								MDDI Bytes (n+1)								MDDI Bytes (n+2)							
	0	1	2	3	4	5	6	7	0	1	2	3	4	5	6	7	0	1	2	3	4	5	6	7
Packet 16bpp	0	1	2	3	4	0	1	2	3	4	5	0	1	2	3	4	0	1	2	3	4	0	1	2
	Pixel 1 Blue					Pixel 1 Green						Pixel 1 Red					Pixel 2 Blue					Pixel 2		
Packet 18bpp	0	1	2	3	4	5	0	1	2	3	4	5	0	1	2	3	4	5	0	1	2	3	4	5
	Pixel 2 Blue						Pixel 2 Green						Pixel 2 Red						Pixel 2 Blue					

Pixel Data Attributes: the image data sent vial Video Stream Packet is recognized as either the data for the main-panel according to the setting in [1:0] bits in this field.

Pixel Data Attributes	Bits[1:0]	Description
0000h	00	Not support sub-panel to ST7796.
0001h	01	Setting Prohibited
0002h	10	Setting Prohibited
0003h	11	The video stream packet data is recognized as the data written to the ST7796. The Video stream packet data is written to the ST7796.
Others		

8.6.2.2 Register Access Packet

Register Access Packet is used when setting instruction to the ST7796.



Note : The parameters colored in gray are not supported by the ST7796.

Read/Write Info: Read or Write information in register access. ST7796 supports the following access setting.

Bits[15:14]	Bits[13:0]	Description
”00”	0001h	Write one register by register access packet
Others	-	Setting Prohibited

Register Address: The index of the register to be accessed is set in Register Address area and the Register Address Packet is directed to the ST7796 by the setting in Register Address area.

Bits[31:16]	Description
0000h	The Register Access Packet is directed to the ST7796 via main-display interface.
0001h	Setting disable
0002h~7FFFh	Setting disable

Bits[15:0]	Description
0000h~FFFFh	Bits[15:0] are used as register index [15:0].

Register Data: The data for register access is written in Register Data. The length of Register Data depends on the parameter length of command.

Example of Register Access Packet (e.g. write to the ST7796)

	0	1	2	3	4	5	6	7
1	Packet Length							(12h)
2								(00h)
3	Packet Type							(92h)
4								(00h)
5	bClient ID							(00h)
6								(00h)
7	Read/Write Info.							(01h)
8								(00h)
9	Register Address							(Index ID[7:0])
10								(Index ID[15:8])
11	(00h) -> Main Display (ST7793) (01h) -> Sub Display							
12								(00h)
13	Parameter CRC							
14								
15	Register Data List (Variable Length)							1st Parameter
16								2nd Parameter
17								3rd Parameter
18								(00h)
19	Parameter CRC							
20								

Note: The parameters colored in gray are not supported by the ST7796.

Register Access Packet Restrictions: ST7796's internal RAM is accessible via Video Stream Packet. RAM access data is not included in Register Access Packet.

8.6.2.3 Link Shutdown Packet

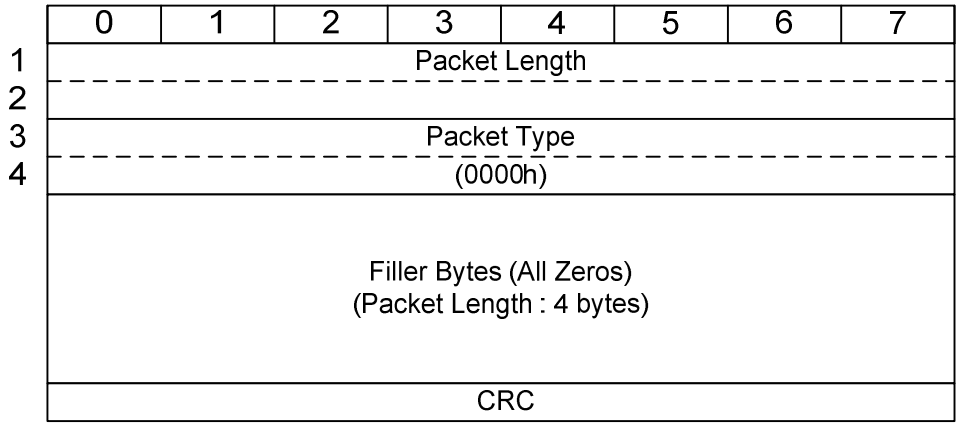
This packet is used to bring Link to the Hibernation state.

	0	1	2	3	4	5	6	7
1	Packet Length							
2	(0014h)							
3	Packet Type							
4	(0045h)							
5	Parameter CRC							
6								
7	All Zeros (Type-I : 16 Bytes)							
22								

Note: The parameters colored in gray are not supported by the ST7796.

8.6.2.4 Filler Packet

The Filler Packet is sent when no other information is available to be sent on the forward or reverse link. It is recommended to send filler packets with minimum length to allow maximum flexibility to send other packets when required. At the very end of a sub-frame or a reverse link encapsulation packet the MDDI link controller shall set the size of the Filler Packet to exactly fill the remaining space to maintain packet integrity. The Filler Packet is required to maintain timing on the link when the host or client have no information to send. Every host and client shall be able to send and receive this packet.



8.6.2.5 Reverse link encapsulation packet

Data is transferred in the reverse direction using the Reverse Link Encapsulation Packet. A forward link packet is sent and the MDDI link is turned around in the middle of this packet so that packets can be sent in the reverse direction. The MDDI_Stb signal is always driven by the host. The host behaves as if it were transmitting a zero for each bit of the Turn-Around, Driver Re-enable, nd

Reverse Data Packets fields of the packet. The result is that the MDDI_Stb toggles at each bit boundary of these portions of the packet. The format of the Reverse Link Encapsulation Packet is as below. For External Mode every host shall be able to generate this packet and receive data, and every client shall be able to receive and send data to the host. Implementation of this packet is optional for Internal Mode, but the Reverse Link Encapsulation Packet is necessary for the host to receive data from the client.

8.6.2.6 Round-trip delay measurement packet

The Round-Trip Delay Measurement Packet is used to measure the propagation delay from the host to the client plus the delay from the client back to the host. This measurement inherently includes all of the delays that exist in the line drivers and receivers and the interconnect subsystem. This measurement is used to set the turn around delay and reverse link rate divisor parameters in the Reverse Link Encapsulation Packet. This packet is most useful when the MDDI link is running at the maximum speed intended for a particular application.

8.6.2.7 Client capability packet

The host needs to know the capability of the client so it can configure the host-to-client link in an optimum manner. It is recommended that the client send a Client Capability Packet to the host after forward link synchronization is acquired, and it is required when requested by the host via the Reverse Link Flags in the Reverse Link Encapsulation Packet. The Client Capability acket is required to inform the host of the capabilities of the client. For External Mode every host shall be able to receive this packet, and every client shall be able to send this packet. Implementation of this packet is optional for Internal Mode.

8.6.2.8 Client request and status packet

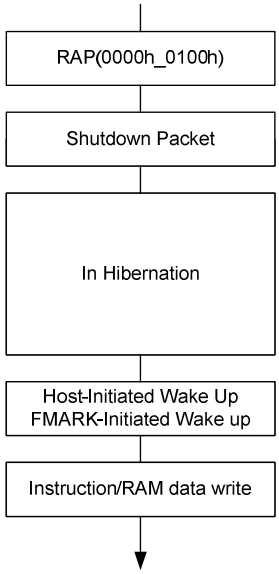
The host needs a small amount of information from the client so it can configure the host-to-client link in an optimum manner. It is recommended that the client send one Client Request and Status Packet to the host each sub-frame. It is recommended that the client send this packet as the first packet in the Reverse Link Encapsulation Packet to ensure that it is delivered reliably to the host, and it is required when requested by the host via the Reverse Link Flags in the Reverse Link Encapsulation Packet. The Client Request and Status Packet is required to report errors and status to the host. For external mode every host shall be able to receive this packet, and every client shall be able to send this packet. It is highly recommended that internal mode hosts and clients also support this packet, but it is not required.

8.6.3 Hibernation Setting

The ST7796 Client MDDI supports Hibernation setting. There are two ways to cancel the Hibernation setting, which can be selected according to the condition of use.

Hibernation Cancellation	
Host-Initiated Wake Up	In Power-Saving Mode Such as Standby
TE-Initiated Wake Up	Save power consumption in transferring moving picture data host-initiated wake up triggered by the output from TE.

The Hibernation setting and cancellation sequence must be compatible with the VESA-MDDI specifications.



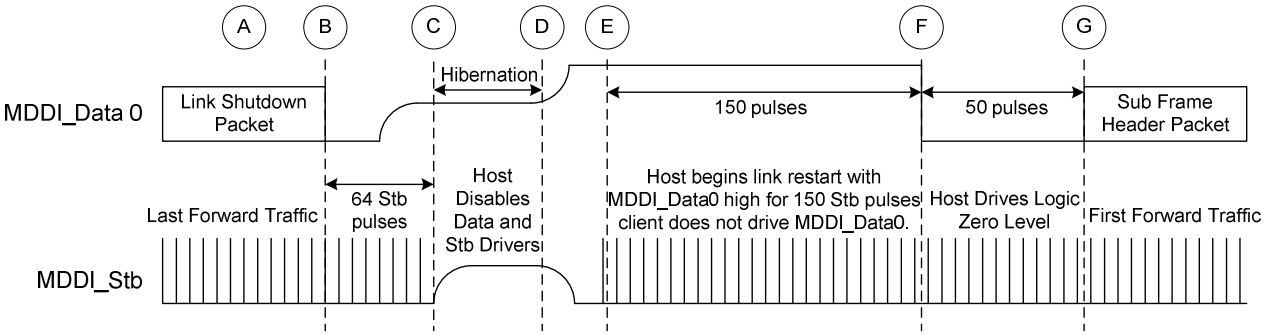
Host-Initiated Wake up from Hibernation

The host initialed wake up is described below without contention from the client trying to wake up at the same time. The following sequence of events is illustrated in the figures below!

- The host sends a Link Shutdown Packet to inform the client that the link will transition to the low power hibernation state.
- Following the CRC of the Link Shutdown Packet the host toggles MDDI_Stb for 64 cycles to allow processing in the client to finish before it stops MDDI_Stb from toggling which stops the recovered clock in the client device. During the interval the host initially sets MDDI_Data0 to a logic zero level, and then disables the MDDI_Data0 output in the range of 16 to 48 MDDI_Stb cycles (including output disable propagation delays) after the CRC. It may be desirable for the client to place its high-speed receivers for MDDI_Data0 and MDDI_Stb into a low power state any time after 48 MDDI_Stb cycles after the CRC and before point C.
- The host enters the low power hibernation state by disabling the MDDI_Data0 and MDDI_Stb drivers and by placing the host controller into a low power hibernation state. It is also allowable for MDDI_Stb to be driven to logic zero level or to continue toggling during hibernation. The client is also in the low power hibernation state.
- After a while, the host begins the line restart sequence by enabling the MDDI_Data0 and MDDI_Stb driver outputs. The host drivers MDDI_Data0 to a logic one level and MDDI_Stb to logic zero level for at least 200ns after MDDI_Data0 reaches a valid logic one level and MDDI_Stb reaches a valid logic zero level before driving pulses on MDDI_Stb. This gives the client sufficient time to prepare to receive high speed pulses on MDDI_Stb. The client first detects the wake up pulse using a low power differential receiver having a +125mV input offset voltage.

The host drivers are fully enabled and MDDI_Data0 is being driven to a logic one level. The host begins to toggle MDDI_Stb in a manner consistent with having a logic zero level on MDDI_Data0 for a duration of 150 MDDI_Stb cycles.

- The host drives MDDI_Data0 to a logic zero level for 50 MDDI_Stb cycles. The client begins to look for the Sub frame Header Packet after MDDI_Data0 is at a logic zero level for 40 MDDI_Stb cycles.
- The host begins to transmit data on the forward link by sending a Sub-frame Header packet. Beginning at point G the MDDI host generates MDDI_Stb based on the logic level on MDDI_Data0 so that proper data-strobe encoding commences form point G.



9 COMMAND

9.1.. Command Table List

COMMAND Table 1														
Instruction	**D/CX**	**WRX**	**RDX**	**D17-8**	**D7**	**D6**	**D5**	**D4**	**D3**	**D2**	**D1**	**D0**	**Hex**	**Function**
NOP	0	↑	1	-	0	0	0	0	0	0	0	0	(00h)	No operation
SWRESET	0	↑	1	-	0	0	0	0	0	0	0	1	(01h)	Software reset
RDDID	0	↑	1	-	0	0	0	0	0	1	0	0	(04h)	Read display ID
	1	1	↑	-	-	-	-	-	-	-	-	-		Dummy read
	1	1	↑	-	ID17	ID16	ID15	ID14	ID13	ID12	ID11	ID10		ID1 read
	1	1	↑	-	ID27	ID26	ID25	ID24	ID23	ID22	ID21	ID20		ID2 read
	1	1	↑	-	ID37	ID36	ID35	ID34	ID33	ID32	ID31	ID30		ID3 read
Read Number of the Errors on DSI	0	↑	1		0	0	0	0	0	1	0	1	(05h)	Read DSI
	1	1	↑	-	-	-	-	-	-	-	-	-		Dummy read
	1	1	↑		D7	D6	D5	D4	D3	D2	D1	D0		
RDDST	0	↑	1	-	0	0	0	0	1	0	0	1	(09h)	Read display status
	1	1	↑	-	-	-	-	-	-	-	-	-		Dummy read
	1	1	↑	-	BSTON	MY	MX	MV	ML	RGB	ST25	ST24		-
	1	1	↑	-	ST23	IFPF2	IFPF1	IFPF0	IDMON	PTLON	SLOUT	NORON		-
	1	1	↑	-	VScroll	ST14	INVON	ALLON	ALLOFF	DISON	TEON	GCS2		-
	1	1	↑	-	GCS1	GCS0	TEM	ST4	ST3	ST2	ST1	ST0		-
RDDPM	0	↑	1	-	0	0	0	0	1	0	1	0	(0Ah)	Read display power
	1	1	↑	-	-	-	-	-	-	-	-	-		Dummy read
	1	1	↑	-	BSTON	IDMON	PTLON	SLPOUT	NORON	DISON	0	0		
RDD MADCTL	0	↑	1	-	0	0	0	0	1	0	1	1	(0Bh)	Read display
	1	1	↑	-	-	-	-	-	-	-	-	-		Dummy read
	1	1	↑	-	MY	MX	MV	ML	RGB	DISDL	0	0		-
RDD Interface Pixel Format	0	↑	1	-	0	0	0	0	1	1	0	0	(0Ch)	Read display pixel
	1	1	↑	-	-	-	-	-	-	-	-	-		Dummy read
	1	1	↑	-	R3	R2	R1	R0	0	D2	D1	D0		-

COMMAND Table 1														
Instruction	**D/CX**	**WRX**	**RDX**	**D17-8**	**D7**	**D6**	**D5**	**D4**	**D3**	**D2**	**D1**	**D0**	**Hex**	**Function**
RDDIM	0	↑	1	-	0	0	0	0	1	1	0	1	(0Dh)	Read display image
	1	1	↑	-	-	-	-	-	-	-	-	-		Dummy read
	1	1	↑	-	VSSON	0	INVON	0	0	GC2	GC1	GC0		-
RDDSM	0	↑	1	-	0	0	0	0	1	1	1	0	(0Eh)	Read display signal
	1	1	↑	-	-	-	-	-	-	-	-	-		Dummy read
	1	1	↑	-	TEON	TEM	HSYN	VSYN	PIXCLK	DATEN	0	DSIER		-
RDDSDR	0	↑	1	-	0	0	0	0	1	1	1	1	(0Fh)	Read display self-diagnostic result
	1	1	↑	-	-	-	-	-	-	-	-	-		Dummy read
	1	1	↑	-	D7	D6	0	0	0	0	0	D0		-
SLPIN	0	↑	1	-	0	0	0	1	0	0	0	0	(10h)	Sleep in
SLPOUT	0	↑	1	-	0	0	0	1	0	0	0	1	(11h)	Sleep out
PTLON	0	↑	1	-	0	0	0	1	0	0	1	0	(12h)	Partial mode on
NORON	0	↑	1	-	0	0	0	1	0	0	1	1	(13h)	Partial off (Normal)
INVOFF	0	↑	1	-	0	0	1	0	0	0	0	0	(20h)	Display inversion off
INVON	0	↑	1	-	0	0	1	0	0	0	0	1	(21h)	Display inversion on
DISPOFF	0	↑	1	-	0	0	1	0	1	0	0	0	(28h)	Display off
DISPON	0	↑	1	-	0	0	1	0	1	0	0	1	(29h)	Display on
CASET	0	↑	1	-	0	0	1	0	1	0	1	0	(2Ah)	Column address set
	1	↑	1	-	XS15	XS14	XS13	XS12	XS11	XS10	XS9	XS8		X address start: 0≦XS≦X
	1	↑	1		XS7	XS6	XS5	XS4	XS3	XS2	XS1	XS0		
	1	↑	1		XE15	XE14	XE13	XE12	XE11	XE10	XE9	XE8		X address start: S≦XE≦X
	1	↑	1		XE7	XE6	XE5	XE4	XE3	XE2	XE1	XE0		

COMMAND Table 1														
Instruction	**D/CX**	**WRX**	**RDX**	**D17-8**	**D7**	**D6**	**D5**	**D4**	**D3**	**D2**	**D1**	**D0**	**Hex**	**Function**
RASET	0	↑	1	-	0	0	1	0	1	0	1	1	(2Bh)	Row address set
	1	↑	1	-	YS15	YS14	YS13	YS12	YS11	YS10	YS9	YS8		Y address start: 0≦YS≦Y
	1	↑	1		YS7	YS6	YS5	YS4	YS3	YS2	YS1	YS0		
	1	↑	1		YE15	YE14	YE13	YE12	YE11	YE10	YE9	YE8		Y address start: S≦YE≦Y
	1	↑	1		YE7	YE6	YE5	YE4	YE3	YE2	YE1	YE0		
RAMWR	0	↑	1	-	0	0	1	0	1	1	0	0	(2Ch)	Memory write
	1	↑	1	-	D7	D6	D5	D4	D3	D2	D1	D0		Write data
RAMRD	0	↑	1	-	0	0	1	0	1	1	1	0	(2Eh)	Memory read
	1	1	↑	-	-	-	-	-	-	-	-	-		Dummy read
	1	1	↑	-	D7	D6	D5	D4	D3	D2	D1	D0		Read data
PTLAR	0	↑	1	-	0	0	1	1	0	0	0	0	(30h)	Partial sart/end address set
	1	↑	1	-	PSL15	PSL14	PSL13	PSL12	PSL11	PSL10	PSL9	PSL8		Partial start address: (0, 1,2, ..P)
	1	↑	1	-	PSL7	PSL6	PSL5	PSL4	PSL3	PSL2	PSL1	PSL0		
	1	↑	1	-	PEL15	PEL14	PEL13	PEL12	PEL11	PEL10	PEL9	PEL8		Partial end address (0, 1,2, 3, , P)
	1	↑	1	-	PEL7	PEL6	PEL5	PEL4	PEL3	PEL2	PEL1	PEL0		
VSCRDEF	0	↑	1	-	0	0	1	1	0	0	1	1	(33h)	Vertical scrolling definition
	1	↑	1	-	TFA15	TFA14	TFA13	TFA12	TFA11	TFA10	TFA9	TFA8		
	1	↑	1	-	TFA7	TFA6	TFA5	TFA4	TFA3	TFA2	TFA1	TFA0		
	1	↑	1	-	VSA15	VSA14	VSA13	VSA12	VSA11	VSA10	VSA9	VSA8		
	1	↑	1	-	VSA7	VSA6	VSA5	VSA4	VSA3	VSA2	VSA1	VSA0		
	1	↑	1	-	BFA15	BFA14	BFA13	BFA12	BFA11	BFA10	BFA9	BFA8		
	1	↑	1	-	BFA7	BFA6	BFA5	BFA4	BFA3	BFA2	BFA1	BFA0		
TEOFF	0	↑	1	-	0	0	1	1	0	1	0	0	(34h)	Tearing effect line off
TEON	0	↑	1	-	0	0	1	1	0	1	0	1	(35h)	Tearing effect line on

COMMAND Table 1														
Instruction	**D/CX**	**WRX**	**RDX**	**D17-8**	**D7**	**D6**	**D5**	**D4**	**D3**	**D2**	**D1**	**D0**	**Hex**	**Function**
	1	↑	1	-	-	-	-	-	-	-	-	TEM		
MADCTL	0	↑	1	-	0	0	1	1	0	1	1	0	(36h)	Memory data access control
	1	↑	1	-	MY	MX	MV	ML	RGB	MH	0	0		-
VSCRSADD	0	↑	1	-	0	0	1	1	0	1	1	1	(37h)	Vertical scrolling start address
	1	↑	1	-	VSP15	VSP14	VSP13	VSP12	VSP11	VSP10	VSP9	VSP8		
	1	↑	1	-	VSP7	VSP6	VSP5	VSP4	VSP3	VSP2	VSP1	VSP0		
IDMOFF	0	↑	1	-	0	0	1	1	1	0	0	0	(38h)	Idle mode off
IDMON	0	↑	1	-	0	0	1	1	1	0	0	1	(39h)	Idle mode on
Interface Pixel Format	0	↑	1	-	0	0	1	1	1	0	1	0	(3Ah)	Interface pixel format
	1	↑	1	-	R3	R2	R1	R0	0	D2	D1	D0		Interface format
RAMWRC	0	↑	1	-	0	0	1	1	1	1	0	0	(3Ch)	Memory write continue
	1	↑	1	-	D17	D16	D15	D14	D13	D12	D11	D10		
	1	↑	1	-	Dx7	Dx6	Dx5	Dx4	Dx3	Dx2	Dx1	Dx0		
	1	↑	1	-	Dn7	Dn6	Dn5	Dn4	Dn3	Dn2	Dn1	Dn0		
RAMRDC	0	↑	1	-	0	0	1	1	1	1	1	0	(3Eh)	Memory read continue
	1	1	↑	-	-	-	-	-	-	-	-	-		Dummy Read
	1	1	↑	-	D17	D16	D15	D14	D13	D12	D11	D10		
	1	1	↑	-	Dx7	Dx6	Dx5	Dx4	Dx3	Dx2	Dx1	Dx0		
	1	1	↑	-	Dn7	Dn6	Dn5	Dn4	Dn3	Dn2	Dn1	Dn0		
TESCAN	0	↑	1	-	0	1	0	0	0	1	0	0	(44h)	Set tear scanline
	1	↑	1	-	N15	N14	N13	N12	N11	N10	N9	N8		
	1	↑	1	-	N7	N6	N5	N4	N3	N2	N1	N0		
RDTESCAN	0	↑	1	-	0	1	0	0	0	1	0	1	(45h)	Get scanline
	1	1	↑	-	-	-	-	-	-	-	-	-		Dummy Read
	1	1	↑	-	N15	N14	N13	N12	N11	N10	N9	N8		
	1	1	↑	-	N7	N6	N5	N4	N3	N2	N1	N0		

COMMAND Table 1														
Instruction	**D/CX**	**WRX**	**RDX**	**D17-8**	**D7**	**D6**	**D5**	**D4**	**D3**	**D2**	**D1**	**D0**	**Hex**	**Function**
WRDISBV	0	↑	1	-	0	1	0	1	0	0	0	1	(51h)	Write display brightness
	1	↑	1	-	DBV7	DBV6	DBV5	DBV4	DBV3	DBV2	DBV1	DBV0		
RDDISBV	0	↑	1	-	0	1	0	1	0	0	1	0	(52h)	Read display brightness value
	1	1	↑	-	-	-	-	-	-	-	-	-		Dummy read
	1	1	↑	-	DBV7	DBV6	DBV5	DBV4	DBV3	DBV2	DBV1	DBV0		
WRCTRLD	0	↑	1	-	0	1	0	1	0	0	1	1	(53h)	Write CTRL display
	1	↑	1	-	0	0	BCTRL	0	DD	BL	0	0		
RDCTRLD	0	↑	1	-	0	1	0	1	0	1	0	0	(54h)	Read CTRL value dsiplay
	1	1	↑	-	-	-	-	-	-	-	-	-		Dummy read
	1	1	↑	-	0	0	BCTRL	0	DD	BL	0	0		
WRCABC	0	↑	1	-	0	1	0	1	0	1	0	1	(55h)	Write content adaptive brightness control
	1	↑	1	-	CECTRL	0	CE1	CE0	0	0	C1	C0		
RDCABC	0	↑	1	-	0	1	0	1	0	1	1	0	(56h)	Read content adaptive brightness control
	1	1	↑	-	-	-	-	-	-	-	-	-		Dummy read
	1	1	↑	-	0	0	0	0	0	0	C1	C0		
WRCABCMB	0	↑	1	-	0	1	0	1	1	1	1	0	(5Eh)	Write CABC minimum brightness
	1	↑	1	-	CMB7	CMB6	CMB5	CMB4	CMB3	CMB2	CMB1	CMB0		
RDCABCMB	0	↑	1	-	0	1	0	1	1	1	1	1	(5Fh)	Read CABC minimum brightness
	1	1	↑	-	-	-	-	-	-	-	-	-		Dummy read
	1	1	↑	-	CMB7	CMB6	CMB5	CMB4	CMB3	CMB2	CMB1	CMB0		

COMMAND Table 1														
Instruction	**D/CX**	**WRX**	**RDX**	**D17-8**	**D7**	**D6**	**D5**	**D4**	**D3**	**D2**	**D1**	**D0**	**Hex**	**Function**
RDFCHKSUM	0	↑	1		1	0	1	0	1	0	1	0	(AAh)	Read First Checksum
	1	1	↑		-	-	-	-	-	-	-	-		Dummy read
	1	1	↑		FCS7	FCS6	FCS5	FCS4	FCS3	FCS2	FCS1	FCS0		
RDCCHKSUM	0	↑	1		1	0	1	0	1	0	1	0	(AFh)	Read Continue Checksum
	1	1	↑		-	-	-	-	-	-	-	-		Dummy read
	1	1	↑		CCS7	CCS6	CCS5	CCS4	CCS3	CCS2	CCS1	CCS0		
RDID1	0	↑	1	-	1	1	0	1	1	0	1	0	(DAh)	Read ID1
	1	1	↑	-	-	-	-	-	-	-	-	-		Dummy read
	1	1	↑	-	ID17	ID16	ID15	ID14	ID13	ID12	ID11	ID10		Read parameter
RDID2	0	↑	1	-	1	1	0	1	1	0	1	1	(DBh)	Read ID2
	1	1	↑	-	-	-	-	-	-	-	-	-		Dummy read
	1	1	↑	-	ID27	ID26	ID25	ID24	ID23	ID22	ID21	ID20		Read parameter
RDID3	0	↑	1	-	1	1	0	1	1	1	0	0	(DCh)	Read ID3
	1	1	↑	-	-	-	-	-	-	-	-	-		Dummy read
	1	1	↑		ID37	ID36	ID35	ID34	ID33	ID32	ID31	ID30		Read parameter

Register List

COMMAND Table 2														
Instruction	**D/CX**	**WRX**	**RDX**	**D17-8**	**D7**	**D6**	**D5**	**D4**	**D3**	**D2**	**D1**	**D0**	**Hex**	**Function**
IFMODE	0	↑	1	-	1	0	1	1	0	0	0	0	(B0h)	Interface Mode Control
	1	↑	1		SDI_EN	0	0	0	VSCP	HSCP	PKP	DEP		
FRMCTR1	0	↑	1	-	1	0	1	1	0	0	0	1	(B1h)	Frame Rate Control（In Normal Mode/Full Colors）
	1	↑	1	-	FRS[3:0]				0	0	DIVA[1:0]			
	1	↑	1	-	0	0	0	RTNA[4:0]						
FRMCTR2	0	↑	1	-	1	0	1	1	0	0	1	0	(B2h)	Frame Rate Control（In Idle Mode/8 colors）
	1	↑	1	-	0	0	0	0	0	0	0	0		
	1	↑	1	-	0	0	0	RTNB[4:0]						
FRMCTR3	0	↑	1	-	1	0	1	1	0	0	1	1	(B3h)	Frame Rate Control（In Partial Mode/Full colors）
	1	↑	1	-	0	0	0	0	0	0	0	0	XX	
	1	↑	1	-	0	0	0	RTNC[4:0]					XX	
INVTR	0	↑	1	-	0	0	0	0	1	0	0	1	(B4h)	Display Inversion Control
	1	↑	1	-	-	-	-	-	-	-	DINV			
PRCTR	0	↑	1	-	1	0	1	1	0	1	0	1	(B5h)	Blanking Porch Control
	1	↑	1	-	VFP[7:0]									
	1	↑	1	-	VBP[7:0]									
	1	↑	1	-	0	0	0	HFP[4:0]						
	1	↑	1	-	HBP[7:0]									
DISCTRL	0	↑	1	-	1	0	1	1	0	1	1	0	(B6h)	Display Function Control
	1	↑	1	-	BYPASS	0	RM	DM	PTG[1:0]		PT[1:0]			
	1	↑	1	-	0	GS	SS	SM	ISC[3:0]					
	1	↑	1	-	0	0	NL[5:0]							
ETMOD	0	↑	1	-	1	0	1	1	0	1	1	1	(B7h)	Entry Mode Set-
	1	↑	1	-	EPF[1:0]		0	0	DSTB	GON	DTE	GAS		
PWR1	0	↑	1	-	1	1	0	0	0	0	0	0	(C0h)	Power Control 1
	1	↑	1	-	AVCLS[3:0]				AVDDS[3:0]					
	1	↑	1	-	0	0	VRHP[5:0]							
	1	↑	1	-	0	0	VRHN[5:0]							
PWR2	0	↑	1	-	1	1	0	0	0	0	0	1	(C1h)	Power Control 2
	1	↑	1	-	VGLS[3:0]				VGHS[3:0]					
VCMPCTL	0	↑	1	-	1	1	0	0	0	1	0	1	(C5h)	Vcom Control
	1	↑	1	-	0	0	VCMP[5:0]							

COMMAND Table 2

Instruction	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	Hex	Function
VCM CTL	0	↑	1	-	1	1	0	0	0	1	1	0	(C6h)	Vcom Control
	1	↑	1	-	VMFSEL	-	-	VMF_REG [4:0]						
RDID4	0	↑	1	-	1	1	0	1	0	0	1	1	(D3h)	Read ID4
	1	1	↑	-	-	-	-	-	-	-	-	-	-	
	1	1	↑	-	ID41[7:0]									
	1	1	↑	-	ID42[7:0]									
	1	1	↑	-	ID41[7:0]									
PGC	0	↑	1	-	1	1	1	0	0	0	0	0	(E0h)	Positive Gamma Control
	1	↑	1	-	-	VRF0P[5:0]								
	1	↑	1	-	-	VOS0P[5:0]								
	1	↑	1	-	-	PK0P[5:0]								
	1	↑	1	-	-	PK1P[5:0]								
	1	↑	1	-	-	PK2P[5:0]								
	1	↑	1	-	-	PK3P[5:0]								
	1	↑	1	-	-	PK4P[5:0]								
	1	↑	1	-	-	PK5P[5:0]								
	1	↑	1	-	-	PK6P[5:0]								
	1	↑	1	-	-	PK7P[5:0]								
	1	↑	1	-	-	PK8P[5:0]								
	1	↑	1	-	-	PK9P[5:0]								
	1	↑	1	-		SELV0P[5:0]								
	1	↑	1	-	-	SELV1P[5:0]								
	1	↑	1	-	-	SELV62P[5:0]								
	1	↑	1	-	-	SELV63P[5:0]								
NGC	0	1	↑	-	1	1	1	0	0	0	0	1	(E1h)	Negative Gamma Control
	1	1	↑	-	-	VRF0N[5:0]								
	1	1	↑	-	-	VOS0N[5:0]								
	1	1	↑	-	-	PK0P[5:0]								
	1	1	↑	-	-	PK1P[5:0]								
	1	1	↑	-	-	PK2P[5:0]								
	1	1	↑	-	-	PK3P[5:0]								
	1	1	↑	-	-	PK4P[5:0]								
	1	1	↑	-	-	PK5P[5:0]								
	1	1	↑	-	-	PK6P[5:0]								

COMMAND Table 2														
Instruction	**D/CX**	**WRX**	**RDX**	**D17-8**	**D7**	**D6**	**D5**	**D4**	**D3**	**D2**	**D1**	**D0**	**Hex**	**Function**
	1	1	↑	-	-		PK7P[5:0]							
	1	1	↑	-	-		PK8P[5:0]							
	1	1	↑	-	-		PK9P[5:0]							
	1	1	↑	-	-		SELV0P[5:0]							
	1	1	↑	-	-		SELV1P[5:0]							
	1	1	↑	-	-		SELV62P[5:0]							
	1	1	↑	-	-		SELV63P[5:0]							
DGC1	0	1	↑	-	1	1	1	0	0	0	1	0	(E2h)	Digital Gamma Control1
	1	1	↑	-	RCA00[3:0]				BCA00[3:0]					
	1	1	↑	-	RCA01[3:0]				BCA01[3:0]					
	1	1	↑	-	:				:					
					:				:					
	1	1	↑	-	RCA62[3:0]				BCA62[3:0]					
	1	1	↑	-	RCA63[3:0]				BCA63[3:0]					
DGC2	0	1	↑	-	1	1	1	0	0	0	1	1	(E3h)	Digital Gamma Control2
	1	1	↑	-	RFA00[3:0]				BFA00[3:0]					
	1	1	↑	-	RFA01[3:0]				BFA01[3:0]					
	1	1	↑	-	:				:					
					:				:					
	1	1	↑	-	RFA62[3:0]				BFA62[3:0]					
	1	1	↑	-	RFA63[3:0]				BFA63[3:0]					

9.2.. Command Table 1

9.2.1 NOP (00h)

00H	NOP (No Operation)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
NOP	0	↑	1	-	0	0	0	0	0	0	0	0	(00h)
Parameter	No Parameter												-
Description	This command is empty command. “-“ Don’t care												
Restriction													

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

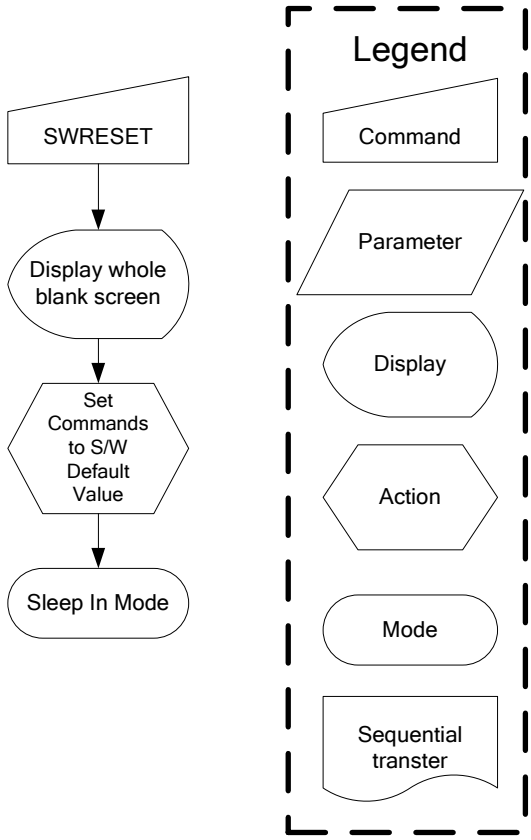
Default

Status	Default Value
Power On Sequence	N/A
S/W Reset	N/A
H/W Reset	N/A

Flow Chart

9.2.2 SWRESET (01h): Software Reset

01H	SWRESET (Software Reset)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
SWRESET	0	↑	1	-	0	0	0	0	0	0	0	1	(01h)
Parameter	No Parameter												-
Description	“-“ Don’t care - When the Software Reset command is written, it causes software reset. It resets the commands and parameters to their S/W Reset default values. - Frame memory contents are unaffected by this command.												
Restriction	It will be necessary to wait 5msec before sending new command following software reset. The display module loads all display suppliers’ factory default values to the registers during this 5msec.												

<table>
<tr><td></td><td>If software reset is sent during sleep in mode, it will be necessary to wait 120msec before sending sleep out command.
Software reset command cannot be sent during sleep out sequence.</td></tr>
<tr><td>Register Availability</td><td>
<table>
<tr><th>Status</th><th>Availability</th></tr>
<tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Sleep In</td><td>Yes</td></tr>
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</td></tr>
<tr><td>Default</td><td>
<table>
<tr><th>Status</th><th>Default Value</th></tr>
<tr><td>Power On Sequence</td><td>N/A</td></tr>
<tr><td>S/W Reset</td><td>N/A</td></tr>
<tr><td>H/W Reset</td><td>N/A</td></tr>
</table>
</td></tr>
<tr><td>Flow Chart</td><td> 
</td></tr>
</table>

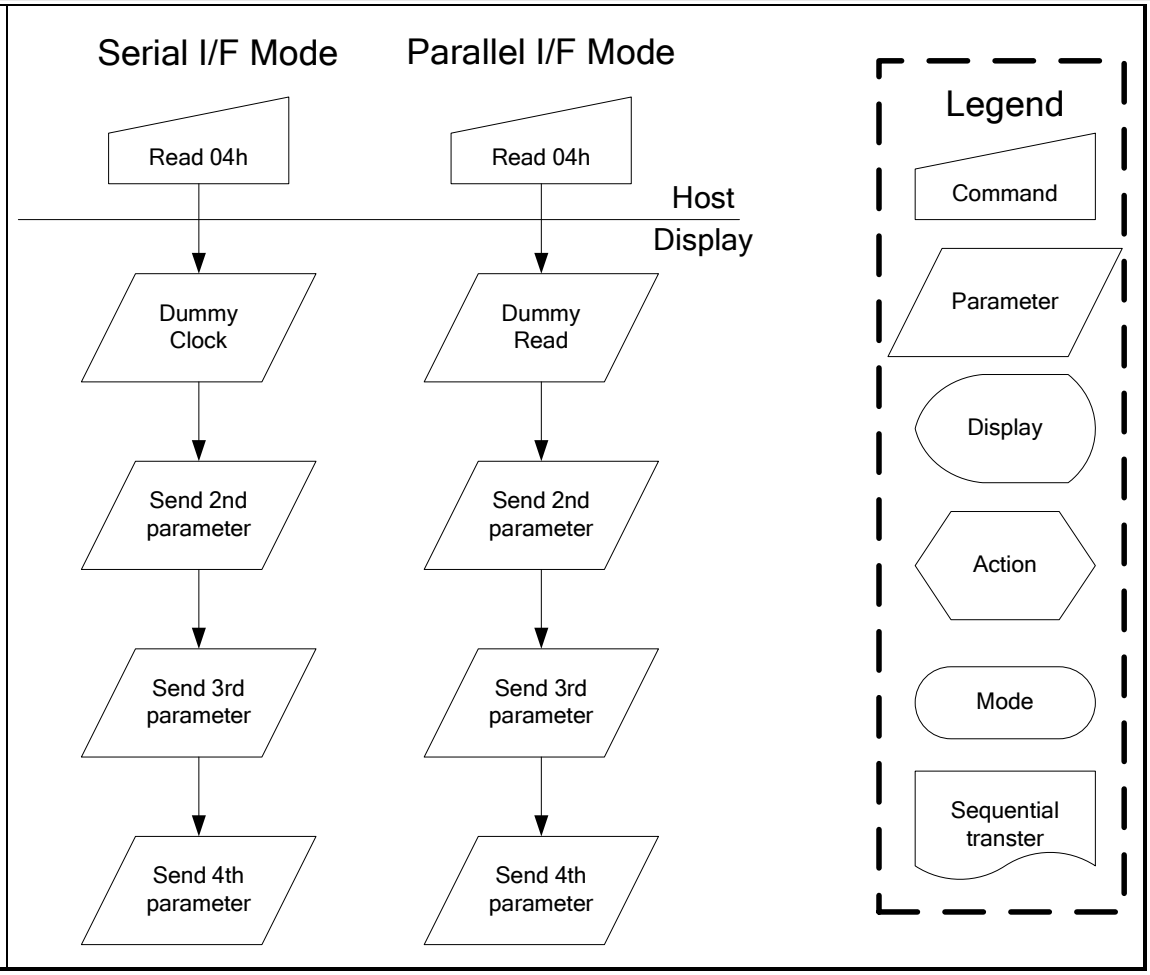
9.2.3 RDDID (04h): Read Display ID

04H	RDDID (Read Display ID)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
RDDID	0	↑	1	-	0	0	0	0	0	1	0	0	(04h)
1st parameter	1	1	↑	-	-	-	-	-	-	-	-	-	-
2nd parameter	1	1	↑	-	ID17	ID16	ID15	ID14	ID13	ID12	ID11	ID10	
3rd parameter	1	1	↑	-	ID27	ID26	ID25	ID24	ID23	ID22	ID21	ID20	
4th parameter	1	1	↑	-	ID37	ID36	ID35	ID34	ID33	ID32	ID31	ID30	
Description	- -The 1st parameter is dummy data -The 2nd parameter (ID17 to ID10): LCD module's manufacturer ID. -The 3rd parameter (ID26 to ID20): LCD module/driver version ID -The 4th parameter (ID37 to ID30): LCD module/driver ID. -Commands RDID1/2/3(DAh, DBh, DCh) read data correspond to the parameters 2,3,4 of the command 04h, respectively. "-" Don't care												
Restriction													
Register availability													
Default													

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Status	Default Value		
	ID1	ID2	ID3
Power On Sequence	See description	See description	See description
S/W Reset	See description	See description	See description
H/W Reset	See description	See description	See description

Flow Chart



9.2.4 Read Number of the Errors on DSI (05h)

00H	RDNUMED (Read Number of the Errors on DSI)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	↑	1	-	0	0	0	0	0	1	0	1	(05h)
1st parameter	1	1	↑	-	-	-	-	-	-	-	-	-	
2nd Parameter	1	1	↑	-	P7	P6	P5	P4	P3	P2	P1	P0	
Parameter	No Parameter												-
Description	The first parameter is telling a number of the errors on DSI. The more detailed description of the bits is below. P[6..0] bits are telling a number of the errors. P[7] is set to '1' if there is overflow with P[6..0] bits. P[7..0] bits are set to '0's (as well as RDDSM(0Eh)'s D0 is set '0' at the same time) after there is sent the second parameter information (= The read function is completed). "-" Don't care												
Restriction	This command is available in MIPI interface. In the other interface, P[7:0] bits are set to "0"s.												

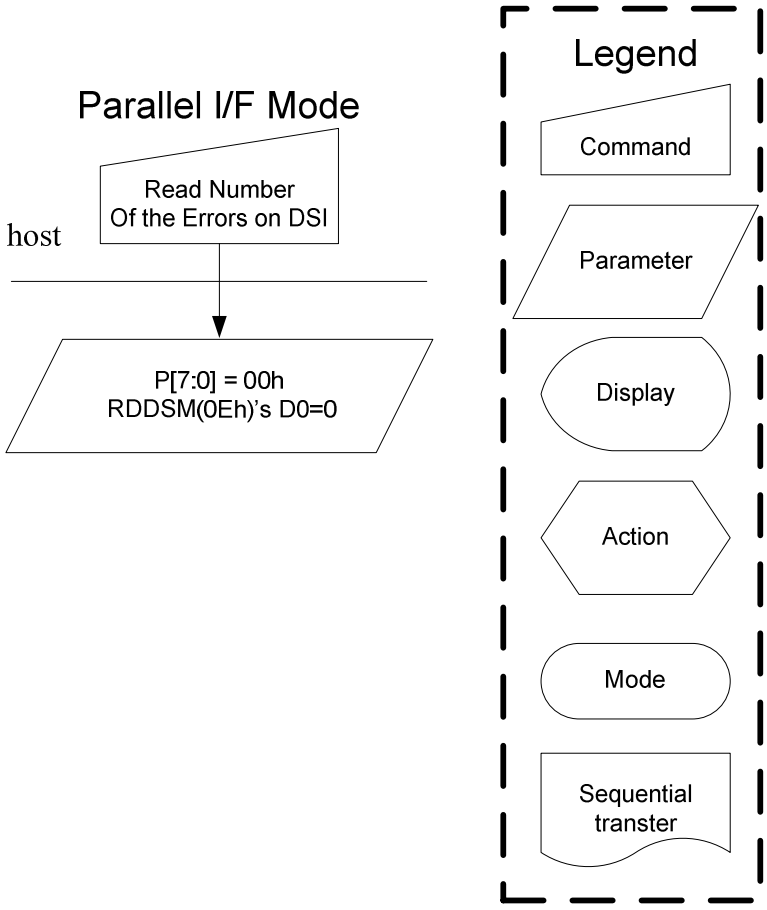
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	00h
S/W Reset	00h
H/W Reset	00h

Flow Chart



9.2.5 RDDST (09h): Read Display Status

09H	RDDST (Read Display Status)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
RDDST	0	↑	1	-	0	0	0	0	1	0	0	1	(09h)
1st parameter	1	1	↑	-	-	-	-	-	-	-	-	-	-
2nd parameter	1	1	↑	-	BSTON	MY	MX	MV	ML	RGB	ST25	ST24	
3rd parameter	1	1	↑	-	ST23	IFPF2	IFPF1	IFPF0	IDMON	PTLON	SLOUT	NORON	
4th parameter	1	1	↑	-	ST15	ST14	INVON	ST12	ST11	DISON	TEON	GCS2	
5th parameter	1	1	↑	-	GCS1	GCS0	TEM	ST4	ST3	ST2	ST1	ST0	

Description

This command indicates the current status of the display as described in the table below:

Bit	Description	Value
BSTON (D31)	Booster Voltage Status	'1' =Booster on, '0' =Booster off
MY(D30)	Row Address Order (MY)	'1' =Decrement, (Bottom to Top, when MADCTL (36h) D7='1') '0' =Increment, (Top to Bottom, when MADCTL (36h) D7='0')
MX(D29)	Column Address Order (MX)	'1' =Decrement, (Right to Left, when MADCTL (36h) D6='1') '0' =Increment, (Left to Right, when MADCTL (36h) D6='0')
MV(D28)	Row/Column Exchange (MV)	'1' = Row/column exchange, (when MADCTL (36h) D5='1') '0' = Normal, (when MADCTL (36h) D5='0'
ML(D27)	Scan Address Order (ML)	'0' =Decrement, (LCD refresh Top to Bottom, when MADCTL (36h) D4='0') '1'=Increment, (LCD refresh Bottom to Top, when MADCTL (36h) D4='1')
RGB(D26)	RGB/ BGR Order (RGB)	'1' =BGR, (When MADCTL (36h) D3='1') '0' =RGB, (When MADCTL (36h) D3='0')
ST245(D25)	For Future Use	'0'
ST24(D24)	For Future Use	'0'
ST23(D23)	For Future Use	'0'
IFPF2(D22)	Interface Color Pixel Format Definition	"011" = 12-bit / pixel, "101" = 16-bit / pixel, "110" = 18-bit / pixel, "111" = 24-bit / pixel, others are no define
IFPF1(D21)		
IFPF0(D20)		
IDMON(D19)	Idle Mode On/Off	'1' = On, "0" = Off
PTLON(D18)	Partial Mode On/Off	'1' = On, "0" = Off

SLPOUT(D17)	Sleep In/Out	'1' = Out, "0" = In
NORON(D16)	Display Normal Mode On/Off	'1' = Normal Display, '0' = Partial Display
ST15(D15)	Vertical Scrolling Status	'1' = Scroll on,"0" = Scroll off
ST14(D14)	Not used	'0'
INVON(D13)	Inversion Status	'1' = On, "0" = Off
ST12(D12)	All Pixels On (Not Used)	'0'
ST11(D11)	All Pixels Off (Not Used)	'0'
DISON(D10)	Display On/Off	'1' = On, "0" = Off
TEON(D9)	Tearing effect line on/off	'1' = On, "0" = Off
GCSEL2(D8)	Gamma Curve Selection	"000" = GC0
GCSEL1(D7)		"001" = GC1
GCSEL0(D6)		"010" = GC2 "011" = GC3 "100" to "111" = Not defined
TEM(D5)	Tearing effect line mode	'0' = mode1, '1' = mode2
ST4(D4)	For Future Use	'0'
ST3(D3)	For Future Use	'0'
ST2(D2)	For Future Use	'0'
ST1(D1)	For Future Use	'0'
ST0(D0)	For Future Use	'0'

"-" Don't care

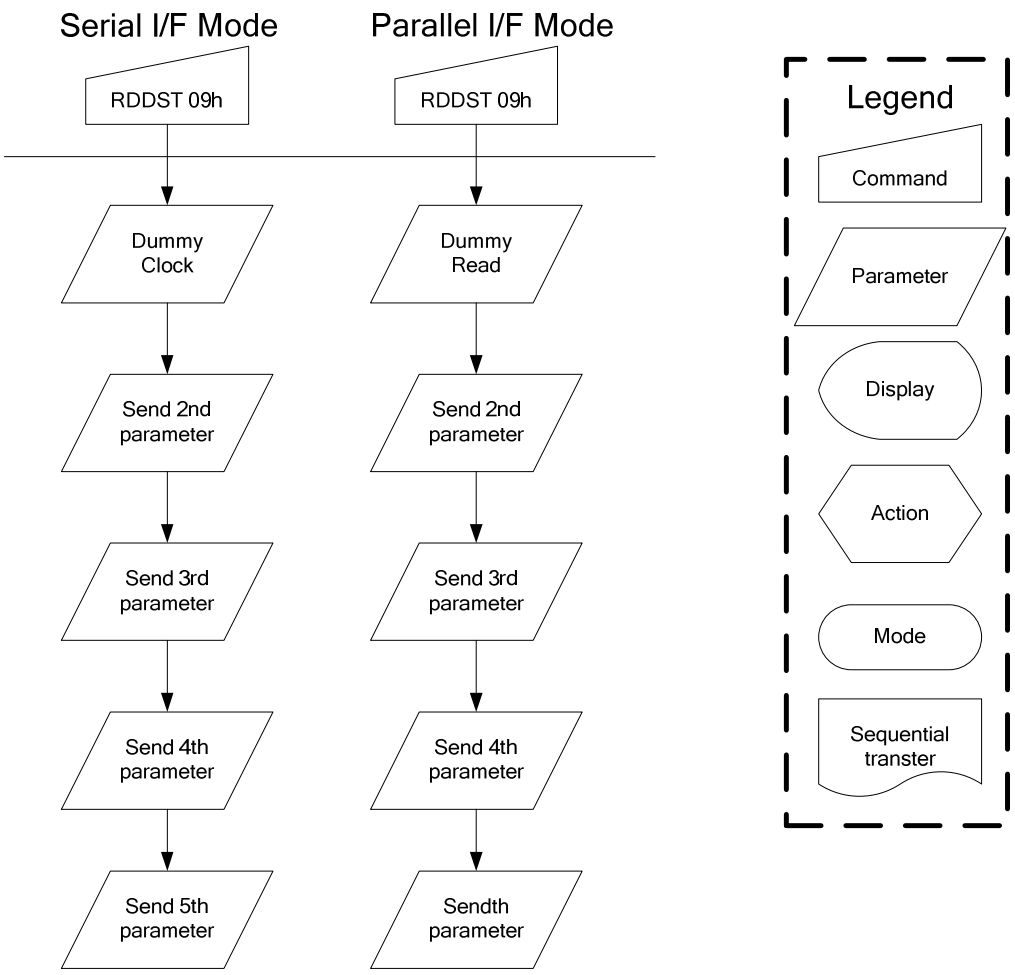
Restriction

Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value (ST31 to ST0)			
	ST[31-24]	ST[23-16]	ST[15-8]	ST[7-0]
Power On Sequence	0000-0000	0110-0001	0000-0000	0000-0000
S/W Reset	0xxx-xx00	0110-0001	0000-0000	0000-0000
H/W Reset	0000-0000	0110-0001	0000-0000	0000-0000

Flow Chart	

Serial I/F Mode
Parallel I/F Mode
RDDST 09h
RDDST 09h
Dummy Clock
Dummy Read
Send 2nd parameter
Send 2nd parameter
Send 3rd parameter
Send 3rd parameter
Send 4th parameter
Send 4th parameter
Send 5th parameter
Sendth parameter
Legend
Command
Parameter
Display
Action
Mode
Sequential transter

9.2.6 RDDPM (0Ah): Read Display Power Mode

0AH	RDDPM (Read Display Power Mode)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
RDDPM	0	↑	1	-	0	0	0	0	1	0	1	0	(0Ah)
1st parameter	1	1	↑	-	-	-	-	-	-	-	-	-	-
2nd parameter	1	1	↑	-	BSTON	IDMON	PTLON	SLPOUT	NORON	DISON	D1	D0	

Description

This command indicates the current status of the display as described in the table below:

Bit	Description	Value
BSTON	Booster Voltage Status	'1' =Booster on, '0' =Booster off
IDMON	Idle mode on/off	'1' = Idle Mode On, '0' = Idle Mode Off
PTLON	Partial mode on/off	'1' =Partial mode on, '0' =Partial mode off,
SLPOUT	Sleep in/out	'1' =Sleep out, '0' =Sleep in,
NORON	Display normal mode on/off	'1' = Normal display, '0' = Partial display,
DISON	Display on/off	'1' =Display on, '0' =Display off,
D1	Not Used	"0"
D0	Not Used	"0"

"-" Don't care

Restriction

There is one dummy parameter when using Parallel interface.

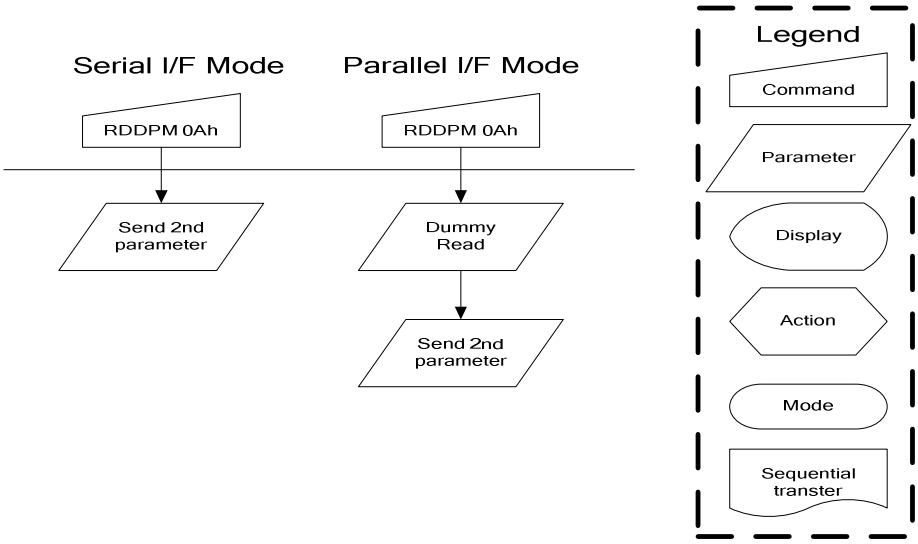
Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value (D7 to D0)
Power On Sequence	0000-1000(08h)
S/W Reset	0000-1000(08h)
H/W Reset	0000-1000(08h)

Flow Chart



9.2.7 RDDMADCTL (0Bh): Read Display MADCTL

0BH	RDDMADCTL (Read Display MADCTL)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
RDDMADCTL	0	↑	1	-	0	0	0	0	1	0	1	1	(0Bh)
1st parameter	1	1	↑	-	-	-	-	-	-	-	-	-	-
2nd parameter	1	1	↑	-	MY	MX	MV	ML	RGB	MH	D1	D0	

Description

This command indicates the current status of the display as described in the table below:

Bit	Description	Value
MY	Row Address Order (MY)	'1' =Decrement, (Bottom to Top, when MADCTL (36h) D7='1') '0' =Increment, (Top to Bottom, when MADCTL (36h) D7='0')
MX	Column Address Order (MX)	'1' =Decrement, (Right to Left, when MADCTL (36h) D6='1') '0' =Increment, (Left to Right, when MADCTL (36h) D6='0')
MV	Row/Column Exchange (MV)	'1' = Row/column exchange, (when MADCTL (36h) D5='1') '0' = Normal, (when MADCTL (36h) D5='0'
ML	Scan Address Order (ML)	'0' =Decrement, (LCD refresh Top to Bottom, when MADCTL (36h) D4='0') '1'=Increment, (LCD refresh Bottom to Top, when MADCTL (36h) D4='1')
RGB	RGB/ BGR Order (RGB)	'1' =BGR, (When MADCTL (36h) D3='1') '0' =RGB, (When MADCTL (36h) D3='0')
MH	Horizontal Order	'0' =Decrement, (LCD refresh Left to Right, when MADCTL (36h) D2='0') '1' =Increment, (LCD refresh Right to Left, when MADCTL (36h) D2='1')
D1	Not used	'0'
D0	Not used	'0'

"-" Don't care

Restriction: There is one dummy parameter when using Parallel interface.

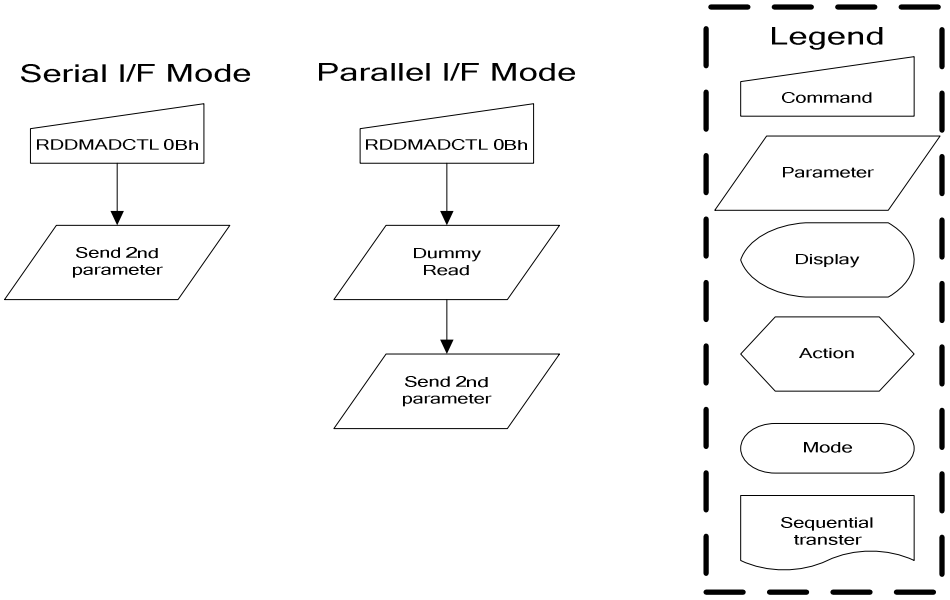
Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value (D7 to D0)
Power On Sequence	0000-0000 (00h)
S/W Reset	No change
H/W Reset	0000-0000 (00h)

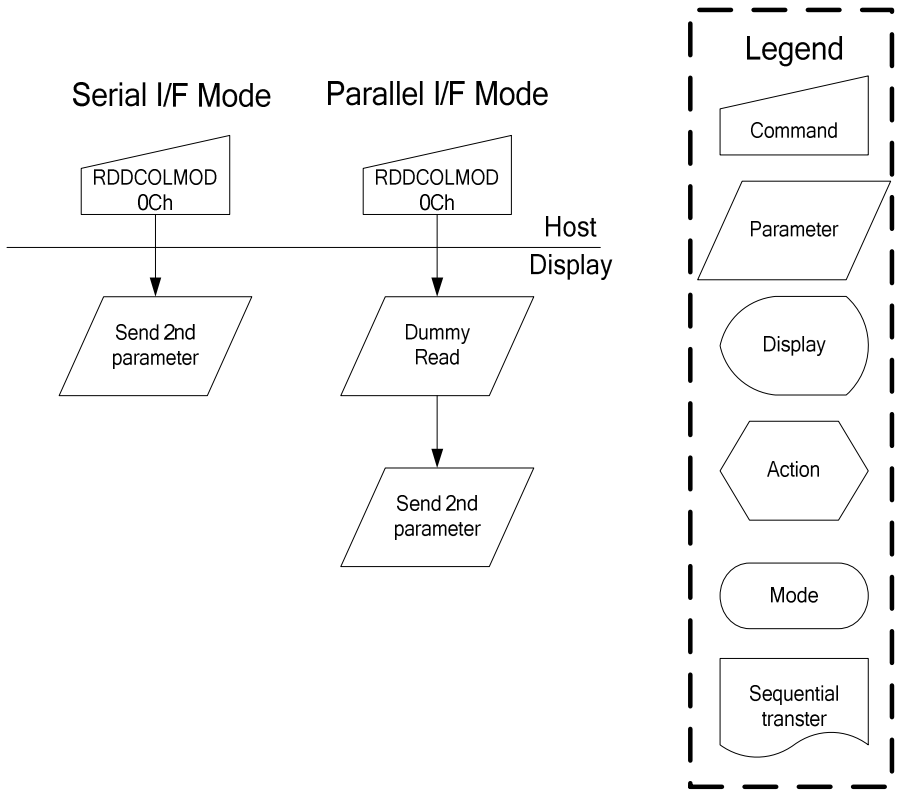
Flow Chart



9.2.8 RDDCOLMOD (0Ch): Read Display Pixel Format

<table>
<tr><td>0CH</td><td colspan="13">RDDCOLMOD (Read Display Pixel Format)</td></tr>
<tr><td>Inst / Para</td><td>D/CX</td><td>WRX</td><td>RDX</td><td>D17-8</td><td>D7</td><td>D6</td><td>D5</td><td>D4</td><td>D3</td><td>D2</td><td>D1</td><td>D0</td><td>HEX</td></tr>
<tr><td>RDDCOLMOD</td><td>0</td><td>↑</td><td>1</td><td>-</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td><td>(0Ch)</td></tr>
<tr><td>1st parameter</td><td>1</td><td>1</td><td>↑</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td></tr>
<tr><td>2nd parameter</td><td>1</td><td>1</td><td>↑</td><td>-</td><td>0</td><td>D6</td><td>D5</td><td>D4</td><td>0</td><td>D2</td><td>D1</td><td>D0</td><td></td></tr>
<tr><td>Description</td><td colspan="13">This command indicates the current status of the display as described in the table below:
<table>
<tr><th>Bit</th><th>Description</th><th>Value</th></tr>
<tr><td>D7</td><td>-</td><td>Set to ‘0’</td></tr>
<tr><td>D6</td><td rowspan="3">RGB interface color format</td><td rowspan="3">‘101’ = 16 bit/pixel
‘110’ = 18 bit/pixel</td></tr>
<tr><td>D5</td></tr>
<tr><td>D4</td></tr>
<tr><td>D3</td><td>-</td><td>Set to ‘0’</td></tr>
<tr><td>D2</td><td rowspan="3">Control interface color format</td><td rowspan="3">‘101’ = 16 bit/pixel
‘110’ = 18 bit/pixel
‘111’ = 24 bit/pixel</td></tr>
<tr><td>D1</td></tr>
<tr><td>D0</td></tr>
</table>
Others are no define and invalid
“-“ Don’t care</td></tr>
<tr><td>Restriction</td><td colspan="13">There is one dummy parameter when using Parallel interface.</td></tr>
<tr><td>Register availability</td><td colspan="13">
<table>
<tr><th>Status</th><th>Availability</th></tr>
<tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Sleep In</td><td>Yes</td></tr>
</table>
</td></tr>
<tr><td>Default</td><td colspan="13">
<table>
<tr><th>Status</th><th>Default Value</th></tr>
<tr><td>Power On Sequence</td><td>0000-0110 (18 bit/pixel)</td></tr>
<tr><td>S/W Reset</td><td>No change</td></tr>
<tr><td>H/W Reset</td><td>0000-0110 (18 bit/pixel)</td></tr>
</table>
</td></tr>
</table>

Flow Chart


Serial I/F Mode
Parallel I/F Mode
RDDCOLMOD
0Ch
RDDCOLMOD
0Ch
Host
Display
Send 2nd
parameter
Dummy
Read
Send 2nd
parameter
Legend
Command
Parameter
Display
Action
Mode
Sequential
transter

9.2.9 RDDIM (0Dh): Read Display Image Mode

0DH	RDDIM (Read Display Image Mode)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
RDDIM	0	↑	1	-	0	0	0	0	1	1	0	1	(0Dh)
1st parameter	1	1	↑	-	-	-	-	-	-	-	-	-	-
2nd parameter	1	1	↑	-	VSSON	0	INVON	0	0	GC2	GC1	GC0	

Description

This command indicates the current status of the display as described in the table below:

-VSSON: Vertical scrolling on/off

-INVON: Inversion on/off

Gamma Curve Selection	GC2	GC1	GC0	Gamma set (26h) Parameter
Gamma curve 1	0	0	0	GC0
Gamma curve 2	0	0	1	GC1
Gamma curve 3	0	1	0	GC2
Gamma curve 4	0	1	1	GC3
Not Defined	1	0	0	Not Defined
Not Defined	1	0	1	Not Defined
Not Defined	1	1	0	Not Defined
Not Defined	1	1	1	Not Defined

Others are no define and invalid

"-" Don't care

Restriction

There is one dummy parameter when using Parallel interface.

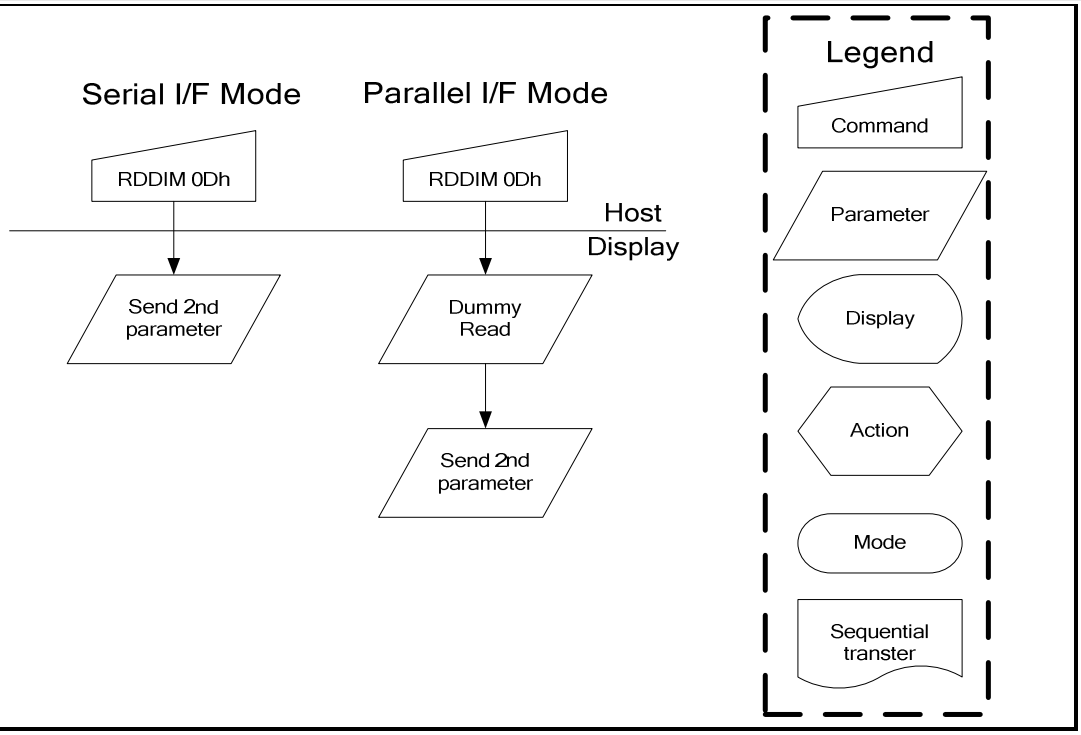
Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	0000-0000
S/W Reset	0000-0000
H/W Reset	0000-0000

Flow Chart



9.2.10 RDDSM (0Eh): Read Display Signal Mode

0EH	RDDSM (Read Display Signal Status)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
RDDSM	0	↑	1	-	0	0	0	0	1	1	1	0	(0Eh)
1st parameter	1	1	↑	-	-	-	-	-	-	-	-	-	-
2nd parameter	1	1	↑	-	TEON	TEM	HS	VS	PixelClk	DataEn	0	ErrorDSI	-

Description

This command indicates the current status of the display as described in the table below:

Bit	Description	Value
TEON	Tearing effect line on/off	'1' = ON, '0' = OFF,
TEM	Tearing effect line mode	'1' = mode2, '0' = mode1,
HS	Horizontal Sync (RGB interface)	'1' = ON, '0' = OFF,
VS	Vertical Sync (RGB interface)	'1' = ON, '0' = OFF,
PixelClk	Pixel Clock (DOTCLK, RGB interface)	'1' = ON, '0' = OFF,
DataEn	Data Enable (DE, RGB interface)	'1' = ON, '0' = OFF,
ErrorDSI	Error On DSI (MIPI Interface)	'1' = Error, '0' = No Error

"-" Don't care

Restriction	There is one dummy parameter when using Parallel interface.

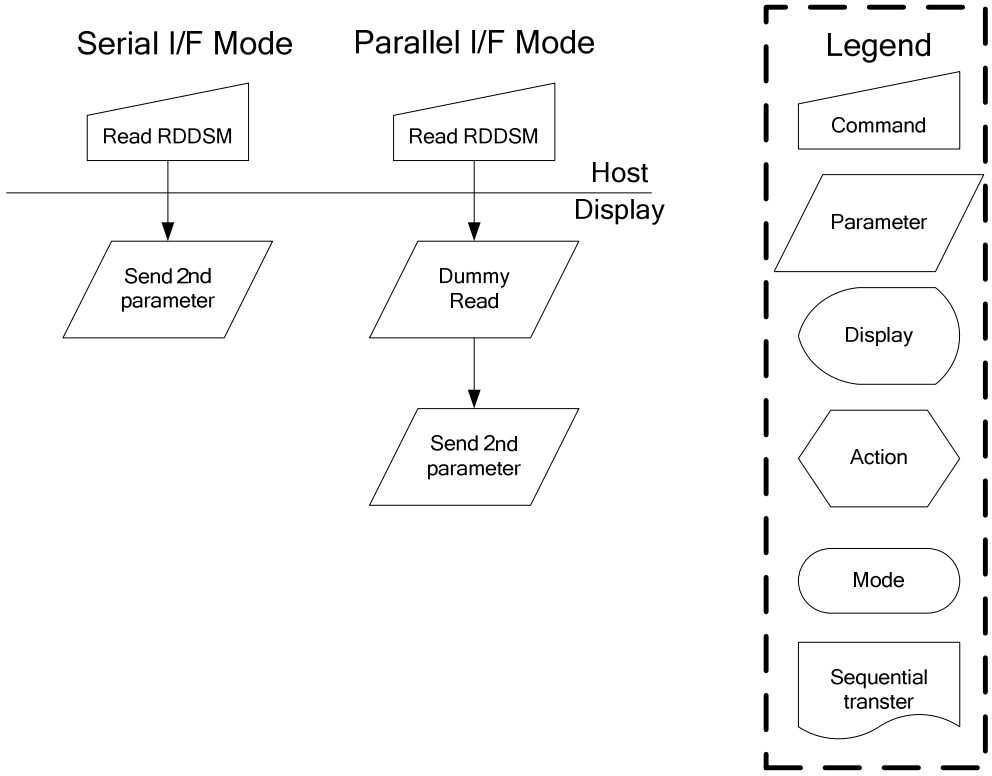
Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	0000-0000
S/W Reset	0000-0000
H/W Reset	0000-0000

Flow Chart



9.2.11 RDDSDR (0Fh): Read Display Self-Diagnostic Result

0FH	RDDSDR (Read Display Self-Diagnostic Result)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
RDDSDR	0	↑	1	-	0	0	0	0	1	1	1	1	(0Fh)

1st parameter	1	1	↑	-	-	-	-	-	-	-	-	-	-
2nd parameter	1	1	↑	-	D7	D6	0	0	0	0	0	D0	-

Description

This command indicates the current status of the display self-diagnostic result after sleep out command as described below:

-D7: Register loading detection

-D6: Functionality detection

-D0: Checksums Comparison, '0' = Checksums are same , '1' = Checksums are not same

See sections: "Read First Checksum (AAh)" and "Read Continue Checksum (AFh)"

"-" Don't care

Restriction

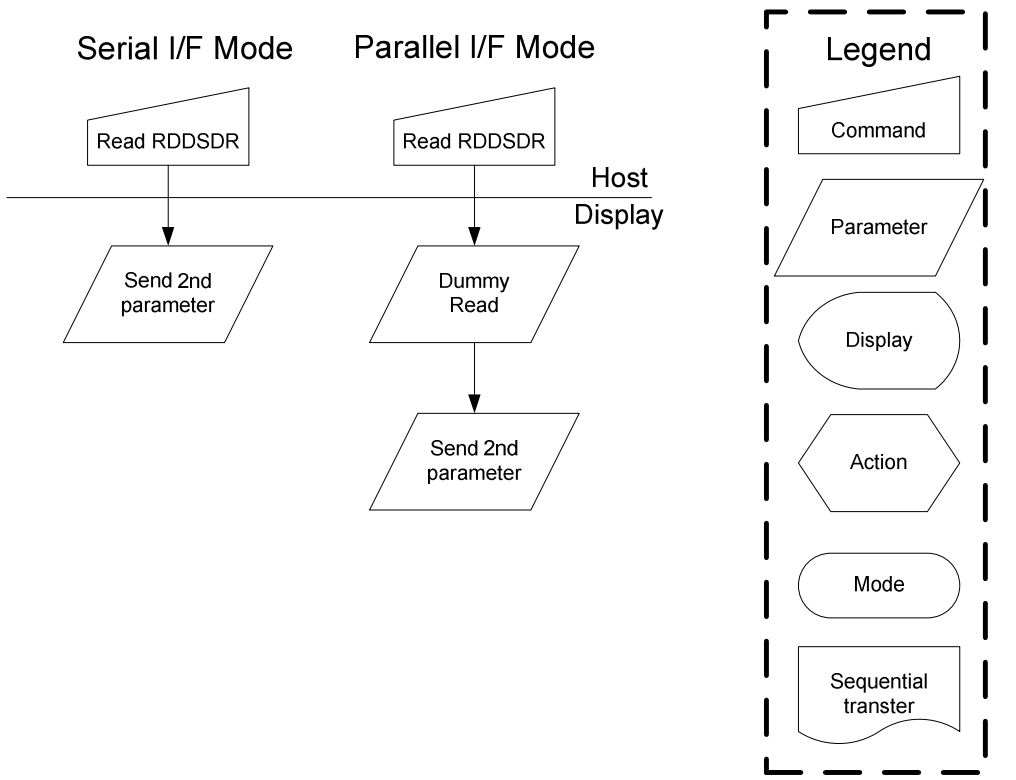
There is one dummy parameter when using Parallel interface.

Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	0000-0000
S/W Reset	0000-0000
H/W Reset	0000-0000

Flow Chart	

9.2.12 SLPIN (10h): Sleep in

10H	SLPIN (Sleep In)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
SLPIN	0	↑	1	-	0	0	0	1	0	0	0	0	(10h)
parameter	No Parameter												
Description	-This command causes the LCD module to enter the minimum power consumption mode. -In this mode the DC/DC converter is stopped, internal oscillator is stopped, and panel scanning is stopped. -MCU interface and memory are still working and the memory keeps its contents. -Dimming function does not work when there is changing mode from Sleep OUT to Sleep IN. "-" Don't care												
Restriction	-This command has no effect when module is already in sleep in mode. Sleep in mode can only be left by the sleep out command (11h). -It will be necessary to wait 5msec before sending any new commands to a display module following this command to allow time for the supply voltages and clock circuits to stabilize. -It will be necessary to wait 120msec after sending sleep out command (when in sleep in mode) before sending an sleep in command.												
Register													

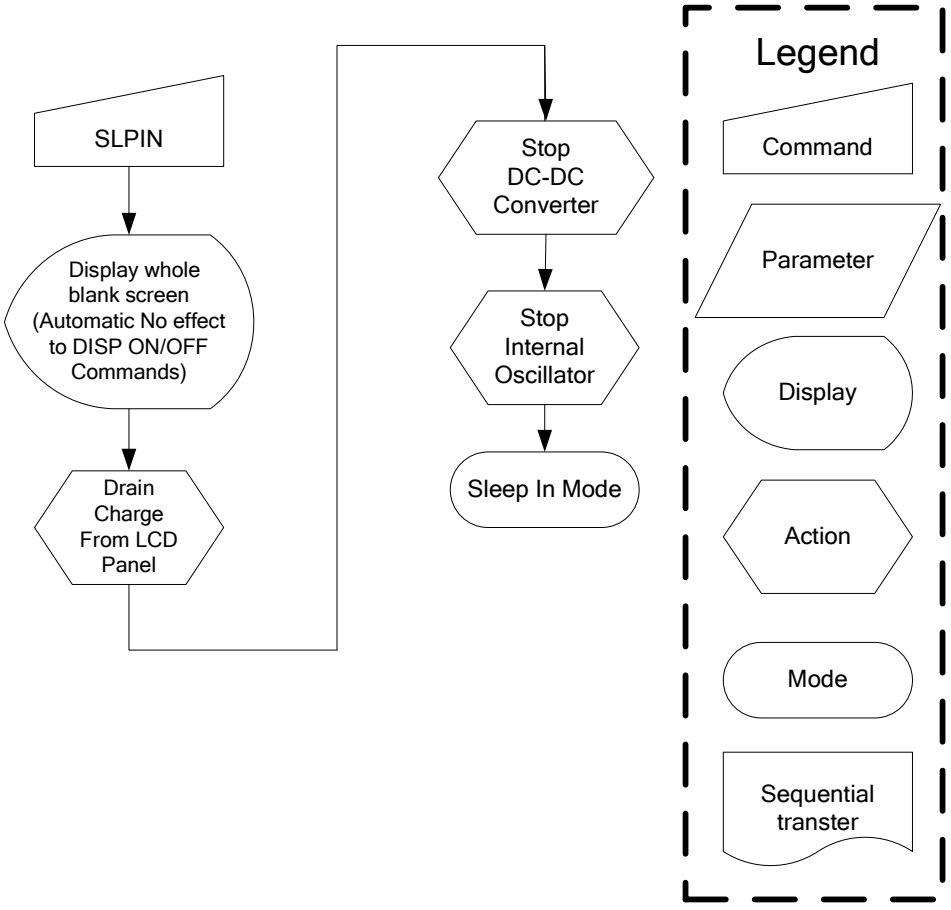
availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	Sleep in mode
S/W Reset	Sleep in mode
H/W Reset	Sleep in mode

Flow Chart

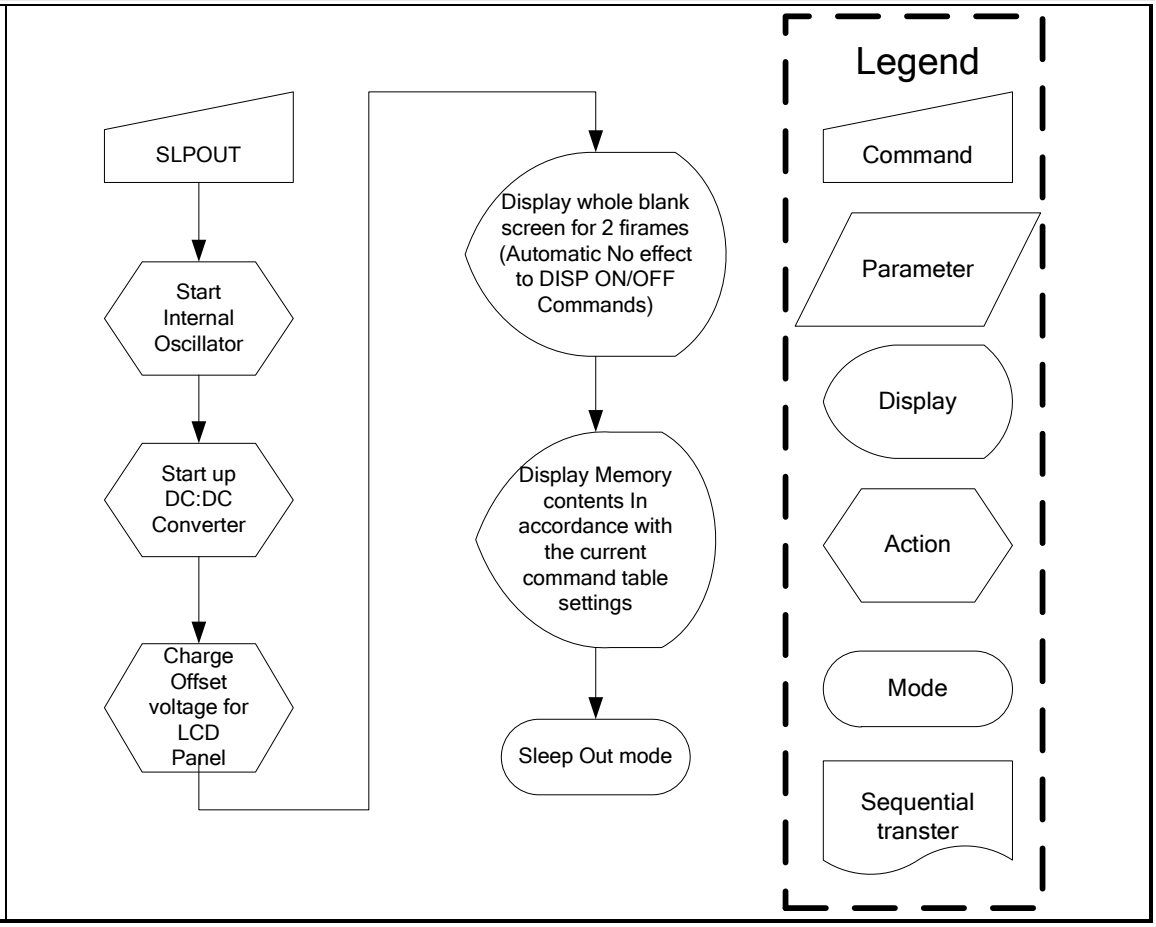


9.2.13 SLPOUT (11h): Sleep Out

11H	SLPOUT (Sleep Out)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
SLPOUT	0	↑	1	-	0	0	0	1	0	0	0	1	(11h)

<table>
<tr><td>parameter</td><td>No Parameter</td></tr>
<tr><td>Description</td><td>-This command turn off sleep mode.
-In this mode the DC/DC converter is enable, internal display oscillator is started, and panel scanning is started.</td></tr>
<tr><td>Restriction</td><td>-This command has no effect when module is already in sleep out mode. Sleep out mode can only be left by the sleep in command (10h).
-It will be necessary to wait 5msec before sending any new commands to a display module following this command to allow time for the supply voltages and clock circuits to stabilize.
-It will be necessary to wait 120msec after sending sleep out command (when in sleep in mode) before sending an sleep in command.
-The display module runs the self-diagnostic functions after this command is received.</td></tr>
<tr><td>Register availability</td><td>
<table>
<tr><th>Status</th><th>Availability</th></tr>
<tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Sleep In</td><td>Yes</td></tr>
</table>
</td></tr>
<tr><td>Default</td><td>
<table>
<tr><th>Status</th><th>Default Value</th></tr>
<tr><td>Power On Sequence</td><td>Sleep in mode</td></tr>
<tr><td>S/W Reset</td><td>Sleep in mode</td></tr>
<tr><td>H/W Reset</td><td>Sleep in mode</td></tr>
</table>
</td></tr>
</table>

Flow Chart


SLPOUT
Start Internal Oscillator
Start up DC:DC Converter
Charge Offset voltage for LCD Panel
Display whole blank screen for 2 firames (Automatic No effect to DISP ON/OFF Commands)
Display Memory contents In accordance with the current command table settings
Sleep Out mode
Legend
Command
Parameter
Display
Action
Mode
Sequential transter

9.2.14 PTLON (12h): Partial Display Mode On

12H	PTLON (Partial Display Mode On)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
PTLON	0	↑	1	-	0	0	0	1	0	0	1	0	(12h)
parameter	No Parameter												
Description	-This command turns on Partial mode. The partial mode window is described by the Partial Area command (30h) -To leave Partial mode, the Normal Display Mode On command (13h) should be written. "-" Don't care												
Restriction	This command has no effect when partial mode is active.												
Register availability													

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

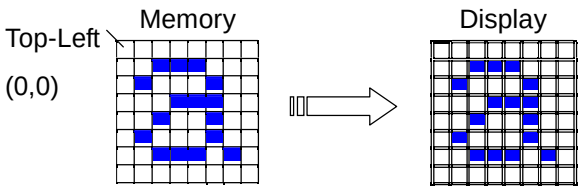
Status	Default Value
Power On Sequence	Normal display mode on
S/W Reset	Normal display mode on
H/W Reset	Normal display mode on

Flow Chart	See Partial Area (30h)

9.2.15 NORON (13h): Normal Display Mode On

<table>
<tr><td>12H</td><td colspan="13">NORON (Normal Display Mode On)</td></tr>
<tr><td>Inst / Para</td><td>D/CX</td><td>WRX</td><td>RDX</td><td>D17-8</td><td>D7</td><td>D6</td><td>D5</td><td>D4</td><td>D3</td><td>D2</td><td>D1</td><td>D0</td><td>HEX</td></tr>
<tr><td>NORON</td><td>0</td><td>↑</td><td>1</td><td>-</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td><td>1</td><td>(13h)</td></tr>
<tr><td>parameter</td><td colspan="13">No Parameter</td></tr>
<tr><td>Description</td><td colspan="13">-This command turns the display to normal mode.
-Normal display mode on means partial mode off.
-Exit from NORON by the partial mode on command.
“-“ Don’t care</td></tr>
<tr><td>Restriction</td><td colspan="13">This command has no effect when normal display mode is active.</td></tr>
<tr><td>Register availability</td><td colspan="13"><table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table></td></tr>
<tr><td>Default</td><td colspan="13"><table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>Normal display mode on</td></tr><tr><td>S/W Reset</td><td>Normal display mode on</td></tr><tr><td>H/W Reset</td><td>Normal display mode on</td></tr></table></td></tr>
<tr><td>Flow Chart</td><td colspan="13">See partial area description for details of when to use this command.</td></tr>
</table>

9.2.16 INVOFF (20h): Display Inversion Off

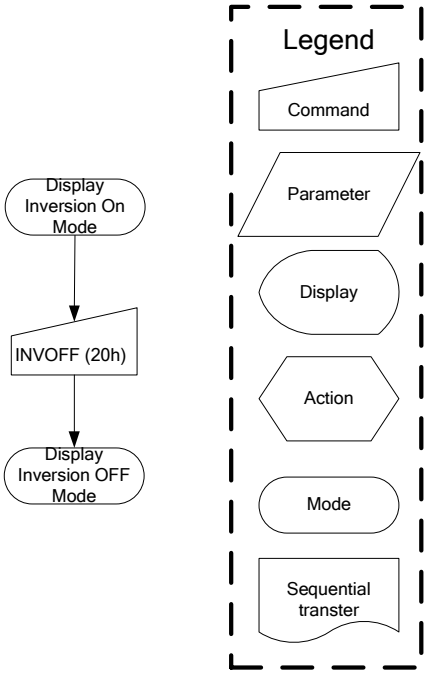
20H	INVOFF (Display Inversion Off)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
INVOFF	0	↑	1	-	0	0	1	0	0	0	0	0	(20h)
parameter	No Parameter												
Description	-This command is used to recover from display inversion mode. "-" Don't care (Example)  												
Restriction	This command has no effect when module is already in inversion off mode.												

Register availability

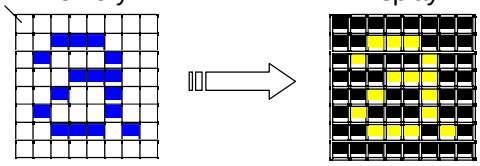
Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	Display inversion off
S/W Reset	Display inversion off
H/W Reset	Display inversion off

Flow Chart	

9.2.17 INVON (21h): Display Inversion On

21H	INVON (Display Inversion On)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
INVON	0	↑	1	-	0	0	1	0	0	0	0	1	(21h)
parameter	No Parameter												
Description	-This command is used to recover from display inversion mode. "-" Don't care (Example) Top-Left (0,0) Memory Display 												
Restriction	This command has no effect when module is already in inversion on mode.												
Register availability													

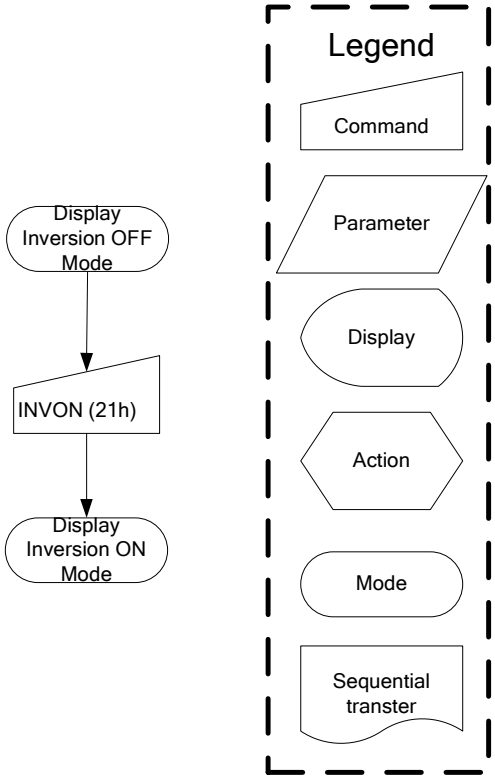
Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes

		Sleep In	Yes	

Default

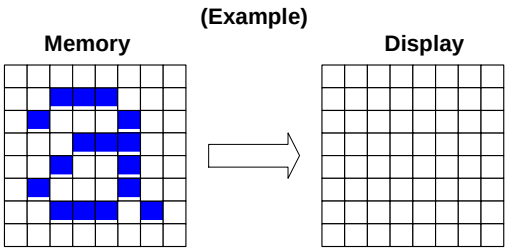
Status	Default Value
Power On Sequence	Display inversion off
S/W Reset	Display inversion off
H/W Reset	Display inversion off

Flow Chart



9.2.18 DISPOFF (28h): Display Off

28H	DISPOFF (Display Off)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
DISPOFF	0	↑	1	-	0	0	1	0	1	0	0	0	(28h)
parameter	No Parameter												
Description	- This command is used to enter into DISPLAY OFF mode. In this mode, the output from Frame Memory is disabled and blank page inserted. - This command makes no change of contents of frame memory. - This command does not change any other status. - There will be no abnormal visible effect on the display. - Exit from this command by Display On (29h)												



Restriction	This command has no effect when module is already in display off mode.

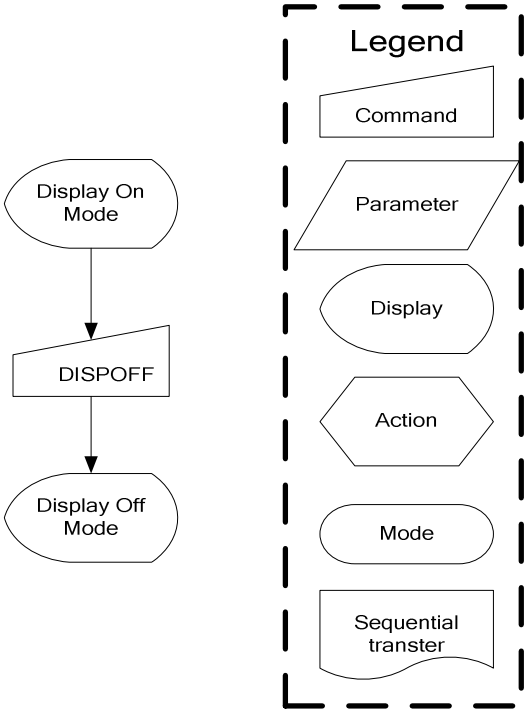
Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

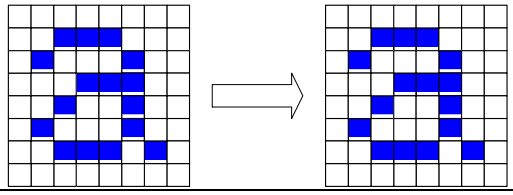
Default

Status	Default Value
Power On Sequence	Display off
S/W Reset	Display off
H/W Reset	Display off

Flow Chart



9.2.19 DISPON (29h): Display On

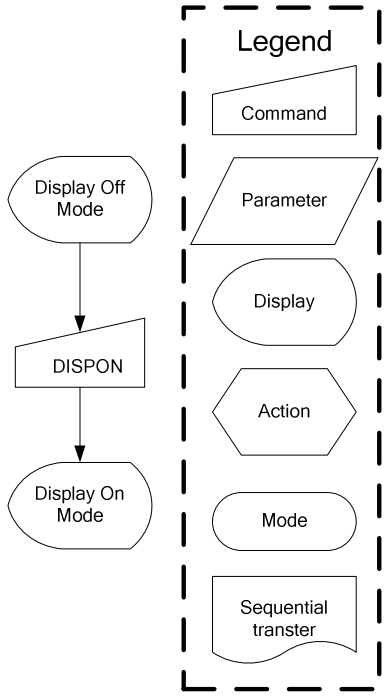
29H	DISPON (Display On)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
DISPO N	0	↑	1	-	0	0	1	0	1	0	0	1	(29h)
parameter	No Parameter												
Description	- This command is used to recover from DISPLAY OFF mode. - Output from the Frame Memory is enabled. - This command makes no change of contents of frame memory. - This command does not change any other status. **(Example)** **Memory** **Display** 												
Restriction	This command has no effect when module is already in display on mode.												

Register availability

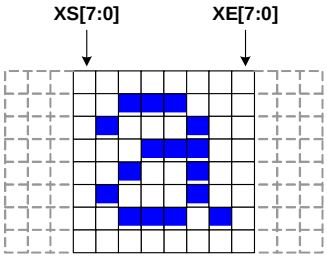
Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	Display off
S/W Reset	Display off
H/W Reset	Display off

Flow Chart	

9.2.20 CASET (2Ah): Column Address Set

2AH	CASET (Column Address Set)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
CASET	0	↑	1	-	0	0	1	0	1	0	1	0	(2Ah)
1st parameter	1	↑	1	-	XS15	XS14	XS13	XS12	XS11	XS10	XS9	XS8	
2nd parameter	1	↑	1	-	XS7	XS6	XS5	XS4	XS3	XS2	XS1	XS0	
3rd parameter	1	↑	1	-	XE15	XE14	XE13	XE12	XE11	XE10	XE9	XE8	
4th parameter	1	↑	1	-	XE7	XE6	XE5	XE4	XE3	XE2	XE1	XE0	
1. Description	-The value of XS [7:0] and XE [7:0] are referred when RAMWR command comes. -Each value represents one column line in the Frame Memory. 												
Restriction	XS [15:0] always must be equal to or less than XE [15:0] When XS [15:0] or XE [15:0] is greater than maximum address like below, data of out of range will be ignored. (Parameter range: 0 < XS [15:0] < XE [15:0] < =(013Fh)): MV="0") (Parameter range: 0 < XS [15:0] < XE [15:0] < =(01DFh)): MV="1")												

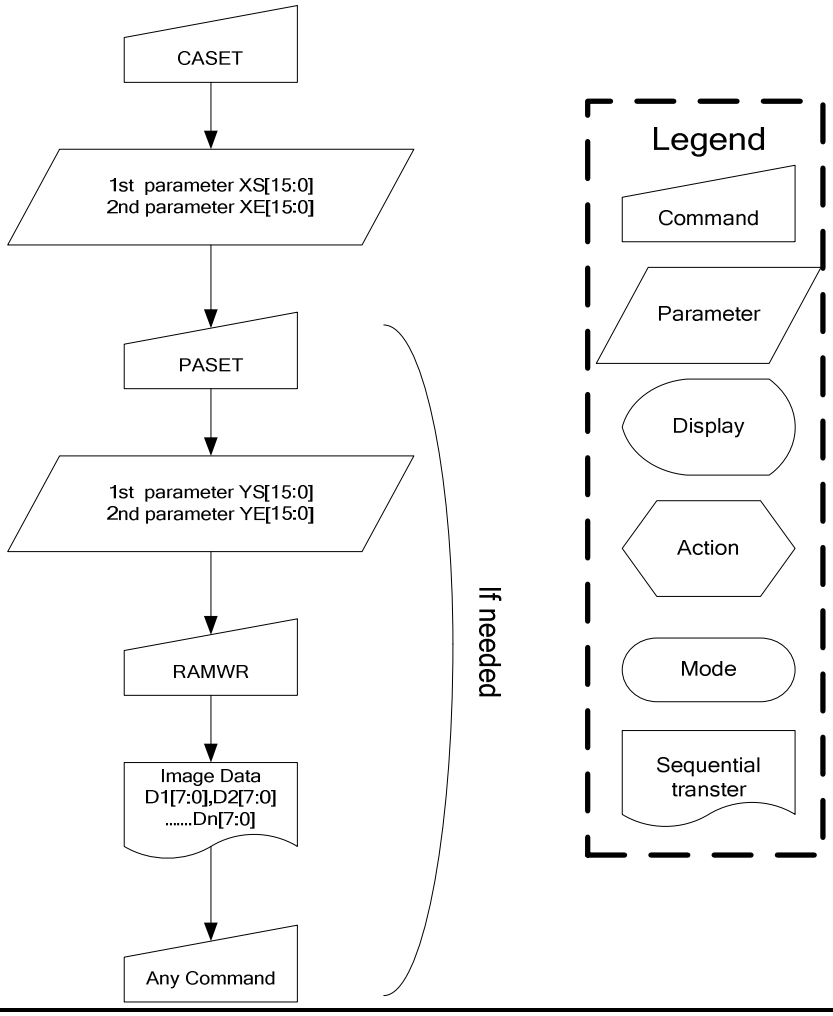
Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value	
Power On Sequence	XS[15:0]=0x00	XE[15:0]=013F
S/W Reset	XS[15:0]=0x00	When MV=0: XE[15:0]=013Fh, When MV=1: XE[15:0]=01DFh
H/W Reset	XS[15:0]=0x00	XE[15:0]=013F

Flow Chart



9.2.21 RASET (2Bh): Row Address Set

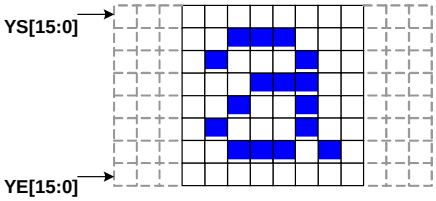
2BH	RASET (Row Address Set)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
RASET	0	↑	1	-	0	0	1	0	1	0	1	1	(2Bh)
1st parameter	1	↑	1	-	YS15	YS14	YS13	YS12	YS11	YS10	YS9	YS8	
2nd parameter	1	↑	1	-	YS7	YS6	YS5	YS4	YS3	YS2	YS1	YS0	
3rd parameter	1	↑	1	-	YE15	YE14	YE13	YE12	YE11	YE10	YE9	YE8	
4th parameter	1	↑	1	-	YE7	YE6	YE5	YE4	YE3	YE2	YE1	YE0	

2. Description

-This command is used to defined area of frame memory where MCU can access.

-The value of YS [15:0] and YE [15:0] are referred when RAMWR command comes.

-Each value represents one page line in the Frame Memory.



Restriction

YS [15:0] always must be equal to or less than YE [15:0]

When YS [15:0] or YE [15:0] is greater than maximum address like below, data of out of range will be ignored.

(Parameter range: 0 < YS [15:0] < YE [15:0] < (01DFh)): MV="0")

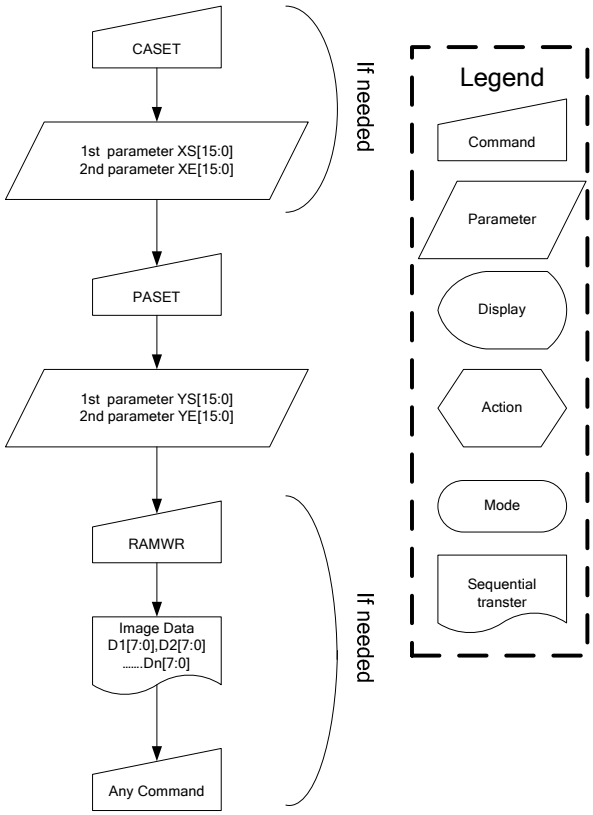
(Parameter range: 0 < YS [15:0] < YE [15:0] < (013Fh)): MV="1")

Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value	
Power On Sequence	YS[15:0]=0000h	YE[15:0]=01DFh
S/W Reset	YS[15:0]=0000h	When MV=0: YE[15:0]=01DFh, When MV=1: YE[15:0]=013Fh

	H/W Reset | YS[15:0]=0000h | YE[15:0]=01DFh
Flow Chart	

9.2.22 RAMWR (2Ch): Memory Write

2CH	RAMWR (Memory Write)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
RAMWR	0	↑	1	-	0	0	1	0	1	1	0	0	(2Ch)
1st parameter	1	↑	1	-	D[17]	D[16]	D[15]	D[14]	D[13]	D[12]	D[11]	D[10]	
…	1	↑	1	-	Dx[7]	Dx[6]	Dx[5]	Dx[4]	Dx[3]	Dx[2]	Dx[1]	Dx[0]	
N parameter	1	↑	1	-	Dn[7]	Dn[6]	Dn[5]	Dn[4]	Dn[3]	Dn[2]	Dn[1]	Dn[0]	
Description	-This command is used to transfer data from MCU to frame memory. -When this command is accepted, the column register and the page register are reset to the start column/start page positions. -The start column/start page positions are different in accordance with MADCTL setting. -Sending any other command can stop frame write.												

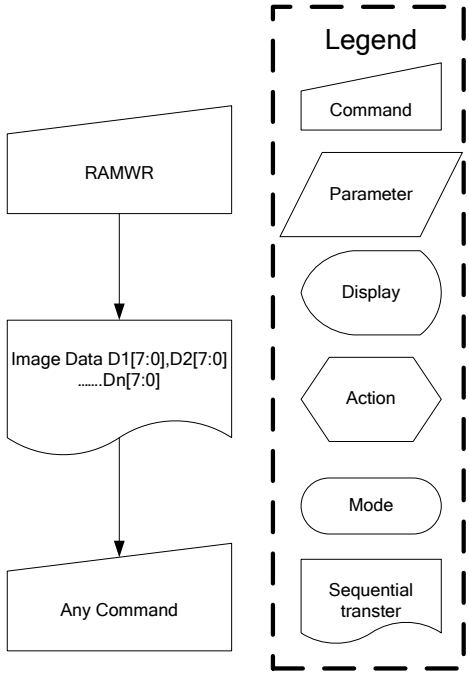
Restriction	In all color modes, there is no restriction on length of parameters.
Register availability	

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	Contents of memory is set randomly
S/W Reset	Contents of memory is not cleared
H/W Reset	Contents of memory is not cleared

Flow Chart



9.2.23 RAMRD (2Eh): Memory Read

2EH	RAMRD (Memory Read)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
RAMRD	0	↑	1	-	0	0	1	0	1	1	1	0	(2Eh)
1st parameter	1	1	↑	-	-	-	-	-	-	-	-	-	
2nd parameter	1	1	↑	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	

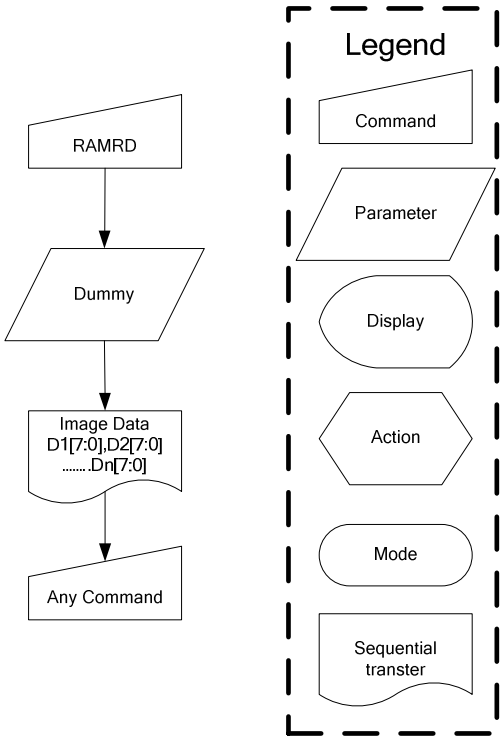
:	1	1	↑	:	:	:	:	:	:	:	:	:	
(N+1)th parameter	1	1	↑	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	

Description	-This command is used to transfer data from frame memory to MCU. -When this command is accepted, the column register and the row register are reset to the Start Column/Start Row positions. -The Start Column/Start Row positions are different in accordance with MADCTL setting. -Then D[17:0] is read back from the frame memory and the column register and the row register incremented -Frame Read can be cancelled by sending any other command.
Restriction	There is no restriction on length of parameters.
Register availability	

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	Contents of memory is set randomly
S/W Reset	Contents of memory is not cleared
H/W Reset	Contents of memory is not cleared

Flow Chart	 RAMRD Dummy Image Data D1[7:0],D2[7:0] .....Dn[7:0] Any Command Legend Command Parameter Display Action Mode Sequential transter

9.2.24 PTLAR (30h): Partial Area

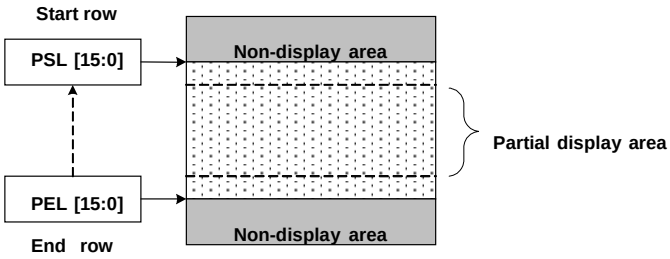
30H	PTLAR (Partial Area)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
PTLAR	0	↑	1	-	0	0	1	1	0	0	0	0	(30h)
1st parameter	1	↑	1	-	PSL15	PSL14	PSL13	PSL12	PSL11	PSL10	PSL9	PSL8	
2nd parameter	1	↑	1	-	PSL7	PSL6	PSL5	PSL4	PSL3	PSL2	PSL1	PSL0	
3rd parameter	1	↑	1	-	PEL15	PEL14	PEL13	PEL12	PEL11	PEL10	PEL9	PEL8	
4th parameter	1	↑	1	-	PEL7	PEL6	PEL5	PEL4	PEL3	PEL2	PEL1	PEL0	

Description

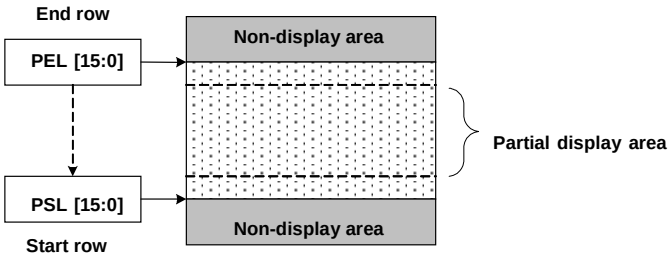
-This command defines the partial mode's display area.

-There are 4 parameters associated with this command, the first defines the Start Row (PSL) and the second the End Row (PEL), as illustrated in the figures below. PSL and PEL refer to the Frame Memory row address counter.

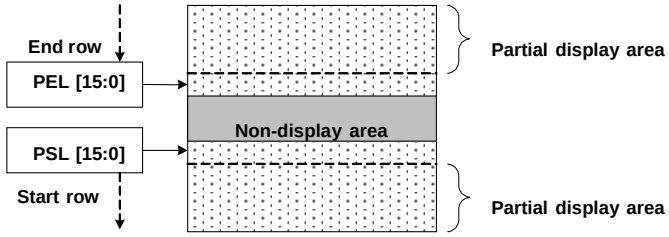
-If End Row > Start Row, when MADCTL ML='0'



-If End Row > Start Row, when MADCTL ML='1'



-If End Row < Start Row, when MADCTL ML='0'



-If End Row = Start Row then the Partial Area will be one row deep.

Restriction	Each detail initial value by the display resolution will be updated.

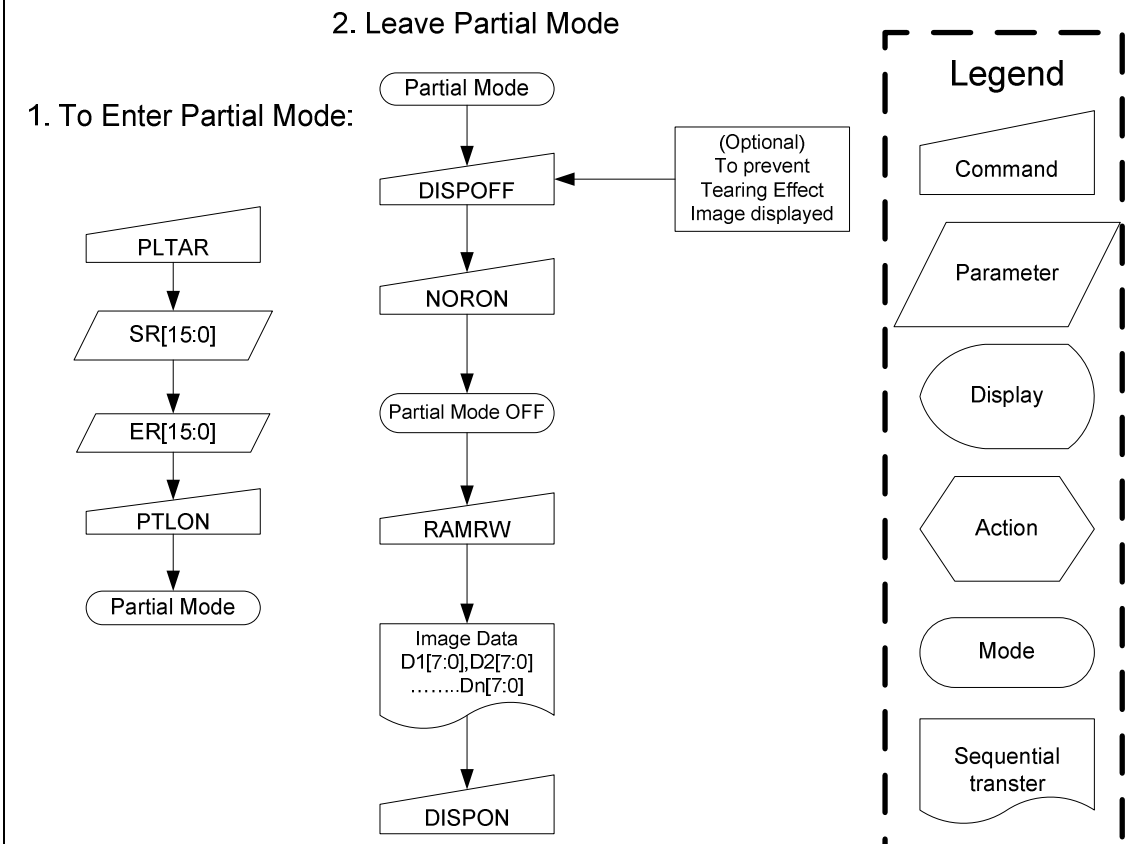
Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	PSL[15:0]=0000h, PEL=01DFh
S/W Reset	PSL[15:0]=0000h, PEL=01DFh
H/W Reset	PSL[15:0]=0000h, PEL=01DFh

Flow Chart



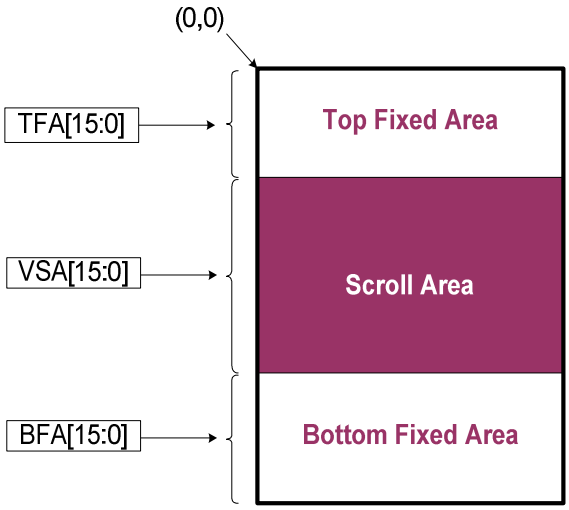
9.2.25 VSCRDEF (33h): Vertical Scrolling Definition

33H	(Vertical Scrolling Definition)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
VSCRDEF	0	↑	1	-	0	0	1	1	0	0	1	1	(33h)
1st parameter	1	↑	1	-	TFA15	TFA14	TFA13	TFA12	TFA11	TFA10	TFA9	TFA8	
2nd parameter	1	↑	1	-	TFA7	TFA6	TFA5	TFA4	TFA3	TFA2	TFA1	TFA0	
3rd parameter	1	↑	1	-	VSA15	VSA14	VSA13	VSA12	VSA11	VSA10	VSA9	VSA8	
4th parameter	1	↑	1	-	VSA7	VSA6	VSA5	VSA4	VSA3	VSA2	VSA1	VSA0	
5th parameter	1	↑	1		BFA15	BFA14	BFA13	BFA12	BFA11	BFA10	BFA9	BFA8	
6th parameter	1	↑	1		BFA7	BFA6	BFA5	BFA4	BFA3	BFA2	BFA1	BFA0	
Description	-This command just defines the Vertical Scrolling Area of the display and not performs vertical scroll -When MADCTL MV=0 -The 1st & 2nd parameter TFA [15:0] describes the Top Fixed Area (in No. of lines from Top of the Frame Memory and Display). -The 3rd & 4th parameter VSA [15:0] describes the height of the Vertical Scrolling Area (in No. of lines of the Frame Memory [not the display] from the Vertical Scrolling Start Address) The first line appears immediately after the bottom												

most line of the Top Fixed Area.

-The 4th & 5th parameter BFA [6:0] describes the Bottom Fixed Area (in No. of lines from Bottom of the Frame Memory and Display).

TFA, VSA and BFA refer to the Frame Memory Line Pointer

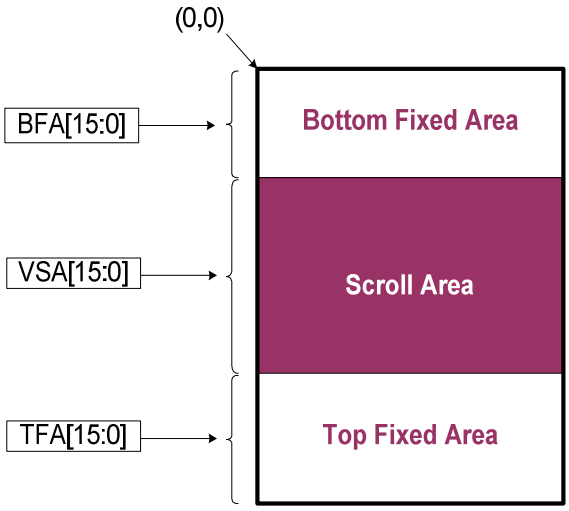


-When MADCTL MV=1

-The 1st & 2nd parameter TFA [15:0] describes the Top Fixed Area (in No. of lines from Bottom of the Frame Memory and Display)

-The 3rd & 4th parameter VSA [15:0] describes the height of the Vertical Scrolling Area (in No. of lines of the Frame Memory [not the display] from the Vertical Scrolling Start Address) The first line appears immediately after the bottom most line of the Top Fixed Area.

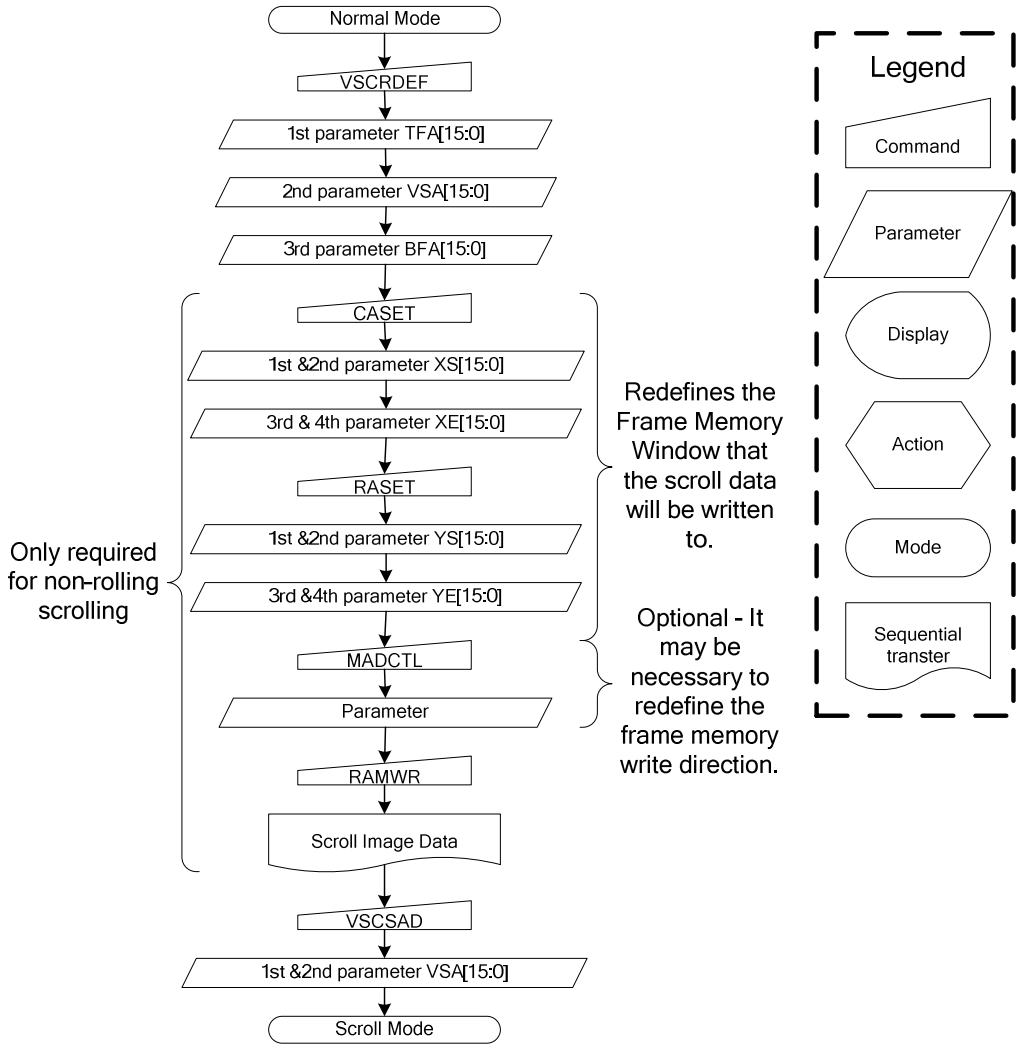
-The 4th & 5th parameter BFA [6:0] describes the Top Fixed Area (in No. of lines from Bottom of the Frame Memory and Display).



<table>
<tr><td>Restriction</td><td>The sum of TFA, VSA and BFA must equal the number of the display device's horizontal lines (pages), otherwise Scrolling mode is undefined.
In Vertical Scrolling Mode, MADCTL parameter MV should be set to '0' – this only affects the Frame Memory write.</td></tr>
<tr><td>Register availability</td><td>
<table>
<tr><th>Status</th><th>Availability</th></tr>
<tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Sleep In</td><td>Yes</td></tr>
</table>
</td></tr>
<tr><td>Default</td><td>
<table>
<tr><th>Status</th><th colspan="3">Default Value</th></tr>
<tr><td>Power On Sequence</td><td>TFA[15:0] = 0000h</td><td>VSA[0:15] = 01E0h</td><td>BFA[15:0] = 0000h</td></tr>
<tr><td>S/W Reset</td><td>TFA[15:0] = 0000h</td><td>VSA[0:15] = 01E0h</td><td>BFA[15:0] = 0000h</td></tr>
<tr><td>H/W Reset</td><td>TFA[15:0] = 0000h</td><td>VSA[0:15] = 01E0h</td><td>BFA[15:0] = 0000h</td></tr>
</table>
</td></tr>
</table>

Flow Chart

1. TO Enter Vertical Scroll Mode:



9.2.26 TEOFF (34h): Tearing Effect Line OFF

34H	TEOFF (Tearing Effect Line OFF)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
TEOFF	0	↑	1	-	0	0	1	1	0	1	0	0	(34h)
parameter	No Parameter												
Description	-This command is used to turn OFF (Active Low) the Tearing Effect output signal from the TE signal line.												
Restriction	This command has no effect when tearing effect output is already off..												
Register availability													

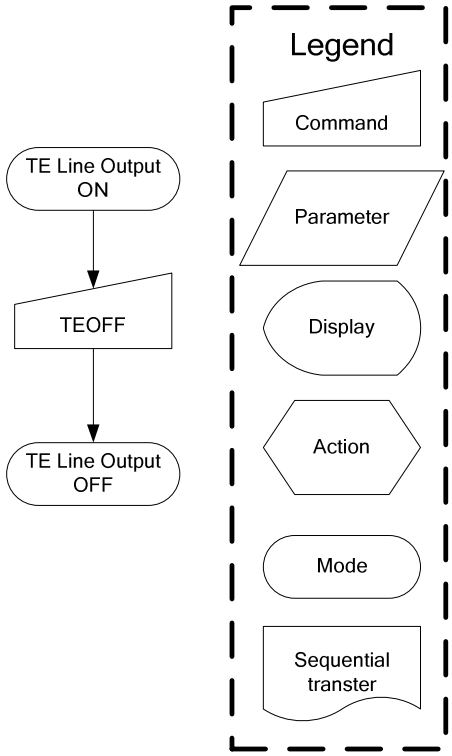
Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes

		Normal Mode On, Idle Mode On, Sleep Out	Yes
		Partial Mode On, Idle Mode Off, Sleep Out	Yes
		Partial Mode On, Idle Mode On, Sleep Out	Yes
		Sleep In	Yes

Default

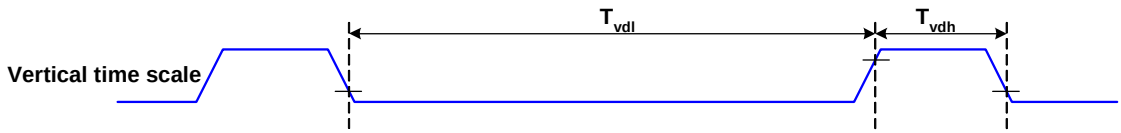
Status	Default Value
Power On Sequence	Off
S/W Reset	Off
H/W Reset	Off

Flow Chart

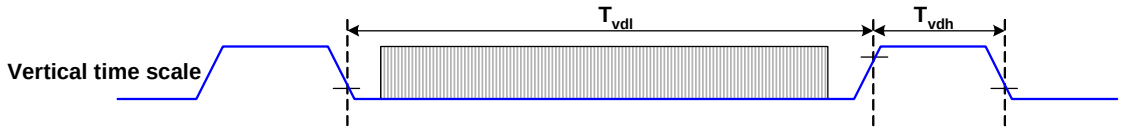


9.2.27 TEON (35h): Tearing Effect Line On

35H	TEON (Tearing Effect Line On)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
TEON	0	↑	1	-	0	0	1	1	0	1	0	1	(35h)
parameter	1	↑	1	-	0	0	0	0	0	0	0	TEM	
Description	-This command is used to turn ON the Tearing Effect output signal from the TE signal line. -This output is not affected by changing MADCTL bit ML. -The Tearing Effect Line On has one parameter, which describes the mode of the Tearing Effect Output Line: -When TEM ='0': The Tearing Effect output line consists of V-Blanking information only												



-When TEM ='1': The Tearing Effect output Line consists of both V-Blanking and H-Blanking information



Note: During Sleep In Mode with Tearing Effect Line On, Tearing Effect Output pin will be active Low.

Restriction

This command has no effect when tearing effect output is already on.

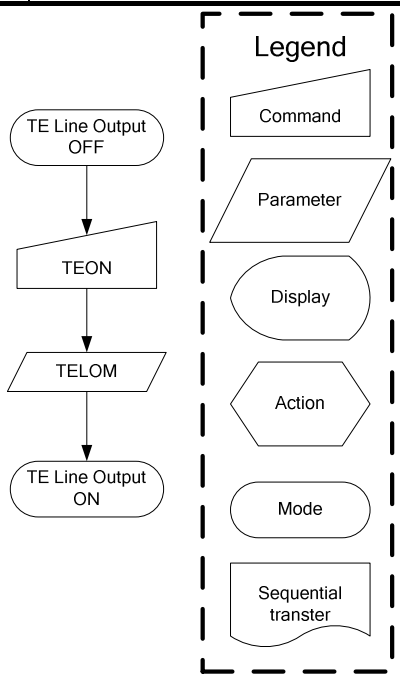
Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	Off
S/W Reset	Off
H/W Reset	Off

Flow Chart



9.2.28 MADCTL (36h): Memory Data Access Control

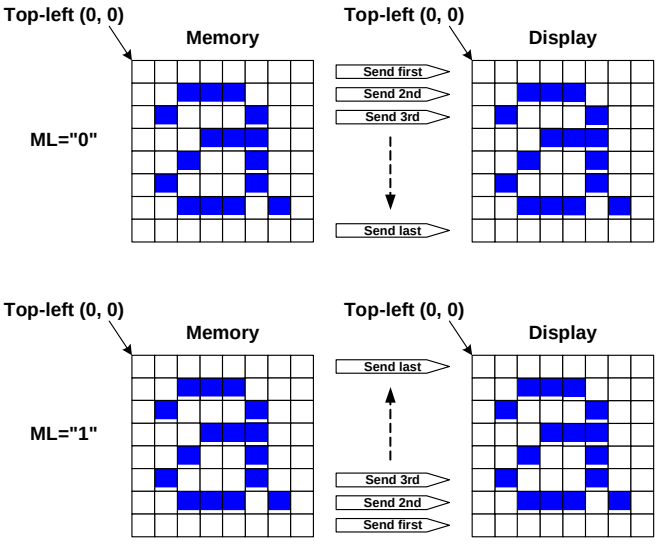
36H	MADCTL (Memory Data Access Control)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
MADCTL	0	↑	1	-	0	0	1	1	0	1	1	0	(36h)
parameter	1	↑	1	-	MY	MX	MV	ML	RGB	-	-	-	

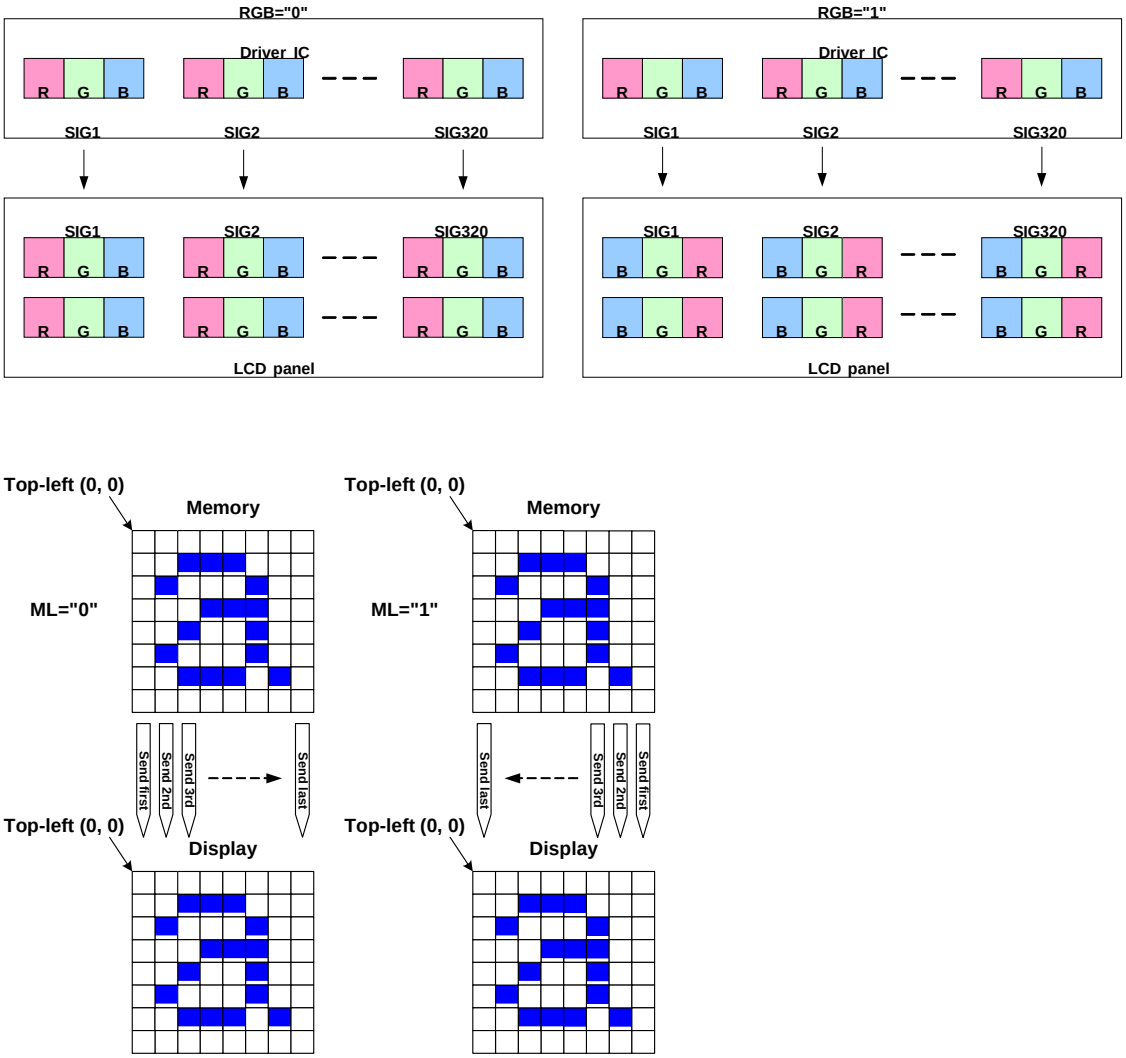
Description

-This command defines read/ write scanning direction of frame memory.

Bit	NAME	DESCRIPTION
MY	Row Address Order	These 3bits controls MCU to memory write/read direction.
MX	Column Address Order	
MV	Row/Column Exchange	
ML	Vertical Refresh Order	LCD vertical refresh direction control '0' = LCD vertical refresh Top to Bottom '1' = LCD vertical refresh Bottom to Top
RGB	RGB-BGR ORDER	Color selector switch control '0' =RGB color filter panel, '1' =BGR color filter panel)

-Bit Assignment





Restriction

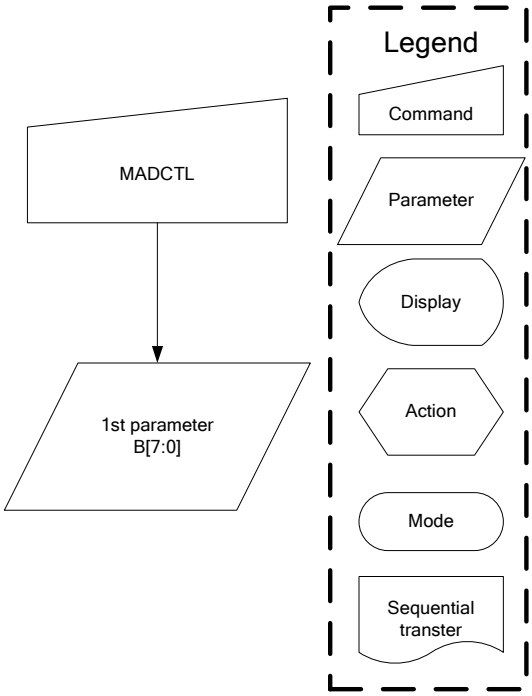
Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

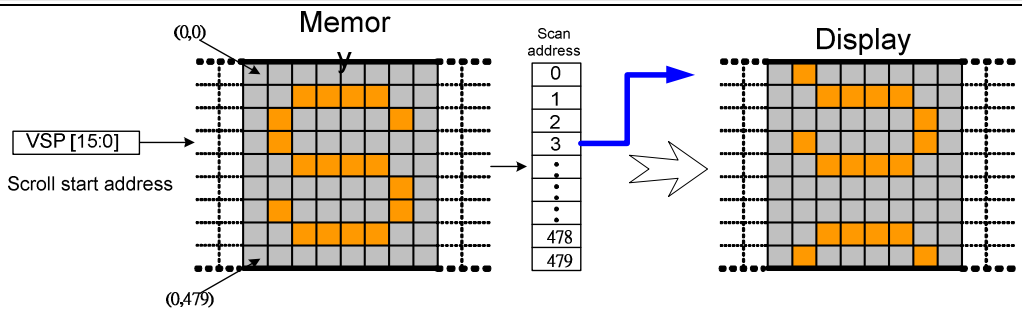
Status	Default Value
Power On Sequence	0000h
S/W Reset	No change
H/W Reset	0000h

Flow Chart



9.2.29 VSCSAD (37h): Vertical Scroll Start Address of RAM

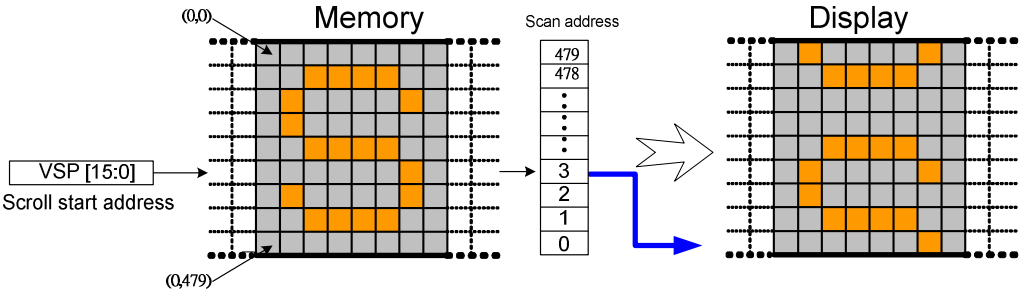
37H	VSCSAD (Vertical Scroll Start Address of RAM)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
VSCSAD	0	↑	1	-	0	0	1	1	0	1	1	1	(37h)
1ST parameter	1	↑	1	-	VSP15	VSP14	VSP13	VSP12	VSP11	VSP10	VSP9	VSP8	
2ND parameter	1	↑	1	-	VSP7	VSP6	VSP5	VSP4	VSP3	VSP2	VSP1	VSP0	
Description	-This command is used together with Vertical Scrolling Definition (33h). -These two commands describe the scrolling area and the scrolling mode. -The Vertical Scrolling Start Address command has one parameter which describes which line in the Frame Memory will be written as the first line after the last line of the Top Fixed Area on the display as illustrated below: When ML=0 Example: When Top Fixed Area = Bottom Fixed Area = 00, vertical Scrolling Area = 480 and VSP = '3'												



When ML=1

Example:

When Top Fixed Area = Bottom Fixed Area = 00, vertical Scrolling Area = 480 and VSP = '3'



NOTE: When new Pointer position and Picture Data are sent, the result on the display will happen at the next Panel Scan to avoid tearing effect.

VSP refers to the Frame Memory line Pointer

Restriction

Since the value of the vertical scrolling start address is absolute (with reference to the frame memory), it must not enter the fixed area (defined by Vertical Scrolling Definition (33h)- otherwise undesirable image will be displayed on the panel)

Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	0000h
S/W Reset	0000h
H/W Reset	0000h

Flow Chart

See Vertical Scrolling Definition (33h) description

9.2.30 IDMOFF (38h): Idle Mode Off

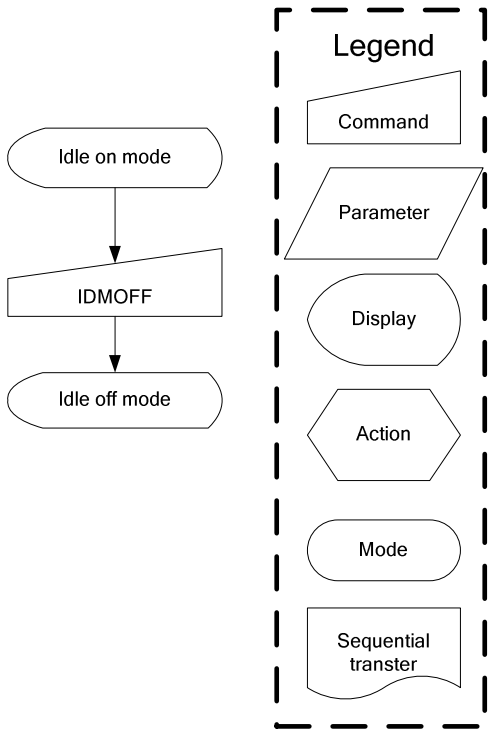
38H	IDMOFF (Idle Mode Off)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
IDMOFF	0	↑	1	-	0	0	1	1	1	0	0	0	(38h)
parameter	No Parameter												
Description	-This command is used to recover from Idle mode on. -In the idle off mode, 1. LCD can display 65k, 262k or 16M colors. 2. Normal frame frequency is applied.												
Restriction	This command has no effect when module is already in idle off mode												

Register availability

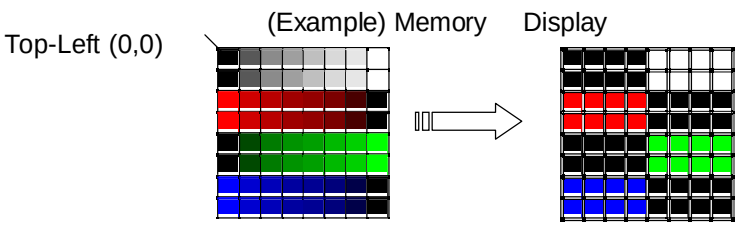
Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	Idle mode off
S/W Reset	Idle mode off
H/W Reset	Idle mode off

Flow Chart	

9.2.31 IDMON (39h): Idle mode on

39H	IDMON (Idle Mode On)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
IDMON	0	↑	1	-	0	0	1	1	1	0	0	1	(39h)
parameter	No Parameter												
Description	-This command is used to enter into Idle mode on. -There will be no abnormal visible effect on the display mode change transition. -In the idle on mode, 1. Color expression is reduced. The primary and the secondary colors using MSB of each R,G and B in the Frame Memory, 8 color depth data is displayed. 2. 8-Color mode frame frequency is applied. 3. Exit from IDMON by Idle Mode Off (38h) command 												

Color	R5 R4 R3 R2 R1 R0	G5 G4 G3 G2 G1 G0	B5 B4 B3 B4 B1 B0
Black	0xxxxx	0xxxxx	0xxxxx
Blue	0xxxxx	0xxxxx	1xxxxx
Red	1xxxxx	0xxxxx	0xxxxx
Magenta	1xxxxx	0xxxxx	1xxxxx
Green	0xxxxx	1xxxxx	0xxxxx
Cyan	0xxxxx	1xxxxx	1xxxxx
Yellow	1xxxxx	1xxxxx	0xxxxx
White	1xxxxx	1xxxxx	1xxxxx

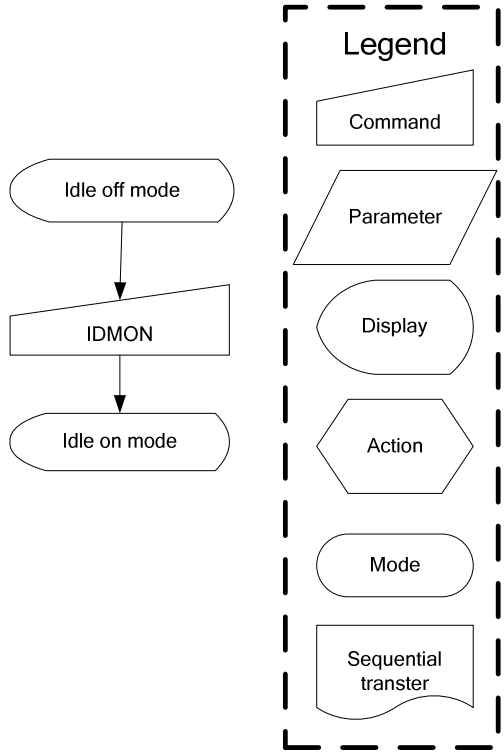
Restriction: This command has no effect when module is already in idle off mode

Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	Idle mode off
S/W Reset	Idle mode off
H/W Reset	Idle mode off

Flow Chart	

9.2.32 COLMOD (3Ah): Interface Pixel Format

3AH	COLMOD (Interface Pixel Format)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
COLMOD	0	↑	1	-	0	0	1	1	1	0	1	0	(3Ah)
1st Parameter	1	↑	1	-	0	D6	D5	D4	0	D2	D1	D0	
Description	This command is used to define the format of RGB picture data, which is to be transferred via the MCU interface. The formats are shown in the table: 1st parameter:												
Restriction													

Bit	Description	
D7	-	Set to ‘0’
D6	RGB interface color format	‘101’ = 16bit/pixel ‘110’ = 18bit/pixel
D5		
D4		
D3	-	Set to ‘0’
D2	Control interface color format	‘101’ = 16bit/pixel ‘110’ = 18bit/pixel ‘111’ = 24 bit/pixel
D1		
D0		

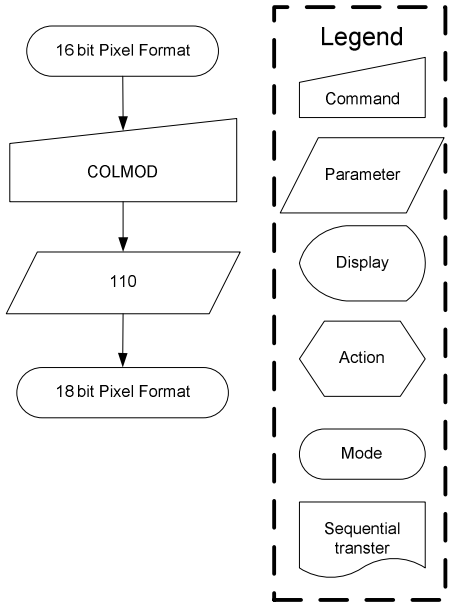
Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	18bit/pixel
S/W Reset	No change
H/W Reset	18bit/pixel

Flow Chart



9.2.33 WRMEMC (3Ch): Write Memory Continue

3CH	WRMEMC (Write Memory Continue)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
WRMEMC	0	↑	1	-	0	0	1	1	1	1	0	0	(3Ch)
1ST parameter	1	↑	1	D[17]-[8]	D[7]	D[6]	D[5]	D[4]	D[3]	D[2]	D[1]	D[0]	
⋮	1	↑	1	Dx[17]-x[8]	Dx[7]	Dx[6]	Dx[5]	Dx[4]	Dx[3]	Dx[2]	Dx[1]	Dx[0]	
Nth parameter	1	↑	1	Dn[17]-Dn[8]	Dn[7]	Dn[6]	Dn[5]	Dn[4]	Dn[3]	Dn[2]	Dn[1]	Dn[0]	
Description	-This command transfers image data from the host processor to the display module's frame memory continuing from												

the pixel location following the previous write memory continue or memory write command.

-If MV=0:

Data is written continuing from the pixel location after the write range of the previous memory write or write memory continue. The column register is then incremented and pixels are written to the frame memory until the column register equals the end column (XE) value. The column register is then reset to XS and the page register is incremented. Pixels are written to the frame memory until the page register equals the end page (YE) value and the column register equals the XE value, or the host processor sends another command. If the number of pixels exceeds (XE-XS+1)*(YE-YS+1) the extra pixels are ignored.

If MV=1:

Data is written continuing from the pixel location after the write range of the previous memory write or write memory continue. The page register is then incremented and pixels are written to the frame memory until the page register equals the end page (YE) value. The page register is then reset to YS and the column register is incremented. Pixels are written to the frame memory until the column register equals the end column (XE) value and the page register equals the YE value, or the host processor sends another command. If the number of pixels exceeds (XE-XS+1)*(YE-YS+1) the extra pixels are ignored.

Condition	Column	Page
Command 2C/2E is accepted	Return to “Start Column”	Return to “Start Page”
Read/Write RAM action	Increment by 1	No change
Column value is large than “End Column”	Return to “Start Column”	Increment by 1
Page value is large than “End Page”	Return to “Start Column”	Return to “Start Page”

Restriction

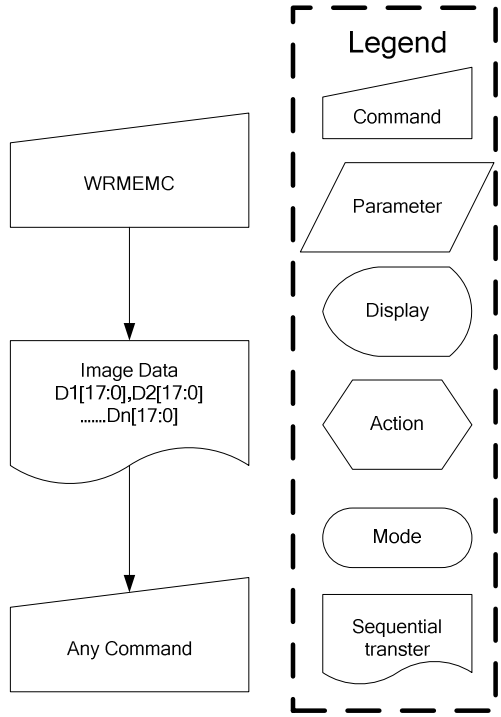
A memory write should follow a column address set or page address set to define the write address. Otherwise, data written with write memory continue is written to undefined addresses.

Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

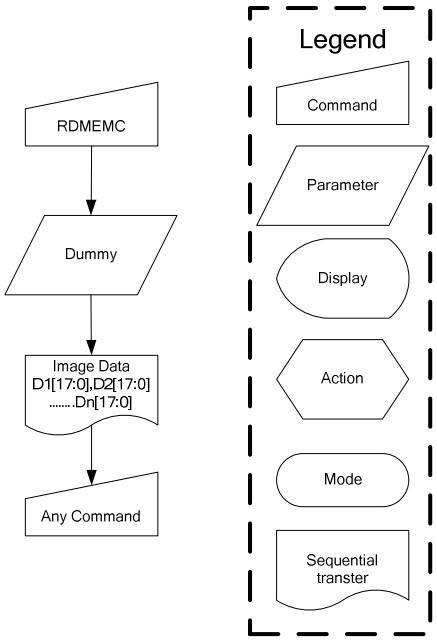
Default

Status	Default Value
Power On Sequence	Contents of memory is set randomly
S/W Reset	Contents of memory is not cleared
H/W Reset	Contents of memory is not cleared

Flow Chart	

9.2.34 RDMEMC (3Eh): Read Memory Continue

3EH	RDMEMC (Read Memory Continue)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
RDMEMC	0	↑	1	-	0	0	1	1	1	1	1	0	(3Eh)
1ST parameter	1	1	↑	-	-	-	-	-	-	-	-	-	
2nd parameter	1	1	↑	D[17]-[8]	D[7]	D[6]	D[5]	D[4]	D[3]	D[2]	D[1]	D[0]	
⋮	1	1	↑	Dx[17]-x[8]	Dx[7]	Dx[6]	Dx[5]	Dx[4]	Dx[3]	Dx[2]	Dx[1]	Dx[0]	
Nth parameter	1	1	↑	Dn[17]-Dn[8]	Dn[7]	Dn[6]	Dn[5]	Dn[4]	Dn[3]	Dn[2]	Dn[1]	Dn[0]	
Description	-This command transfers image data from the host processor to the display module's frame memory continuing from the pixel location following the previous read memory continue or memory read command. -If MV=0: Pixels are read continuing from the pixel location after the read range of the previous memory read or read memory continue. The column register is then incremented and pixels are read from the frame memory until the column register equals the end column (XE) value. The column register is then reset to XS and the page register is incremented. Pixels are read from the frame memory until the page register equals the end page (YE) value and the column register equals the XE value, or the host processor sends another command. If MV=1: Pixels are read continuing from the pixel location after the read range of the previous memory read or read memory continue. The page register is then incremented and pixels are read from the frame memory until the page register												

<table>
<tr><td></td><td>equals the end page (YE) value. The page register is then reset to YS and the column register is incremented. Pixels are read from the frame memory until the column register equals the end column (XE) value and the page register equals the YE value, or the host processor sends another command.
<table>
<tr><th>Condition</th><th>Column</th><th>Page</th></tr>
<tr><td>Command 2C/2E is accepted</td><td>Return to “Start Column”</td><td>Return to “Start Page”</td></tr>
<tr><td>Read/Write RAM action</td><td>Increment by 1</td><td>No change</td></tr>
<tr><td>Column value is large than “End Column”</td><td>Return to “Start Column”</td><td>Increment by 1</td></tr>
<tr><td>Page value is large than “End Page”</td><td>Return to “Start Column”</td><td>Return to “Start Page”</td></tr>
</table></td></tr>
<tr><td>Restriction</td><td>Regardless of the color mode set in interface pixel format, the pixel format returned by read memory continue is always 18-bit so there is no restriction on the length of data</td></tr>
<tr><td>Register availability</td><td>
<table>
<tr><th>Status</th><th>Availability</th></tr>
<tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Sleep In</td><td>Yes</td></tr>
</table></td></tr>
<tr><td>Default</td><td>
<table>
<tr><th>Status</th><th>Default Value</th></tr>
<tr><td>Power On Sequence</td><td>Contents of memory is set randomly</td></tr>
<tr><td>S/W Reset</td><td>Contents of memory is not cleared</td></tr>
<tr><td>H/W Reset</td><td>Contents of memory is not cleared</td></tr>
</table></td></tr>
<tr><td>Flow Chart</td><td> 
</td></tr>
</table>

9.2.35 STE (44h): Set Tear Scanline

44H	STE (Set Tear ScanLine)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
STE	0	↑	1	-	0	1	0	0	0	1	0	0	(44h)
1st parameter	1	↑	1	-	N15	N14	N13	N12	N11	N10	N9	N8	
2nd parameter	1	↑	1	-	N7	N6	N5	N4	N3	N2	N1	N0	

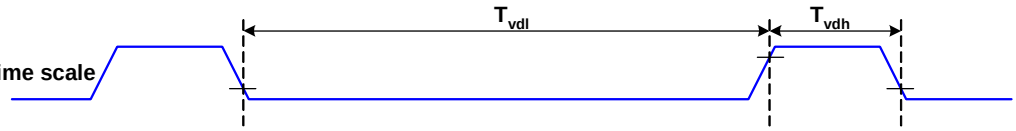
Description

-This command turns on the display module's Tearing Effect output signal on the TE signal line when the display module reaches line N. The TE signal is not affected by changing MV.

-The tearing effect line on has one parameter that describes the tearing effect output line mode.

-The tearing effect output line consist of V-blanking information only.

Vertical time scale



Note that set tear scanline with N=0 is equivalent to tearing effect line on with TEM=0.

The tearing effect output line shall be active low when the display module is in sleep mode

Restriction

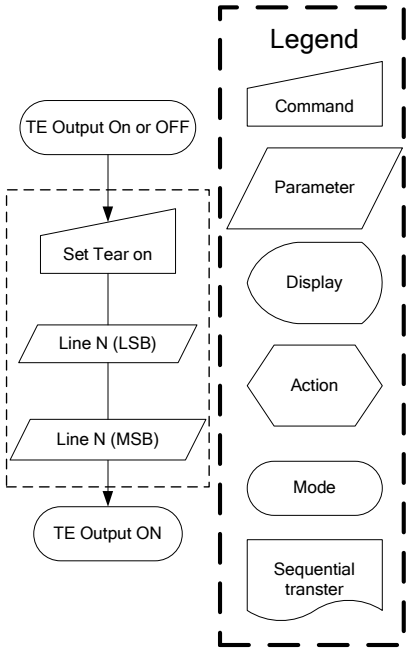
This command takes affect on the frame following the current frame. Therefore, if the tear effect (TE) output is already on, the TE output shall continue to operate as programmed by the previous tearing effect line on or set tear scanline command until the end of the frame

Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

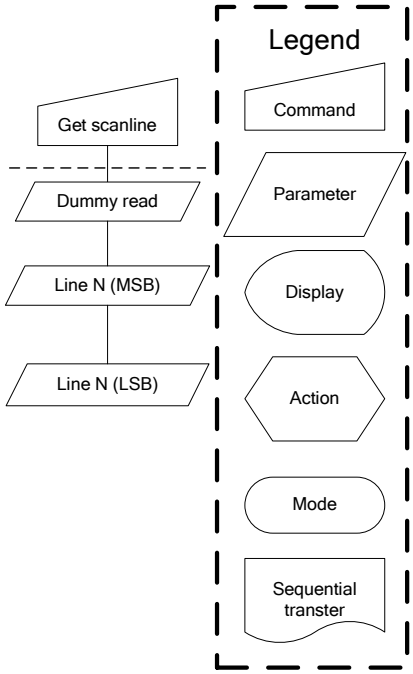
Status	Default Value
Power On Sequence	0000h
S/W Reset	0000h
H/W Reset	0000h

Flow Chart	

9.2.36 GSCAN (45h): Get Scanline

45H	GSCAN (Get ScanLine)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
GSCAN	0	↑	1	-	0	1	0	0	0	1	0	1	(45h)
1st parameter	1	1	↑	-	-	-	-	-	-	-	-	-	
2nd parameter	1	1	↑	-	N15	N14	N13	N12	N11	N10	N9	N8	
3rd parameter	1	1	↑	-	N7	N6	N5	N4	N3	N2	N1	N0	
Description	-The display module returns the current scanline ,N, used to update the display device. The total number of scanlines on a display device is defined as VSYNC+VBP+VACT+VFP. The first scanline is defined as the first line of V Sync and is denoted as Line 0. -When in sleep in mode, the value returned by get scanline is undefined. Note: that Set Tear Scan Line with N = 0 is equivalent to Tearing Effect Line ON with M = 0.												
Restriction	-												
Register availability													

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default	<table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>0000h</td></tr><tr><td>S/W Reset</td><td>0000h</td></tr><tr><td>H/W Reset</td><td>0000h</td></tr></table>
Flow Chart	

9.2.37 WRDISBV (51h): Write Display Brightness

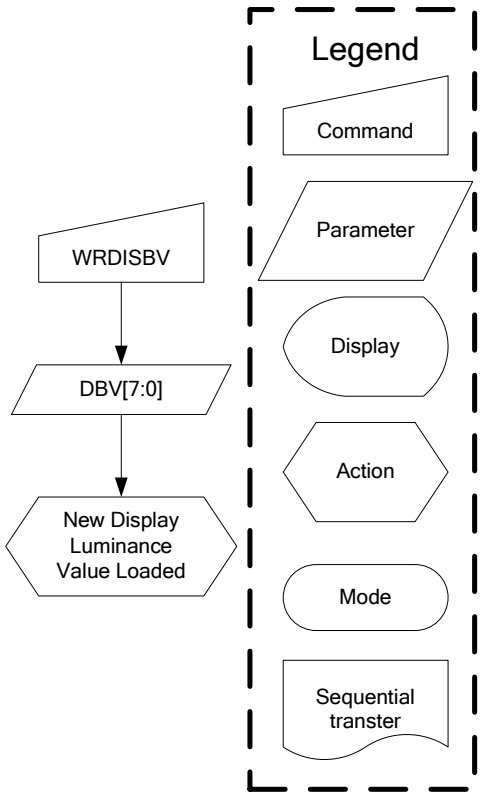
51H	WRDISBV (Write Display Brightness)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
WRDISBV	0	↑	1	-	0	1	0	1	0	0	0	1	(51h)
Parameter	1	↑	1	-	DBV7	DBV6	DBV5	DBV4	DBV3	DBV2	DBV1	DBV0	
Description	-This command is used to adjust the brightness value of the display. -It should be checked what the relationship between this written value and output brightness of the display is. This relationship is defined on the display module specification. -In principle relationship is that 00h value means the lowest brightness and FFh value means the highest brightness.												
Restriction													
Register availability	<table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr></table>												

		Partial Mode On, Idle Mode Off, Sleep Out	Yes
		Partial Mode On, Idle Mode On, Sleep Out	Yes
		Sleep In	Yes

Default

Status	Default Value
Power On Sequence	0000h
S/W Reset	0000h
H/W Reset	0000h

Flow Chart



9.2.38 RDDISBV (52h): Read Display Brightness Value

52H	RDDISBV (Read Display Brightness Value)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
RDDISBV	0	↑	1	-	0	1	0	1	0	0	1	0	(52h)
1st parameter	1	1	↑	-	-	-	-	-	-	-	-	-	
2nd parameter	1	1	↑	-	DBV7	DBV6	DBV5	DBV4	DBV3	DBV2	DBV1	DBV0	
Description	-This command returns the brightness value of the display. -It should be checked what the relationship between this returned value and output brightness of the display. This relationship is defined on the display module specification is.												

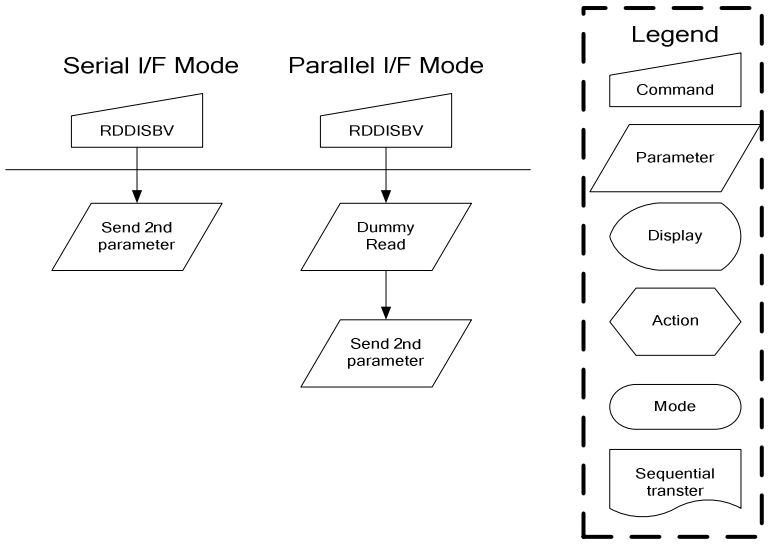
	-In principle the relationship is that 00h value means the lowest brightness and FFh value means the highest brightness. -DBV[7:0] is reset when display is in sleep in mode. -DBV[7:0] is '0' when bit BCTRL of write CTRL display command (53h) is '0' -DBV[7:0] is manual set brightness specified with write CTRL display command (53h) when bit BCTRL is '1'
Restriction	-
Register availability	

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	0000h
S/W Reset	0000h
H/W Reset	0000h

Flow Chart



9.2.39 WRCTRLD (53h): Write CTRL Display

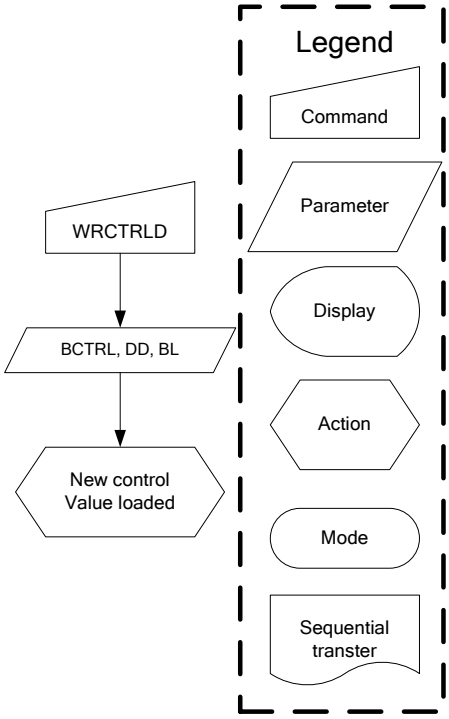
53H	WRCTRLD (Write CTRL Display)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
WRCTRLD	0	↑	1	-	0	1	0	1	0	0	1	1	(53h)
Parameter	1	↑	1	-	0	0	BCTRL	0	DD	BL	0	0	

<table>
<tr><td>Description</td><td>-This command is used to control display brightness.
-BCTRL: Brightness Control Block On/Off, This bit is always used to switch brightness for display.
0 = Off (Brightness register are 00h, DBV[7:0])
1 = On (Brightness register are active, according to the other parameters.)

-DD: Display Dimming (Only for manual brightness setting)
DD = 0: Display Dimming is off.
DD = 1: Display Dimming is on.

-BL: Backlight Control On/Off
0 = Off (Completely turn off backlight circuit. Control lines must be low.)
1 = On

-Dimming function is adapted to the brightness registers for display when bit BCTRL is changed at DD=1.
-When BL bit changed from ‘on’ to ‘off’, backlight is turned off without gradual dimming, even if dimming-on (DD=1) are selected.</td></tr>
<tr><td>Restriction</td><td></td></tr>
<tr><td>Register availability</td><td>
<table>
<tr><th>Status</th><th>Availability</th></tr>
<tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Sleep In</td><td>Yes</td></tr>
</table>
</td></tr>
<tr><td>Default</td><td>
<table>
<tr><th>Status</th><th>Default Value</th></tr>
<tr><td>Power On Sequence</td><td>0000h</td></tr>
<tr><td>S/W Reset</td><td>0000h</td></tr>
<tr><td>H/W Reset</td><td>0000h</td></tr>
</table>
</td></tr>
</table>

Flow Chart	

9.2.40 RDCTRLD (54h): Read CTRL value Display

54H	RDCTRLD (Read CTRL value Display)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
RDCTRLD	0	↑	1	-	0	1	0	1	0	1	0	0	(54h)
1st parameter	1	1	↑	-	-	-	-	-	-	-	-	-	
2nd parameter	1	1	↑	-	0	0	BCTRL	0	DD	BL	0	0	
Description	-This command returns ambient light and brightness control values.. -BCTRL: Brightness Control Block On/Off, This bit is always used to switch brightness for display. 0 = Off 1 = On -DD: Display Dimming (Only for manual brightness setting) DD = 0 DD = 1 -BL: Backlight Control On/Off 0 = Off 1 = On												

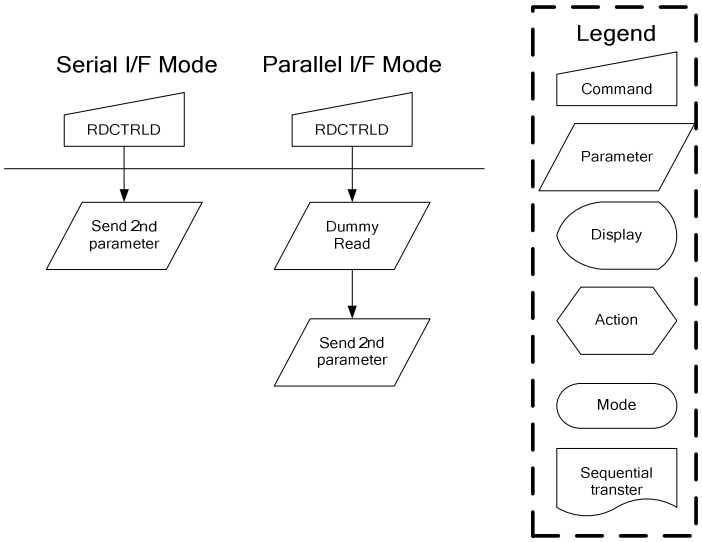
Restriction	-
Register availability	

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	0000h
S/W Reset	0000h
H/W Reset	0000h

Flow Chart



9.2.41 WRCABC (55h): Write Adaptive Brightness Control

55H	WRCABC (Write Adaptive Brightness Control)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
WRCABC	0	↑	1	-	0	1	0	1	0	1	0	1	(55h)
Parameter	1	↑	1	-	CECTRL	0	CE1	CE0	0	0	C1	C0	

Description

-This command is used to set parameters for image content based adaptive brightness control functionality.

-There is possible to used 4 different modes for content adaptive image functionality, which are defined on a table below.

C1	C0	Function

0	0	Off
0	1	User Interface Mode
1	0	Still Picture
1	1	Moving Image

-CECTRL: Color Enhancement Control Bit:

CECTRL=0: Color Enhancement Off.

CECTRL=1: Color Enhancement On.

-There are three color enhancement levels can be set.

CE1	CE0	Color enhancement level
0	0	Low enhancement
0	1	Medium enhancement
1	1	High enhancement

'-': Don't care

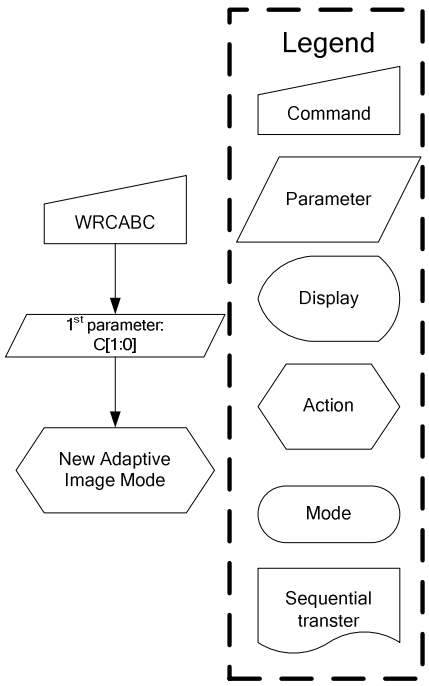
Restriction

Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	0000h
S/W Reset	0000h
H/W Reset	0000h

Flow Chart	

9.2.42 RDCABC (56h): Read Content Adaptive Brightness Control

56H	RDCABC (Read Content Adaptive Brightness Control)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
RDCABC	0	↑	1	-	0	1	0	1	0	1	1	0	(56h)
1st parameter	1	1	↑	-	-	-	-	-	-	-	-	-	
2nd parameter	1	1	↑	-	0	0	0	0	0	0	C1	C0	

Description

-This command is used to read the settings for image content based adaptive brightness control functionality.

-There is possible to used 4 different modes for content adaptive image functionality, which are defined on a table below.

C1	C0	Function
0	0	Off
0	1	User Interface Mode
1	0	Still Picture
1	1	Moving Image

'-': Don't care

Restriction

-

Register availability

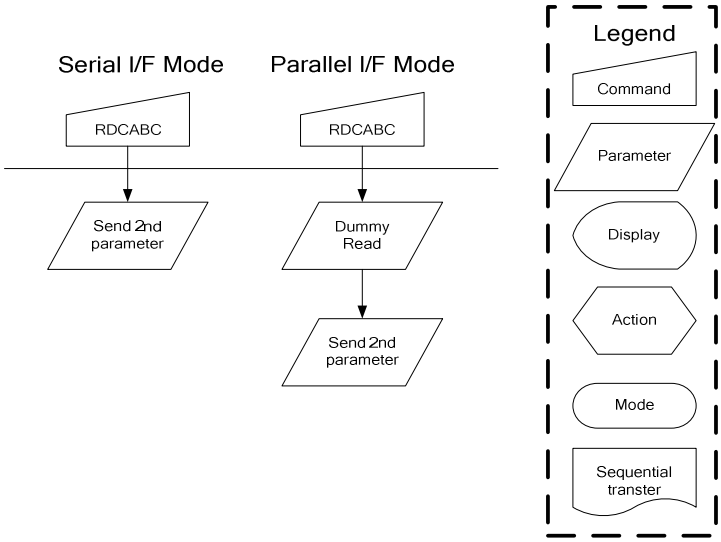
Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes

	Normal Mode On, Idle Mode On, Sleep Out	Yes
	Partial Mode On, Idle Mode Off, Sleep Out	Yes
	Partial Mode On, Idle Mode On, Sleep Out	Yes
	Sleep In	Yes

Default

Status	Default Value
Power On Sequence	0000h
S/W Reset	0000h
H/W Reset	0000h

Flow Chart



9.2.43 WRCABCMB (5Eh): Write CABC Minimum Brightness

5EH	WRCABCMB (Write CABC Minimum Brightness)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
WRCABCMB	0	↑	1	-	0	1	0	1	1	1	1	0	(5Eh)
Parameter	1	↑	1	-	CMB7	CMB6	CMB5	CMB4	CMB3	CMB2	CMB1	CMB0	
Description	-This command is used to set the minimum brightness value of the display for CABC function. -In principle relationship is that 00h value means the lowest brightness for CABC and FFh value means the brightness for CABC. '-': Don't care												
Restriction													

Register availability

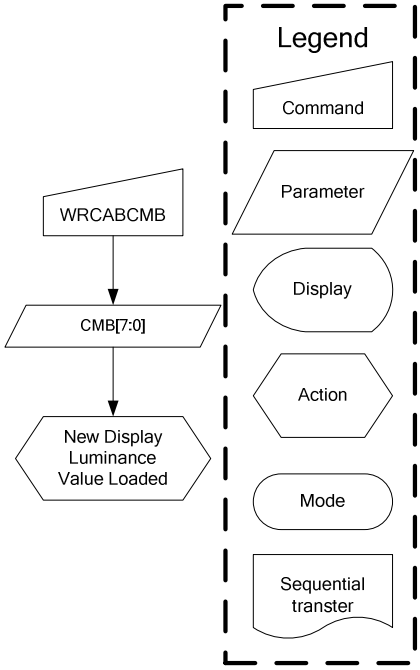
Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes

		Normal Mode On, Idle Mode On, Sleep Out	Yes
		Partial Mode On, Idle Mode Off, Sleep Out	Yes
		Partial Mode On, Idle Mode On, Sleep Out	Yes
		Sleep In	Yes

Default

Status	Default Value
Power On Sequence	0000h
S/W Reset	0000h
H/W Reset	0000h

Flow Chart



9.2.44 RDCABCMB (5Fh): Read CABC Minimum Brightness

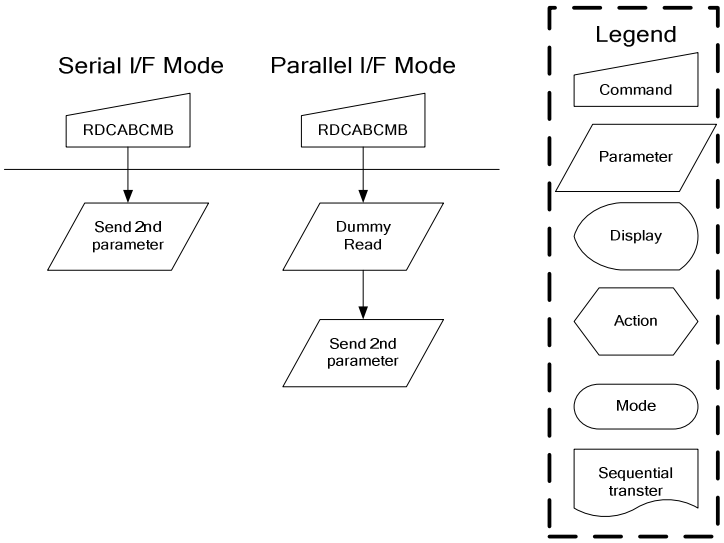
5FH	RDCABCMB (Read CABC Minimum Brightness)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
RDCABCMB	0	↑	1	-	0	1	0	1	1	1	1	1	(5Fh)
1st parameter	1	1	↑	-	-	-	-	-	-	-	-	-	
2nd parameter	1	1	↑	-	CMB7	CMB6	CMB5	CMB4	CMB3	CMB2	CMB1	CMB0	
Description	-This command returns the minimum brightness value of CABC function. -In principle relationship is that 00h value means the lowest brightness for CABC and FFh value means the brightness for CABC. '-': Don't care												
Restriction	-												

Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	0000h
S/W Reset	0000h
H/W Reset	0000h

Flow Chart	

9.2.45 RDFCS (AAh): Read First Checksum

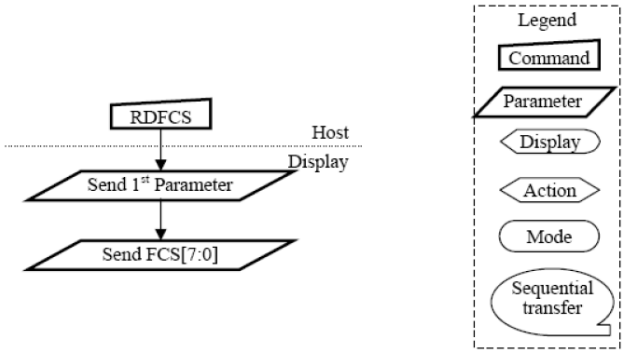
AAH	RDFCS (Read First Checksum)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
RDID1	0	↑	1	-	1	0	1	0	1	0	1	0	(AAh)
1st parameter	1	1	↑	-	-	-	-	-	-	-	-	-	
2nd parameter	1	1	↑	-	FCS[7:0]								
Description	- This command returns the first checksum what has been calculated from User's area registers and the frame memory after the write access to those registers and/or frame memory has been done. X = can be '0' or '1'												
Restriction	- It will be necessary to wait 150ms after there is the last write access on User area registers before there can read this checksum value.												

Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	00h
S/W Reset	00h
H/W Reset	00h

Flow Chart	

9.2.46 RDCFCS (AFh): Read Continue Checksum

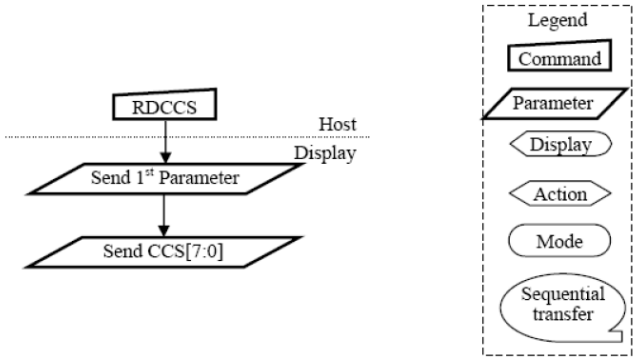
AFH	RDCFCS (Read Continue Checksum)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
RDID1	0	↑	1	-	1	0	1	0	1	1	1	1	(AFh)
1st parameter	1	1	↑	-	-	-	-	-	-	-	-	-	
2nd parameter	1	1	↑		CCS[7:0]								
Description	- This command returns the continue checksum what has been calculated continuously after the first checksum has calculated from User's area registers and the frame memory after the write access to those registers and/or frame memory has been done. X = can be '0' or '1'												
Restriction	- It will be necessary to wait 300ms after there is the last write access on User area registers before there can read this checksum value in the first time..												

Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	00h
H/W Reset	00h

Flow Chart	

9.2.47 RDID1 (DAh): Read ID1

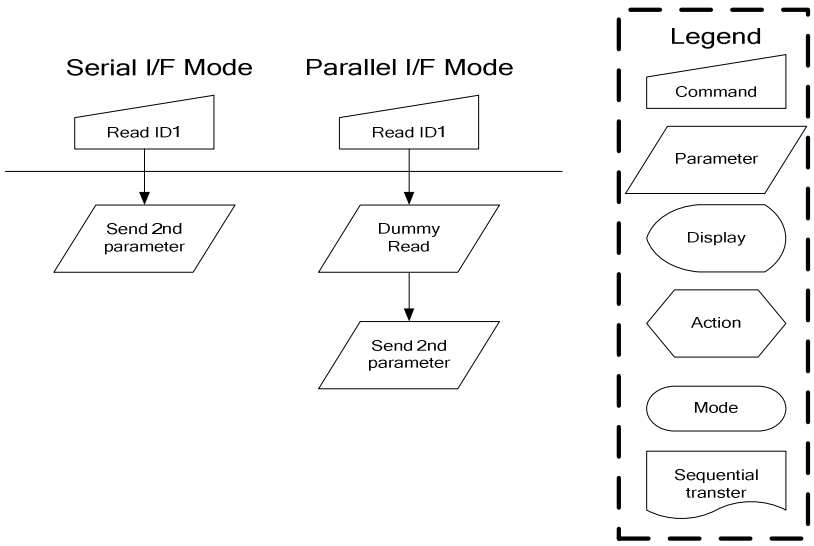
DAH	RDID1 (Read ID1)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
RDID1	0	↑	1	-	1	1	0	1	1	0	1	0	(DAh)
1st parameter	1	1	↑	-	-	-	-	-	-	-	-	-	
2nd parameter	1	1	↑	-	ID17	ID16	ID15	ID14	ID13	ID12	ID11	ID10	
Description	-This read byte identifies the LCD module's manufacturer. '-': Don't care.												
Restriction	-												

Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	00h
S/W Reset	00h
H/W Reset	00h

Flow Chart	

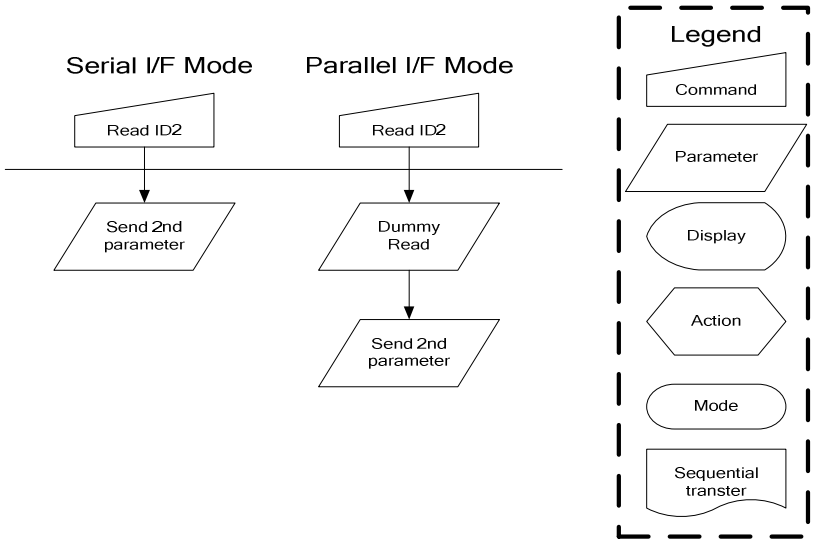
9.2.48 RDID2 (DBh): Read ID2

DBH	RDID2 (Read ID2)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
RDID2	0	↑	1	-	1	1	0	1	1	0	1	1	(DBh)
1st parameter	1	1	↑	-	-	-	-	-	-	-	-	-	
2nd parameter	1	1	↑	-	ID27	ID26	ID25	ID24	ID23	ID22	ID21	ID20	
Description	This read byte is used to track the LCD module/driver IC version. '-': Don't care.												
Restriction	-												
Register availability													

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	00h
S/W Reset	00h
H/W Reset	00h

Flow Chart	

9.2.49 RDID3 (DCh): Read ID3

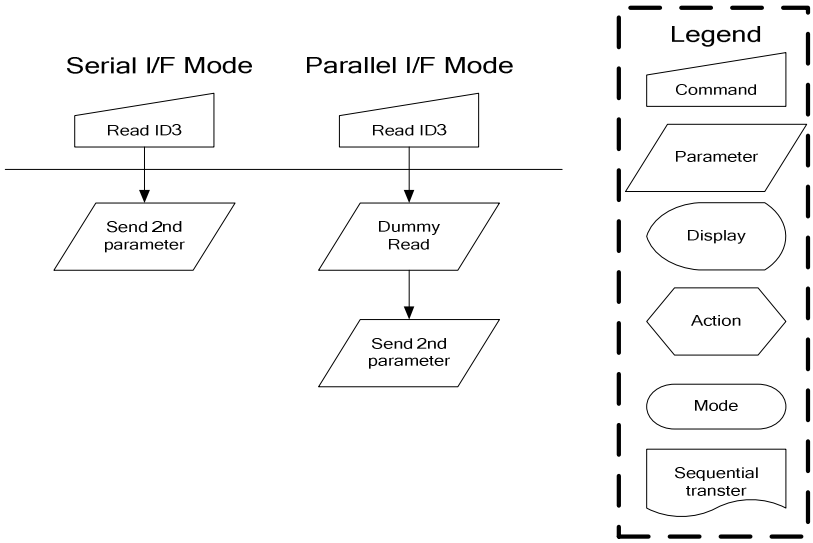
DCH	RDID3 (Read ID3)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
RDID3	0	↑	1	-	1	1	0	1	1	1	0	0	(DCh)
1st parameter	1	1	↑	-	-	-	-	-	-	-	-	-	
2nd parameter	1	1	↑	-	ID37	ID36	ID35	ID34	ID33	ID32	ID31	ID30	
Description	This read byte identifies the LCD module/driver. '-': Don't care.												
Restriction	-												

Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	00h
S/W Reset	00h
H/W Reset	00h

Flow Chart	

9.3.. Command Table 2

9.3.1 IFMODE (B0h): Interface Mode Control

B0H	IFMODE (Interface Mode Control)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
IFMODE	0	↑	1	-	1	0	1	1	0	0	0	0	(B0h)
1st parameter	1	↑	1	-	SPI_EN	0	0	0	VSCP	HSCP	PKP	DEP	
Description	Sets the operation status of the display interface. The setting becomes effective as soon as the command is received. DEP: DE polarity (“0”= High enable for RGB interface, “1”=Low enable for RGB interface) PKP: PCLK polarity set (“0”=data fetched at the rising time, “1”=data fetched at the falling time) HSCP: HSYNC polarity (“0”=Low level sync clock, “1”=High level sync clock) VSCP: VSYNC polarity (“0”= Low level sync clock, “1”= High level sync clock) SPI_EN: 4 wire serial interface selection SPI_EN = “0”, DIN and DOUT pins are used for 4 wire serial interface. SPI_EN = “1”, DIN/SDA pin is used for 4 wire serial interface and DOUT pin is not used. ‘-’: Don’t care.												
Restriction	-												
Register availability													

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes

		Sleep In	Yes

Default

Status	Default Value
Power On Sequence	00h
S/W Reset	00h
H/W Reset	00h

Flow Chart

9.3.2 FRMCTR1 (B1h): Frame Rate Control (In Normal Mode/Full Colors)

B1H	FRMCTR1 (Frame Rate Control In Normal Mode/Full Colors)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
FRMCTR1	0	↑	1	-	1	0	1	1	0	0	0	1	(B1h)
1st parameter	1	↑	1	-	FRS[3:0]				0	0	DIVA[1:0]		
2nd parameter	1	↑	1	-	0	0	0	RTNA[4:0]					

Description

FRS[3:0]: Sets the frame frequency of full color normal mode.

DIVA [1:0] : division ratio for internal clocks when Normal mode.

DIVA [1:0]	Inversion mode
2'b00	Fosc
2'b01	Fosc/2
2'b10	Fosc/4
2'b11	Fosc/8

RTNA [4:0] : RTNA[4:0] is used to set 1H (line) period of Normal mode at CPU interface.

'-': Don't care.

Restriction

-

Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default	<table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>00h</td></tr><tr><td>S/W Reset</td><td>00h</td></tr><tr><td>H/W Reset</td><td>00h</td></tr></table>
Flow Chart	

9.3.3 FRC (B2h): Frame Rate Control (In Idle Mode/8 colors)

B2H	Frame Rate Control (In Idle Mode/8 colors)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
FRC IDLE	0	↑	1	-	1	0	1	1	0	0	1	0	(B2h)
1st parameter	1	↑	1	-	0	0	0	0	0	0	0	0	
2nd parameter	1	↑	1	-	0	0	0	RTNB[4:0]					
Description	Sets the division ratio for internal clocks of Idle mode at CPU interface. RTNB [4:0] : RTNB[4:0] is used to set 1H (line) period of Idle mode at CPU interface. '-': Don't care.												
Restriction	-												
Register availability	<table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table>												
Default	<table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>00h</td></tr><tr><td>S/W Reset</td><td>00h</td></tr><tr><td>H/W Reset</td><td>00h</td></tr></table>												
Flow Chart													

9.3.4 FRC (B3h): Frame Rate Control (In Partial Mode/Full Colors)

B3H	Frame Rate Control (In Partial Mode/Full Colors)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
FRC PTL	0	↑	1	-	1	0	1	1	0	0	1	1	(B3h)
1st parameter	1	↑	1	-	0	0	0	0	0	0	0	0	
2nd parameter	1	↑	1	-	0	0	0	RTNC[4:0]					
Description	Sets the division ratio for internal clocks of Partial mode (Idle mode off) at CPU interface. RTNC [4:0] : RTNC[4:0] is used to set 1H (line) period of Partial mode at CPU interface. '-': Don't care.												
Restriction	-												

Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	00h
S/W Reset	00h
H/W Reset	00h

Flow Chart

9.3.5 DIC (B4): Display Inversion Control

B4H	Display Inversion Control												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
DIC	0	↑	1	-	1	0	1	1	0	1	0	0	(B4h)
1st parameter	1	↑	1	-							DINV		

Description

DINV[1:0] : Set the inversion mode

DINV [1:0]	Inversion mode
2'b00	Column inversion
2'b01	1-dot inversion

	2'b10 | 2-dot inversion 2'b11 | Reserved '-': Don't care.
Restriction	-
Register availability	Status | Availability Normal Mode On, Idle Mode Off, Sleep Out | Yes Normal Mode On, Idle Mode On, Sleep Out | Yes Partial Mode On, Idle Mode Off, Sleep Out | Yes Partial Mode On, Idle Mode On, Sleep Out | Yes Sleep In | Yes
Default	Status | Default Value Power On Sequence | 01h S/W Reset | 01h H/W Reset | 01h
Flow Chart	

9.3.6 BPC(B5): Blanking Porch Control

B5H	Blanking Porch Control												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
BPC	0	↑	1	-	1	0	1	1	0	1	0	1	(B5h)
1st parameter	1	↑	1	-	VFP[7:0]								
2nd parameter	1	↑	1	-	VBP[7:0]								
3rd parameter	1	↑	1		0	0	0	0	0	0	0	0	
4th parameter	1	↑	1	-	HBP[7:0]								
Description	VFP [7:0] / VBP[7:0]: The FP [7:0] and BP [7:0] bits specify the line number of vertical front and back porch period respectively.												

VFP [7:0]	Front porch of Number lines
00h	Reserved
01h	Reserved
02h	2
:	:
:	:
FDh	253
FEh	254
FFh	255

VBP [7:0]	Front porch of Number lines
00h	Reserved
01h	Reserved
02h	2
:	:
:	:
FDh	253
FEh	254
FFh	255

HBP [7:0]: The HBP[7:0] bits specify the dotclk number of horizontal back porch period.

HBP [7:0]	Back porch of Number lines
00h	Reserved
01h	Reserved
02h	2
:	:
:	:
FDh	253
FEh	254
FFh	255

'-': Don't care.

Restriction: -

Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value		
	VFP [7:0]	VBP [7:0]	HBP [7:0]
Power On Sequence	02h	02h	04h
S/W Reset	02h	02h	04h

	H/W Reset	02h	02h	04h
Flow Chart				

9.3.7 DFC(B6): Display Function Control

B6H	Blanking Porch Control												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
DFC	0	↑	1	-	1	0	1	1	0	1	1	0	(B6h)
1st parameter	1	↑	1	-	BYPASS	RCM	RM	DM	PTG[1:0]		PT[1:0]		
2nd parameter	1	↑	1	-		GS	SS	SM	ISC[3:0]				
3rd parameter	1	↑	1	-	0	0	NL[5:0]						

Description

DM: Select the display operation mode.

DM	Interface Mode
0	Internal system clock
1	RGB interface

RM: Select the interface to access the GRAM. When RM='0', the driver will write display data to GRAM via system interface and the driver will write display data to GRAM via RGB interface when RM='1'.

RM	Interface for RAM access
0	System Interface
1	RGB interface

RCM: RGB interface selection (refer to the RGB interface section).

RCM	RGB transfer mode
0	DE Mode
1	SYNC Mode

BYPASS: Select the display data path whether memory or direct to shift register when RGB interface is used.

BYPASS	Display data path
0	Memory
1	Direct to shift register

Note: RGB input signal, when set to bypass mode the Hsync low≧3,HBP≧3, HFP≧10.

PTG [1:0]: Set the scan mode in non-display area.

PTG [1:0]		Gate outputs in non-display area	Source outputs in non-display area
0	0	Normal scan	Set with the PT[2:0] bits
0	1	Setting prohibited	--
1	0	Interval scan	Set with the PT[2:0] bits
1	1	Setting prohibited	--

PT [1:0]: Determine source/VCOM output in a non-display area in the partial display mode.

PT[1:0]		Source output on non-display area
0	0	V63
0	1	V0
1	0	--
1	1	--

SS: Select the shift direction of outputs from the source driver.

SS	Source Output Scan Direction
0	S1→ S960
1	S960 → S1

In addition to the shift direction, the settings for both SS and BGR bits are required to change the assignment of R, G, B dots to the source driver pins.

To assign R, G, B dots to the source driver pins from S1 to S960, set SS = 0.

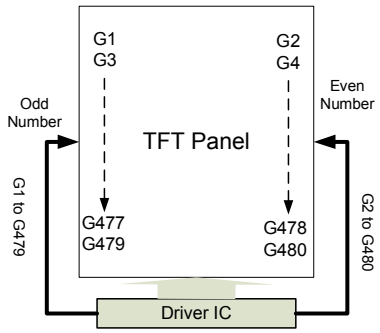
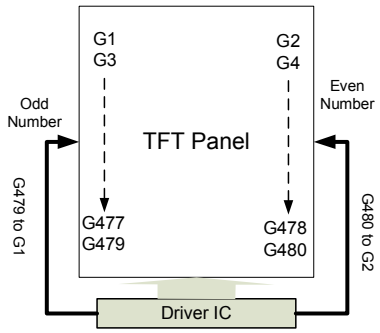
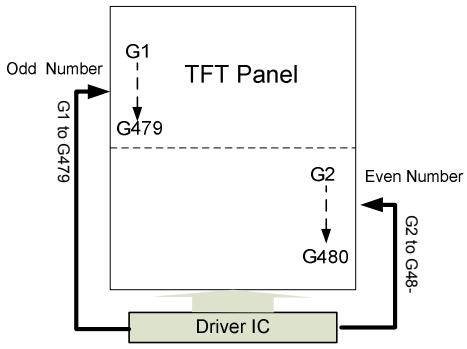
To assign R, G, B dots to the source driver pins from S960 to S1, set SS = 1.

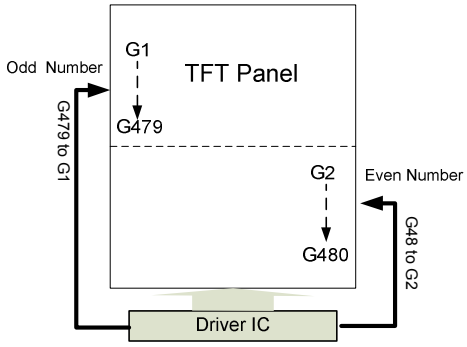
ICS[3:0] : Set the scan cycle when PTG selects interval scan in non-display area drive period. The scan cycle is defined by n frame periods, where n is an odd number from 3 to 31. The polarity of liquid crystal drive voltage from the gate driver is inverted in the same timing as the interval scan cycle.

GS: Sets the direction of scan by the gate driver.

GS	Gate Output Scan Direction
0	G1→ G480
1	G480 → G1

SM: Sets the gate driver pin arrangement in combination with the GS to select the optimal scan mode for the module.

SM	GS	Scan Direction	Gate Output Sequence
0	0		G1,G2,G3,G4…..,G476 G477,G478,G479, G480
0	1		G480,G479,……G8 G7,G6,G5,G4,G3,G2,G1
1	0		G1,G3,G5…….G471 G473,G475,G477,G479 G2,G4,G6…….G472 G474,G476,G478,G480

	1	1		G480,G488,.....G14 G12,G10,G8,G6,G4,G2 G479,G477.....G13 G11,G9,G7,G5,G3,G1

NL [5:0]: Sets the number of lines to drive the LCD at an interval of 8 lines. The GRAM address mapping is not affected by the number of lines set by NL[5:0]. The number of lines must be the same or more than the number of lines necessary for the size of the liquid crystal panel.

NL[5:0]	LCD Drive Line
6'h00 ~ 6'h3B	8 * (NL5:0]+1) lines
other	Setting inhibited

'-': Don't care.

Restriction: -

Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value				
	BYPASS	RCM	RM	PTG[1:0]	PT[1:0]
Power On Sequence	1b	0b	0b	00b	00b
S/W Reset	1b	0b	0b	00b	00b
H/W Reset	1b	0b	0b	00b	00b

Status	Default Value			
	GS	SS	SM	ISC[3:0]
Power On Sequence	0b	0b	0b	0010b

S/W Reset	0b	0b	0b	0010b
H/W Reset	0b	0b	0b	0010b

Status	Default Value
	NL[5:0]
Power On Sequence	3bh
S/W Reset	3bh
H/W Reset	3bh

Flow Chart	

9.3.8 EM(B7): Entry Mode Set

B7H	Entry Mode Set												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
EM	0	↑	1	-	1	0	1	1	0	1	1	1	(B7h)
1st parameter	1	↑	1	-	EPF[1:0]		0	0	DSTB	GON	DTE	0	

Description

DSTB: In Deep Standby mode (**DSTB=1)**, both internal logic power and SRAM power are turn off, the display data stored in the Frame Memory and the instructions are not saved. Rewrite Frame Memory content and instructions after the Deep Standby Mode is exited. Exit the Deep Standby Mode is as below,

1. Exit Deep Standby Mode by pull down CSX to low (“0”) 6 times.
2. Exit Deep Standby Mode by input RESX pulse.

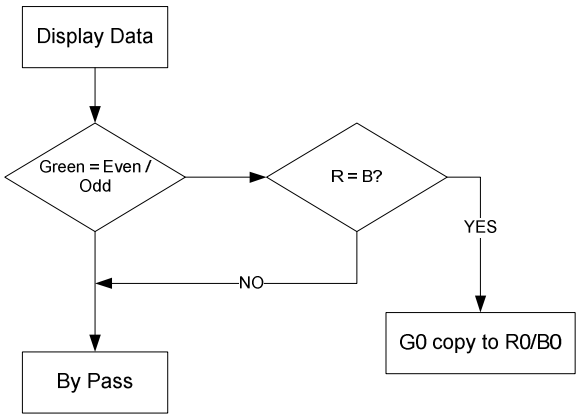
GON/DTE: Set the output level of gate driver G1 ~ G480 as follows

GON	DTE	G1~G480 Gate Output
0	0	VGH
0	1	VGH
1	0	VGL
1	1	Normal display

EPF[1:0] Set the data format when 16bbp (R,G,B) to 18 bbp (r, g, b) is stored in the internal GRAM

EPF[1:0]	65k (R, G, B) to 262k (r, g, b)
00	” 0” is written to the LSB
01	” 1” is written to the LSB.
10	The MSB value is written to the LSB.
11	Setting inhibited

EPF[1:0] = 11b is described as below.



	'-': Don't care.
Restriction	-
Register availability	

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value			
	EPF[1:0]	DSTB	GON	DTE
Power On Sequence	00b	0b	1b	1b
S/W Reset	00b	0b	1b	1b
H/W Reset	00b	0b	1b	1b

Flow Chart

9.3.9 Power Control 1 (C0h)

C0H	PWR1 (Power Control 1)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
	0	↑	1	-	1	1	0	0	0	0	0	0	(C0h)
1st parameter	1	↑	1	-	AVCLS[3:0]				AVDDS[3:0]				
2nd parameter	1	↑	1	-	0	0	VRHP[5:0]						
3rd parameter	1	↑	1	-	0	0	VRHN[5:0]						

Description

AVDDS[3:0]/ AVCLS[3:0] : AVDD/AVCL setting is as below.

AVDDS[3:0]	AVDD	AVDDS[3:0]	AVDD
0	6.91	8	6.10
1	6.80	9	5.99
2	6.70	10	5.89
3	6.60	11	5.79
4	6.50	12	5.69
5	6.40	13	5.59
6	6.30	14	5.49
7	6.20	15	5.39

AVCLS[3:0]	AVCL	AVCLS[3:0]	AVCL
0	-4.93	8	-4.35
1	-4.86	9	-4.28
2	-4.79	10	-4.21
3	-4.71	11	-4.14
4	-4.64	12	-4.06
5	-4.57	13	-3.99
6	-4.50	14	-3.92
7	-4.43	15	-3.85

VRHP/VRHN [5:0]: GVDD/GVCL voltage setting, One step is 20mV.

VRHP[5:0]	GVDD	VRHP[5:0]	GVDD
0	5.40	32	4.76
1	5.38	33	4.74
2	5.36	34	4.72
3	5.34	35	4.70
4	5.32	36	4.68
5	5.30	37	4.66
6	5.28	38	4.64
7	5.26	39	4.62
8	5.24	40	4.60
9	5.22	41	4.58
10	5.20	42	4.56
11	5.18	43	4.54
12	5.16	44	4.52
13	5.14	45	4.50
14	5.12	46	4.48
15	5.10	47	4.46
16	5.08	48	4.44
17	5.06	49	4.42
18	5.04	50	4.40
19	5.02	51	4.38
20	5.00	52	4.36
21	4.98	53	4.34
22	4.96	54	4.32
23	4.94	55	4.30
24	4.92	56	4.28
25	4.9	57	4.26
26	4.88	58	4.24
27	4.86	59	4.22
28	4.84	60	4.20
29	4.82	61	4.18
30	4.80	62	4.16
31	4.78	63	4.14

VRHN[5:0]	GVCL	VRHN[5:0]	GVCL
0	-5.40	32	-4.76
1	-5.38	33	-4.74
2	-5.36	34	-4.72
3	-5.34	35	-4.70
4	-5.32	36	-4.68
5	-5.30	37	-4.66
6	-5.28	38	-4.64
7	-5.26	39	-4.62
8	-5.24	40	-4.60
9	-5.22	41	-4.58
10	-5.20	42	-4.56
11	-5.18	43	-4.54
12	-5.16	44	-4.52
13	-5.14	45	-4.50
14	-5.12	46	-4.48
15	-5.10	47	-4.46
16	-5.08	48	-4.44
17	-5.06	49	-4.42
18	-5.04	50	-4.40
19	-5.02	51	-4.38
20	-5.00	52	-4.36
21	-4.98	53	-4.34
22	-4.96	54	-4.32
23	-4.94	55	-4.30
24	-4.92	56	-4.28
25	-4.90	57	-4.26
26	-4.88	58	-4.24
27	-4.86	59	-4.22
28	-4.84	60	-4.20
29	-4.82	61	-4.18
30	-4.80	62	-4.16
31	-4.78	63	-4.14

'-': Don't care.

Restriction: -

Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes

		Sleep In	Yes

Default

Status	Default Value			
	AVDDS[3:0]	AVCLS[3:0]	VRHP[5:0]	VRHN [5:0]
Power On Sequence	--	--	100000b	100000b
S/W Reset	--	--	100000b	100000b
H/W Reset	--	--	100000b	100000b

Flow Chart

9.3.10 Power Control 2 (C1h)

C1H	PWR2 (Power Control 2)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
	0	↑	1	-	1	1	0	0	0	0	0	1	(C1h)
1st parameter	1	↑	1	-	VGLS[3:0]				VGHS[3:0]				

Description

VGLS[3:0]/VGHS[3:0] ; VGH/VGL Voltage level selection.

VGHS [3:0]	VGH	VGHS [3:0]	VGH
0	12.7	8	14.3
1	12.9	9	14.5
2	13.1	10	14.7
3	13.3	11	14.9
4	13.5	12	15.1
5	13.7	13	15.3
6	13.9	14	15.5
7	14.1	15	15.7

VGLS [3:0]	VGL	VGLS [3:0]	VGL
0	-13.47	8	-10.77
1	-13.13	9	-10.44
2	-12.79	10	-10.1
3	-12.46	11	-9.76
4	-12.12	12	-9.42
5	-11.78	13	-9.09
6	-11.45	14	-8.75
7	-11.11	15	-8.41

'-': Don't care.

Restriction: -

Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default	
Flow Chart	

Status	Default Value	
	VGHS [3:0]	VGLS [3:0]
Power On Sequence	1011b	0100b
S/W Reset	1011b	0100b
H/W Reset	1011b	0100b

9.3.11 VCMP Control (C5h)

C5H	VCMPCTL (Vcom Control)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
	0	↑	1	-	1	1	0	0	0	1	0	1	(C5h)
1st parameter	1	↑	1	-	-	-	VCMP[5:0]						

Description

VCMP[5:0]: Set the relative VCOM offset.

	VCMP[5:0]	VCOM		VCMP[5:0]	VCOM		VCMP[5:0]	VCOM		VCMP[5:0]	VCOM
0	000000b	0.42	16	010000b	0.74	32	100000	1.06	48	110000	1.38
1	000001b	0.44	17	010001b	0.76	33	100001	1.08	49	110001	1.4
2	000010b	0.46	18	010010b	0.78	34	100010	1.1	50	110010	1.42
3	000011b	0.48	19	010011b	0.8	35	100011	1.12	51	110011	1.44
4	000100b	0.5	20	010100b	0.82	36	100100	1.14	52	110100	1.46
5	000101b	0.52	21	010101b	0.84	37	100101	1.16	53	110101	1.48
6	000110b	0.54	22	010110b	0.86	38	100110	1.18	54	110110	1.5
7	000111b	0.56	23	010111b	0.88	39	100111	1.2	55	110111	1.52
8	001000b	0.58	24	011000b	0.9	40	101000	1.22	56	111000	1.54
9	001001b	0.6	25	011001b	0.92	41	101001	1.24	57	111001	1.56
10	001010b	0.62	26	011010b	0.94	42	101010	1.26	58	111010	1.58
11	001011b	0.64	27	011011b	0.96	43	101011	1.28	59	111011	1.6
12	001100b	0.66	28	011100b	0.98	44	101100	1.3	60	111100	1.62
13	001101b	0.68	29	011101b	1	45	101101	1.32	61	111101	1.64
14	001110b	0.7	30	011110b	1.02	46	101110	1.34	62	111110	1.66
15	001111b	0.72	31	011111b	1.04	47	101111	1.36	63	111111	1.68

'-': Don't care.

Restriction	-

Register availability	<table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table>
Default	<table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>011101b</td></tr><tr><td>S/W Reset</td><td>011101b</td></tr><tr><td>H/W Reset</td><td>011101b</td></tr></table>
Flow Chart	

9.3.12 VCMF Control (C6h)

C6H	VCM CTL (Vcom Control)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
	0	↑	1	-	1	1	0	0	0	1	1	0	(C6h)
1st parameter	1	↑	1	-	VMFSEL	-	-	VMF_REG [4:0]					
Description	VMFSEL: '0' from NV memory; '1': from VMF_REG[4:0] setting. VMF_REG[4:0]: add an offset to VCMP(for optimum display quality), One step is 20mV.												

VMF_REG [4:0]	Description
01111	VCMP + 15 Step
01110	VCMP + 14 Step
:	:
00010	VCMP + 2 Step
00001	VCMP + 1 Step
00000	VCMP
11111	VCMP – 1 Step
11110	VCMP – 2 Step
:	:
10010	VCMP – 14 Step
10001	VCMP – 15 Step
10000	VCMP – 16 Step

'-': Don't care.

Restriction	-
Register availability	

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value	
	VMFSEL	VCMP[4:0]
Power On Sequence	0b	00000b
S/W Reset	0b	00000b
H/W Reset	0b	00000b

Flow Chart

9.3.13 NVM Address/Data Write(D0h)

D0H	NVMADW (NVM Address/Data Write)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
	0	↑	1	-	1	1	0	1	0	0	0	0	(D0h)
1st parameter	1	↑	1	-	-	-	-	PROG_ADDR [4:0]					
2nd parameter	1	↑	1	-	PROG_DATA [7:0]								

Description

- This command is used to program the NVM data.
- PROG_ADDR [4:0]: The select bits of ID1, ID2, ID3, VMF[6:0] and MDDI V1.0 programming.

PROG_ADDR [4:0]	Description
00000	ID1 programming
00001	ID2 programming
00010	ID3 programming
00011	VMF programming
00100	MDDI V1.0 programming.
Others	Not allowed

- PROG_DATA [7:0]: PROG_DATA is set by user for NVM data.

'-': Don't care.

Restriction	-
Register availability	

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value	
	PROG_DATA [7:0]	PROG_ADDR [4:0]
Power On Sequence	--	00000b
S/W Reset	--	00000b
H/W Reset	--	00000b

Flow Chart

9.3.14 NVM Byte Program(D1h)

D1H	NVMBPROG (NVM Byte Program)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
	0	↑	1	-	1	1	0	1	0	0	0	1	(D1h)
1st parameter	1	↑	1	-	PROGCODE[23:16]								
2nd parameter	1	↑	1		PROGCODE[15:8]								
3rd parameter	1	↑	1	-	PROGCODE[7:0]								
Description	PROGCODE[23:0:] NVM program enable code. C3AA3Ch to start an auto byte program operation. '-': Don't care.												
Restriction	-												
Register availability													

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default	Status / Default Value Power On Sequence: 0 S/W Reset: 0 H/W Reset: 0
Flow Chart	

9.3.15 NVM Status Read(D2h)

D2H	NVMSTRD (NVM Status Read)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
	0	↑	1	-	1	1	0	1	0	0	0	1	(D2h)
1st parameter	1	1	↑	-	-	-	-	-	-	-	-	-	
2nd parameter	1	1	↑	-	ID2CNT[3:0]				ID1CNT[3:0]				
3rd parameter	1	1	↑	-	VMFCNT[3:0]				ID3CNT[3:0]				
4th parameter	1	1	↑	-	BUSY	-	-	-	-	-	-	MDDI10	
5th parameter	1	1	↑	-	-	-	-	VMF[4:0]					

Description

ID1CNT[3:0], ID2CNT[3:0], ID3CNT[3:0], VMFCNT[3:0]: Programmed times status.

ID2CNT/ ID1CNT[3:0] VMFCNT/ ID3CNT[3:0]	Description
0000b	No Programmed
0001b	1 time
0011b	2 times
0111b	3 times
1111b	4 times

BUSY: This bit '1' means NVM is busy in programming. '0' is idle.

MDDI10: MDDI version identification. '0' → MDDI V1.2 , '1' →MDDI V1.0.

VMF[4:0]: This byte returns the VMF[4:0] value, which was programmed to NVM by command D0h.

'-': Don't care.

Restriction: -

Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes

	Partial Mode On, Idle Mode Off, Sleep Out	Yes
	Partial Mode On, Idle Mode On, Sleep Out	Yes
	Sleep In	Yes

Default

Status	Default Value	
	PROG_DATA [7:0]	PROG_ADDR [4:0]
Power On Sequence	--	00000b
S/W Reset	--	00000b
H/W Reset	--	00000b

Flow Chart

9.3.16 Read ID4 (D3h)

D3H	RDID4(Read ID4)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
	0	↑	1	-	1	0	1	1	0	1	1	1	(D3h)
1st parameter	1	1	↑	-	X	X	X	X	X	X	X	X	
2nd parameter	1	1	↑	-	0	0	0	0	0	0	0	0	
3rd parameter	1	1	↑	-	ID4_15	ID4_14	ID4_13	ID4_12	ID4_11	ID4_10	ID4_9	ID4_8	
4th parameter	1	1	↑	-	ID4_7	ID4_6	ID4_5	ID4_4	ID4_3	ID4_2	ID4_1	ID4_0	

Description

Read IC device code.

The 1st parameter is dummy read period.

The 2nd parameter means the IC version.

The 3rd and 4th parameter mean the IC model name.

'-': Don't care.

Restriction

-

Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default	<table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>--</td></tr><tr><td>S/W Reset</td><td>--</td></tr><tr><td>H/W Reset</td><td>--</td></tr></table>
Flow Chart	

9.3.17 Positive Gamma Control (E0h)

E0H	PGC (Positive Gamma Control)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
	0	↑	1	-	1	1	1	0	0	0	0	0	(E0h)
1st parameter	1	↑	1	-	X	X	VRF0P[5:0]						
2nd parameter	1	↑	1	-	X	X	VOS0P[5:0]						
3rd parameter	1	↑	1	-	X	X	PK0P[5:0]						
4th parameter	1	↑	1	-	X	X	PK1P[5:0]						
5th parameter	1	↑	1	-	X	X	PK2P[5:0]						
6th parameter	1	↑	1	-	X	X	PK3P[5:0]						
7th parameter	1	↑	1	-	X	X	PK4P[5:0]						
8th parameter	1	↑	1	-	X	X	PK5P[5:0]						
9th parameter	1	↑	1	-	X	X	PK6P[5:0]						
10th parameter	1	↑	1	-	X	X	PK7P[5:0]						
11th parameter	1	↑	1	-	X	X	PK8P[5:0]						
12th parameter	1	↑	1	-	X	X	PK9P[5:0]						
13th parameter	1	↑	1	-	X	X	SELV0P[5:0]						
14th parameter	1	↑	1	-	X	X	SELV1P[5:0]						
15th parameter	1	↑	1	-	X	X	SELV62P[5:0]						
16th parameter	1	↑	1	-	X	X	SELV63P[5:0]						
Description	Set the gray scale voltage to adjust the Positive gamma of the TFT panel. '-': Don't care.												
Restriction	-												
Register availability	<table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr></table>												

		Normal Mode On, Idle Mode On, Sleep Out	Yes
		Partial Mode On, Idle Mode Off, Sleep Out	Yes
		Partial Mode On, Idle Mode On, Sleep Out	Yes
		Sleep In	Yes

Default	Status	Default Value
	Power On Sequence	--
	S/W Reset	--
	H/W Reset	--

Flow Chart	

9.3.18 Negative Gamma Control (E1h)

E1H	NGC (Negative Gamma Control)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
	0	↑	1	-	1	1	1	0	0	0	0	1	(E1h)
1st parameter	1	↑	1	-	X	X	VRF0N[5:0]						
2nd parameter	1	↑	1	-	X	X	VOS0N[5:0]						
3rd parameter	1	↑	1	-	X	X	PK0N[5:0]						
4th parameter	1	↑	1	-	X	X	PK1N[5:0]						
5th parameter	1	↑	1	-	X	X	PK2N[5:0]						
6th parameter	1	↑	1	-	X	X	PK3N[5:0]						
7th parameter	1	↑	1	-	X	X	PK4N[5:0]						
8th parameter	1	↑	1	-	X	X	PK5N[5:0]						
9th parameter	1	↑	1	-	X	X	PK6N[5:0]						
10th parameter	1	↑	1	-	X	X	PK7N[5:0]						
11th parameter	1	↑	1	-	X	X	PK8N[5:0]						
12th parameter	1	↑	1	-	X	X	PK9N[5:0]						
13th parameter	1	↑	1	-	X	X	SELV0N[5:0]						
14th parameter	1	↑	1	-	X	X	SELV1N[5:0]						
15th parameter	1	↑	1	-	X	X	SELV62N[5:0]						
16th parameter	1	↑	1	-	X	X	SELV63N[5:0]						
Description	Set the gray scale voltage to adjust the Negative gamma of the TFT panel. . '-': Don't care.												

Restriction	-
Register availability	

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	--
S/W Reset	--
H/W Reset	--

Flow Chart

9.3.19 Digital Gamma Control 1 (E2h)

E2H	DGC1 (Digital Gamma Control 1)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
	0	↑	1	-	1	1	1	0	0	0	1	0	(E2h)
1st parameter	1	↑	1	-	RDG00[7:0]								
2nd parameter	1	↑	1	-	RDG 01[7:0]								
	1	↑	1	-	:								
62nd parameter	1	↑	1	-	RDG 62[7:0]								
63rd parameter	1	↑	1	-	RDG 63[7:0]								
Description	RDG xx [7:0]: Digital Gamma adjustment register for red gamma curve. '-': Don't care.												
Restriction	-												
Register availability													

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes

		Sleep In	Yes
Default		Status / Default Value (RDGXX [7:0]): Power On Sequence: --; S/W Reset: --; H/W Reset: --	
Flow Chart			

9.3.20 Digital Gamma Control 2 (E3h)

E3H	DGC2 (Digital Gamma Control 2)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
	0	↑	1	-	1	1	1	0	0	0	1	0	(E3h)
1st parameter	1	↑	1	-	BDG00[7:0]								
2nd parameter	1	↑	1	-	BDG 01[7:0]								
	1	↑	1	-	:								
62nd parameter	1	↑	1	-	BDG 62[7:0]								
63rd parameter	1	↑	1	-	BDG 63[7:0]								
Description	BDGxx [7:0]: Digital Gamma adjustment register for blue gamma curve. '-': Don't care.												
Restriction	-												
Register availability	Status / Availability: Normal Mode On, Idle Mode Off, Sleep Out: Yes; Normal Mode On, Idle Mode On, Sleep Out: Yes; Partial Mode On, Idle Mode Off, Sleep Out: Yes; Partial Mode On, Idle Mode On, Sleep Out: Yes; Sleep In: Yes												
Default	Status / Default Value (BDGXX [7:0]): Power On Sequence: --; S/W Reset: --; H/W Reset: --												

Flow Chart	

9.3.21 SPI Read Control (FBh)

FBh	SPIRC (SPI Read Control)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
	0	↑	1	-	1	1	1	1	1	0	1	1	(FBh)
1st parameter	1	↑	1					SPI_REN	SPI_CNT[3:0]				
Description	SPI_REN: SPI read enable SPI_CNT [3:0]: SPI read parameter number '-': Don't care.												
Restriction	-												
Register availability													

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value	
	SPI_REN	SPI_CNT [3:0]
Power On Sequence	0b	0000b
S/W Reset	0b	0000b
H/W Reset	0b	0000b

Flow Chart	

10 Function Description

10.1.. Display Data RAM

10.1.1 Configuration

The display module has an integrated 320x480x18-bit graphic type static RAM. This 2,764,800-bit memory allows storing on-chip a 320xRGBx480 image with an 18-bpp resolution (262K-color). There will be no abnormal visible effect on the display when there is a simultaneous Panel Read and Interface Read or Write to the same location of the Frame Memory.

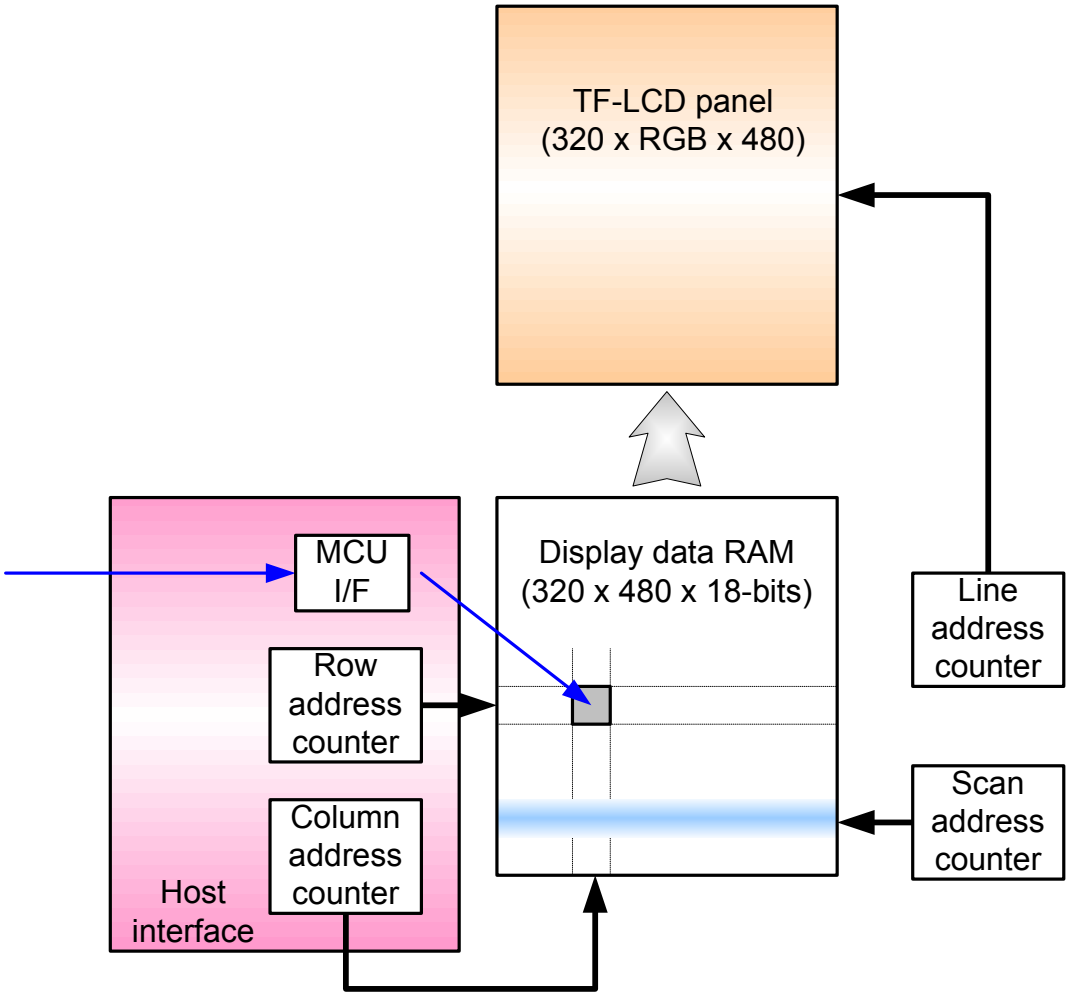
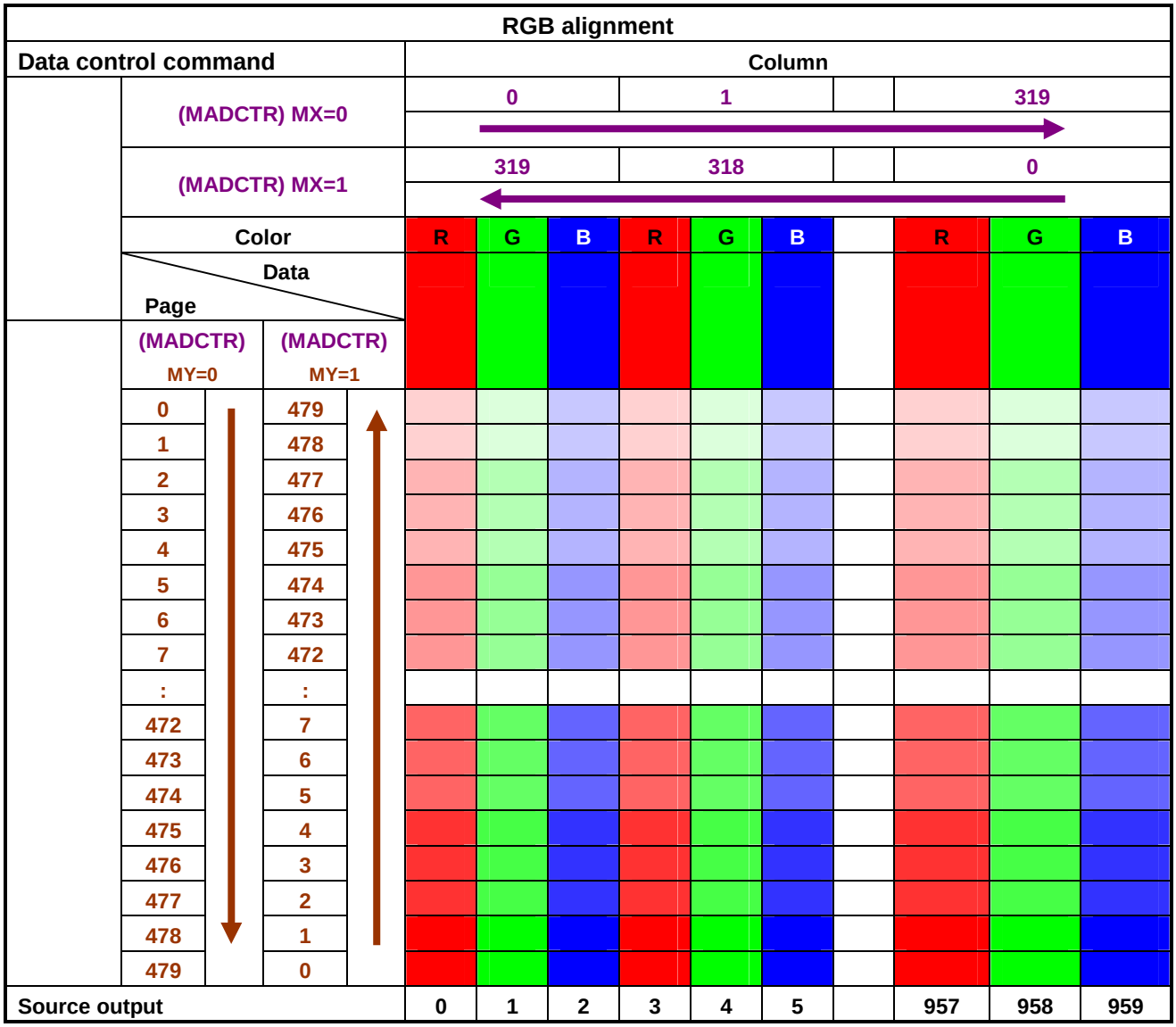


Figure Display data RAM organization

10.1.2 Memory to display address mapping



10.2.. Address Control

The address counter sets the addresses of the display data RAM for writing and reading. Data is written pixel-wise into the RAM matrix of DRIVER. The data for one pixel or two pixels is collected (RGB 6-6-6-bit), according to the data formats. As soon as this pixel-data information is complete the “Write access” is activated on the RAM. The locations of RAM are addressed by the address pointers. The address ranges are X=0 to X=319 and Y=0 to Y=479. Addresses outside these ranges are not allowed. Before writing to the RAM, a window must be defined that will be written. The window is programmable via the command registers XS, YS designating the start address and XE, YE designating the end address.

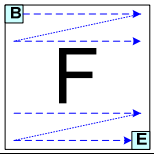
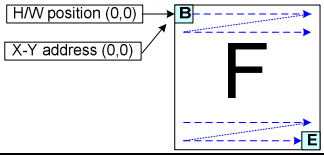
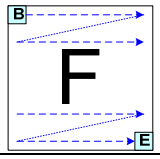
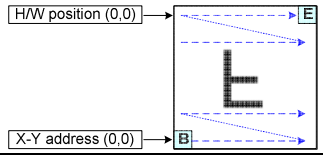
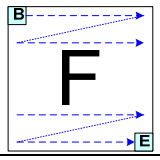
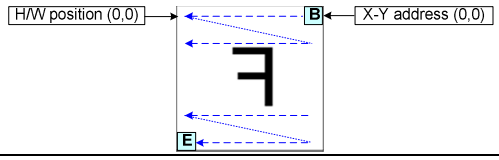
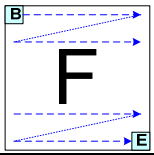
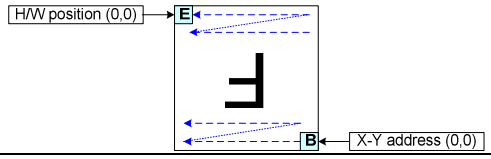
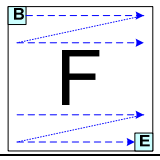
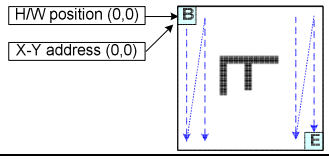
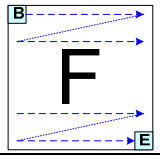
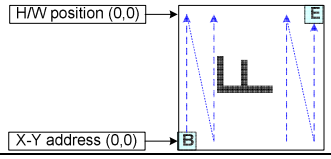
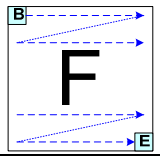
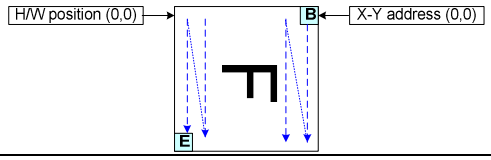
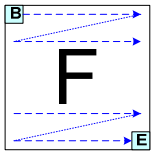
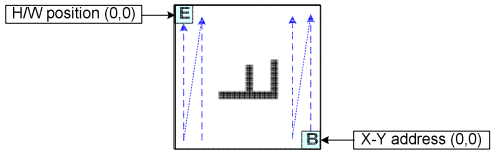
For example the whole display contents will be written, the window is defined by the following values: XS=0 (0h) YS=0 (0h) and XE=319, YE=479.

In vertical addressing mode (MV=1), the Y-address increments after each byte, after the last Y-address (Y=YE), Y wraps around to YS and X increments to address the next column. In horizontal addressing mode (V=0), the X-address increments after each byte, after the last X-address (X=XE), X wraps around to XS and Y increments to address the next row. After the every last address (X=XE and Y=YE) the address pointers wrap around to address (X=XS and Y=YS).

For flexibility in handling a wide variety of display architectures, the commands “CASET, RASET and MADCTL”, define flags MX and MY, which allows mirroring of the X-address and Y-address. All combinations of flags are allowed.

For each image condition, the controls for the column and row counters apply as below

Condition	Column	Page
Command 2C/2E is accepted	Return to “Start Column”	Return to “Start Page”
Read/Write RAM action	Increment by 1	No change
Column value is large than “End Column”	Return to “Start Column”	Increment by 1
Page value is large than “End Page”	Return to “Start Column”	Return to “Start Page”

Display Data Direction	MADCTR Parameter			Image in the Host (MPU)	Image in the Driver (DDRAM)
	MV	MX	MY		
Normal	0	0	0		
Y-Mirror	0	0	1		
X-Mirror	0	1	0		
X-Mirror Y-Mirror	0	1	1		
X-Y Exchange	1	0	0		
X-Y Exchange Y-Mirror	1	0	1		
X-Y Exchange X-Mirror	1	1	0		
X-Y Exchange X-Mirror Y-Mirror	1	1	1		

10.3.. Data Transfer Break and Recovery

If there is a break in data transmission by RESX pulse, while transferring a command or frame memory data or multiple parameter command data, before Bit D0 of the byte has been completed, then driver will reject the previous bits and have reset the interface such that it will be ready to receive command data again when the chip select line (CSX) is next activated after RESX have been HIGH state.

If there is a break in data transmission by CSX pulse, while transferring a command or frame memory data or multiple parameter command data, before Bit D0 of the byte has been completed, then driver will reject the previous bits and have reset the interface such that it will be ready to receive the same byte re-transmitted when the chip select line (CSX) is next activated.

If 1, 2 or more parameter commands are being sent and a break occurs while sending any parameter before the last one and if the host then sends a new command rather than re-transmitting the parameter that was interrupted, then the parameters that were successfully sent are stored and the parameter where the break occurred is rejected. The interface is ready to receive next byte as shown below.

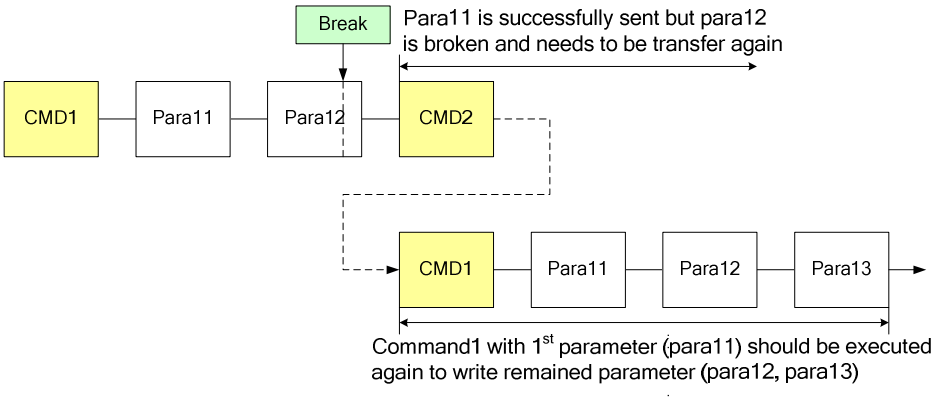


Figure Write interrupts recovery (serial interface)

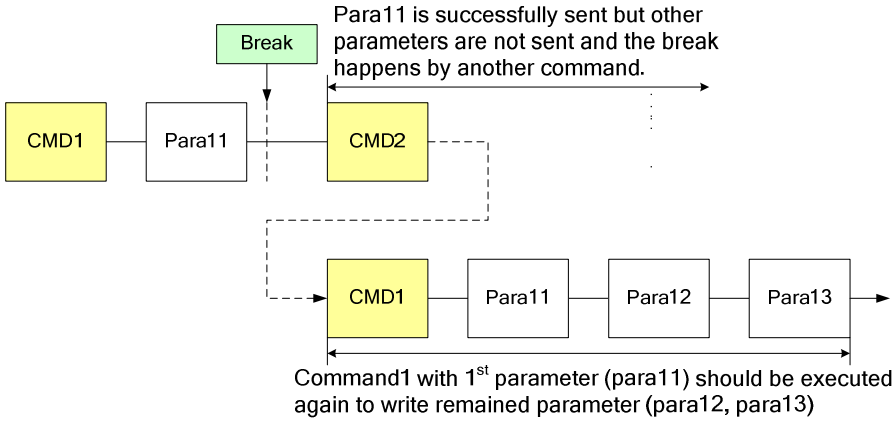


Figure Write interrupts recovery (both serial and parallel Interface)

10.4.. Data Transfer Pause

It will be possible when transferring a command, frame memory data or multiple parameter data to invoke a pause in the data transmission. If the chip select line is released after a whole byte of a frame memory data or multiple parameter data has been completed, then driver will wait and continue the frame memory data or parameter data transmission from the point where it was paused. If the chip select Line is released after a whole byte of a command has been completed, then the display module will receive either the command‘s parameters (if appropriate) or a new command when the chip select line is next enabled as shown below.

This applies to the following 4 conditions:

1) Command-Pause-Command

2) Command-Pause-Parameter

3) Parameter-Pause-Command

4) Parameter-Pause-Parameter

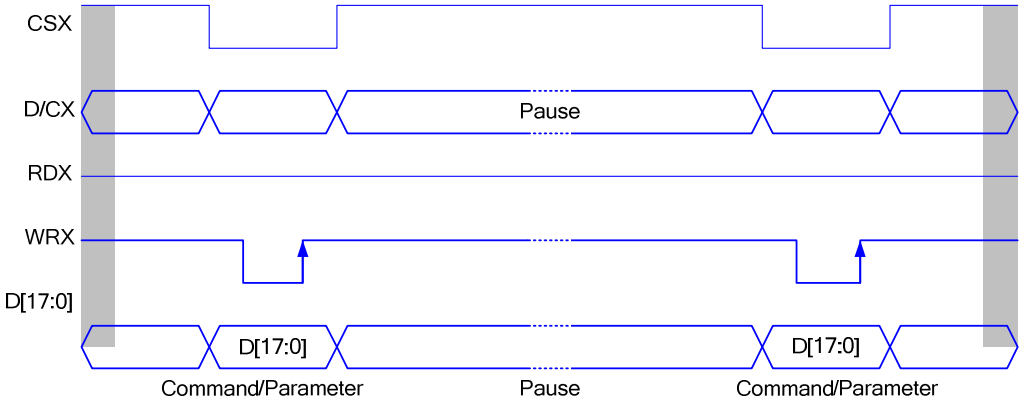


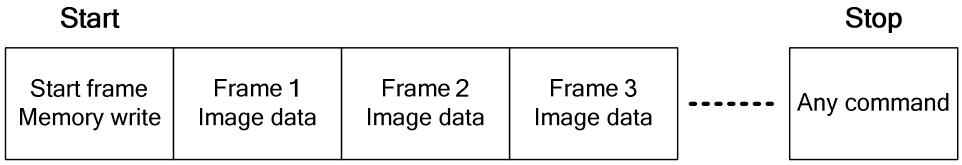
Figure Parallel bus pause protocol (paused by CSX)

10.5.. Data Transfer Mode

The module has three kinds color modes for transferring data to the display RAM. These are 12-bit color per pixel, 16-bit color per pixel and 18-bit color per pixel. The data format is described for each interface. Data can be downloaded to the frame memory by 2 methods.

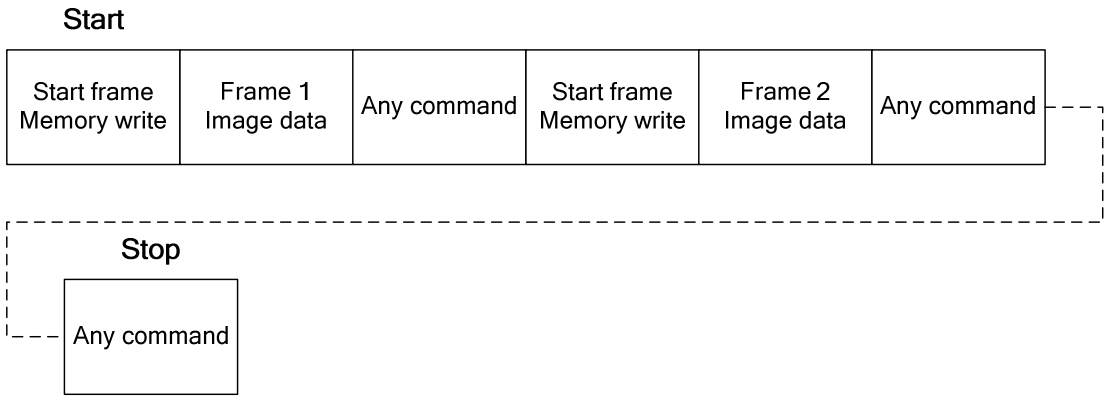
10.5.1 Method 1

The image data is sent to the frame memory in successive frame writes, each time the frame memory is filled, the frame memory pointer is reset to the start point and the next frame is written.



10.5.2 Method 2

The image data is sent and at the end of each frame memory download, a command is sent to stop frame memory write. Then start memory write command is sent, and a new frame is downloaded.



Note 1: These apply to all data transfer Color modes on both serial and parallel interfaces.

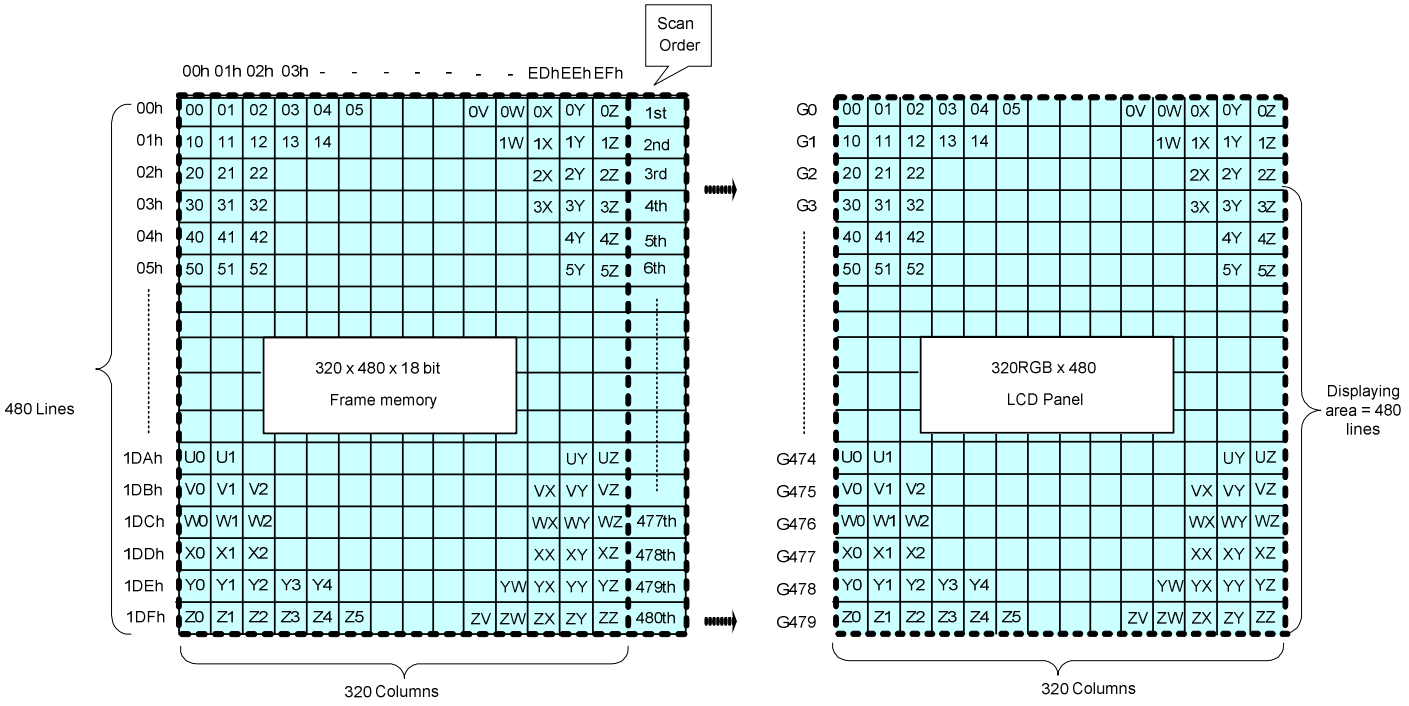
Note 2: The frame memory can contain both odd and even number of pixels for both methods. Only complete pixel data will be stored in the frame memory.

10.6.. Normal Display On or Partial Mode On, Vertical Scroll Off

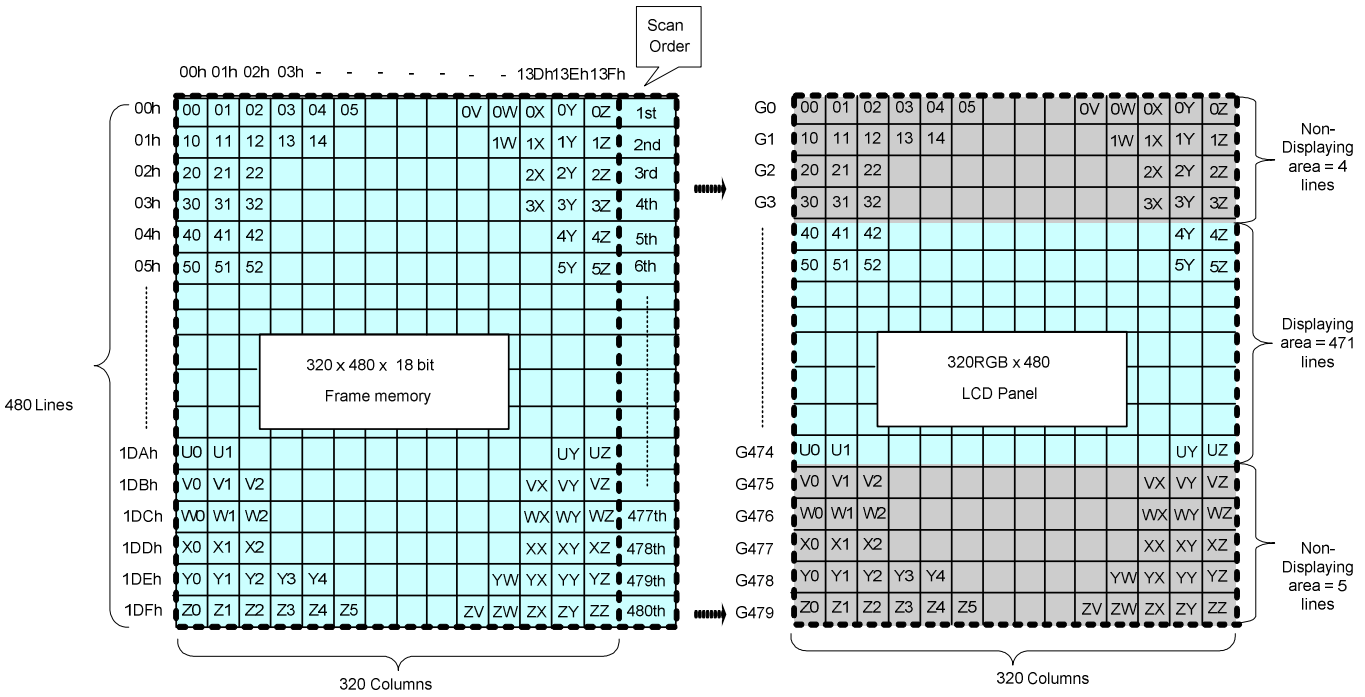
In this mode, contents of the frame memory within an area where column address is 00h to 83h and row address is 00h to 83h is displayed.

To display a dot on leftmost top corner, store the dot data at (column address, row address) = (0,0).

Example1: Normal Display On



Example2) Partial Display On: PSL[15:0] = 0004h, PEL[15:0] = 01DBh, MADCTR (ML)=0



10.7.. Vertical Scroll Mode

10.7.1 Rolling scroll

There is just one types of vertical scrolling, which are determined by the commands “Vertical Scrolling Definition” (33h) and “Vertical Scrolling Start Address” (37h).

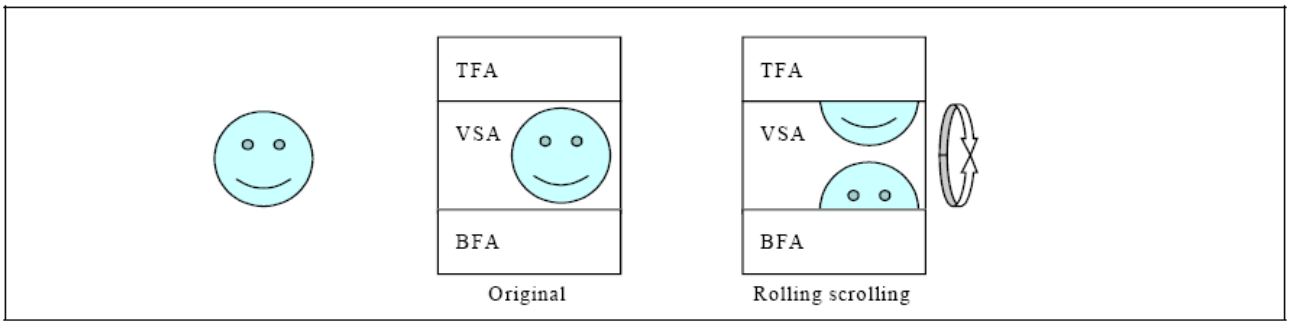
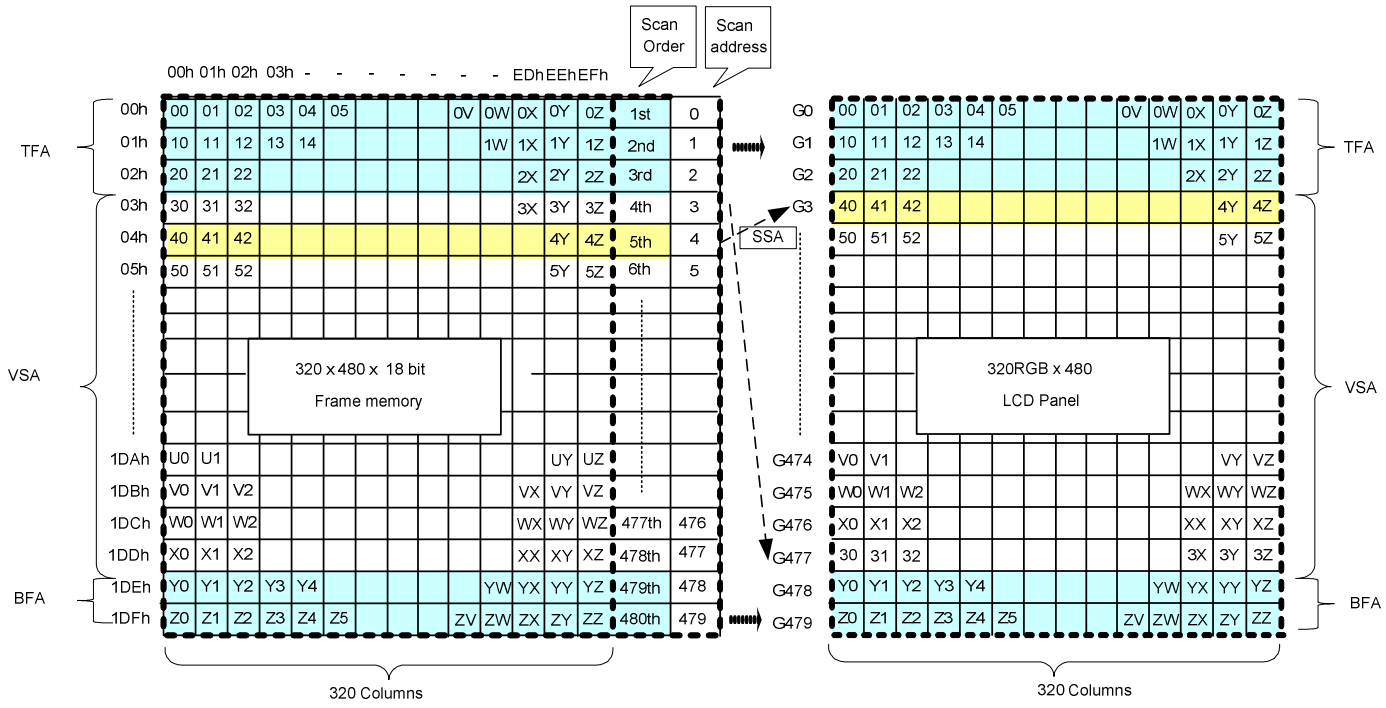


Figure Rolling Scroll Definition

When Vertical Scrolling Definition Parameters (TFA+VSA+BFA) =480. In this case, ‘rolling’ scrolling is applied as shown below. All the memory contents will be used.

Example: Panel size=320 x 480, TFA =3, VSA=475, BFA=2, SSA=4, MADCTR ML=0: Rolling Scroll



10.7.2 Vertical Scroll Example

There are 2 types of vertical scrolling, which are determined by the commands “Vertical Scrolling Definition” (33h) and “Vertical Scrolling Start Address” (37h).

Case 1: TFA + VSA + BFA≠Panel total scan lines. In this case, scrolling is applied as shown below.

N/A. Do not set TFA + VSA + BFA≠Panel total scan lines. In that case, unexpected picture will be shown.

Case 2: TFA + VSA + BFA= Panel total scan lines

Example1) When MADCTR parameter ML=”0”, TFA=0, VSA=480, BFA=0 and VSCSAD=40.

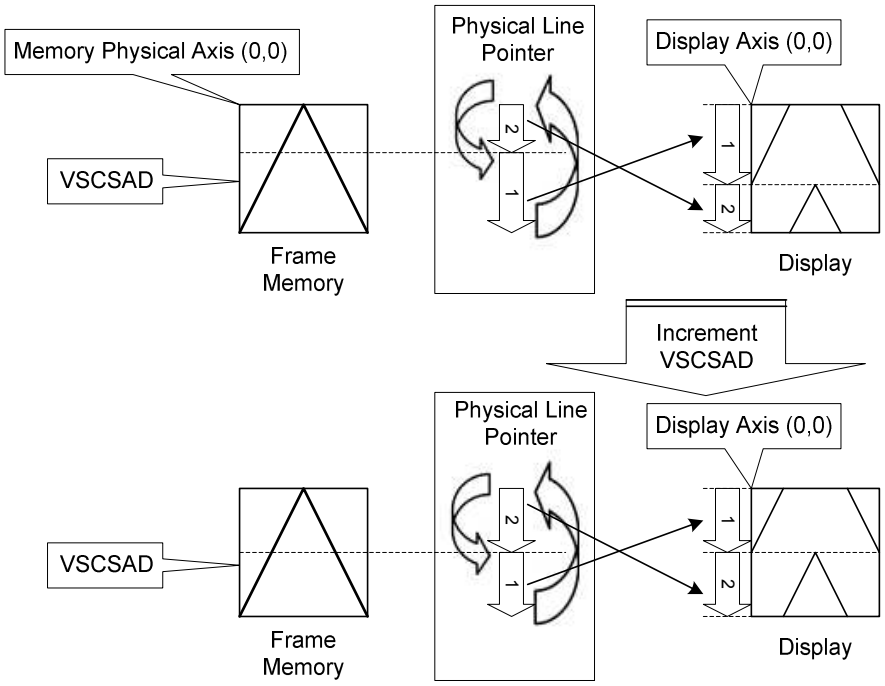


Figure Display of Vertical Scroll Example 1

Example2) When MADCTR parameter ML=”1”, TFA=60, VSA=420, BFA=0 and VSCSAD=160.

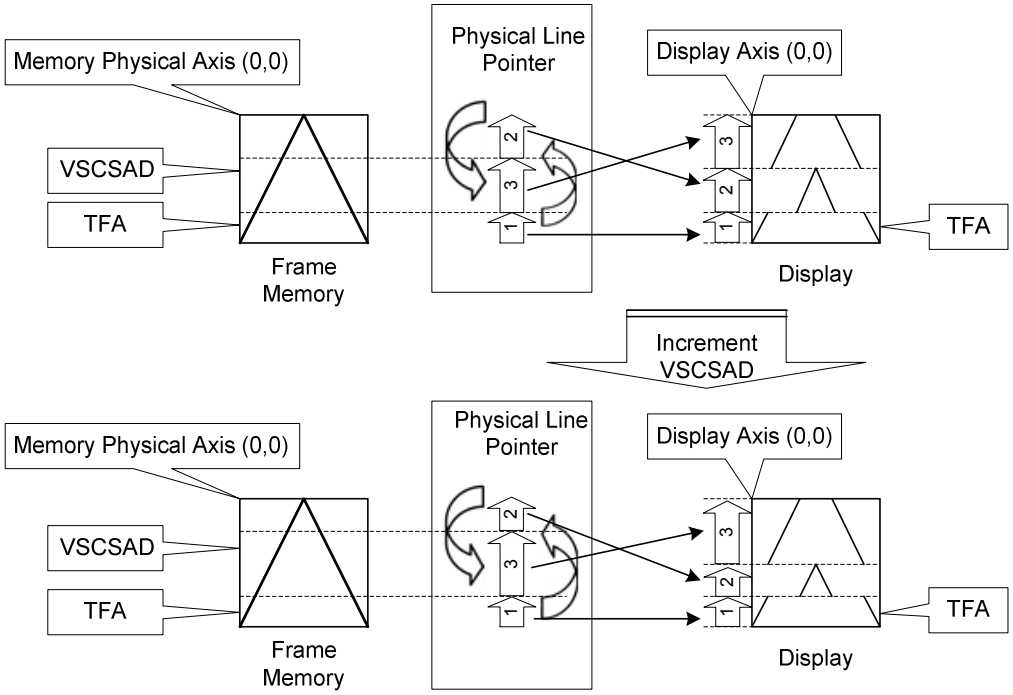


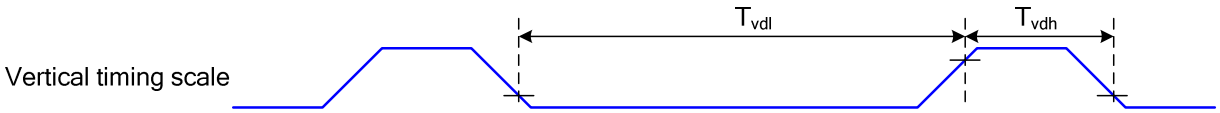
Figure Display of Vertical Scroll Example 2

10.8.. Tearing Effect

The Tearing Effect output line supplies to the MPU a Panel synchronization signal. This signal can be enabled or disabled by the Tearing Effect Line Off & On commands. The mode of the Tearing Effect signal is defined by the parameter of the Tearing Effect Line On command. The signal can be used by the MPU to synchronize Frame Memory Writing when displaying video images.

10.8.1 Tearing effect line modes

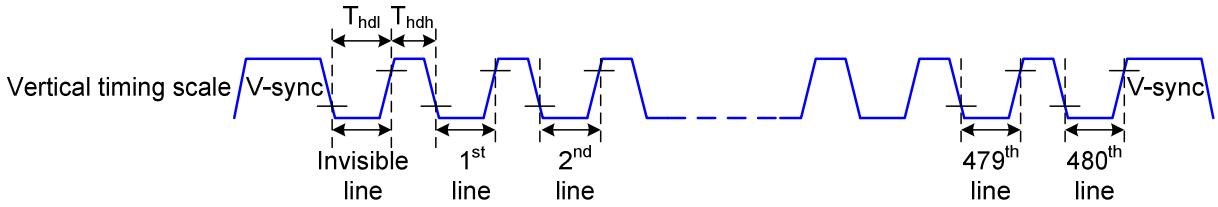
Mode 1, the Tearing Effect Output signal consists of V-Blanking Information only:



tvdh= The LCD display is not updated from the Frame Memory

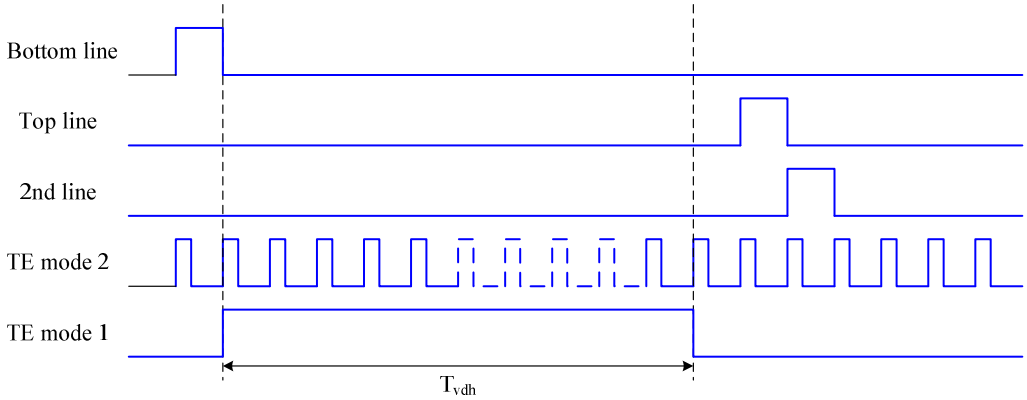
tvdl= The LCD display is updated from the Frame Memory (except Invisible Line – see above)

Mode 2, the Tearing Effect Output signal consists of V-Blanking and H-Blanking Information, there is one V-sync and 480 H-sync pulses per field.



thdh= The LCD display is not updated from the Frame Memory

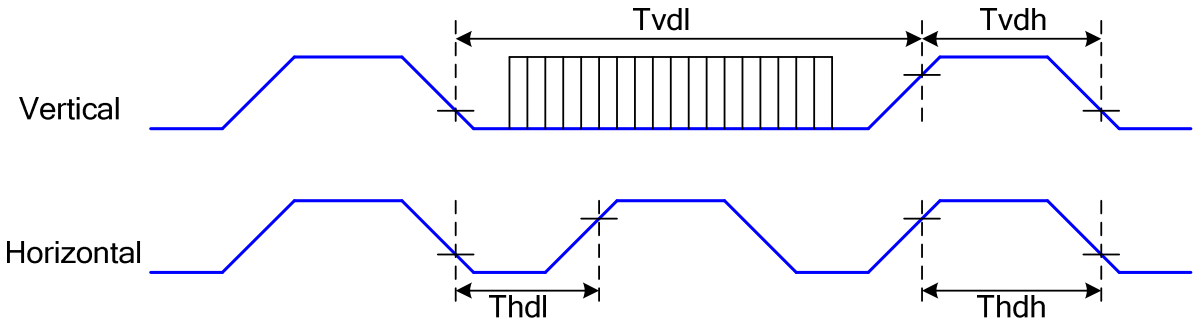
thdl= The LCD display is updated from the Frame Memory (except Invisible Line – see above)



Note: During Sleep In Mode, the Tearing Output Pin is active Low.

10.8.2 Tearign effect line timings

The Tearing Effect signal is described below:

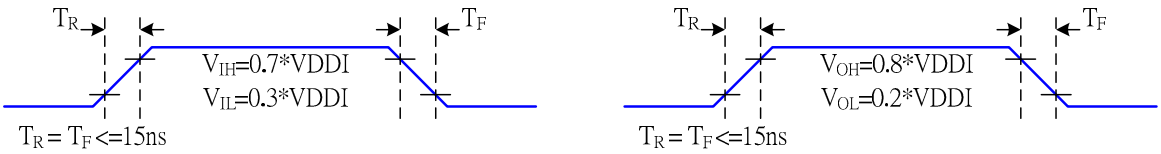


Symbol	Parameter	min	max	unit	description
tvdl	Vertical Timing Low Duration	13	-	ms	
tvdh	Vertical Timing High Duration	1000	-	µs	
thdl	Horizontal Timing Low Duration	33	-	µs	
thdh	Horizontal Timing Low Duration	25	500	µs	

Table AC characteristics of Tearing Effect Signal Idle Mode Off (Frame Rate = 60 Hz, Ta=25°C)

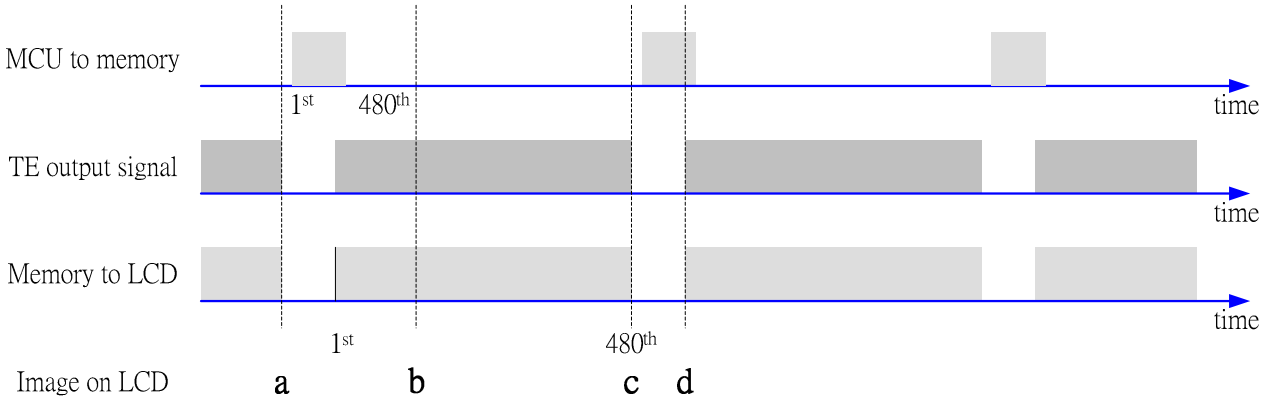
Note: The timings in Table 15 apply when MADCTL ML=0 and ML=1

The signal's rise and fall times (tf, tr) are stipulated to be equal to or less than 15ns.

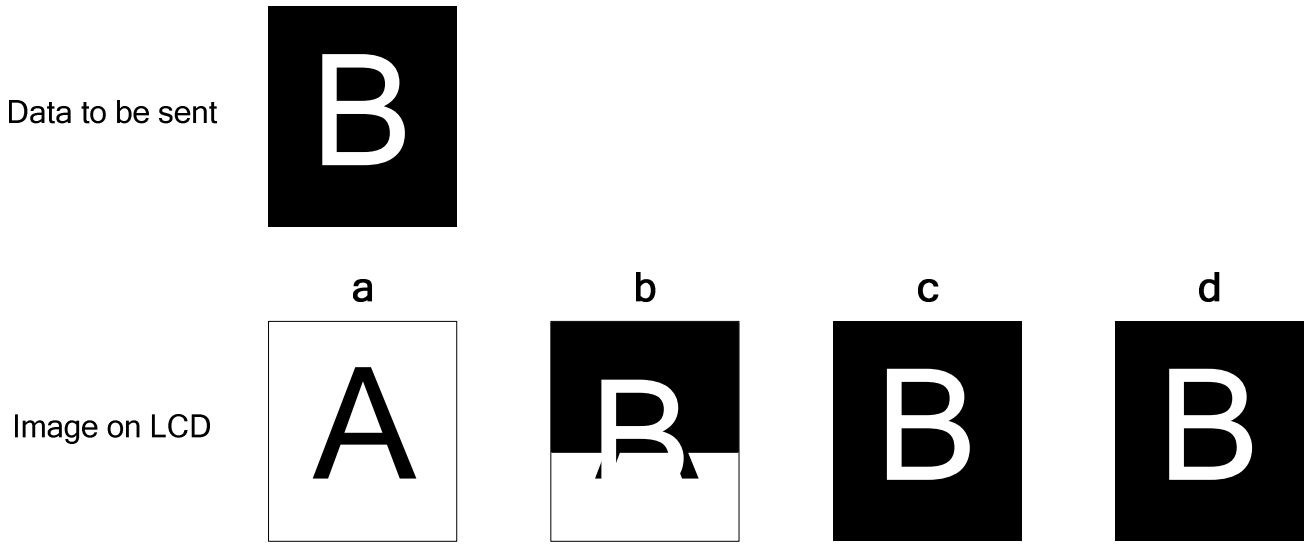


The Tearing Effect Output Line is fed back to the MPU and should be used as shown below to avoid Tearing Effect:

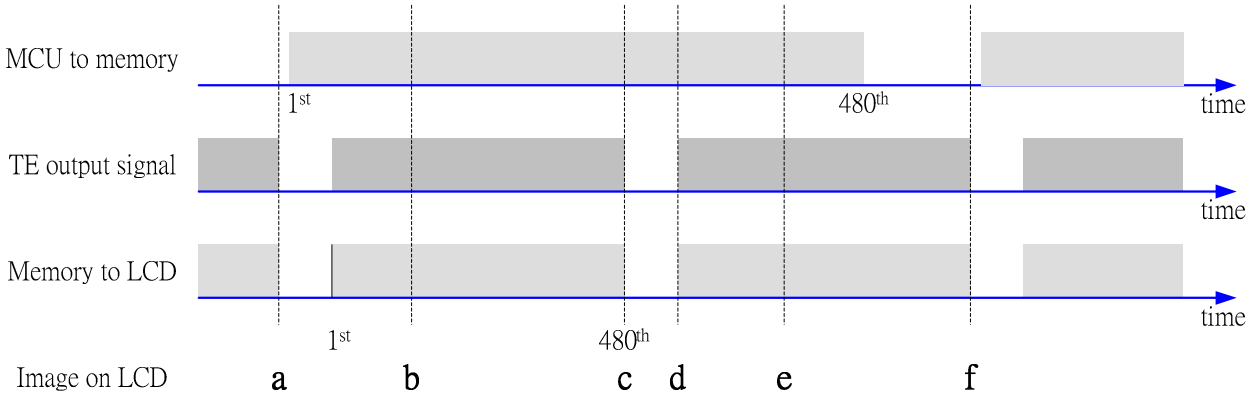
10.8.3 Example 1: MPU Write is faster than panel read



Data write to Frame Memory is now synchronized to the Panel Scan. It should be written during the vertical sync pulse of the Tearing Effect Output Line. This ensures that data is always written ahead of the panel scan and each Panel Frame refresh has a complete new image:



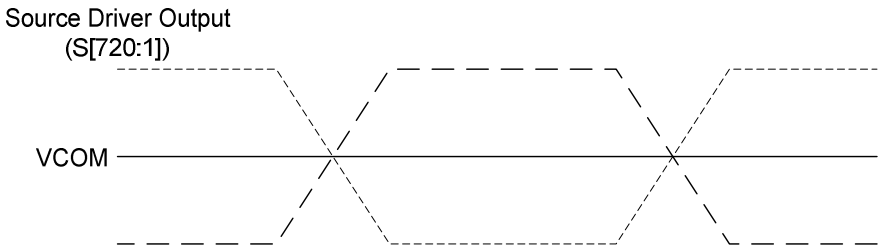
10.8.4 Example 2: MPU write is slower than panel read

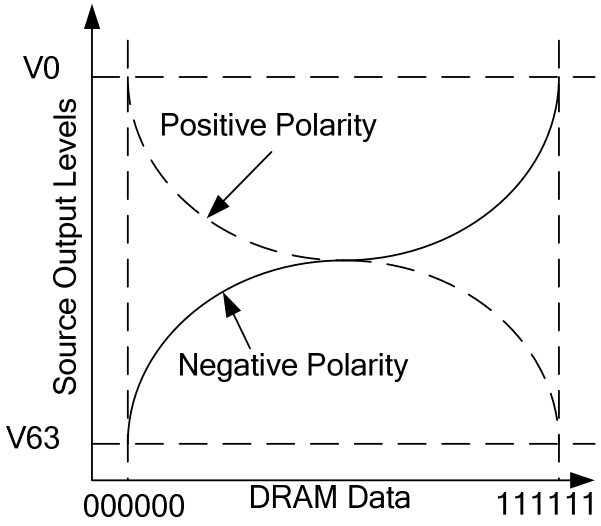


The MPU to Frame Memory write begins just after Panel Read has commenced i.e. after one horizontal sync pulse of the Tearing Effect Output Line. This allows time for the image to download behind the Panel Read pointer and finishing download during the subsequent Frame before the Read Pointer “catches” the MPU to Frame memory write position.

10.9.. Gamma Correction

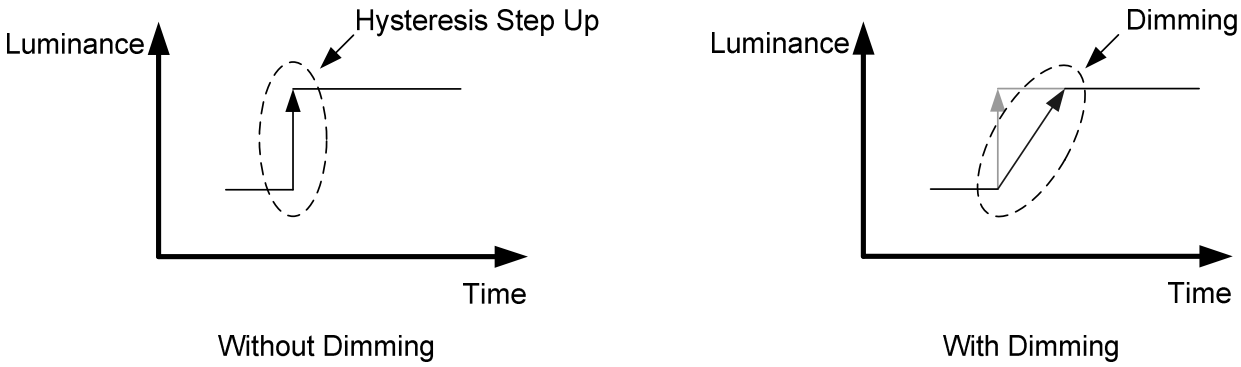
ST7796 incorporate the gamma correction function to display 262,244 colors for the LCD panel. The gamma correction is performed with 3 groups of registers, which are gradient adjustment, contrast adjustment and fine- adjustment registers for positive and negative polarities, and RGB can be adjusted individually.





10.10.. Display Dimming

A dimming function (how fast to change the brightness from old to new level and what are brightness levels during the change) is used when changing from one brightness level to another. This dimming function curve is the same in increment and decrement. The basic idea is described below.



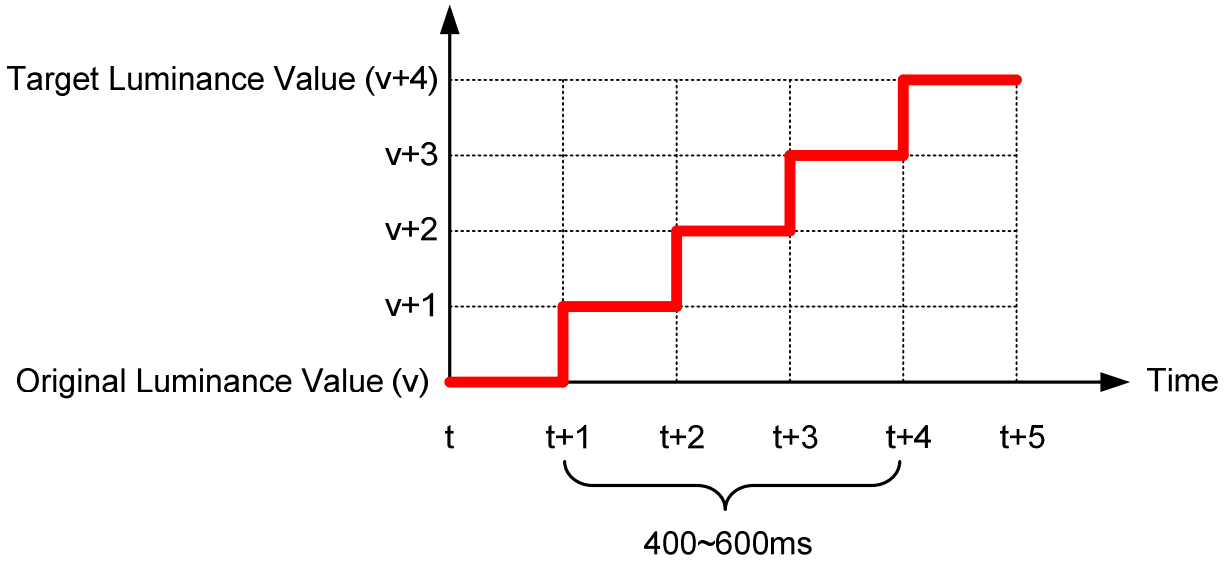
Dimming function can be enable and disable. See “Write CTRL Display (53h)” (bit DD) for more information.

10.10.1 Dimming Requirement

Dimming function in the display module should be implemented so that 400-600ms is used for the transition between the original brightness value and the target brightness value. The transferring time steps between these two brightness values are equal making the transition linear.

The dimming function is working similarly in both upward and downward directions.

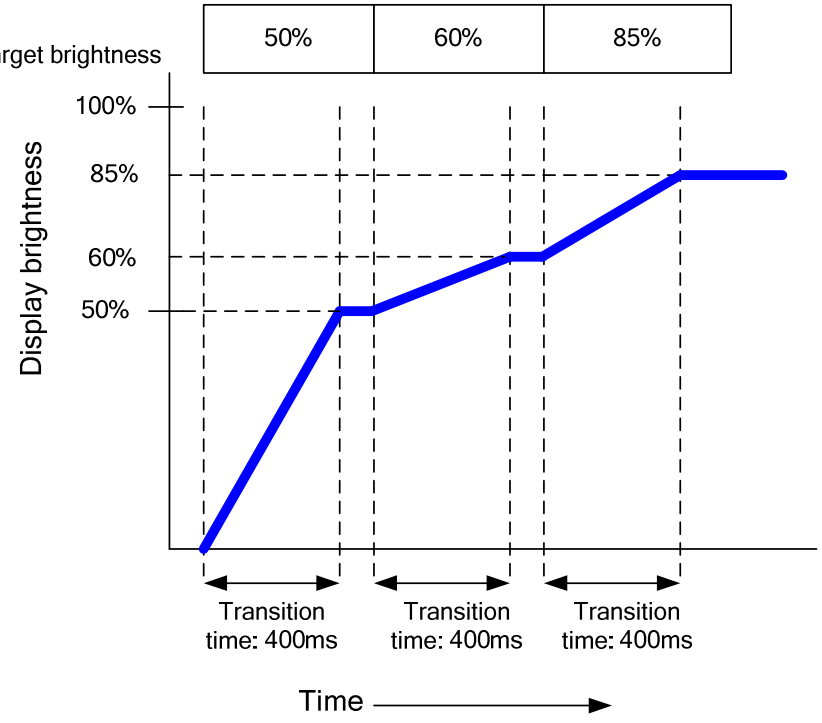
An upward example is illustrate below



10.10.2 Definition of brightness transition time

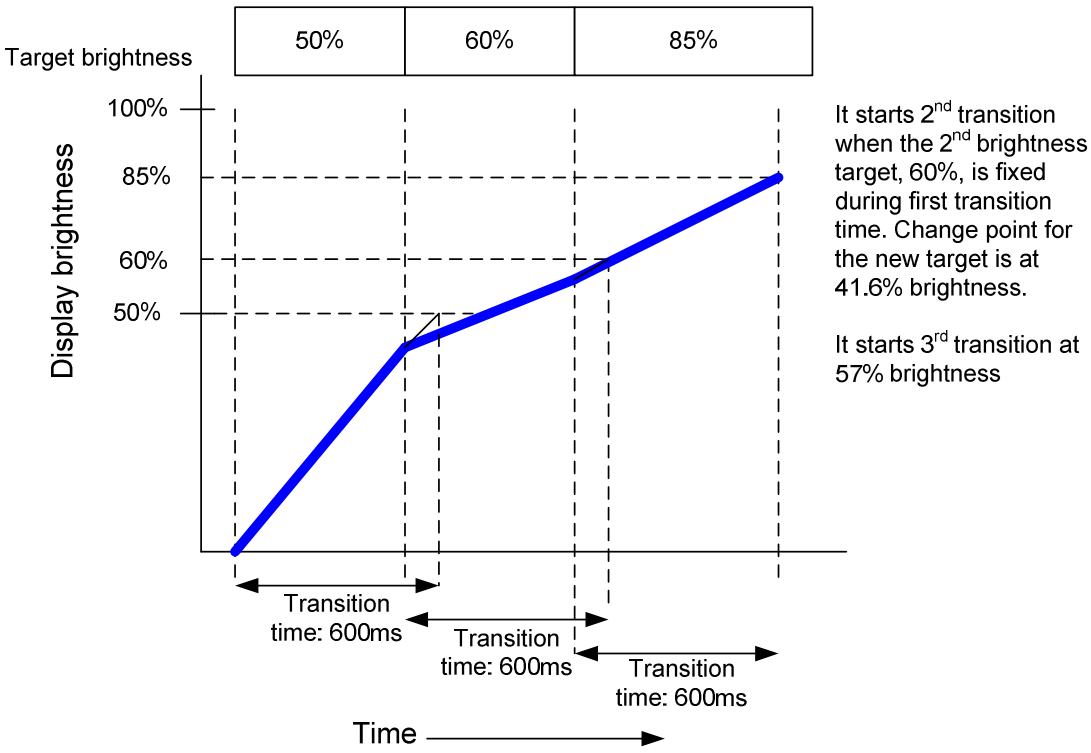
- Shorter transition time than 500ms.

There is some stable time between transitions. Below drawing is for transition time: 400ms.



- Longer transition time than 500ms

There is no any stable time between transitions. Below drawing is for transition time: 600ms.



It starts 2nd transition when the 2nd brightness target, 60%, is fixed during first transition time. Change point for the new target is at 41.6% brightness.

It starts 3rd transition at 57% brightness

10.11.. Content Adaptive Brightness Control (CABC)

10.11.1 Definition of CABC

A Content Adaptive Brightness Control function can be used to reduce the power consumption of the luminance source. Content adaptation means that content gray level scale can be increased while simultaneously lowering brightness of the backlight to achieve same perceived brightness. The adjusted gray level scale and thus the power consumption reduction

Definition of Modes and target power reduction ratio:

- Off mode: Content Adaptive Brightness Control functionality is totally off.
- UI [User interface] image mode: Optimized for UI image. It is kept image quality as much as possible. Target power consumption reduction ratio: 10% or less.
- Still picture mode: Optimized for still picture. Some image quality degradation would be acceptable. Target power consumption reduction ratio: more than 30%.
- Moving image mode: Optimized for moving image. It is focused on the biggest power reduction with image quality degradation. Target power consumption reduction ratio: more than 30%.

Note 1: Updating partial area of the image data should be supported by CABC functionality.
Note 2: Processing power consumption of CABC should be minimized.

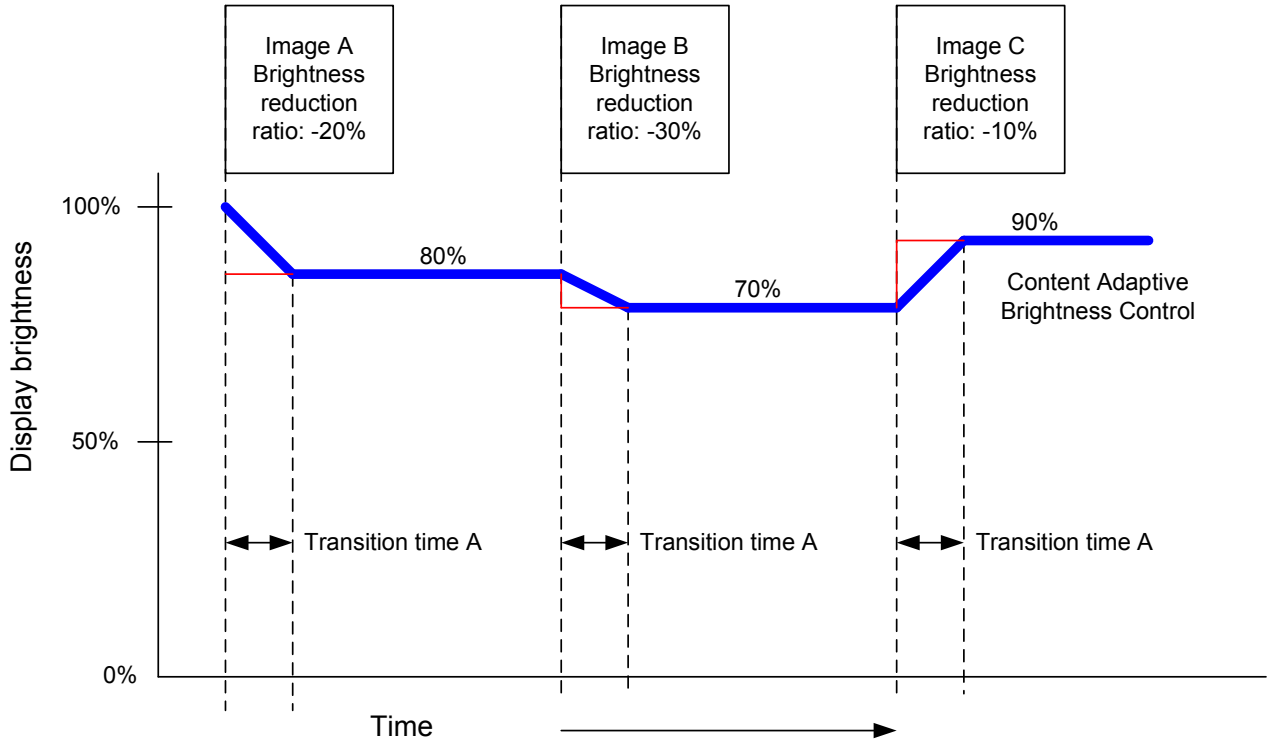
The transition time for dimming function is illustrated below.

- Content Adaptive Brightness Control

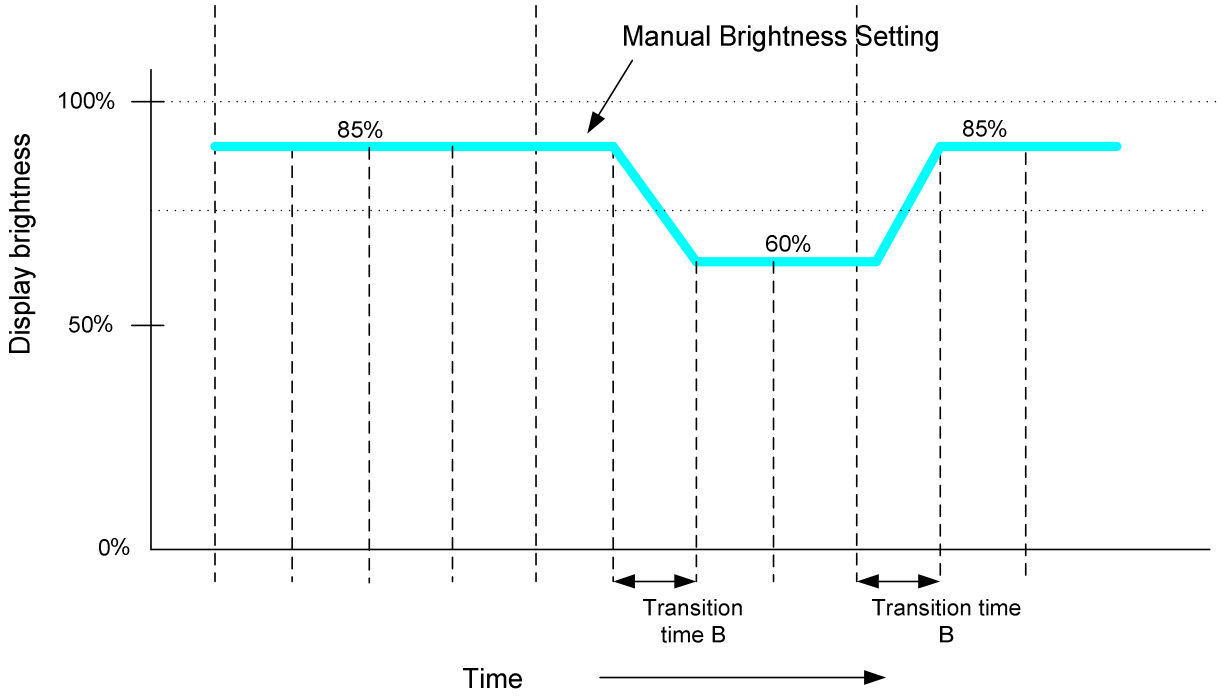
Display brightness is changed, according to the image contents. The following graph mentions the case of displaying three different images.

- Image A: -20% brightness reduction
- Image B: -30% brightness reduction
- Image C: -30% brightness reduction

Transition time from the previous image to the current displayed image is “transition time A”.



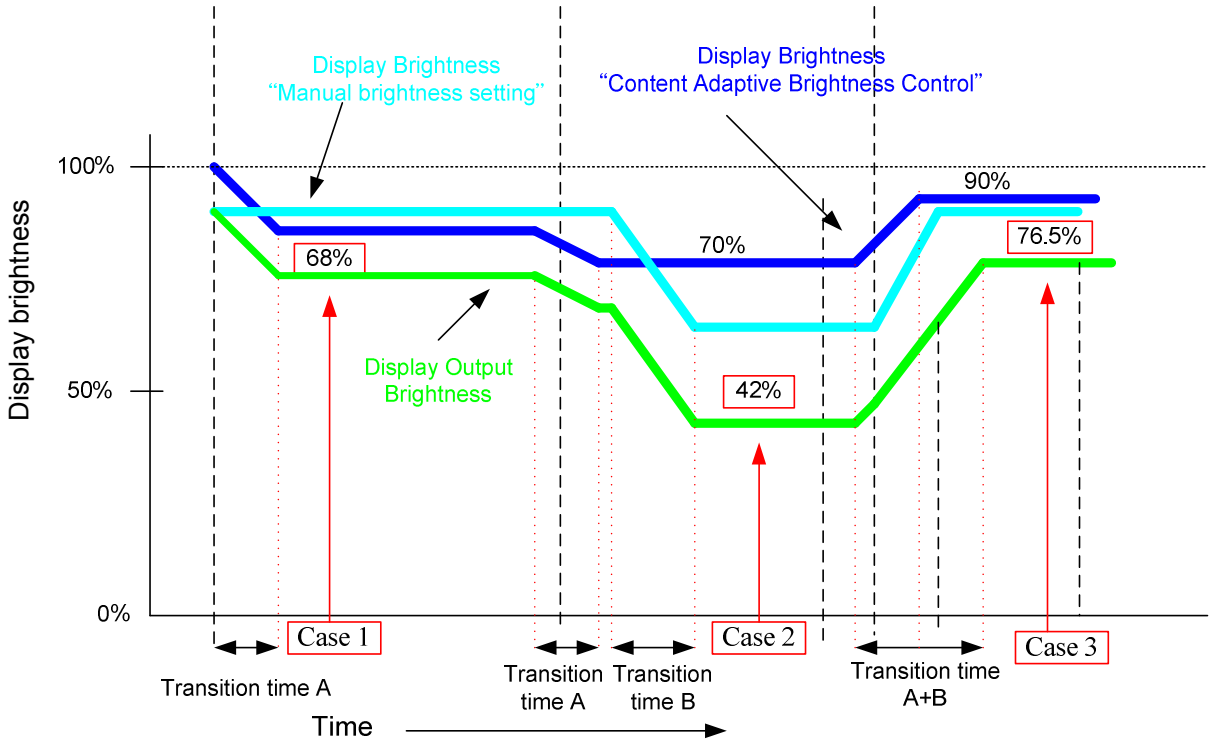
- Manual brightness setting and Dimming function



- Combine Display brightness

Green line in the following graph is for the output brightness of display. It is combined with both display brightness, which are defined in the above graphs.

Maximum transition time is transition time A+B.



Brightness level calculates with the following formula.

Display Output brightness = Manual Brightness setting * CABC brightness ratio

	Manual Brightness setting	Brightness ratio [CABC]	Display Output brightness
Case 1	85%	80%	68%
Case 2	60%	70%	42%
Case 3	85%	90%	76.5%

Transition time from the current brightness to target brightness is A+B in the worst case.

10.11.2 Minimum brightness setting of CABC function

CABC function is automatically reduced backlight brightness based on image contents. In the case of the combination with the LABC or manual brightness setting, display brightness is too dark. It must affect to image quality degradation. CABC minimum brightness setting is to avoid too much brightness reduction. When CABC is active, CABC can not reduce the display brightness to less than CABC minimum brightness setting. If CABC algorithm works without any abnormal visual effect, image processing function can operate even when the brightness can not be changed.

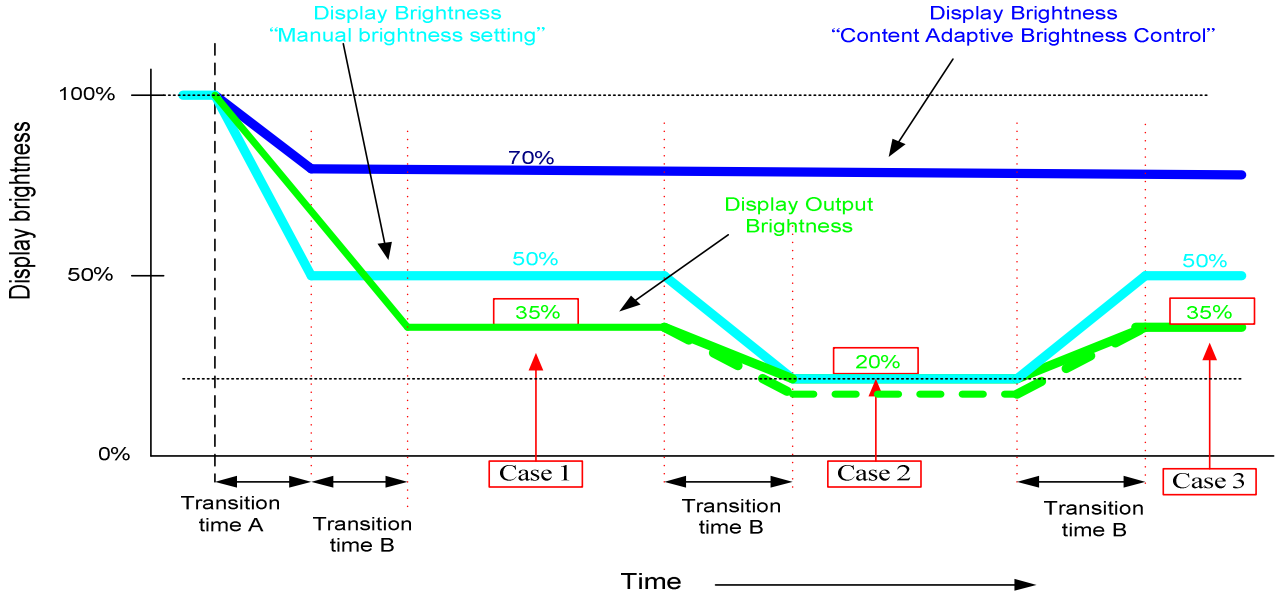
This function does not affect to the other function, manual brightness setting. Manual brightness can be set the display brightness to less than CABC minimum brightness. Smooth transition and dimming function can be worked as normal.

When display brightness is turned off (BCTRL=0 of “9.2.39 Write CTRL Display (53h)”), CABC minimum brightness setting is ignored. “9.2.44 Read CABC minimum brightness (5Fh)” always read the setting value of “9.2.43 Write CABC minimum brightness (5Eh)”.

	WRCABC (55h)	Function	RDCABCMB (5Fh)	Image
Sleep-in		NA	WRCABCMB (5Eh)	
CABC off	00b	Disable	WRCABCMB (5Eh)	Original
CABC on	01b/10b/11b	Enable	WRCABCMB (5Eh)	CABC modified

Brightness level calculates with the following formula.
Display Output Brightness = Manual brightness setting * CABC brightness ratio

Below drawing is for the explanation of the CABC minimum brightness setting.



CABC minimum brightness value = 51 (33h: 20% display brightness)

	Display Brightness [manual setting]	Brightness ratio [CABC]	Calculation result of the display brightness formula	Display Output Brightness	Image
Case 1	50%	70%	35%	35%	CABC modified
Case 2	20%	70%	14%	20%	CABC modified
Case 3	50%	70%	35%	35%	CABC modified

At the case 2, the calculation result of the display brightness is 14%. CABC minimum brightness value is set to 20% brightness. Actual display brightness is 20% as the CABC minimum brightness setting.

11 Power Definition

11.1.. Power Level

6 level modes are defined they are in order of Maximum Power consumption to Minimum Power Consumption

1. Normal Mode On (full display), Idle Mode Off, Sleep Out.

In this mode, the display is able to show maximum 262,144 colors.

2. Partial Mode On, Idle Mode Off, Sleep Out.

In this mode part of the display is used with maximum 262,144 colors.

3. Normal Mode On (full display), Idle Mode On, Sleep Out.

In this mode, the full display area is used but with 8 colors.

4. Partial Mode On, Idle Mode On, Sleep Out.

In this mode, part of the display is used but with 8 colors.

5. Sleep In Mode

In this mode, the DC: DC converter, internal oscillator and panel driver circuit are stopped. Only the MCU interface and memory works with VDDI power supply. Contents of the memory are safe.

6. Power Off Mode

In this mode, both VDD and VDDI are removed.

Note: Transition between modes 1-5 is controllable by MCU commands. Mode 6 is entered only when both Power supplies are removed.

11.2.. Power ON/OFF Sequence

VDD must be powered on before the VDDI.

VDDI must be powered off before the VDD.

During power off, if LCD is in the Sleep Out mode, VDD and VDDI must be powered down minimum 120msec after RESX has been released.

During power off, if LCD is in the Sleep In mode, VDDI or VDD can be powered down minimum 0msec after RESX has been released.

CSX can be applied at any timing or can be permanently grounded. RESX has priority over CSX.

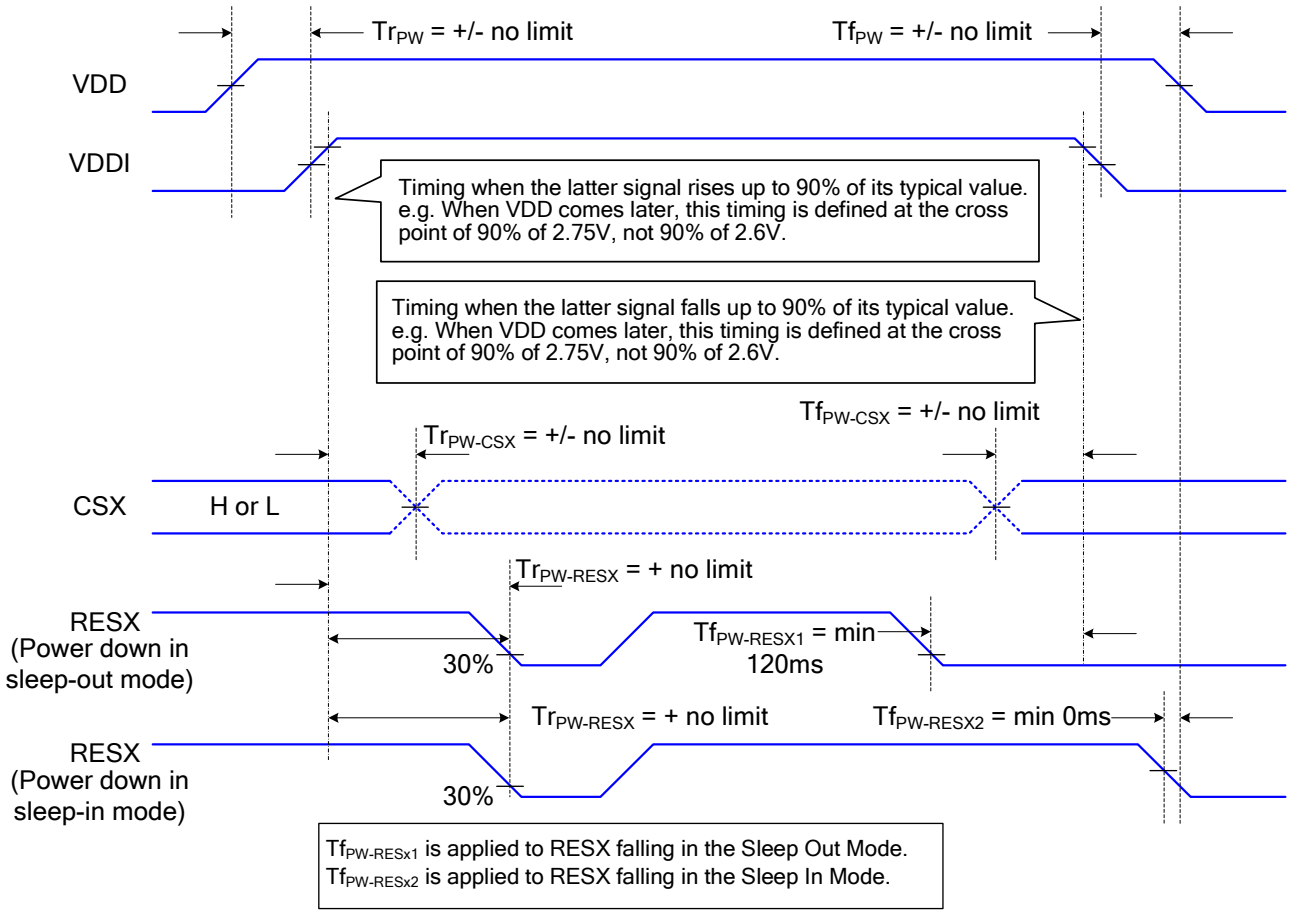
Note 1: There will be no damage to the display module if the power sequences are not met.

Note 2: There will be no abnormal visible effects on the display panel during the Power On/Off Sequences.

Note 3: There will be no abnormal visible effects on the display between end of Power On Sequence and before receiving Sleep Out command. Also between receiving Sleep In command and Power Off Sequence.

Note 4: If RESX line is not held stable by host during Power On Sequence as defined in the sequence below, then it will be necessary to apply a Hardware Reset (RESX) after Host Power On Sequence is complete to ensure correct operation. Otherwise function is not guaranteed.

The power on/off sequence is illustrated below

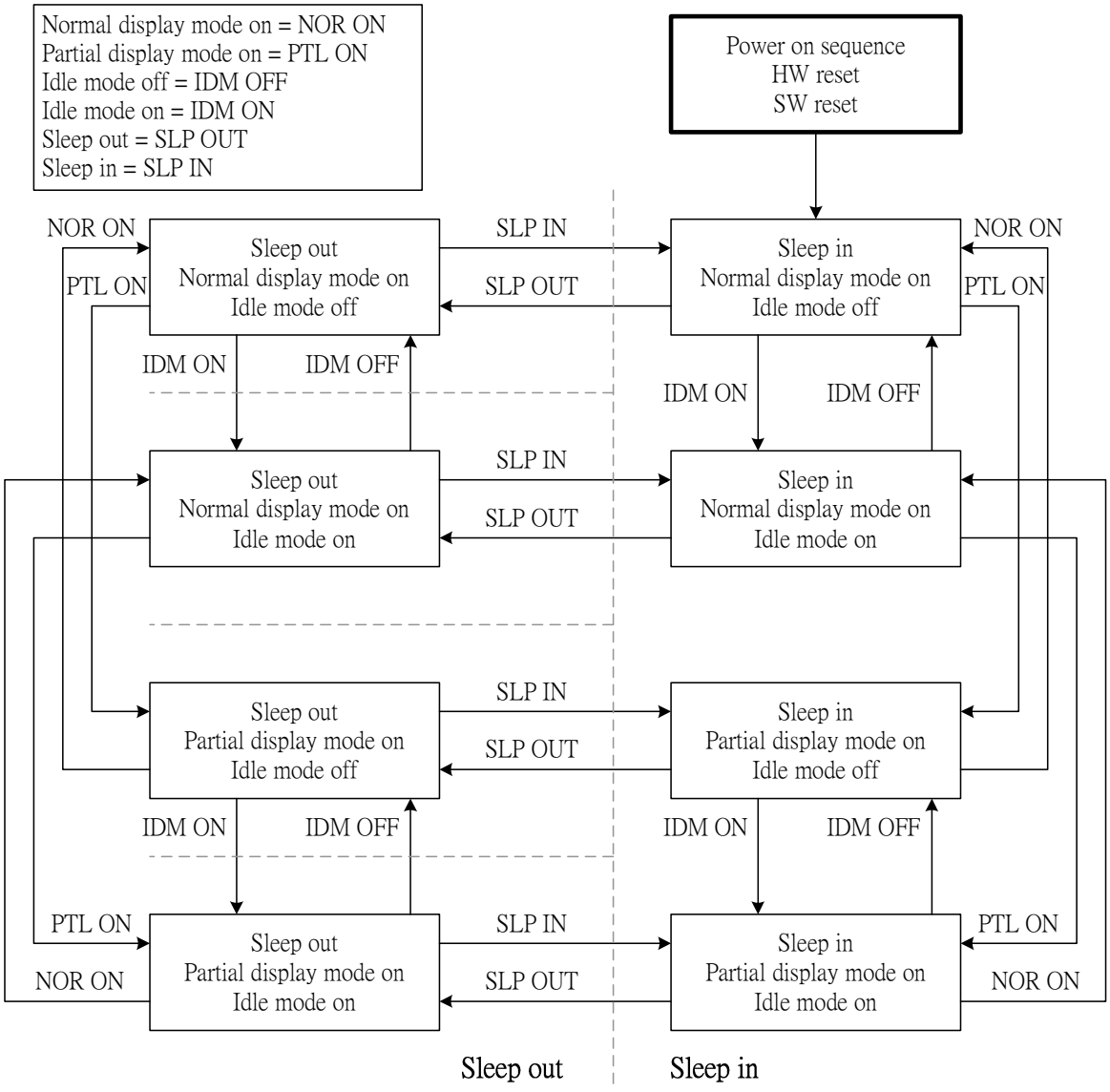


11.3.. Uncontrolled Power OFF

The uncontrolled power-off means a situation which removed a battery without the controlled power off sequence. It will neither damage the module or the host interface.

If uncontrolled power-off happened, the display will go blank and there will not any visible effect on the display (blank display) and remains blank until “Power On Sequence” powers it up.

11.4.. Power Flow Chart



11.5.. Applied Voltage to the TFT panel

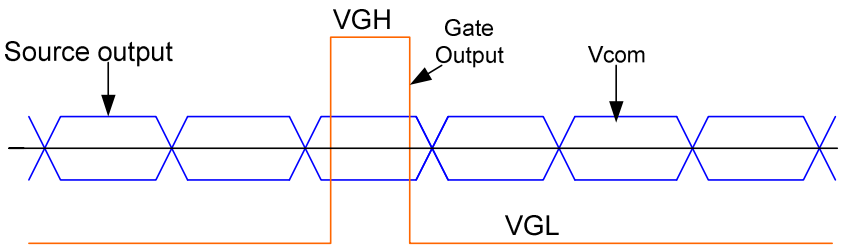
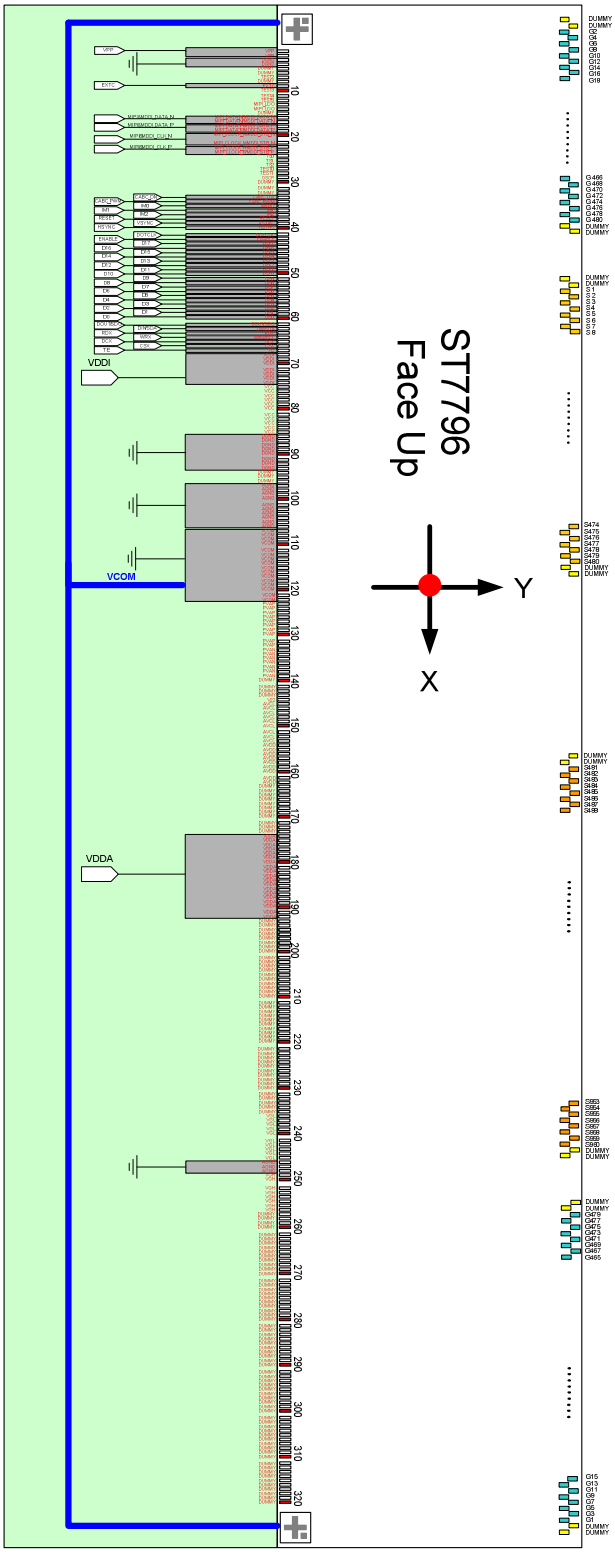


Figure 2 Voltage Output to TFT LCD Panel

12 Application Note

12.1.. FPC Suggestion



12.2.. Layout Resistance Suggestion

Pin Name	Type	Maximum Resistance	
VDDI, VDDA, AGND, DGND	Power supply	10	Ω
VPP	Power supply	10	Ω
VCOM	Common Electrode	10	Ω
MIPI_CLK_P/ MDDI_STB_P MIPI_CLK_N/ MDDI_STB_N MIPI_DATA_P/ MDDI_DATA_P MIPI_DATA_N/ MDDI_DATA_N	MDDI/MIPI	10	Ω
IM[2:0], RESET, CSX, DCX, RDX, WRX, VSYNC, HSYNC, ENABLE, DOTCLK	I	100	Ω
TE, CABC_PWM, CABC_ON, SDO	O	100	Ω
DB[17:0], SDA	I/O	100	Ω

13 REVISION HISTORY

Version	Date	Description
V1.0	2012/12	First Issue